Doherty Cars

Write-up

Candidate number: 2097
Centre number: 71605

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Background and planning

Introduction

Since the advent of the ubiquitous integration of technology - particularly database systems - into all facets of commerce, the processing and storing of business data has become a task more streamlined than ever: the validation of data; abstraction of the underlying storage medium; and seamless accessibility all offer a cohesive and highly effective solution to a challenge that has long confronted enterprises. However, this comes at a cost to those who remain tethered to antiquated methodologies, often leaving them unable to keep accurate and consistent records commensurate with their growth, impeding operational efficiency and constraining sustainable growth.

Unfortunately, this is the predicament that has beset Patrick Doherty, proprietor of the local, family-operated car dealership Doherty Cars. Over two decades of consistent customer-focused, high-quality service, they have established a firm local reputation that has contributed to their gradual yet notable influx in customer traffic. However, the dealership's continued dependence on manual processes - namely written memoranda, disparate spreadsheets and fragmented, interpersonal emails - has rendered it incapable of managing its schedules, customer interactions and inventory with the precision and consistency now required.

While such methods may have sufficed during the company's formative years, in the present day it creates critical - and more importantly, preventable - problems, including delays (often as a result of underestimating the time taken for activities such as appointments, inspections and deliveries to be carried out), lost paperwork which leads to uncertainty and ultimately customer dissatisfaction, and human errors such as miscommunication. These all stem from the current-day difficulty in ensuring that all employees have consistent, up-to-date information when it comes to appointments, records, schedules, queries, etc. Furthermore, the administrative workload provided by the dealership's expansion into territories such as new car sales, servicing, financing and trading-in has imposed unmanageable strain on the current system, urging for a universal, centralised replacement better-equipped to handle this strain. This urgency is underscored by Mr Doherty's desire to open a second branch, further pressing for a system that would be able to compartmentalise each branch's actions while ensuring that they are accessible by him.

In response, Mr Doherty envisions the incremental implementation of an integrated digital administrative system, with modular components addressing discrete domains before converging on the final centralised framework. Such a framework would mitigate any errors and inefficiencies by ensuring the synchronisation of data across all sectors of the enterprise, ensuring organisational coherence.

Target audience

Unlike any of my previous projects, the final system is intended to be an internal tool - one used solely by the company for its operation - and so the target audience for the system will only be staff members. However, due to the inherent hierarchy present in the enterprise (with staff split into sectors such as technical, financial and managerial, each of which have their own lead staff members), the actions carried out by each role and consequently their interaction with the system will differ. For example, mechanics will be able to access and modify data relating to appointments, vehicles and parts, whereas finance staff will be able to access reports to gain information on profits and performance.

Taking into account the habits of staff members, attempts will be made to ensure the system is intuitive to improve ease of transition. Minimal experience with technology will be assumed, and this will be reflected in the simple design and verbal explanations implemented. Due to the internal nature of the system, the ability to register an account will not be provided to prevent unauthorised access.

Primary issues addressed

Due to limitations of time and solo development, the solution developed will only be able to address issues with the system deemed most critical. Hence, the solution will predominantly address service appointments, staffing, reporting and the implicit issue of logging. However, this is not to say that other issues will not be partially addressed, as many - such as stock management - are interdependent on those chosen.

For service appointments, a subsystem will be implemented which displays each of the 3 mechanics' appointments on a particular day alongside the services, parts and extras required for each appointment, ensuring that mechanics can easily modify and see the latest changes to them. This should ensure consistency in schedules due to the centralised nature of the system and, consequently, mitigate any problems arising from appointments being made when mechanics or parts are unavailable, ultimately making the whole process a lot smoother. By extension, this will accommodate reporting, as a separate form will be provided which will allow staff to view the performance of mechanics, total profits from appointments etc. based on this data, as well as that of any adjacent tables implemented.

For staffing and logging, a system will be implemented which keeps track of the user currently logged in, externally writing their actions to a text file and preventing them from accessing forms that are not relevant to their role, thus providing full traceability and accountability which will prevent the mistakes which plague Mr Doherty.

System dev. methodology

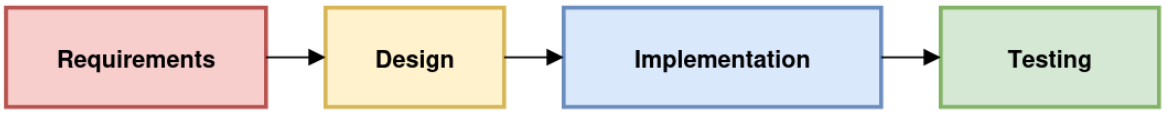
Without defined structure in the development of the system, it is unlikely that a cohesive, functional solution would be delivered within time and budget constraints. Thus, a suitable system development methodology is to be chosen prior to any design, implementation or testing.

Agile (in this case denoting more specifically the XP, DSDM and Scrum approaches) is one of the most widely used modern methodologies. An incremental and iterative approach, development is split into multiple releases, with each release going through the SDLC (software development life cycle) before being released to the customer as a functional subset of the system that they can then use and provide feedback which informs future iterations. Similarly, Rapid Application Application (RAD) emphasises rapid prototyping and frequent user feedback. RAD may involve throwaway prototyping, where prototypes are discarded after validation, or evolutionary prototyping, where prototypes are incrementally refined into the final system. Both methodologies promote adaptability and continuous improvement, which are valuable in projects with evolving requirements or highly engaged stakeholders.

Therein lies their unsuitability for the project, despite their advantages. For this project, the requirements consist of a fixed and clearly defined list of issues, reducing the need for evolving requirements. Furthermore, ongoing stakeholder involvement is limited due to my distance from the company, preventing regular feedback. The overhead associated with Agile practices, such as sprint planning, backlog management, and retrospectives, would also introduce unnecessary complexity for a project of relatively small scope executed by a solo developer. In a similar vein, RAD relies heavily on repeated user feedback and rapid prototyping, which are not required here due to the stability and clarity of the project requirements.

Instead, the Waterfall approach has been selected. Within the Waterfall approach, the development life cycle is split into stages such as design, implementation, and testing, and each stage is followed sequentially. Given that the requirements provided by *Doherty Cars* are unambiguous, clear and unchanging, a linear approach is appropriate, as the client is only required at the start to define these requirements and at the end to test the system - accommodating our difficulty in communication. Additionally, the structured and sequential nature of Waterfall provides a clear framework for managing the project effectively. The constraints of the project - its small, stable scope, limited stakeholder availability, and single-developer environment - make Waterfall an efficient, low-risk choice.

Although only a subset of the system will be developed in this cycle, the Waterfall process can be applied to future development if more functionality is deemed required.

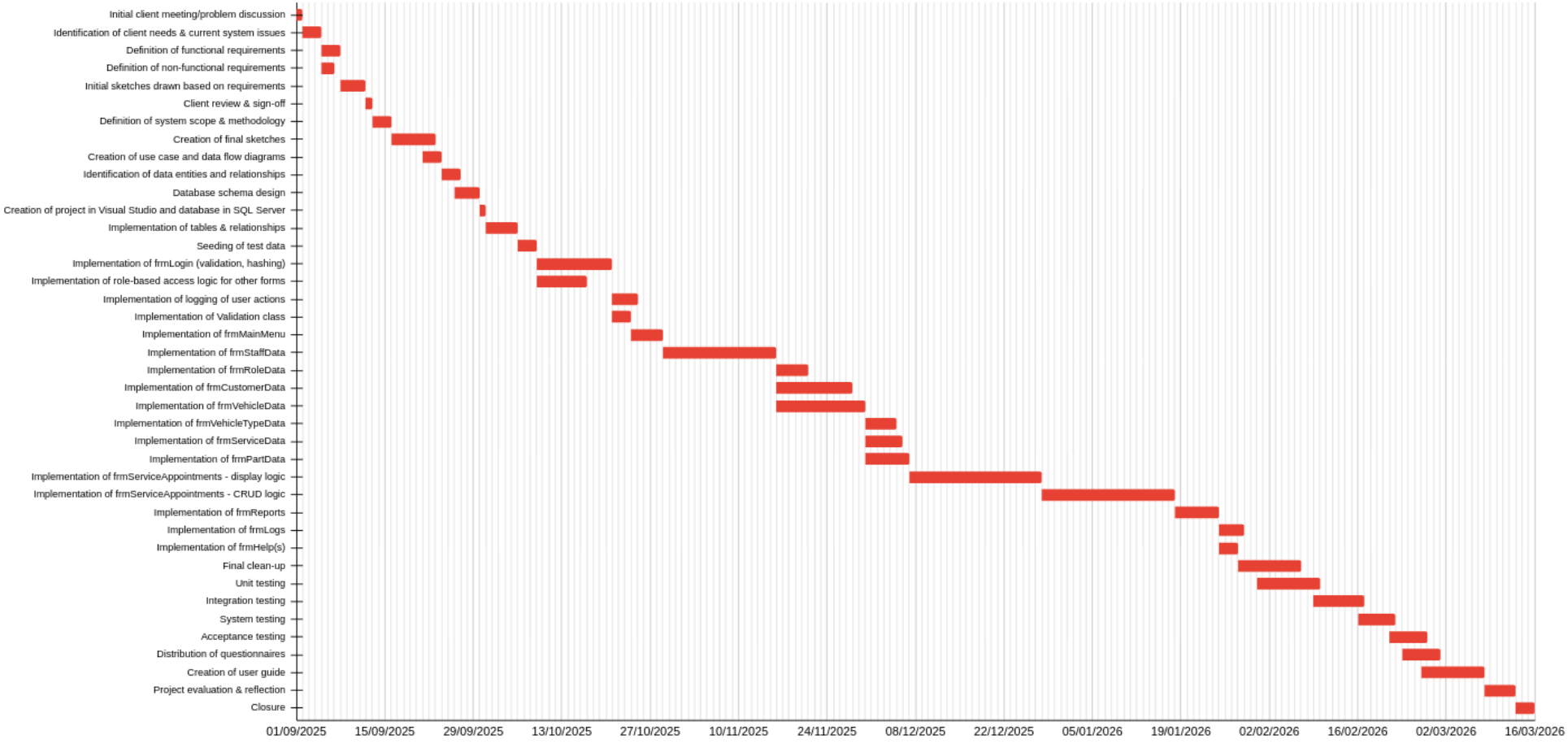


The stages followed during the system development lifecycle will be requirements, followed by design, implementation, and testing, with the deployment of the program left to Doherty Cars. Testing will include white box, black box and ultimately acceptance to ensure a suitable solution has been developed, but will be further discussed in the testing section

Project management

Gantt chart

To provide the project with a tentative timescale, a Gantt chart has been constructed as seen below. An attempt will be made to adhere to the allocated time to each task, but the delivery time may fluctuate as a result of underestimating the complexity of tasks, additional tasks being introduced or - hopefully - a task being completed sooner than expected. Ultimately, it will serve to guide the project lifecycle and monitor progress. Note that forms are planned to be developed in batches where noticeable overlap in functionality is expected to improve efficiency.

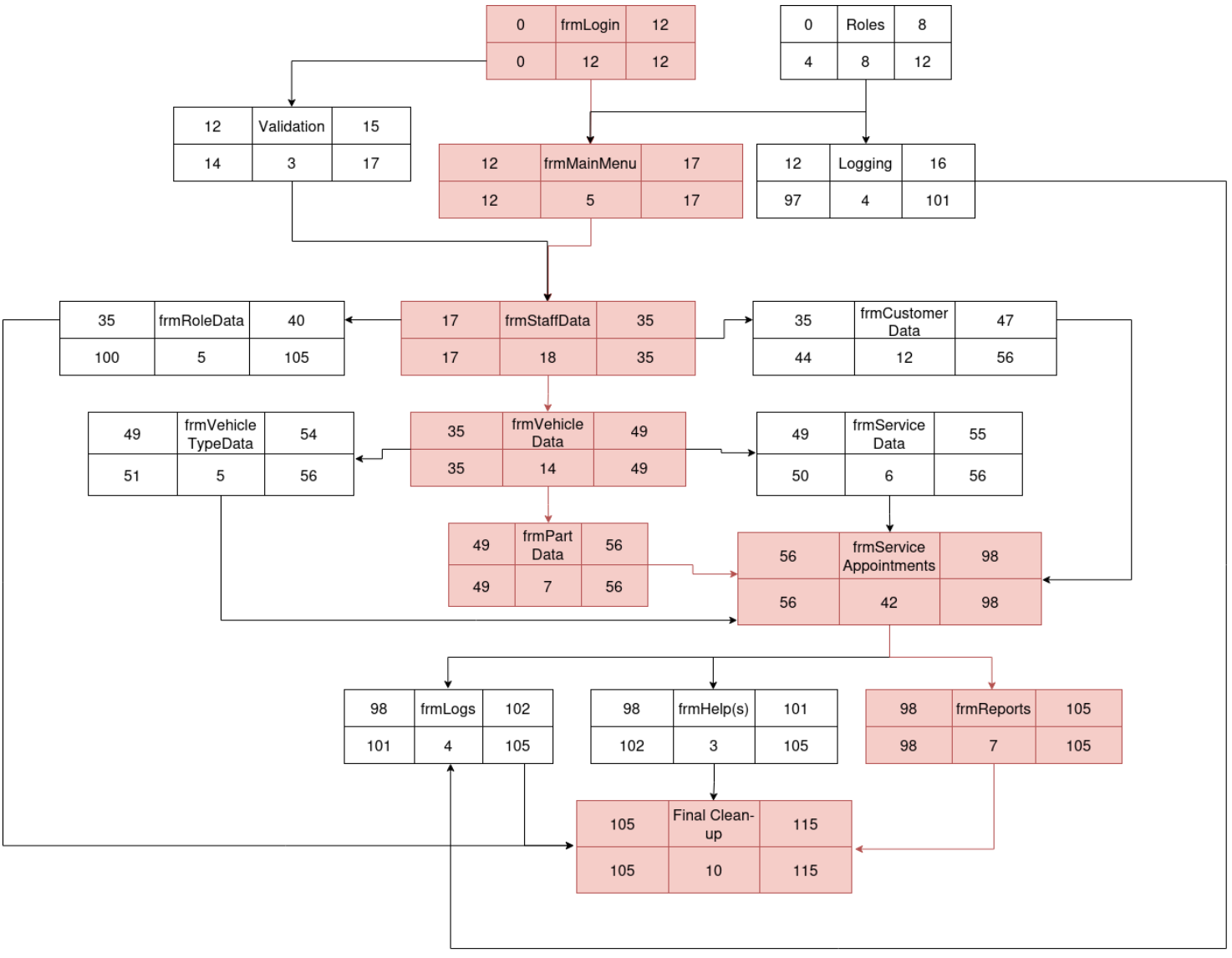


PERT chart

As the majority of the tasks in the project are sequential in nature, a PERT chart for the entire project is unnecessary. However, many of the tasks within the implementation phase of the project can be carried out concurrently, and a Gantt chart alone may not properly illustrate the interdependencies between the modules of the system or the impact delaying a component may have on others.

For this reason, a PERT chart has been constructed for the implementation phase, depicting the early start and early finish at the top of each node, representing the earliest points at which a task can begin and conclude, and the late start and late finish at the bottom, indicating the latest a task may commence and complete without delaying subsequent tasks.

The critical path - the longest sequence of tasks that must be completed on time for the entire project to be delivered on time - has been marked in red. As illustrated, the majority of effort will have to be spent addressing the handling of service appointments. Other modules exhibit measurable float, meaning limited delays in their completion would not immediately impact the overall time of completion.



Project plan

Based on the tasks defined in creating the initial Gantt chart, a rough project plan has been created as follows, defining the resources (both human and physical) required at each stage. Note that the developer - I - and the workstation on which the system is developed are involved in all stages but excluded for brevity. Also, the start date denotes that the task starts at the start of that day, while the end date denotes that it ends at the end of that date, e.g. task 1 starts the morning of the 1st of September and finishes that evening.

Task No.	Task	Start Date	Duration (days)	End Date	Resources
Analysis					
1	Initial client meeting/problem discussion	1-Sep	1	1-Sep	Client
2	Identification of client needs & current system issues	2-Sep	3	4-Sep	Client
3	Definition of functional requirements	5-Sep	3	7-Sep	—
4	Definition of non-functional requirements	5-Sep	2	6-Sep	—
5	Initial sketches based on requirements	8-Sep	4	11-Sep	—
6	Client review & sign-off	12-Sep	1	12-Sep	Client
Design					
7	Definition of system scope & methodology	13-Sep	3	15-Sep	—
8	Creation of final sketches	16-Sep	7	22-Sep	—
9	Use case and data flow diagrams	21-Sep	3	23-Sep	—
10	Identification of data entities & relationships	24-Sep	3	26-Sep	—
11	Database schema design	26-Sep	4	29-Sep	SQL Server

Implementation					
12	Project setup in Visual Studio and database setup	30-Sep	1	30-Sep	Visual Studio, SQL Server
13	Implementation of tables & relationships	1-Oct	5	5-Oct	Visual Studio, SQL Server
14	Seeding of test data	6-Oct	3	8-Oct	SQL Server
15	Implementation of frmLogin (validation, hashing)	9-Oct	12	20-Oct	Visual Studio, SQL Server
16	Role-based access logic	9-Oct	8	16-Oct	Visual Studio
17	Logging of user actions	21-Oct	4	24-Oct	Visual Studio, SQL Server
18	Validation class implementation	21-Oct	3	23-Oct	Visual Studio
19	Implementation of frmMainMenu	24-Oct	5	28-Oct	Visual Studio
20	Implementation of frmStaffData	29-Oct	18	15-Nov	Visual Studio, SQL Server
21	Implementation of frmRoleData	16-Nov	5	20-Nov	Visual Studio, SQL Server
22	Implementation of frmCustomerData	16-Nov	12	27-Nov	Visual Studio, SQL Server
23	Implementation of frmVehicleData	16-Nov	14	29-Nov	Visual Studio, SQL Server
24	Implementation of frmVehicleTypeData	30-Nov	5	4-Dec	Visual Studio, SQL Server
25	Implementation of frmServiceData	30-Nov	6	5-Dec	Visual Studio, SQL Server
26	Implementation of frmPartData	30-Nov	7	6-Dec	Visual Studio, SQL Server
27	ServiceAppointments (display logic)	7-Dec	21	27-Dec	Visual Studio, SQL Server
28	ServiceAppointments (CRUD logic)	28-Dec	21	17-Jan	Visual Studio, SQL Server
29	Implementation of frmReports	18-Jan	7	24-Jan	Visual Studio, SQL Server

30	Implementation of frmLogs	25-Jan	4	28-Jan	Visual Studio, SQL Server
31	Implementation of frmHelp(s)	25-Jan	3	27-Jan	Visual Studio
32	Final clean-up	28-Jan	10	6-Feb	Visual Studio
Testing					
33	Unit testing	31-Jan	10	9-Feb	Visual Studio, SQL Server
34	Integration testing	9-Feb	8	16-Feb	Visual Studio, SQL Server
35	System testing	16-Feb	6	21-Feb	Visual Studio, SQL Server
36	Acceptance testing	21-Feb	6	26-Feb	Client, Test Users
Evaluation					
37	Distribution of questionnaires	23-Feb	6	28-Feb	Test Users
38	Creation of user guide	26-Feb	10	7-Mar	—
39	Project evaluation & reflection	8-Mar	5	12-Mar	—
40	Closure	13-Mar	3	15-Mar	—

User requirements

The solution must…

---------- **Functional requirements**

1. …implement a login system that authenticates staff members using their username and password, preventing unauthorised system access
2. …keep a record of the logged-in user in memory, enabling dynamic modification of the system and its functionality using their information
3. …allow users to log out and log back in with different credentials
4. …assign access to certain forms based on the user's role, ensuring they only modify data that is relevant to them
5. …allow users to easily access relevant forms through the use of a centralised main menu form
6. …maintain records relevant to the business externally in a database, including tables for:
    a. Customers
    b. Staff
    c. Vehicles
    d. Vehicle types
    e. Parts
    f. Roles
    g. Services
    h. Service appointments
7. …allocate certain permissions - namely read, restricted write and unrestricted write - based on the user's role's authorisation level, determining the actions they can take on the system
8. …enable users to access pertinent records in the database by providing a form for each table and displaying it in a tabular, interactive format
9. …validate any data input to the system before making changes to the database to ensure data is consistent, accurate and correct
10. …enable users to create new records in the database by inputting the data into the respective fields provided within the table's form
11. …allow users to easily select records that they wish to modify or delete, either through the use of controls or mouse interaction
12. …enable users to update a selected record in the database by making changes to the data automatically populated into the form
13. …allow users to soft delete a selected record, ensuring that it is ignored by the system during data display and processing, but still exists on the database
14. …clear a form's fields after a CRUD action to accommodate new data entry
15. …prevent data entry upon a system error, ensuring data cannot become corrupted or lost due to a non-existent or broken database

16. …properly handle entity relationships, ensuring referential integrity through foreign keys. In particular, when soft-deleting, the following rules will apply:
    a. Deleting a customer will delete any of their vehicles and service appointments related to those vehicles
    b. Deleting a vehicle will delete any of its service appointments
    c. Deleting a staff member will not be permitted while they have existing service appointments
    d. Deleting a vehicle type will not be permitted while its being used by existing vehicles
    e. Deleting a part will not be permitted while its being used in an existing service appointment
    f. Deleting a role will not be permitted while it applies to existing staff
    g. Deleting a service will not be permitted while its being used in an existing service appointment
17. …ensure that new users added to the system by administrators can log in using the administrators' determined username and password
18. …allow mechanics to easily view, modify, create and delete appointments using a drag and drop system, displaying each day's available and taken timeslots
19. …enable mechanics to select the services and parts used for each appointment
20. …ensure that all appointments created are valid i.e. don't conflict with other appointments, fit within one work day, and don't occur on a weekend
21. …generate reports and invoices using data stored on the database to analyse the business' performance and assist with customer service
22. …automatically determine an appointment's length and cost based on the services carried out, parts used and the type of vehicle serviced
23. …capture any error messages resulting from invalid data or system errors and display them to the user so they can respond accordingly
24. …provide relevant feedback through confirmation and success messages to provide certainty
25. …log user actions externally with timestamps, activity type and user, ensuring that no critical actions can be taken on the system without accountability
26. …offer help forms to ensure that new users of the system are able to navigate and make use of it effectively, mitigating any risk of error and thus data loss
27. …hash passwords before storing them to ensure the security of the system

—---------- Non-functional requirements

28. …implement an intuitive and visually appealing interface to maximise staff efficiency
29. …ensure users can easily navigate between their required forms, and that navigation is fluid
30. …implement icons where applicable to graphically denote the purpose of each section and aid in navigation
31. …have designs consistent throughout the program. In particular, all menus will:
    a. follow a similar colour scheme
    b. implement the same fonts and and similar font sizes throughout
32. …respond to user input within acceptable time frames (i.e less than 2 seconds)

Hardware and Software Requirements

As with all systems, the environment on which the solution will be executed must be taken into consideration before beginning development. While most imagine a “program” to be a Windows-native GUI-based application, many solutions exist solely for Linux or BSD and live within the terminal. Therefore, it is worthwhile explicitly defining what we want the system to run on.

Software requirements:

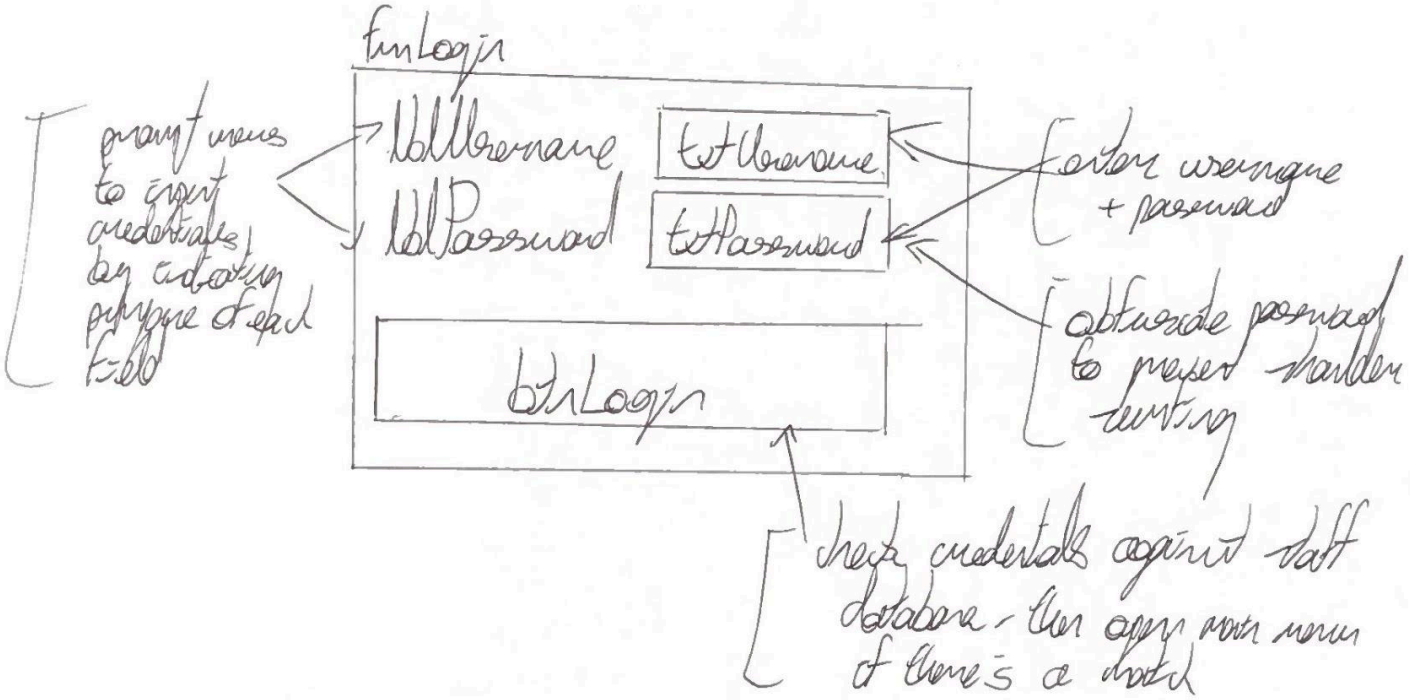
- **Operating system:**
  - Windows 10 or later (64-bit)
  - OR Windows Server
  - OR if the user is sufficiently technologically capable, execution on Linux through a compatibility layer such as WINE may be possible. However, as the program will be executed from Visual Studio, it is unlikely this is viable
- **Runtime/framework**
  - .NET 8 (Windows Desktop Runtime)
- **Dependencies**
  - Microsoft SQL Server
  - Visual Studio 2019
  - Relevant NuGet packages

Hardware requirements:

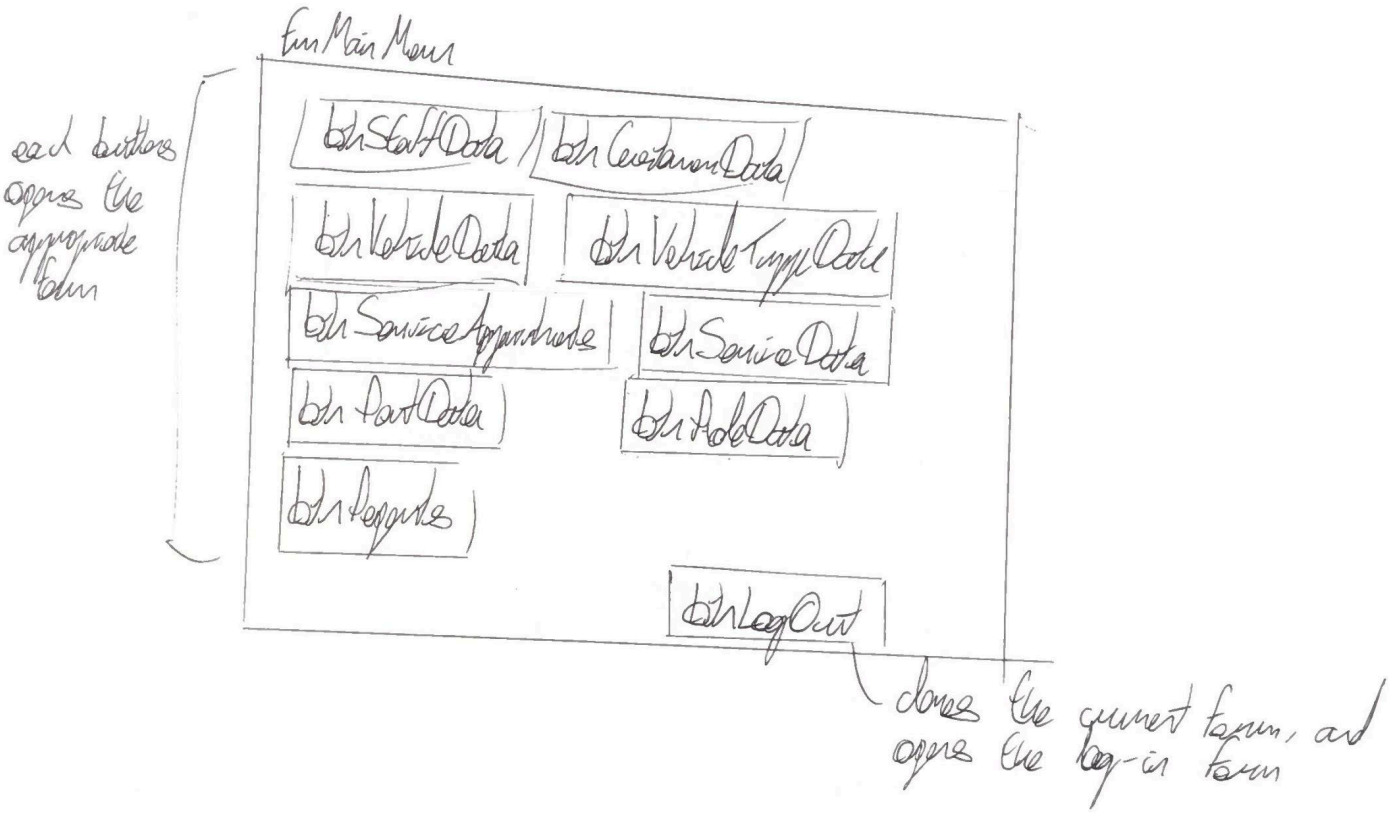
- **Minimum:**
  - Dual-core 1.6 GHz CPU
  - 4GB RAM
  - 500MB-2GB free disk space
  - 1366x768 resolution display
  - Stable LAN connection
- **Recommended:**
  - Intel i5/Ryzen 5 or equivalent
  - 8GB+ RAM
  - 2-5GB free SSD disk space
  - 1920x1080 resolution display

Draft storyboards

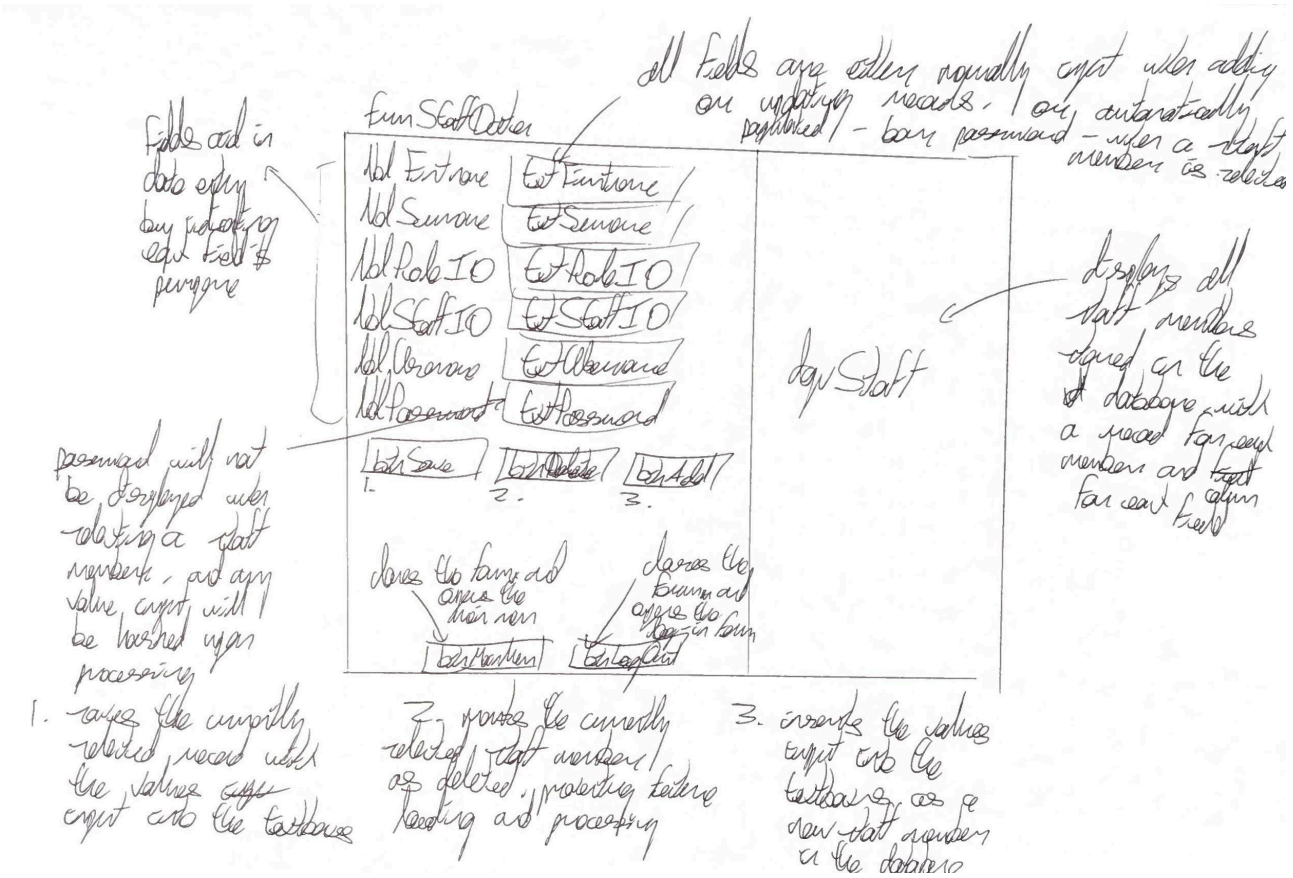
frmLogin



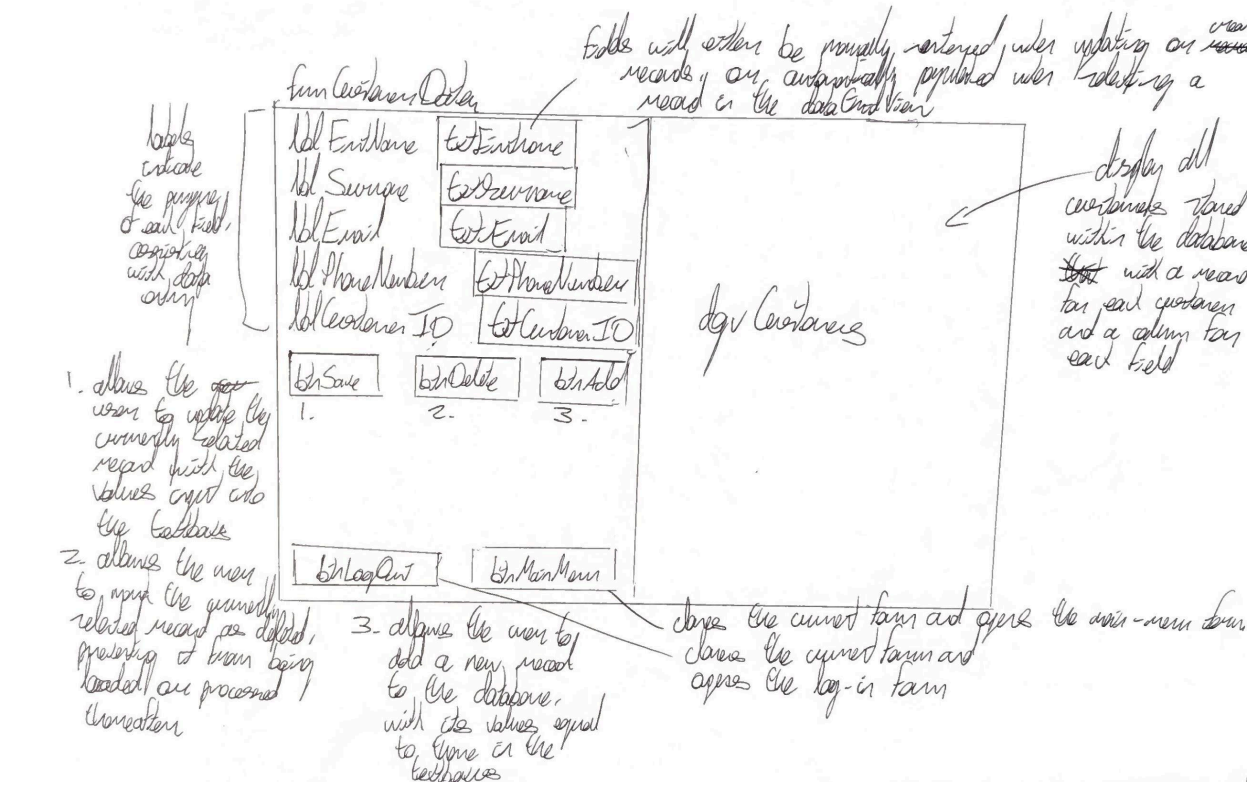
frmMainMenu



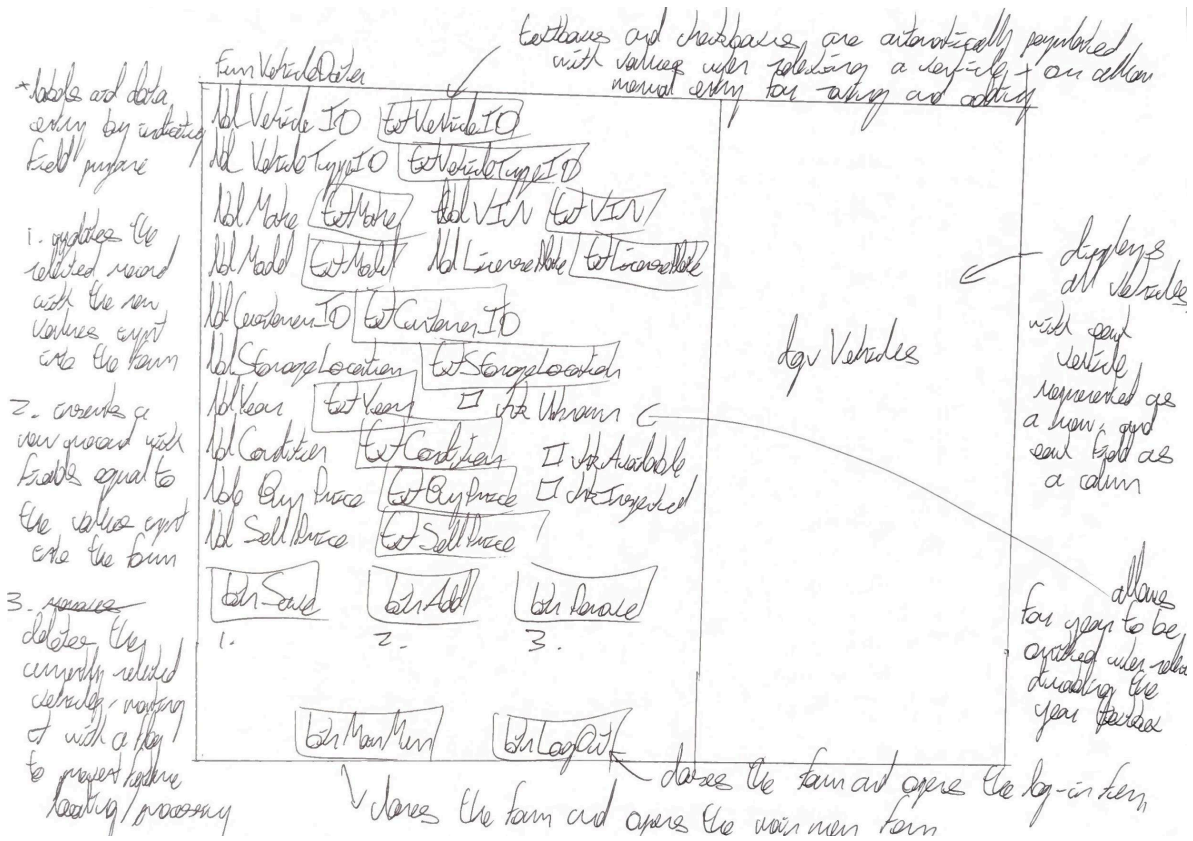
frmStaffData



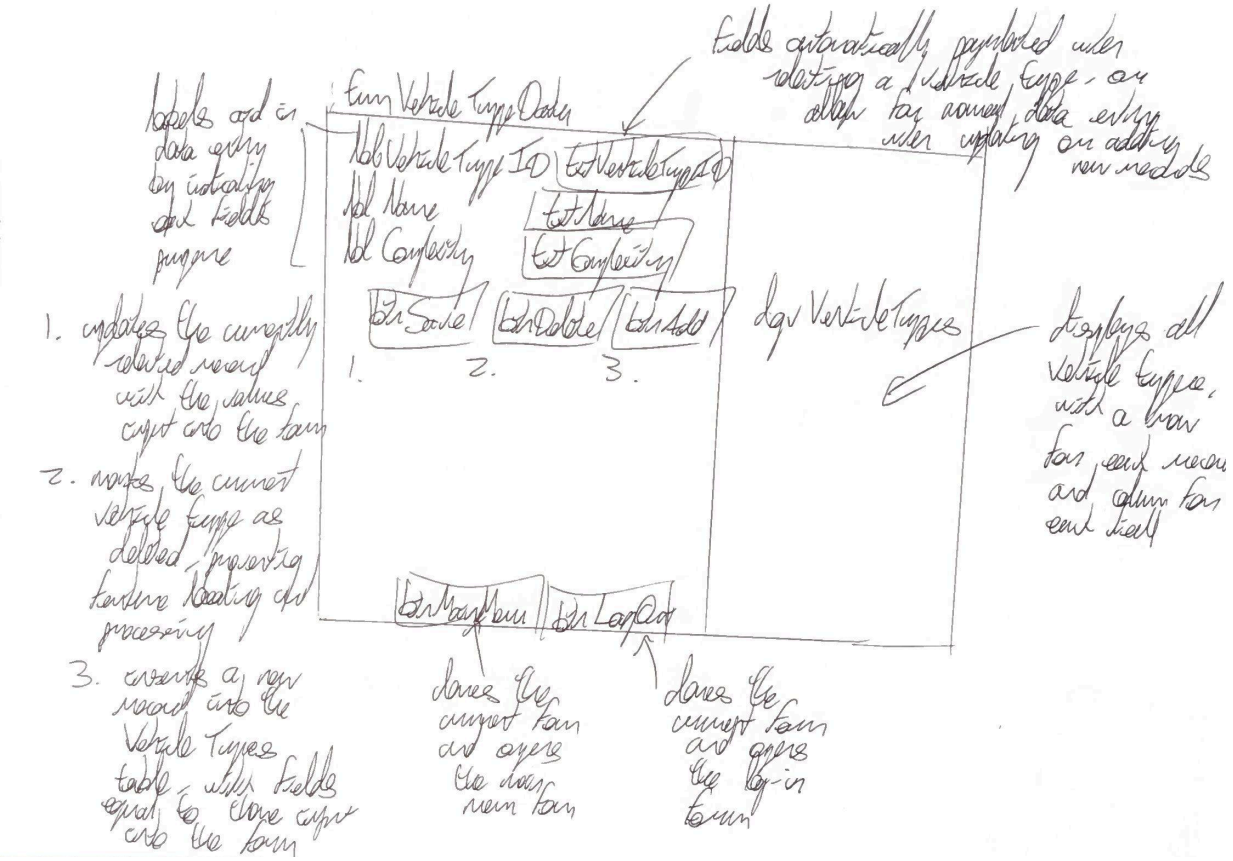
frmCustomerData



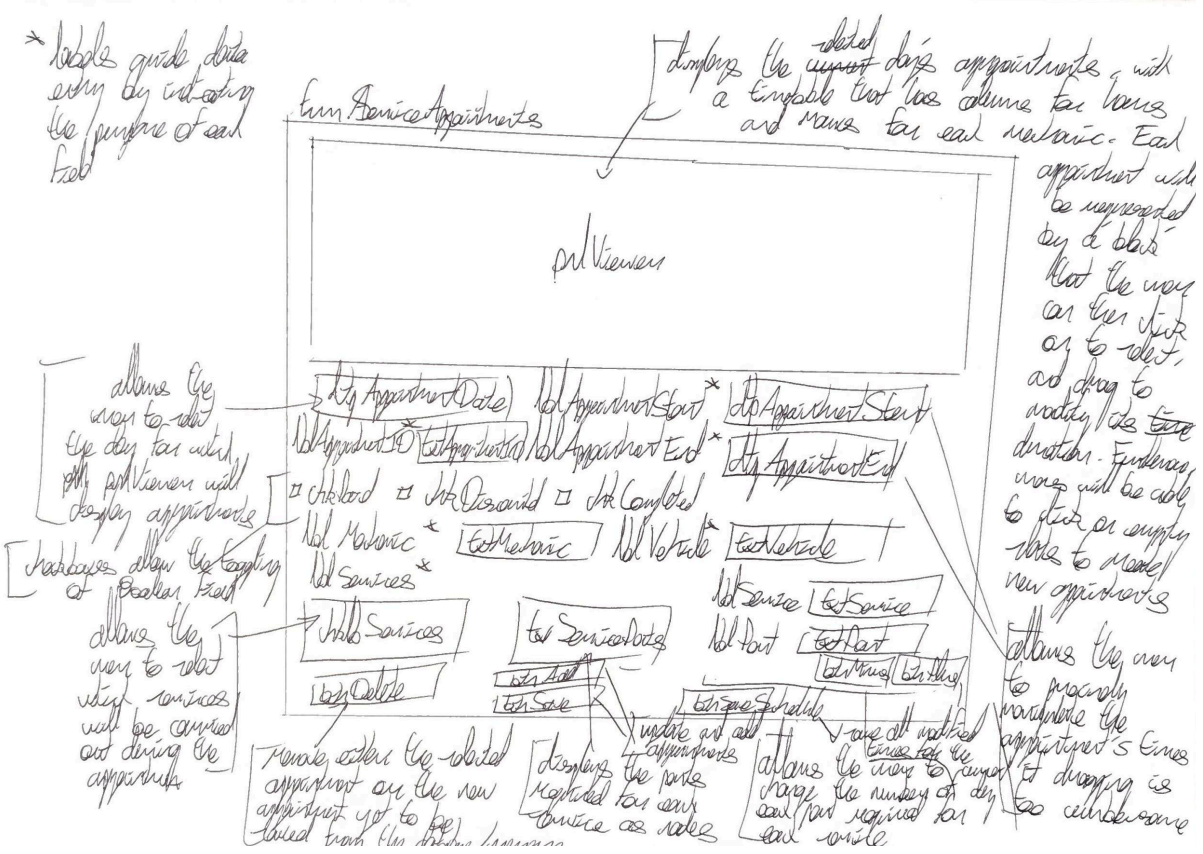
frmVehicleData



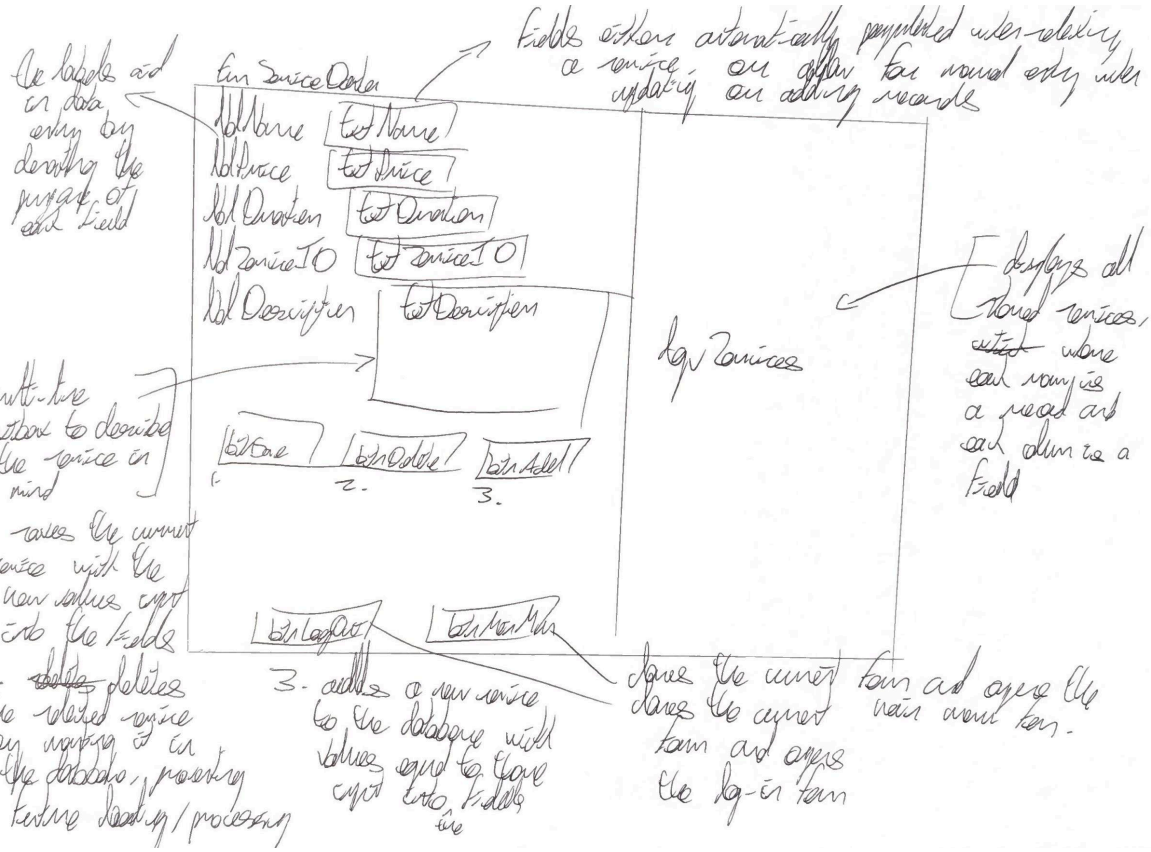
frmVehicleTypeData



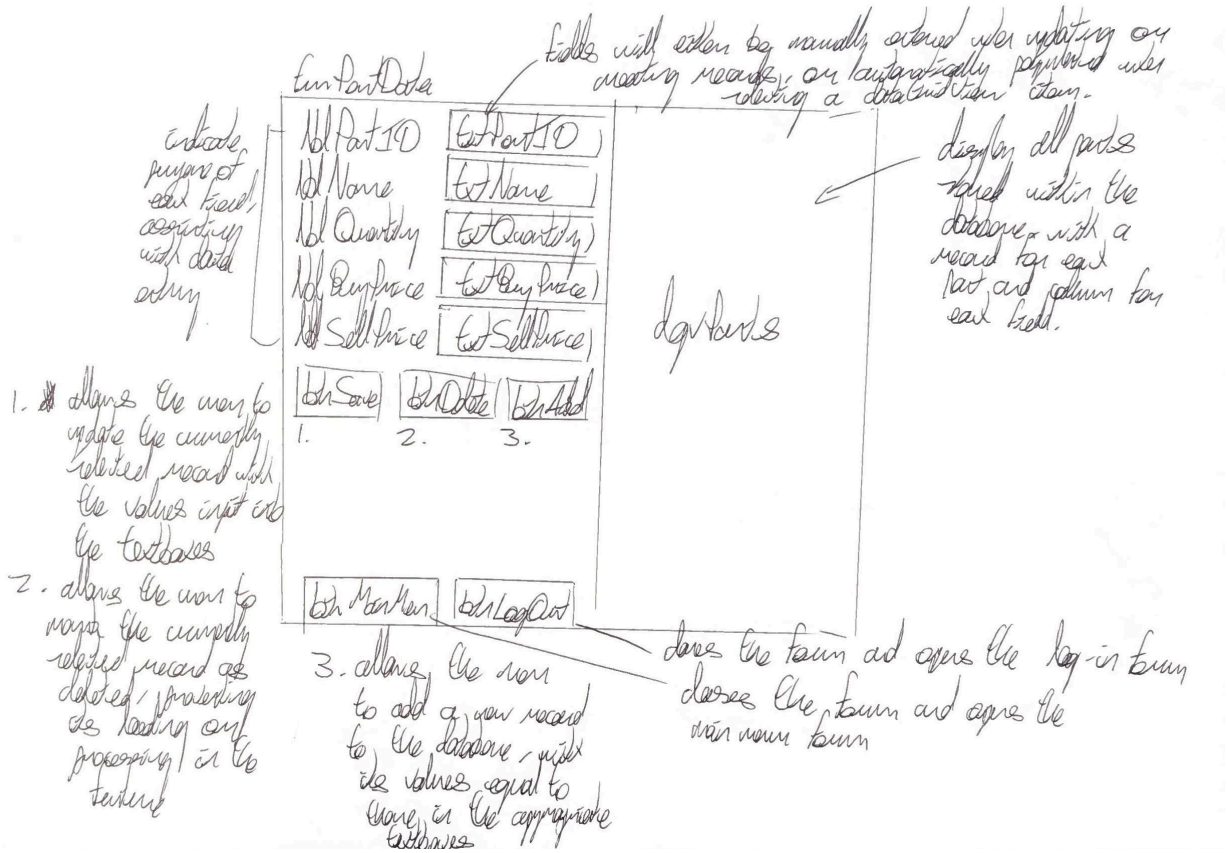
frmServiceAppointments



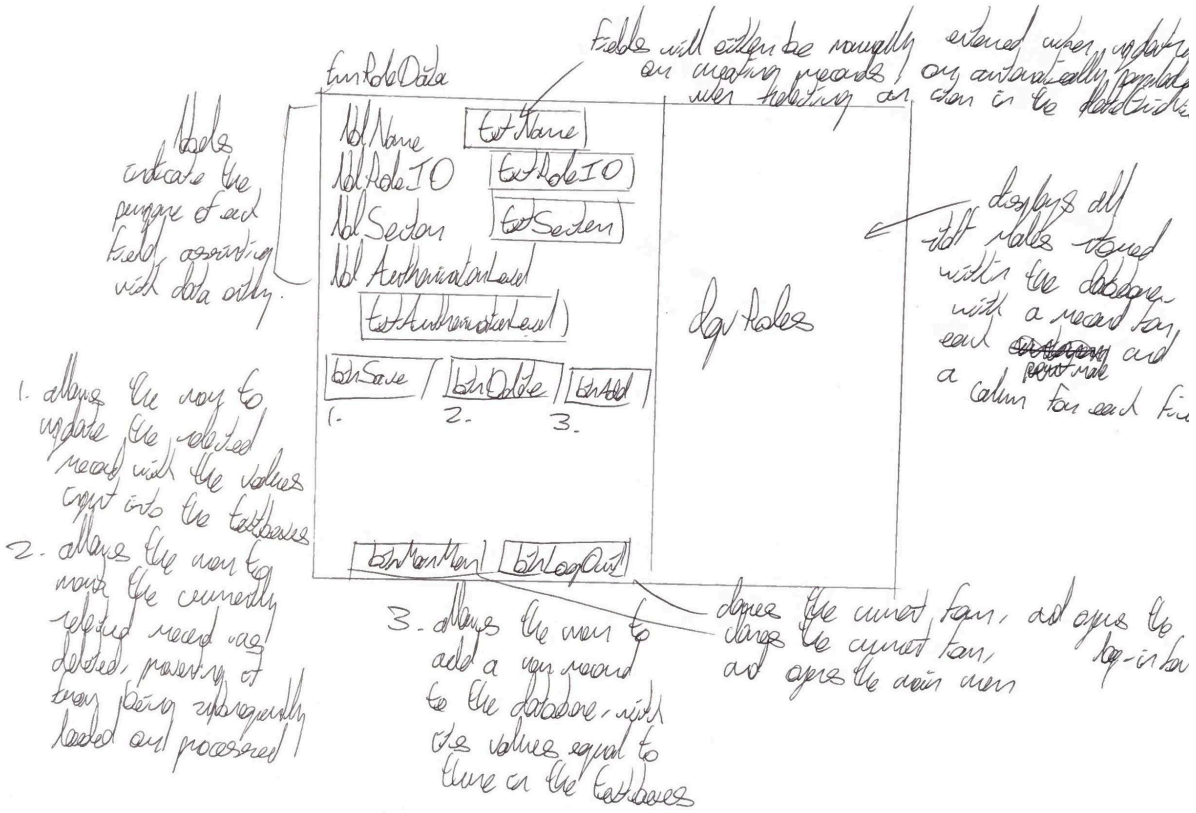
frmServiceData



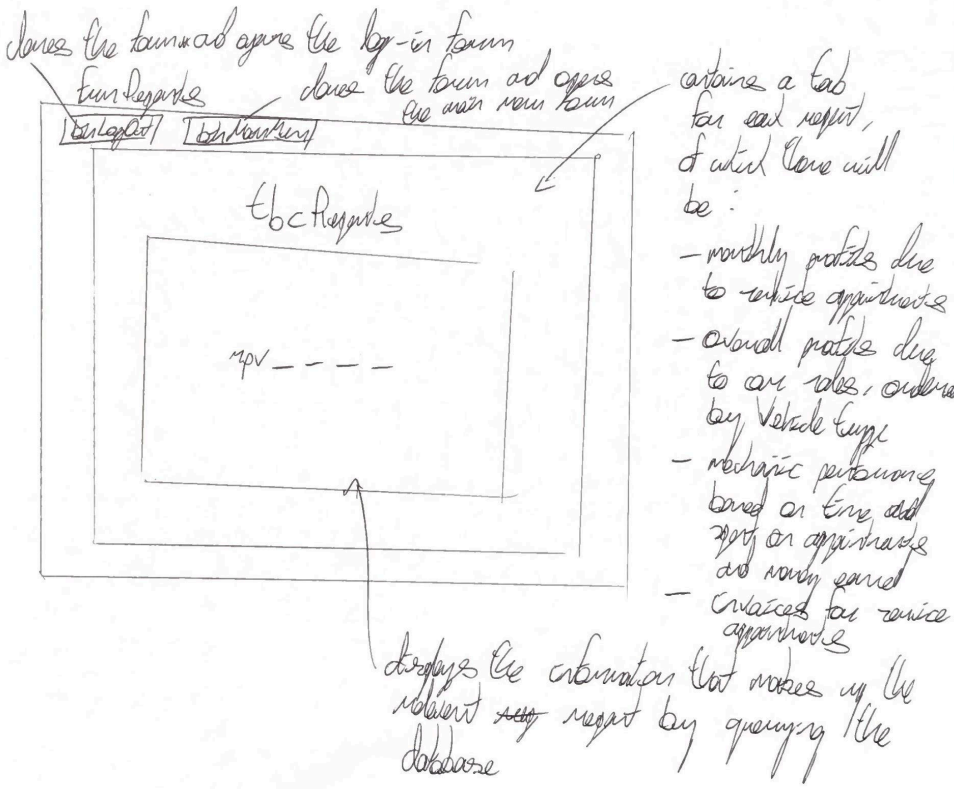
frmPartData



frmRoleData



frmReports



User feedback

After presenting each of these sketches to the client, they requested me to implement the following changes:

- frmLogin
    - Add a “show password” feature to ensure that users are able to verify they’ve input the correct credentials, improving the accessibility of the program and assisting those who struggle to type accurately
    - Add the company logo at the top of the form to add aesthetic value
    - Visually separate the log-in functionality from the rest of the form by adding a panel behind the textboxes and buttons
- frmMainMenu
    - Place all buttons in a sidebar on the left
    - Implement hover logic on the sidebar so that buttons display only their icon when idle, but expand to show text as well when hovered over
    - Add a help button to explain how to start using the program
    - Instead of having each button open its respective form separately, display each form within a panel in the main menu for continuity
    - Add a home form which displays within this panel upon logging in which displays the user’s name, the date, and the user’s role, as well as the company’s logo

    - Add home and log out buttons
- All frm_____Data
    - Change the layout of the forms so that the dataGridViews are horizontal instead of vertical
    - Add search functionality in each form, allowing the user to select the field to search, type of search and search text
    - Add a reset button which clears each field in a form without having to carry out a CRUD action first
    - Ensure that the add, update and delete buttons are appropriately and pleasantly coloured
    - Visually separate groups of related fields by adding lines as dividers within the form
    - Visually separate the search and CRUD functionality by adding panels to house it
    - Ensure that primary key fields are read-only to maintain data integrity
    - Ensure that all controls in related groups are visually aligned with each other i.e. have the same or very similar lengths and the same end-point vertically
    - Add a help form which explains any restrictions to the data that can be input or the purpose of fields
    - Remove the main menu and log out buttons
- frmStaffData
    - Store passwords as hashes instead of plain text to improve security
    - Do not display the password hash when selecting a user, instead leaving it blank, allowing the user to input a new plain text password to be hashed upon updating or adding a record
    - Change the role field to a ComboBox control which is automatically populated with valid records
- frmVehicleData
    - Change the vehicle type, customer, storage location and condition fields to be ComboBox controls automatically populated with valid records/values
    - Change the buy and sell price fields to be NumericUpDown controls to prevent the entry of invalid data
- frmVehicleTypeData
    - Change the complexity field to be a NumericUpDown control to prevent the entry of invalid data
- frmServiceAppointments
    - Add a toggleable “insert mode” that determines whether the user can edit existing appointments or create new ones, ensuring that users do not accidentally add unnecessary appointments during editing or edit valid ones during creation
    - Ensure that regular mechanics are unable to edit the schedules of their colleagues, reserving this right to the lead mechanic
    - Dynamically update and display the time required and cost of the appointment as changes are made to its vehicle, parts and services

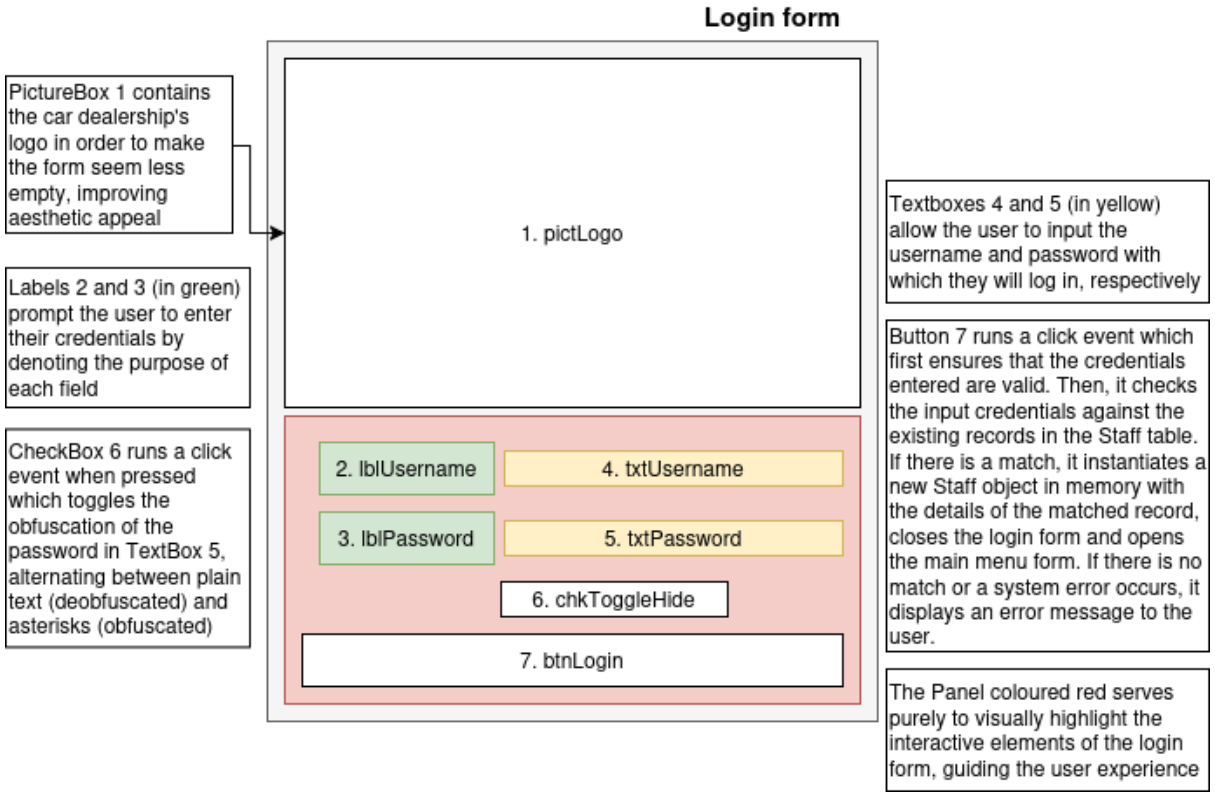
    - Add a help button which explains exactly how the user can add, delete, insert and update appointments
    - Change the mechanic, vehicle, service and service part fields to ComboBox controls which are automatically populated with valid records
    - Add search functionality which allows the user to filter by vehicle and mechanic
    - Visually separate groups of related fields by adding lines as dividers within the form
    - Visually separate the search and CRUD functionality by adding panels to house it
    - Ensure that primary key fields are read-only to maintain data integrity
    - Ensure that all controls in related groups are visually aligned with each other i.e. have the same or very similar lengths and the same end-point vertically
    - Remove the main menu and logout buttons
- frmServiceData
    - Add a character count to description to ensure that users know exactly how much more they can use
    - Change the price and duration fields to NumericUpDown controls to prevent the entry of invalid data
- frmPartData
    - Change the buy price, sell price and quantity field to NumericUpDown controls to prevent the entry of invalid data
- frmRoleData
    - Change the sector and authorisation level fields to ComboBox controls with predetermined values
- frmReports
    - Add the following reports:
        - Vehicle stock
        - Customer interest by vehicle type
        - Missed appointments
    - Add a help button to explain what each report tells the user and how to use the form
- frmLogs
    - Add a logs form which parses and displays the logs that have been stored externally
    - Add functionality to filter these logs by date, type of action and staff member
    - Add a help button to explain how the log system works, including the log types

Design

Final sketches

frmLogin

Drawing



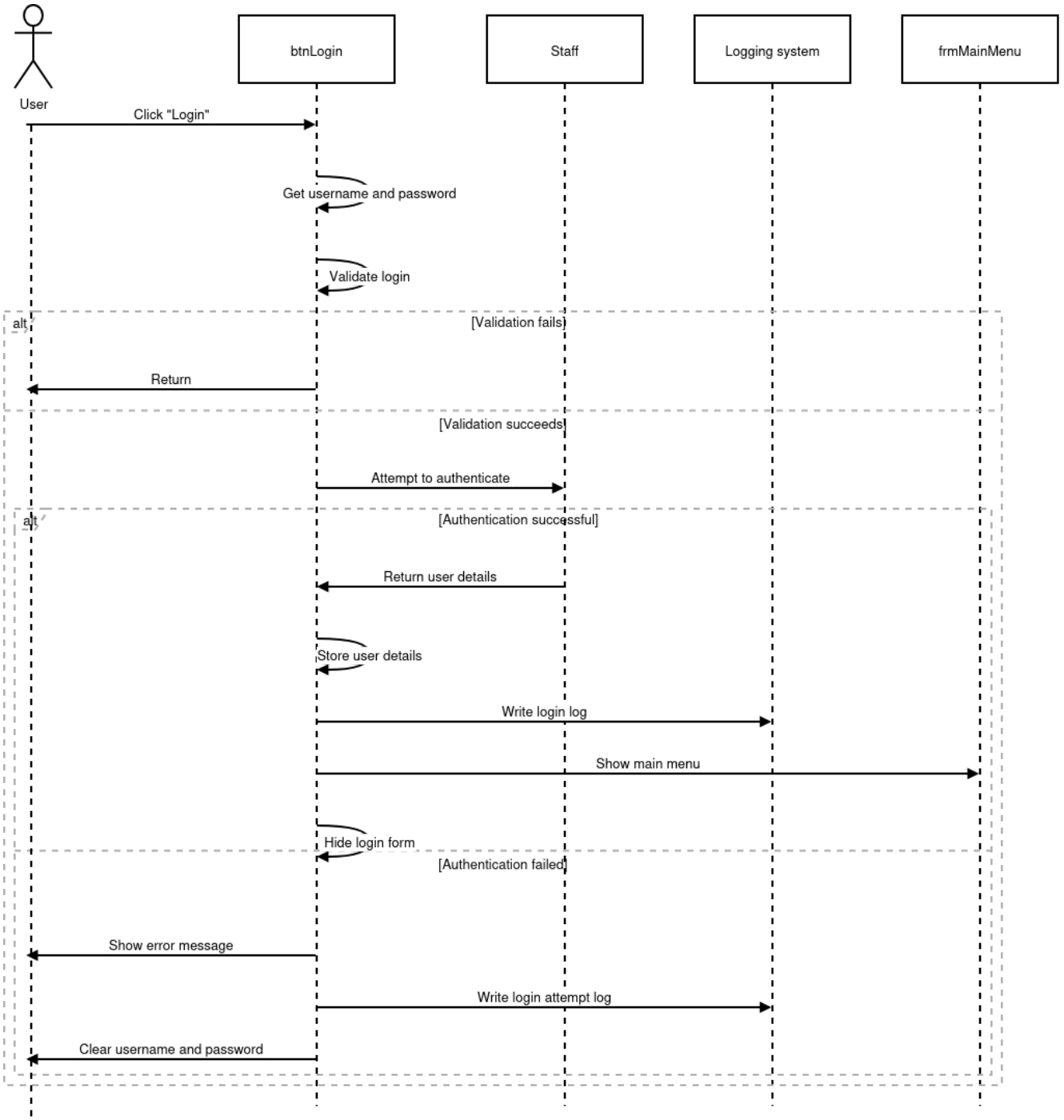
Explanation

As the first form seen upon opening the application, this form allows staff members to validate their identity by inputting their username and password. Through this process, it is ensured that the details relating to that particular staff member are loaded into memory. By doing so, the actions the staff member takes on the system can now be attributed to them - introducing a form of traceability - and the system can dynamically update its state, particularly in the main menu when populating the starting main menu form and determining which forms the user is able to access. Consequently, the user details in memory are always set to null when loading this form to ensure any details from a staff member who has previously logged out are not carried over. Notably, no register button is present to prevent unauthorised access to the system, and users must be added manually by an administrator. If any issues arise as a result of a login attempt, an appropriate error MessageBox is displayed, and if the user wishes to see their password before logging in, they may use the respective checkbox.

Pseudocode and diagrams

Form event listener	Pseudocode

chkToggleHide_CheckedChanged	`IF checked IS true`
	`  SET txtPassword.HideChar TO null`
	`ELSE`
	`  SET txtPassword.HideChar TO "*"`



Validation and error-trapping

- If statements will be used to ensure the following:
    - Usernames input are between 3-50 characters
    - Passwords input are between 8-128 characters
    - Usernames only alphanumeric characters or underscores
    - Password or username fields are not empty
- Try-catch statements will surround the loading of staff objects from the database when authenticating a user, ensuring that a system error does not cause a crash

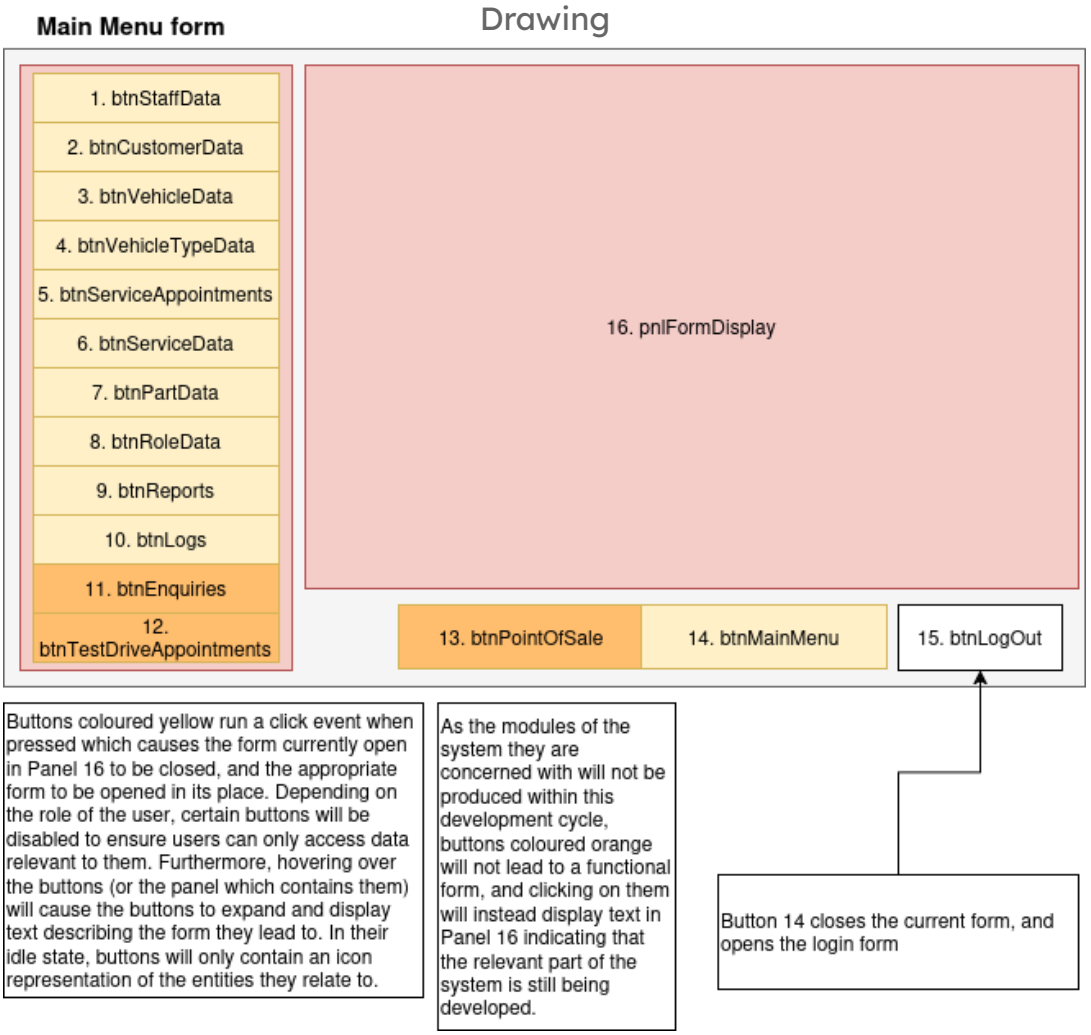
Input specifications

Field/Control	Data Type	Validation Rules	Purpose
txtUsername	Text	Required, 3-50 characters, alphanumeric, _ or . allowed, trimmed, converted to lowercase	User enters login username
txtPassword	Text	Required, 8-128 characters, trimmed, case-sensitive	User enters login password

User requirements met: 1, 2, 3, 9, 23, 24

frmMainMenu

Drawing



Explanation

As the form seen upon logging in, frmMainMenu serves as the central navigation point to the rest of the forms on the system. Using the details stored in memory after a user logs in, their role's sector is determined and the appropriate buttons are enabled, with all others being disabled by default. Important to note is the method by which forms available in the sidebar are displayed: rather than opening a separate form and closing the main menu, these forms are instantiated and docked within the appropriate display panel, preventing any confusion or delay caused by navigating between various disjointed forms. Furthermore, the sidebar on the left is collapsed by default, with only icons visible to the user, but will expand smoothly to reveal text on the buttons, aiding in navigation. Due to the centralised nature of the program, the log-out button will only be present on this form as it will always be visible, thus preventing the cluttering of the forms displayed in the panel which already have limited space available. Lastly, some buttons are planned to eventually lead to functionality that has not been addressed in this development cycle (this being enquiries, test drive appointments and point of sale) but at the moment will simply display text indicating they are under development.

Form event listener	Pseudocode

frmMainMenu_Load	`IF NO USER IS LOGGED IN`
	`  OUTPUT "Session expired. Please log in again."`
	`  REDIRECT TO frmLogin`
	`END IF`
	
	`OPEN frmMainMenuStartingForm`
	
	`RETRIEVE USER'S Role`
	`IF NO Role FOUND`
	`  OUTPUT "Role configuration error"`
	`  STOP LOADING`
	`END IF`
	
	`FOREACH Button IN frmMainMenu`
	`  IF USER'S Role IS PERMITTED`
	`  ENABLE Button`
	`  END IF`
	`END FOREACH`

tmrTransition_Tick	`WHILE EXPANDING`
	`  IF THE MOUSE ENTERED THE PANEL`
	`  INCREMENT THE BUTTON AND PANEL WIDTHS`
	`  IF THE BUTTONS AND PANEL MEET A PARTICULAR LENGTH`
	`  DISPLAY BUTTON TEXT`
	`  END IF`
	`  ELSE IF THE MOUSE LEFT THE PANEL`
	`  DECREMENT THE BUTTON AND PANEL WIDTHS`
	`  HIDE BUTTON TEXT`
	`  END IF`
	`END WHILE`

btn———_Click	`LOG ACCESS ACTION`
	`REMOVE EXISTING FORM IN PANEL`
	`INSTANTIATE NEW FORM`
	`ADD IT TO THE PANEL`
	`DOCK TO THE PANEL`
	`SHOW THE NEW FORM`

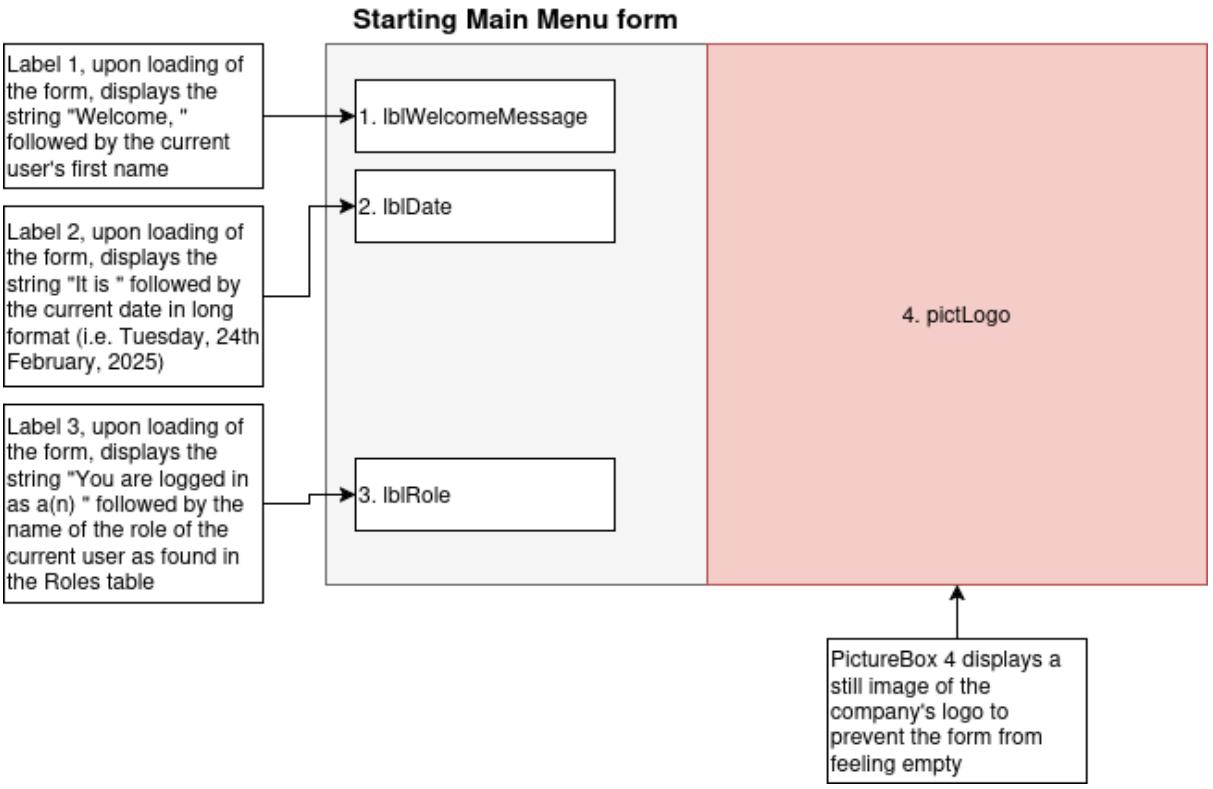
Validation and error-trapping

- If statements will be implemented to ensure that the current user is valid i.e. their object in memory is not null. If it is, they will be redirected back to the login form
- If statements will be implemented to ensure that the user's role has been successfully loaded and that their sector exists within the list of sector-form associations
- If statements will be implemented to ensure that the referenced form exists before attempting to load it within the viewing panel

User requirements met: 2, 3, 4, 5, 9, 15, 23, 24

frmMainMenuStartingForm

Drawing



Explanation

The first form docked within the main menu panel upon opening the program, this form serves no functional purpose, aiming instead to prevent the main menu from looking incomplete or barren. However, there is still a dynamic element present, as each label's text is determined at runtime. The first label seen - the welcome message - will simply access the Staff object stored in memory, retrieve their first name and display it, adding visual confirmation that the user has logged in with the correct credentials. Furthermore, the second label will contain the current date, ultimately serving as a reminder to ensure that the user accesses and edits records pertaining to the correct date. Lastly, the role label states the user's role's name as stored on the system, aiming to explain why access to particular forms is disabled.

Pseudocode and diagrams

Form event listener	Pseudocode

frmStartingMainMenuForm_Load	IF A USER IS LOGGED IN
	  DISPLAY "Welcome, " FOLLOWED BY THEIR FIRST NAME
	ELSE
	  OUTPUT "Your name could not be retrieved. Please ensure you have logged in correctly."
	END IF
	
	DISPLAY “It is “ FOLLOWED BY THE CURRENT DATE
	
	IF THE USER HAS A VALID ROLE
	  DISPLAY “You are logged in as a(n) “ + FOLLOWED BY THEIR FIRST NAME
	ELSE
	  OUTPUT “Your role could not be retrieved”
	END IF

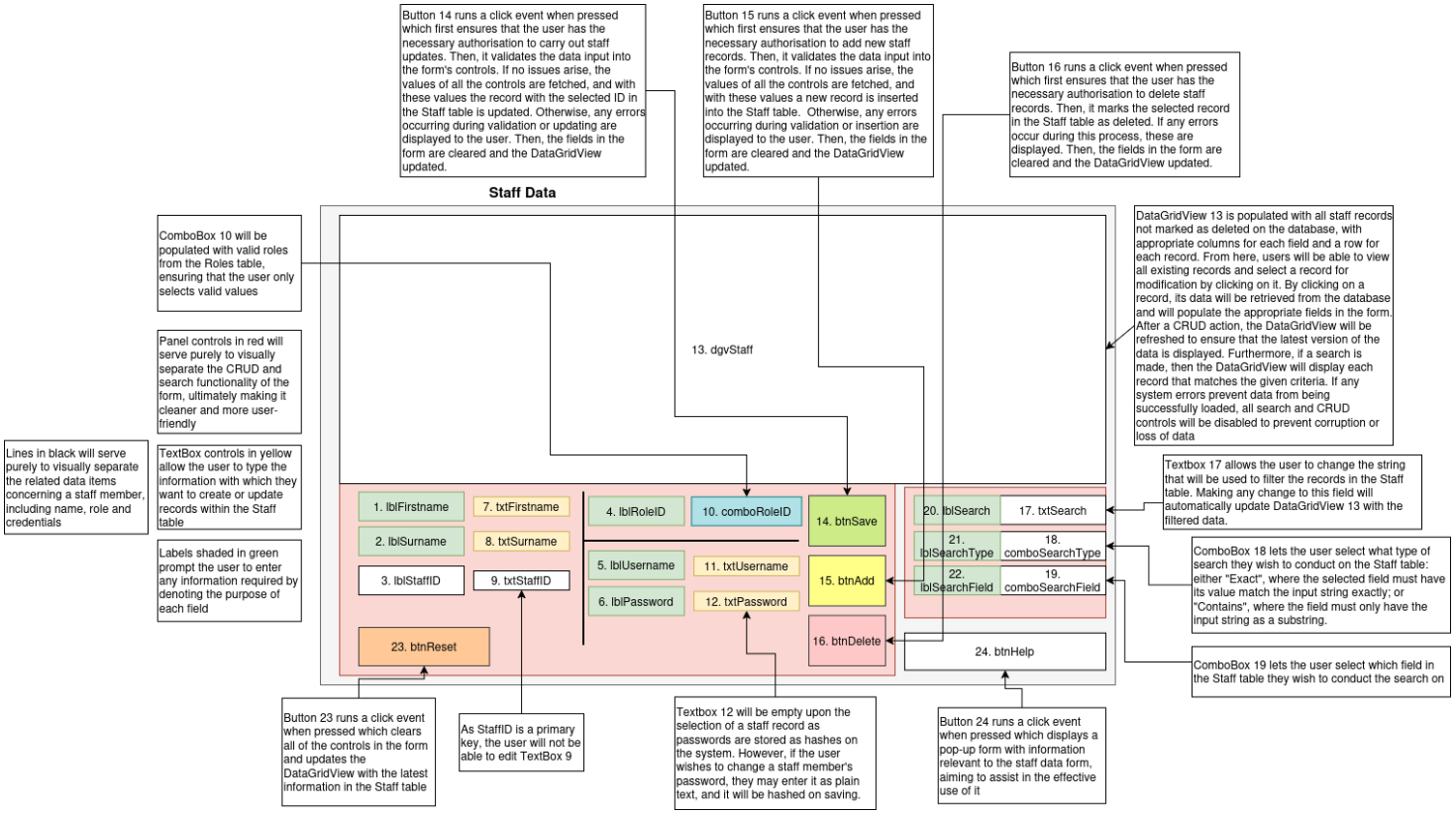
Validation and error-trapping

- If statements will ensure that the current user has a valid staff object in memory before displaying their name
- If statements will ensure that the current user has a valid role associated with them before displaying its name

User requirements met: 2, 15

frmStaffData

Drawing



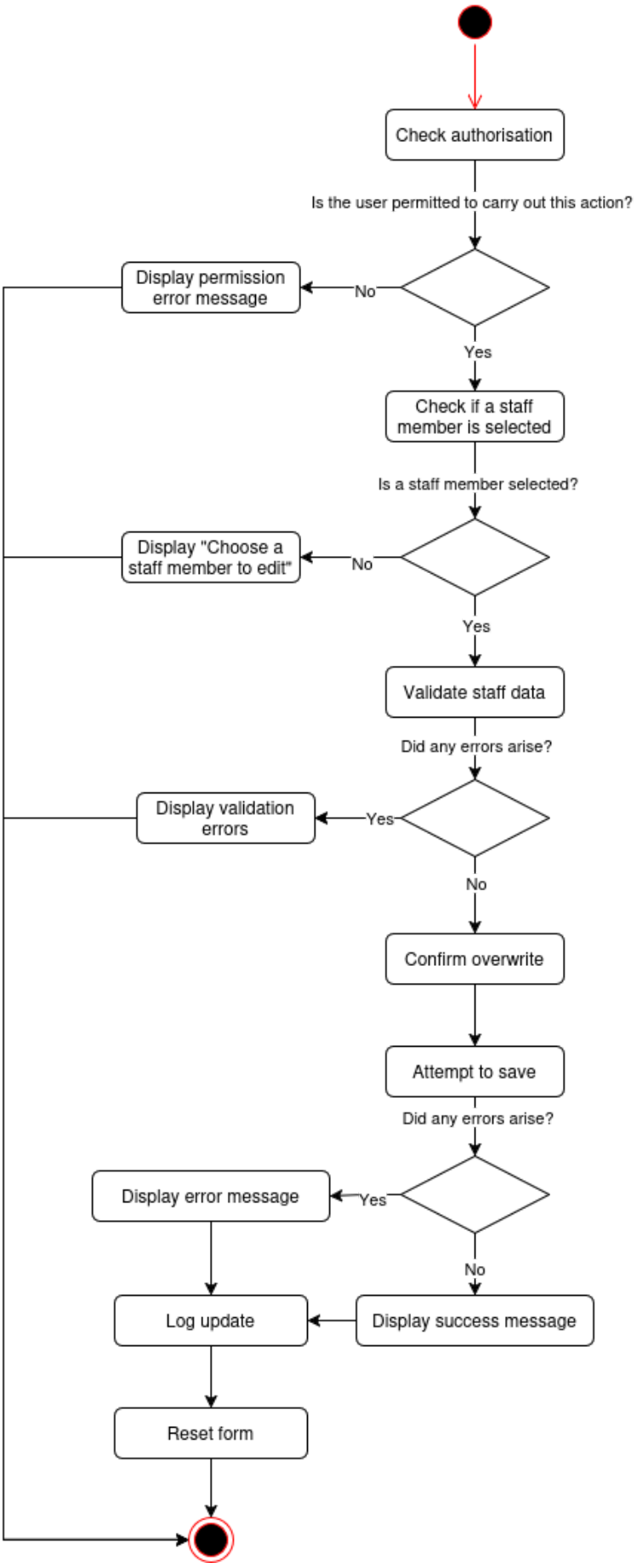
Explanation

Currently planned to be accessible only by administrators, the staff data form allows for the viewing and manipulation of records stored on the Staff table. Upon loading, all records in the table will be instantiated on the system as Staff objects and displayed in the dataGridView. If any errors arise during this process, an error will be displayed and the form locked down to prevent further damage to the system. Furthermore, checks will be made to ensure that usernames input are duplicates which could compromise the login system. Important to note is the nature of the password field - users can see the hash stored in the Staff table on the dataGridView, but this will not be loaded into the appropriate textbox when selecting a record. Instead, the textbox will purely be for the input of new passwords, which are hashed before being used to update the table. Furthermore, the password field will not be available within the search functionality. Upon modifying or adding a record, preliminary checks are made to ensure that the fields input are valid for storage in the table, before a Staff object is instantiated with the fields to ensure that it can exist on the system. If both checks pass, the appropriate action will be carried out with the fields. Column names will be manually set upon displaying data from the table to ensure that the purpose of each field remains clear. Furthermore, all values in the role ComboBox will be fetched from the Roles table at runtime. It is important to note that deleting a staff member will not be possible if they have at least one associated service appointment

Form event listener	Pseudocode/diagram
dgvStaff_SelectionChanged	RETRIEVE THE STAFF BY THE SELECTED RECORD'S ID IF THE STAFF MEMBER EXISTS POPULATE THE APPROPRIATE FIELDS IN THE FORM WITH THEIR DATA END IF

btnAdd_Click	CHECK PERMISSION
	IF permissionError IS NOT null THEN
	        DISPLAY permissionError
	        EXIT process
	ENDIF
	VALIDATE fields
	
	IF password IS empty OR whitespace THEN
	        ADD "Password cannot be empty." TO validationErrors
	ENDIF
	
	IF validationErrors IS NOT empty THEN
	        DISPLAY all validationErrors
	        EXIT process
	ENDIF
	
	TRY
	        CREATE newStaff WITH FIELDS
	        IF IsUsernameUnique(username, 0) IS FALSE THEN
	            DISPLAY "Username already exists. Choose a different username."
	            EXIT process
	        ENDIF
	        ATTEMPT TO ADD newStaff, OUTPUTTING ERROR MESSAGE TO addingMessage
	        IF addingMessage IS null THEN
	            DISPLAY "Staff member has been added successfully."
	            LOG ACTIOn
	        ELSE
	            DISPLAY "Staff member not added: " + addingMessage
	        ENDIF
	CATCH exception
	        DISPLAY exception message
	        EXIT process
	ENDTRY
	    CALL LoadFormDefaults()

btnSave_Click



btnDelete_Click	CHECK PERMISSION
	IF permissionError IS NOT null THEN
	DISPLAY permissionError
	EXIT process
	ENDIF
	
	CHECK A STAFF MEMBER HAS BEEN SELECTED AND IT IS NOT THE ONE LOGGED IN
	
	IF CHECK PASSES
	TRY
	CHECK IF THE SELECTED STAFF MEMBER HAS ANY APPOINTMENTS
	IF THEY DO
	OUTPUT ERROR MESSAGE
	END PROCESS
	ELSE
	MARK STAFF MEMBER AS DELETED
	END IF
	CATCH
	OUTPUT ERROR MESSAGE
	END CATCH
	ELSE
	OUTPUT ERROR MESSAGE
	END IF
	
	LOG DELETE
	RESET FIELDS

txtSearch_TextChanged	RETRIEVE THE TYPE OF SEARCH SELECTED RETRIEVE THE FIELD SELECTED FOR SEARCHING FIND ALL RECORDS ON THE DATABASE WHICH MATCH THE SEARCH TYPE AND STRING IN THAT PARTICULAR FIELD

Validation and error-trapping

- If statements will be used to ensure:
  - StaffID is greater than 0
  - RoleID is greater than 0
  - Firstname and surname fields are not empty
  - Firstname and surname are no longer than 50 characters
  - Username is not empty
  - Username length is between 3–50 characters
  - Username contains only letters, numbers, underscores, or periods
  - Username does not contain spaces
  - Password is not empty
  - Password length is between 8–255 characters

- Try-catch statements will surround all database operations to handle errors and provide user feedback
- Try-catch statements executed during loading of the form will ensure that the form is locked down in the event of a database error
- If statements will be used to ensure that a valid staff member has been selected for manipulation
- If statements will be used to ensure that, if roles fail to load, the form is locked down

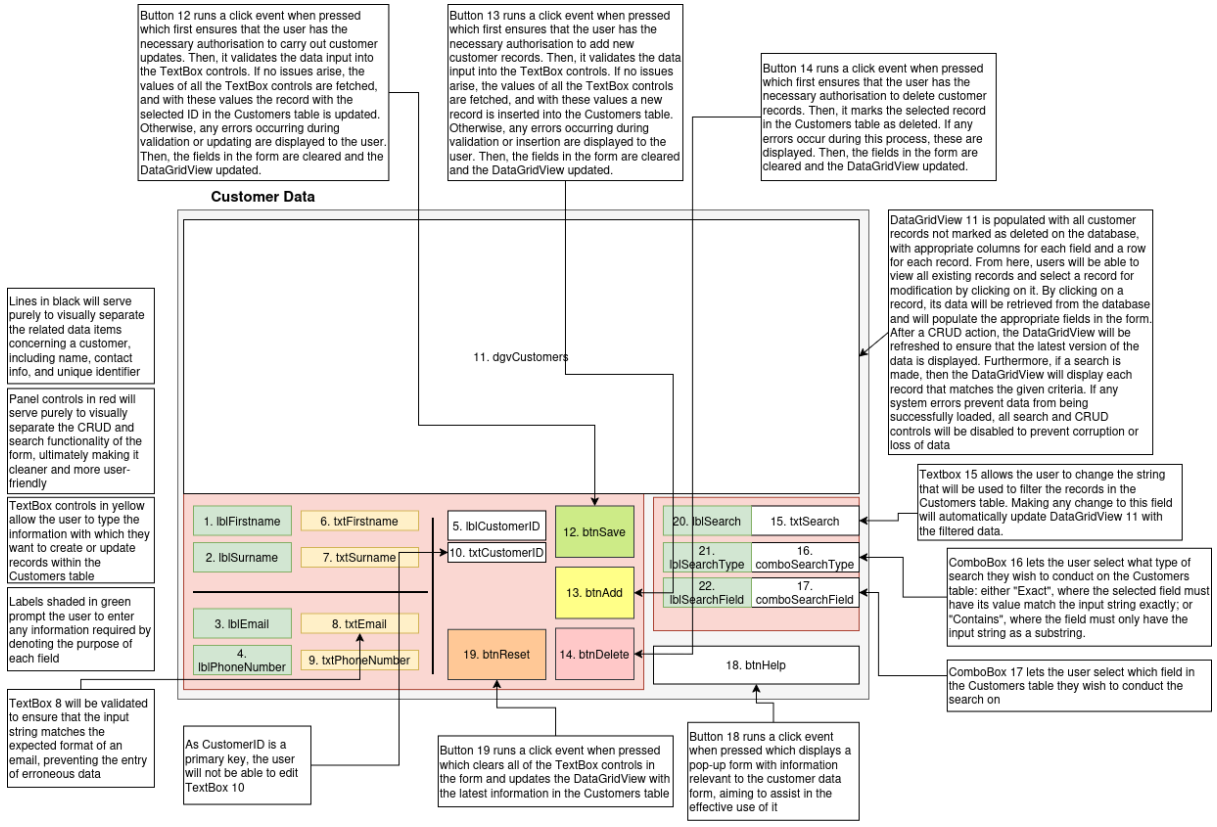
Input specifications

Field/Control	Data Type	Validation Rules	Purpose
txtStaffID	Integer	Must be >= 0; auto-populated for existing staff	Represents the StaffID of the staff member being edited; used for updates
txtFirstname	Text	Required; cannot be blank; max 50 characters	First name of the staff member
txtSurname	Text	Required; cannot be blank; max 50 characters	Surname of the staff member
txtUsername	Text	Required; cannot be blank; 3–50 characters; letters, numbers, _ or . only; must be unique	Username for staff login
txtPassword	Text	Optional for updates; if provided, 8-128 characters	Password for staff login; hashed before storing
comboRoleID	ComboBox (Integer)	Required; value > 0	Role assigned to the staff member (used for authorisation)

User requirements met: 6b, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16c, 17, 23, 24, 26, 27

frmCustomerData

Drawing



Explanation

Planned to be accessible by both administrative and customer service staff, the customer data form will allow for the manipulation and viewing of records on the Customers table. Similarly to frmStaffData, all records stored in the table are instantiated as Customer objects to be displayed in the dataGridView, and in the event of an error the form will be locked down to prevent further input. Saving and adding customers will only require “restricted write” access due to the rather harmless nature of these actions, but deleting will require “unrestricted write” to prevent the loss of pertinent data. Once again, column names will be manually set rather than using the names of the public properties of the Customer class. Unique to this form will be the validation of emails - to ensure that valid contact information is stored in the table, it will be ensured that the value within the email field matches the expected format before carrying out updates or adds. However, no such validation will exist for the phone number field due to the relatively unpredictable nature of phone numbers, and so only length checks will be implemented for it. Much of the logic for this form carries over from frmStaffData, and as such will not be repeated for brevity. However, it is important to note that deleting a customer will also cause any associated vehicles and appointments to be deleted.

Pseudocode and diagrams

Note: most of the CRUD logic carries over from frmStaffData, so only new or significantly different functionality will be represented in future CRUD forms

Form event listener	Pseudocode

btnDelete_Click	`CHECK PERMISSION`
	`IF permissionError IS NOT null THEN`
	`  DISPLAY permissionError`
	`  EXIT process`
	`ENDIF`
	
	`CHECK A CUSTOMER HAS BEEN SELECTED`
	`IF NOT`
	`  OUTPUT “Select a customer to delete.”`
	`  END PROCESS`
	`END IF`
	
	`DISPLAY A CONFIRMATION PROMPT`
	`IF CONFIRMED`
	`  TRY`
	`  DELETE ALL RECORDS IN ServiceAppointments CONTAINING A VEHICLE OWNED BY THE CUSTOMER`
	`  DELETE ALL RECORDS IN Vehicles THAT ARE OWNED BY THE CUSTOMER`
	`  DELETE THE CUSTOMER`
	`  CATCH`
	`  OUTPUT ERROR MESSAGE`
	`  END TRY`
	`END IF`
	
	`LOG THE DELETE`
	`RESET FORM`

Validation and error-trapping

- If statements will be used to ensure
  - CustomerID is greater than 0
  - First name and surname fields are not empty
  - First name and surname are no longer than 50 characters
  - Email, if provided, is properly formatted
  - Email is no longer than 50 characters
  - Phone number is no longer than 15 characters
- Try-catch statements will surround all database operations to handle errors and provide user feedback
- Try-catch statements executed during loading of the form will ensure that the form is locked down in the event of a database error

- If statements will be used to ensure that a valid customer has been selected for manipulation

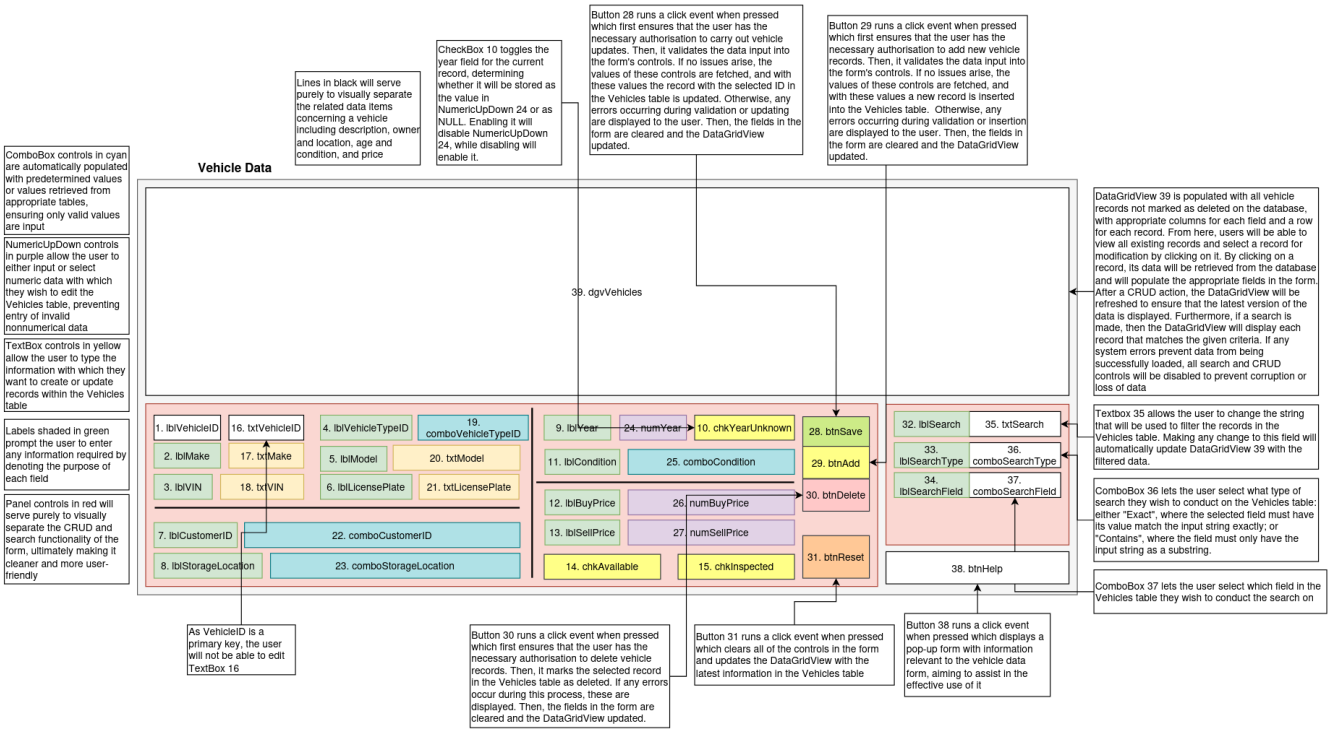
Input specifications

Field/Control	Data Type	Validation Rules	Purpose
txtCustomerID	Integer	Must be >= 0; auto-populated for existing customers	Represents the CustomerID; used for identifying the customer in updates
txtFirstname	Text	Required; cannot be blank; max 50 characters	Customer's first name
txtSurname	Text	Required; cannot be blank; max 50 characters	Customer's surname
txtEmail	Text	Optional; must be valid email format if provided; max 50 characters	Customer's email address
txtPhoneNumber	Text	Optional; max 15 characters	Customer's phone number

User requirements met: 6a, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16a, 23, 24, 26

frmVehicleData

Drawing



Explanation

Planned to be accessible by administrative and technical staff, this form will serve as an interface to the Vehicles table. Once again, much of the functionality carries over from the previous CRUD forms, and as such only differences will be highlighted. Notably, two ComboBox controls exist for records existing in other tables (these being VehicleTypes and Customers), and consequently all records in those tables will be stored as a list containing the record ID followed by an human-readable identifier such as a name. Furthermore, the condition and storage location ComboBox controls will contain hard-coded valid values. Though not scalable, this is due to the fact that these fields are not used in any calculations within the subsystem being developed, and only contain a fixed number of values. A cleaner, reusable solution may be implemented in the future, but this will suffice for now. Important to note is the CheckBox for unknown year, which prevents the year from being fetched from its field, instead being stored as null. Furthermore, as the VIN of a car is unique, checks will be made to ensure that there are no duplicate values and thus that each car has an identifier that links it to the physical world. There are no checks to ensure that the vehicle is being sold at a profit to prevent compromising the system in the event that a vehicle must be sold at below retail value, such as when clearing old stock. Lastly, deleting a vehicle will delete all related service appointments.

Form event listener	Pseudocode

frmVehicleData_Load	FETCH ALL CUSTOMERS FROM THE Customers TABLE
	IF AN ERROR PREVENTS CUSTOMERS FROM BEING LOADED
	  OUTPUT “No customers could be loaded.”
	  LOCK DOWN THE FORM
	  END PROCESS
	END IF
	
	FOR EACH CUSTOMER
	  ADD A NEW OPTION TO THE COMBOBOX CONTAINING THE CUSTOMER’S ID FOLLOWED BY THEIR FULL NAME
	END FOR
	
	FETCH ALL VEHICLE TYPES FROM THE VehicleTypes TABLE
	IF AN ERROR PREVENTS VEHICLE TYPES FROM BEING LOADED
	  OUTPUT “No vehicle types could be loaded.”
	  LOCK DOWN THE FORM
	  END PROCESS
	END IF
	
	FOR EACH VEHICLE TYPE
	  ADD A NEW OPTION TO THE COMBOBOX CONTAINING THE VEHICLE TYPE’S ID FOLLOWED BY ITS NAME
	END FOR
	
	ATTEMPT TO LOAD VEHICLES
	
	IF AN ERROR OCCURS
	  OUTPUT "There was an error when attempting to load vehicle data. Please amend erroneous data in the database before proceeding."
	  LOCK DOWN FORM
	END IF

btnDelete_Click	

```
CHECK PERMISSION
IF permissionError IS NOT null THEN
        DISPLAY permissionError
        EXIT process
ENDIF

CHECK A VEHICLE HAS BEEN SELECTED
IF NOT
       OUTPUT “Select a vehicle to delete.”
       END PROCESS
END IF

DISPLAY A CONFIRMATION PROMPT
IF CONFIRMED
   TRY
       DELETE ALL RECORDS IN ServiceAppointments CONTAINING THE
       VEHICLE
       DELETE THE VEHICLE
   CATCH
       OUTPUT ERROR MESSAGE
   END TRY
END IF

LOG THE DELETE
RESET FORM
```

Validation and error-trapping

- If statements will be used to ensure the following:
    - Vehicle ID is greater than 0
    - Vehicle Type ID is greater than 0
    - Customer ID, if provided, is greater than 0
    - Make and Model fields are not empty and do not exceed 50 characters
    - Year, if provided, is between 1800 and 2050
    - VIN field is not empty, does not exceed 50 characters, and is unique
    - Condition field is not empty and does not exceed 50 characters
    - Buy price and Sell price, if provided, are not negative
    - License plate, if provided, does not exceed 20 characters.
    - Storage location, if provided, does not exceed 50 characters.
- Try-catch statements will surround all database operations to handle errors and provide user feedback
- Try-catch statements executed during loading of the form will ensure that the form is locked down in the event of a database error
- If statements will be used to ensure that a valid vehicle has been selected for manipulation
- If statements will be used to ensure that the form is locked down if no customers or vehicle types could be loaded

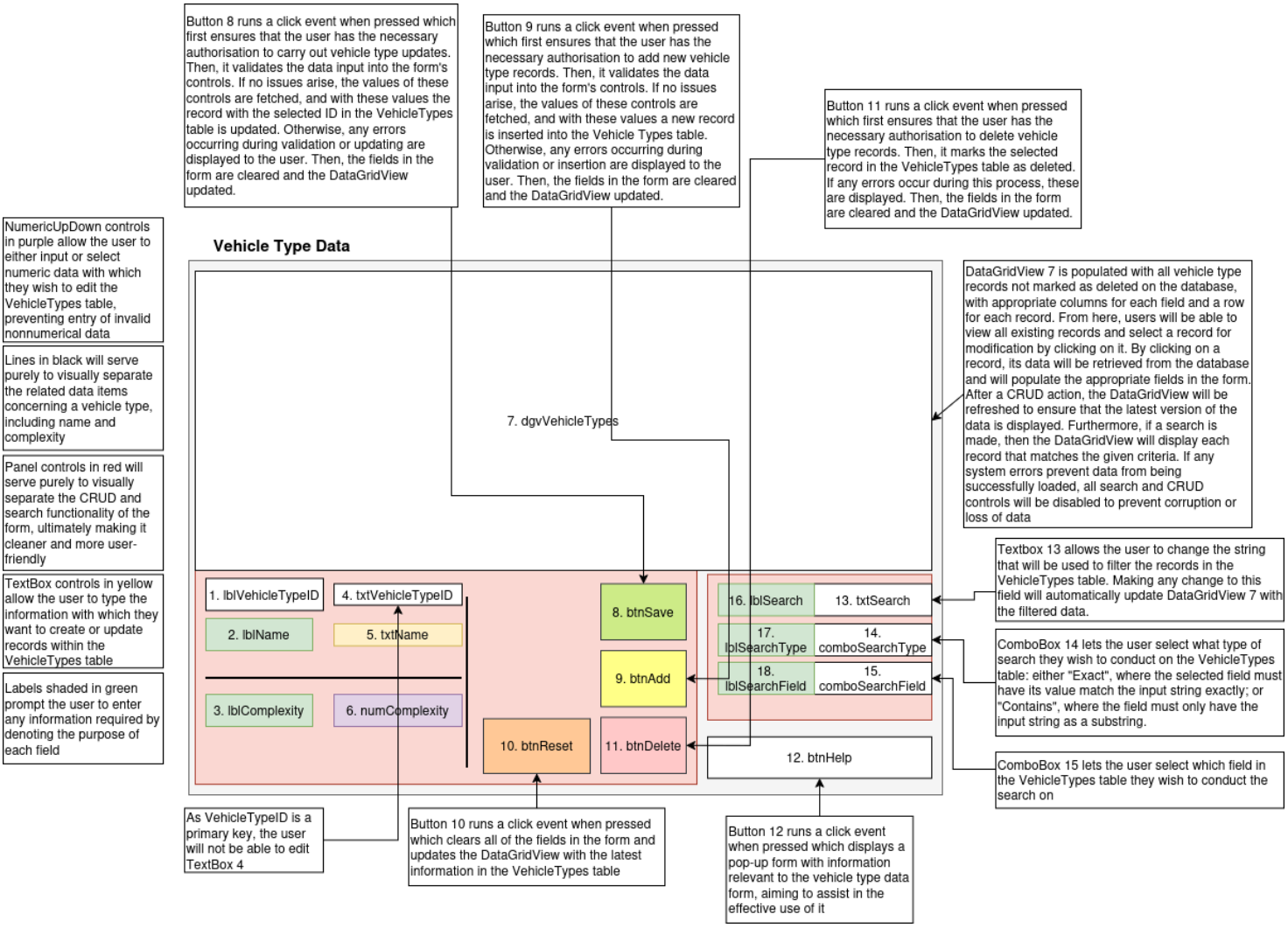
Input specifications

Field / Control	Data Type	Validation Rules	Purpose
txtVehicleID	Integer	Must be >= 0; 0 for new records, >0 for existing vehicles	Identifies the vehicle in the database
comboVehicleTypeID	ComboBox (integer)	Must select a valid VehicleTypeID (>0)	Specifies the vehicle type
comboCustomerID	ComboBox (integer)	Nullable; if selected, must be >= 0	Associates a customer with the vehicle
txtMake	Text	Required; cannot be blank; max 50 characters	Vehicle make
txtModel	Text	Required; cannot be blank; max 50 characters	Vehicle model
numYear / chkYearUnknown	Integer / Checkbox	Year must be 1800–2050 or null if unknown	Vehicle manufacture year
txtVIN	Text	Required; cannot be blank; max 50 characters; must be unique	Vehicle VIN
txtLicensePlate	Text	Optional; max 20 characters	Vehicle license plate
comboCondition	ComboBox (text)	Required; cannot be blank; max 50 characters	Vehicle condition
chkAvailable	Checkbox	Boolean	Vehicle availability
chkInspected	Checkbox	Boolean	Vehicle inspection status
numBuyPrice	NumericUpDown (decimal)	Optional; >= 0 if provided	Purchase price
numSellPrice	NumericUpDown (decimal)	Optional; >= 0 if provided	Selling price
comboStorageLocation	ComboBox (text)	Optional; max 50 characters	Vehicle storage location

User requirements met: 6b, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16b, 23, 24, 26

frmVehicleTypeData

Drawing



Explanation

Planned to be accessible by technical and administrative staff, this form will serve as an interface for the VehicleTypes table. Most notably, there is only one field other than name which is complexity. Each record in the Vehicles table will have at least one vehicle type associated with it. When calculating the duration of an appointment, the duration of the services carried out will be summed, then the appointment's duration will be multiplied by the vehicle's complexity. It is important to note deleting a vehicle type will not be possible is there are vehicles of that type

Pseudocode and diagrams

Form event listener	Pseudocode

btnDelete_Click	CHECK PERMISSION
	IF permissionError IS NOT null THEN
	DISPLAY permissionError
	EXIT process
	ENDIF
	
	CHECK A VEHICLE TYPE HAS BEEN SELECTED
	IF NOT
	OUTPUT “Select a vehicle type to delete.”
	END PROCESS
	END IF
	
	DISPLAY A CONFIRMATION PROMPT
	IF CONFIRMED
	CHECK IF ANY VEHICLES IN Vehicles ARE OF THAT VEHICLE TYPE
	
	IF THERE ARE
	OUTPUT “Vehicle Type is in use and cannot be deleted.”
	END PROCESS
	END IF
	
	TRY
	DELETE THE VEHICLE TYPE
	CATCH
	OUTPUT ERROR MESSAGE
	END TRY
	END IF
	
	LOG THE DELETE
	RESET FORM

Validation and error-trapping

- If statements will be used to ensure:
  - Vehicle Type ID is positive
  - Name is not empty
  - Name does not exceed 50 characters
  - Complexity is zero or positive but does not exceed 5
- Try-catch statements will surround all database operations to handle errors and provide user feedback
- Try-catch statements executed during loading of the form will ensure that the form is locked down in the event of a database error
- If statements will be used to ensure that a valid VehicleType has been selected for manipulation

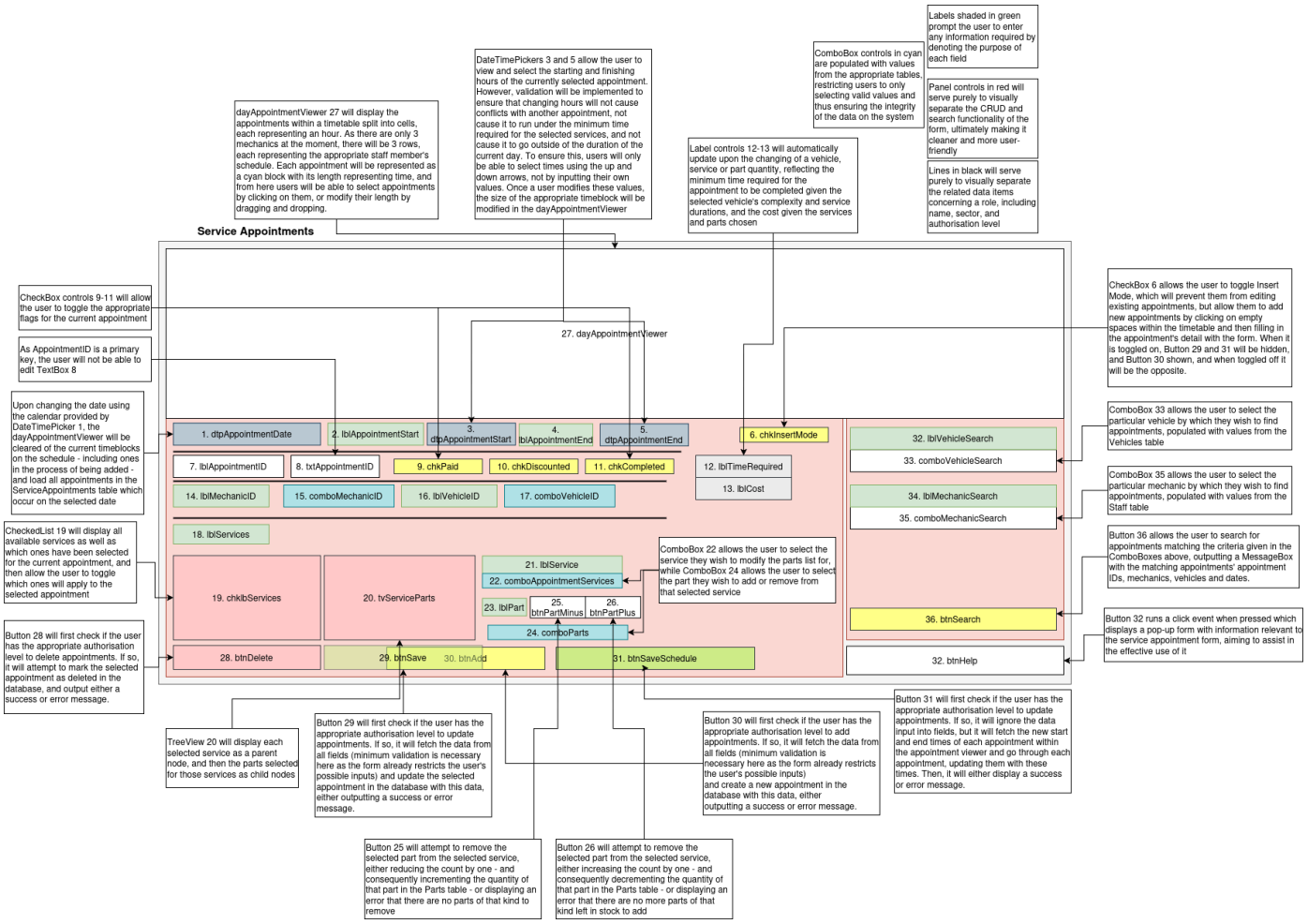
Input specifications

Field / Control	Data Type	Validation Rules	Purpose
txtVehicleTypeID	Integer (TextBox)	Must be >= 0; 0 for new records, >0 for existing records	Identifies the vehicle type in the database
txtVehicleTypeName	Text	Required; cannot be blank; max 50 characters	Name of the vehicle type
numComplexity	NumericUpDown (double)	Required; >= 0; cannot exceed 5	Complexity rating of the vehicle type (0–5)

User requirements met: 6d, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16d, 23, 24, 26

frmServiceAppointments

Drawing



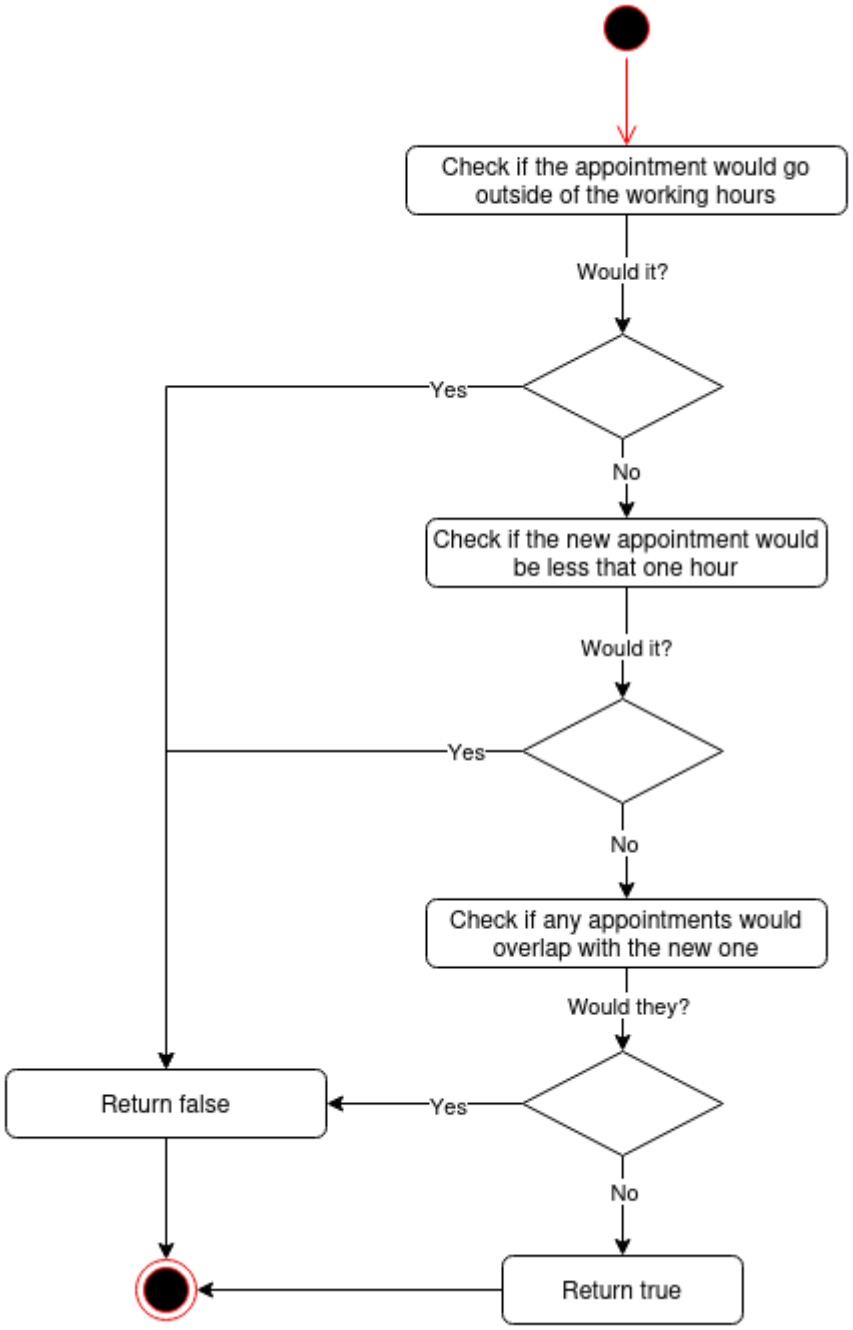
Explanation

The main focus of the program, this form - which will be accessible by administrative and technical staff - allows for the modification and creation of service appointments. Most notably, a custom control will have to be implemented for the actual appointment viewer, as much of the drawing and mouse logic would be significantly easier to manage through its encapsulation. A substantial amount of validation will have to be implemented to ensure that only valid appointments can be created, and as such the effects of a change on a booking - such as the removal of a service or the servicing of a different car - will have to be immediately calculated upon attempting to change an appointment. Furthermore, most of the fields within the form limit the possible input of the user through the use of ComboBox controls populated with the existing values from other tables, ensuring referential integrity within a table that would contain so many foreign keys. Lastly, the search functionality aims to make it easier for the user to find a particular booking when only one day can be viewed at a time, thus making the user's experience less tedious.

Form event listener	Pseudocode
DayAppointments_Load	DETERMINE THE HEIGHT AND WIDTH OF THE ENTIRE SCHEDULE AND EACH CELL IN PIXELS RETRIEVE ALL APPOINTMENTS, AND GROUP BY MECHANIC FOR STORAGE IN MEMORY DRAW THE APPOINTMENTS FOR THE CURRENT DAY

DrawAppointment	DRAW THE LABELS DENOTING THE START OF EACH WORKING HOUR AT THE TOP OF THE VIEWING PANEL, SEPARATED BY THE WIDTH OF A CELL
	
	DRAW THE TIME SLOTS FOR EACH MECHANIC, REPRESENTED AS EQUALLY SIZED RECTANGLES
	
	DRAW EACH MECHANICS NAME AT THE START OF THEIR ALLOCATED ROW OF TIME SLOTS
	
	CLEAR THE OLD LIST OF CALCULATED TIMEBLOCK COORDINATES
	
	FOR EACH APPOINTMENT OCCURRING ON THE SELECTED DAY
	  CALCULATE THE WIDTH OF THE APPOINTMENT'S TIMEBLOCK USING THE DIFFERENCE IN START AND END TIME
	  DETERMINE THE POSITION OF THE START OF THE TIMEBLOCK
	  IF THE CURRENT TIMEBLOCK IS BEING MOVED, ADJUST ITS STARTING POSITION AND LENGTH ACCORDINGLY
	  DRAW THE TIMEBLOCK USING THE INFORMATION CALCULATED
	END FOR

SetQueryDate	UPDATE THE SELECTED DATE IN MEMORY CLEAR ALL SCHEDULE DATA IN MEMORY, AND RETRIEVE NEW DATA FROM APPROPRIATE TABLES RERUN DrawAppointment

IsValidTimeSet	

dtpAppointmentDate_ValueChanged	CHECK IF THE CHOSEN DATE OCCURS DURING THE WEEKEND
	
	IF IT DOES
	    OUTPUT "We're closed during the weekend!"
	    SET THE DATE BACK TO ITS ORIGINAL VALUE
	    END PROCESS
	END IF
	
	RUN DrawAppointments WITH THE NEW DATE

frmServiceAppointments_Load	FETCH ALL STAFF FROM THE Staff TABLE
	VALIDATE THAT NO SYSTEM ERRORS OCCURRED DURING FETCHING
	FILTER STAFF BY ROLE TO RETRIEVE ONLY MECHANICS
	ADD EACH MECHANIC TO THE MECHANIC COMBOBOX CONTAINING THE MECHANIC'S ID AND NAME
	
	FETCH ALL VEHICLES FROM THE Vehicles TABLE
	VALIDATE THAT NO SYSTEM ERRORS OCCURRED DURING FETCHING
	FOREACH VEHICLE
	      IF THE VEHICLE HAS AN OWNER
	         ADD THE VEHICLE'S ID, NAME AND THE OWNER'S NAME TO THE VEHICLE COMBOBOX
	      ELSE
	         ADD THE VEHICLE'S ID AND NAME TO THE VEHICLE COMBOBOX
	      END IF
	END FOR
	
	FETCH ALL SERVICES FROM THE Services TABLE
	VALIDATE THAT NO SYSTEM ERRORS OCCURRED DURING FETCHING
	ADD EACH SERVICE AND ITS PRICE TO THE CHECKEDLISTBOX
	
	FETCH ALL PARTS FROM THE Parts TABLE
	VALIDATE THAT NO SYSTEM ERRORS OCCURRED DURING FETCHING
	ADD EACH PART, ITS PRICE AND QUANTITY TO THE PART COMBOBOX

PopulateAppointmentDetails	RETRIEVE APPOINTMENT FROM ServiceAppointments TABLE USING ITS ID POPULATE BASIC FIELDS DIRECTLY (Appointment Start, Appointment End, Appointment ID, Paid, Completed, Discounted) FIND THE APPROPRIATE MECHANIC IN THE COMBOBOX AND SET THEM AS THE SELECTED ITEM FIND THE APPROPRIATE VEHICLE IN THE COMBOBOX AND SET IT AS THE SELECTED ITEM GO THROUGH EACH SERVICE AND CHECK IT IN THE CHECKEDLISTBOX, THEN ADD EACH SERVICE TO THE SERVICE COMBOBOX

btnSaveScheduleOnly_Click	DISPLAY A CONFIRMATION MESSAGE
	
	TRY
	  FOR EACH APPOINTMENT IN DayAppointmentViewer
	  FETCH ITS ID
	  UPDATE ONLY THE START AND END DATE OF THE APPOINTMENT IN ServiceAppointments
	  END FOR
	CATCH
	  OUTPUT ERROR MESSAGE
	END TRY

btnPartMinus_Click	IF NO SERVICE HAS BEEN CHOSEN IN THE COMBOBOX
	  OUTPUT "No service has been chosen to remove a part from."
	  END PROCESS
	END IF
	
	IF NO PART HAS BEEN CHOSEN IN THE COMBOBOX
	  OUTPUT "No part has been chosen to remove from the service."
	  END PROCESS
	END IF
	
	IF THE SERVICE-PARTS LIST DOES NOT CONTAIN ANY PARTS FOR THE SERVICE
	  OUTPUT "The current service doesn't have any of the selected item!"
	  END PROCESS
	END IF
	
	INCREMENT THE NUMBER OF PARTS OF THE CHOSEN KIND ASSOCIATED WITH THAT SERVICE

btnPartPlus_Click	IF NO SERVICE HAS BEEN CHOSEN IN THE COMBOBOX
	    OUTPUT "No service has been chosen to remove a part from."
	    END PROCESS
	END IF
	
	IF NO PART HAS BEEN CHOSEN IN THE COMBOBOX
	    OUTPUT "No part has been chosen to remove from the service."
	    END PROCESS
	END IF
	
	IF THE Parts TABLE HAS A Quantity FOR THAT PART LESS THAN THE NEW PART COUNT WOULD BE
	    OUTPUT "You don't have enough of this part in stock!"
	    END PROCESS
	END IF
	
	DECREMENT THE NUMBER OF PARTS OF THE CHOSEN KIND ASSOCIATED WITH THAT SERVICE

Validation and error-trapping

- If statements will be used to ensure:
    - Vehicle ID is greater than zero
    - Staff ID is greater than zero
    - Appointment end is after appointment start
    - Appointment starts and ends on the same day
    - The appointment duration is sufficient for the selected services
- Try-catch statements will surround all database operations to handle errors and provide user feedback
- Try-catch statements executed during loading of the form will ensure that the form is locked down in the event of a database error
- If statements will be used to ensure that a valid service and part have been selected for manipulation
- If statements will check that mechanic and vehicle selections do not cause overlapping appointments
- If statements will check stock availability when adding parts to services
- If statements will ensure weekend appointments are blocked

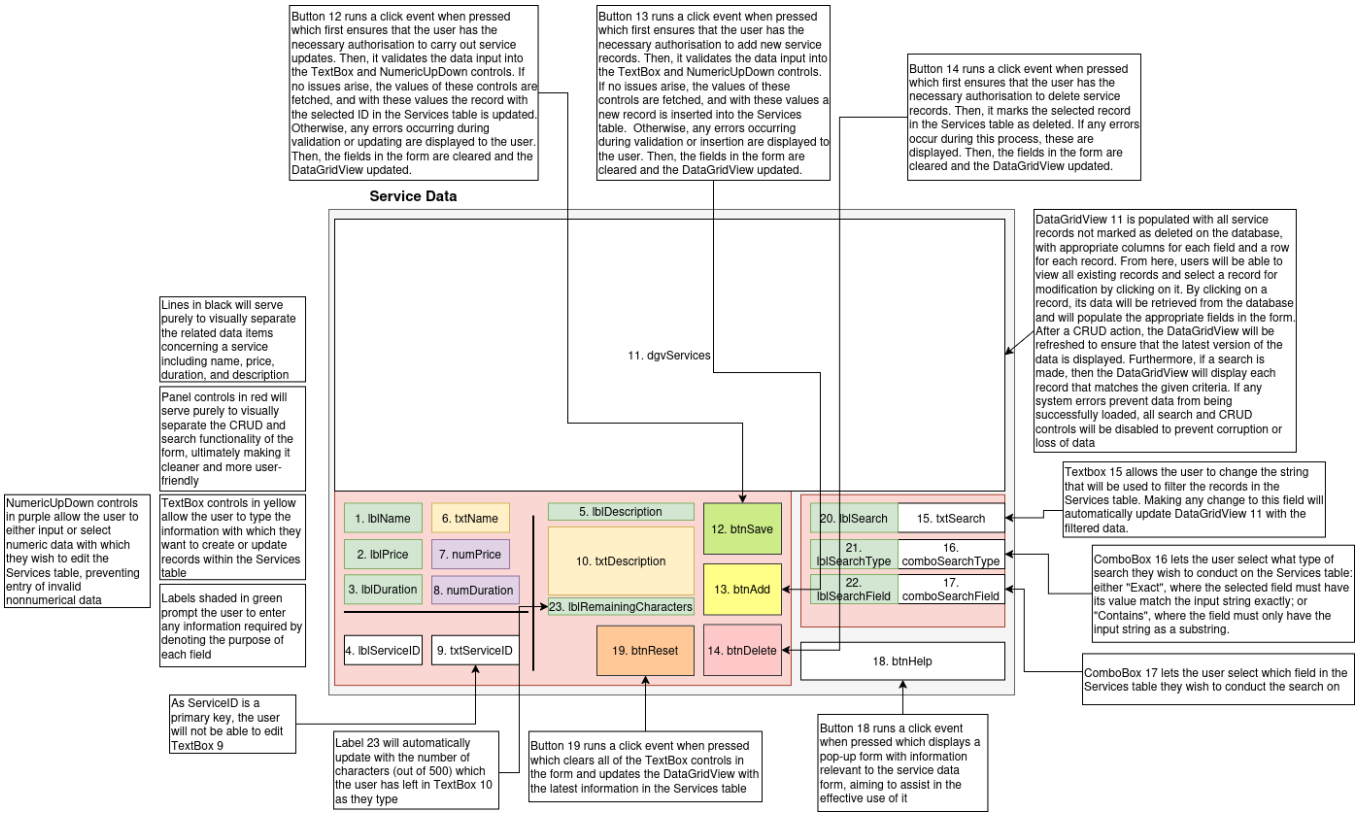
Input specifications

Field / Control	Data Type	Validation Rules	Purpose
txtAppointmentID	Integer (TextBox)	Must be >= 0; 0 for new records, >0 for existing records	Identifies the appointment in the database
dtpAppointmentDate	DateTime	Must be not fall on a weekend	Day on which an appointment falls
dtpAppointmentStart	DateTime	Must be less than end; must be on same date as end; must be greater than the start of day	Time at which appointment starts
dtpAppointmentEnd	DateTime	Must be greater than start; must be on same date as start; must be less than end of day	Time at which appointment ends
chkPaid	Checkbox	Boolean	If appointment has been paid for
chkDiscounted	Checkbox	Boolean	If appointment has been discounted
chkCompleted	Checkbox	Boolean	If appointment has been completed
comboMechanicID	ComboBox (integer)	Must be >= 0; must exist in Staff table	Mechanic carrying out appointment
comboVehicleID	ComboBox (integer)	Must be >= 0; must exist in Vehicles table	Vehicle being serviced
chklbServices	CheckedListBox (integer)	Must be >= 0; must exist in Services table; sum of times must not exceed allocated time	Services carried out during appointment
comboParts	ComboBox (integer)	Must be >= 0; must exist in Parts table; total amount of part used must not exceed amount in stock	Parts used in services during appointment

User requirements met: 6h, 7, 8, 9, 10, 11, 12, 13, 14, 15, 18, 19, 20, 22, 23, 24, 26

frmServiceData

Drawing



Explanation

Accessible by administrative and technical staff, this form will allow users to modify the values within the Services table. Little is unique to this form bar the description fields, which allows users to write the specific details of a service in prose. Provided with this field is a remaining character count, which ensures that users do not exceed the limit provided to them. The details of a service will be used to calculate the cost and duration of an appointment. It is important to note deleting services is not possible if they are used within any appointment.

Form event listener	Pseudocode

btnDelete_Click	CHECK PERMISSION
	IF permissionError IS NOT null THEN
	DISPLAY permissionError
	EXIT process
	ENDIF
	
	CHECK A SERVICE HAS BEEN SELECTED
	IF NOT
	OUTPUT “Select a service to delete.”
	END PROCESS
	END IF
	
	DISPLAY A CONFIRMATION PROMPT
	IF CONFIRMED
	CHECK IF ANY SERVICE APPOINTMENTS IN ServiceAppointments USE THAT SERVICE
	IF THEY DO
	OUTPUT “Service is currently used in service appointments and cannot be deleted”.
	END PROCESS
	END IF
	
	TRY
	DELETE THE SERVICE
	CATCH
	OUTPUT ERROR MESSAGE
	END TRY
	END IF
	
	LOG THE DELETE
	RESET FORM

txtDescription_TextChanged	CALCULATE 500 - THE NEW LENGTH OF THE STRING IN txtDescription SET THE TEXT OF lblRemainingCharacters EQUAL TO “Remaining characters: “ APPENDED WITH THE CALULCATED VALUE

Validation and error-trapping

- If statements will be used to ensure:
    - Service ID is positive
    - Service name is not empty and does not exceed 50 characters
    - Description does not exceed 500 characters
    - Duration is between 1 and 10 hours
    - Price is zero or positive
- Try-catch statements will surround all database operations to handle errors and provide user feedback
- Try-catch statements executed during loading of the form will ensure that the form is locked down in the event of a database error
- If statements will be used to ensure that a valid part has been selected for manipulation

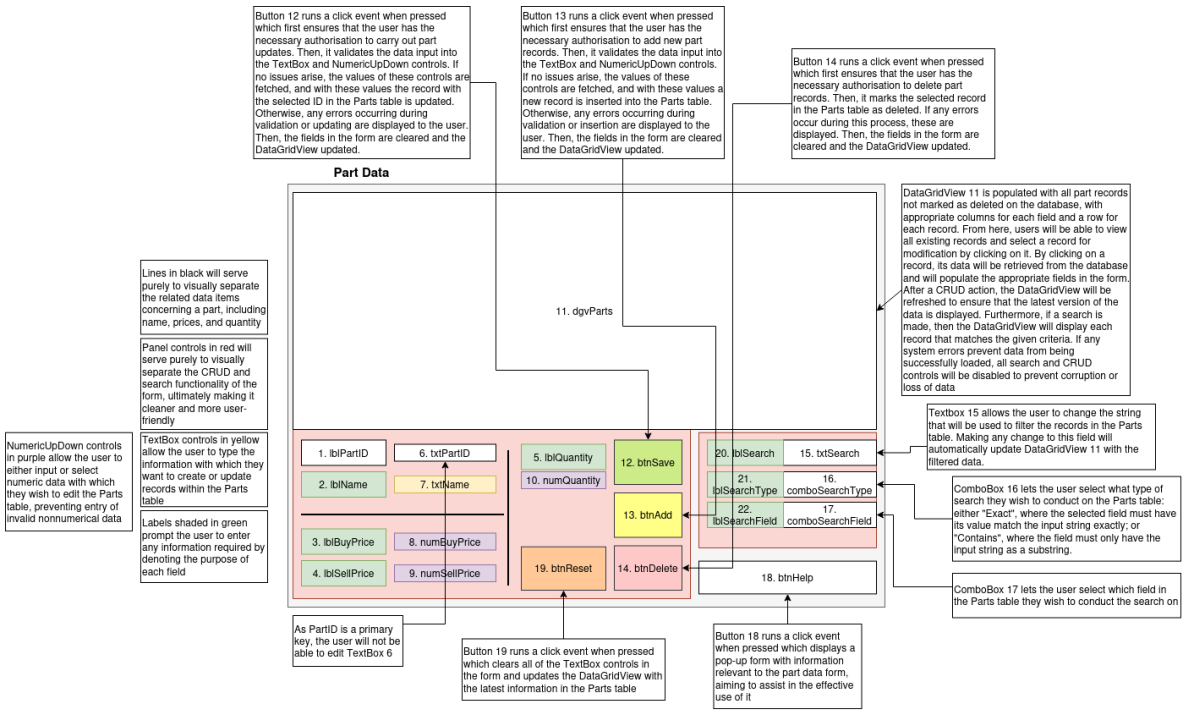
Input specifications

Field / Control	Data Type	Validation Rules	Description / Purpose
txtServiceID	Integer (TextBox)	Required; >= 0; auto-populated when selecting a service	Unique identifier for the service
txtName	String (TextBox)	Required; non-empty; max 50 characters	Name of the service
txtDescription	String (TextBox)	Optional; max 500 characters	Description of the service
numDuration	Integer (NumericUpDown)	Required; 1-10	Duration of the service in hours
numPrice	Decimal (NumericUpDown)	Required; >= 0	Price of the service

User requirements met: 6g, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16g, 23, 24, 26

frmPartData

Drawing



Explanation

Accessible by administrative and technical staff, this form will allow users to modify the Parts table. The cost of a part will make up part of the total cost of an appointment. It is important to note that deleting a part will not be possible if it is part of an existing appointment.

Form event listener	Pseudocode

btnDelete_Click	CHECK PERMISSION
	IF permissionError IS NOT null THEN
	DISPLAY permissionError
	EXIT process
	ENDIF
	
	CHECK A SERVICE HAS BEEN SELECTED
	IF NOT
	OUTPUT “Select a service to delete.”
	END PROCESS
	END IF
	
	DISPLAY A CONFIRMATION PROMPT
	IF CONFIRMED
	CHECK IF ANY SERVICE APPOINTMENTS IN ServiceAppointments USE THAT SERVICE
	IF THEY DO
	OUTPUT “Service is currently used in service appointments and cannot be deleted”.
	END PROCESS
	END IF
	
	TRY
	DELETE THE SERVICE
	CATCH
	OUTPUT ERROR MESSAGE
	END TRY
	END IF
	
	LOG THE DELETE
	RESET FORM

Validation and error-trapping

- If statements will be used to ensure:
    - Part ID is positive
    - Part name is not empty and does not exceed 50 characters
    - Buy price is zero or positive
    - Sell price is zero or positive
    - Quantity is zero or positive
- Try-catch statements will surround all database operations to handle errors and provide user feedback
- Try-catch statements executed during loading of the form will ensure that the form is locked down in the event of a database error
- If statements will be used to ensure that a valid part has been selected for manipulation

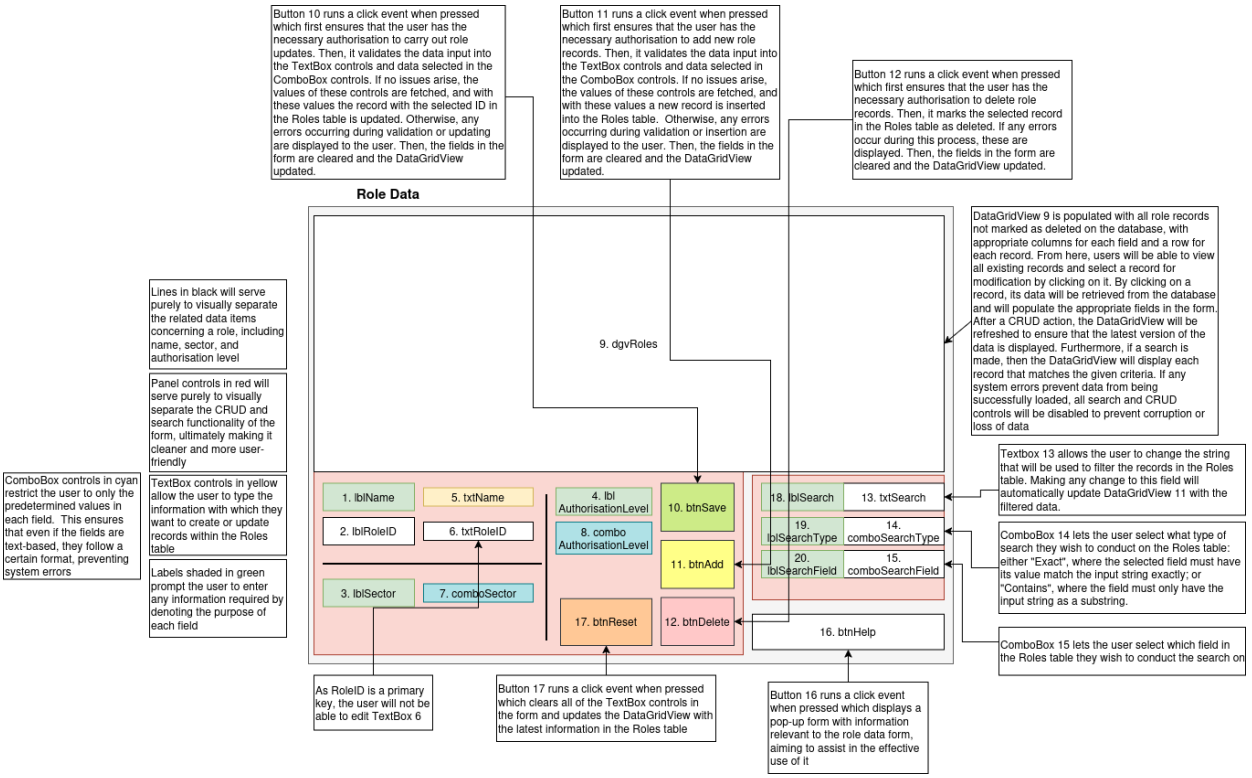
Input specifications

Field / Control	Data Type	Validation Rules	Purpose
txtPartID	Integer (TextBox)	Must be >= 0; auto-populated for new parts	Identifies the part; used for edits or deletes
txtName	Text	Required; cannot be blank; max 50 characters	Part name
numBuyPrice	NumericUpDown (decimal)	>= 0	Purchase price of the part
numSellPrice	NumericUpDown (decimal)	>= 0	Sale price of the part
numQuantity	NumericUpDown (integer)	>= 0	Quantity in stock

User requirements met: 6e, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 23, 24, 26

frmRoleData

Drawing



Explanation

This form will allow users to modify the Roles table. Due to the inherent risk of unauthorised users gaining access to key parts of the system by modifying the details of their role, only administrative staff will be able to access this form. At the moment, there are a limited number of sectors (these being administrative, HR, technical, customer Service, and finance) and as such the sector ComboBox will be hard-coded with these values. When the business eventually expands and requires a more sophisticated sector system, a separate table could be implemented and the same logic used that populates ComboBox controls in other forms. Similarly, there are only 3 authorisation levels - 0 being read-only, 1 being restricted write, and 2 being unrestricted write - and these will be hardcoded as well. Roles will not be stored as objects on the system when a user logs in, and instead the table will be queried when role data is required to ensure that only the latest permitted forms and permissions are displayed. It is important to note that deleting roles will not be possible if there are staff members of that role

Pseudocode and diagrams

Form event listener	Pseudocode

btnDelete_Click	CHECK PERMISSION
	IF permissionError IS NOT null THEN
	DISPLAY permissionError
	EXIT process
	ENDIF
	
	CHECK A ROLE HAS BEEN SELECTED
	IF NOT
	OUTPUT “Select a role to delete.”
	END PROCESS
	END IF
	
	DISPLAY A CONFIRMATION PROMPT
	IF CONFIRMED
	CHECK IF ANY STAFF MEMBERS IN Staff ARE OF THAT ROLE
	IF THEY DO
	OUTPUT “Role is assigned to staff members and cannot be deleted”
	END PROCESS
	END IF
	
	TRY
	DELETE THE ROLE
	CATCH
	OUTPUT ERROR MESSAGE
	END TRY
	END IF
	
	LOG THE DELETE
	RESET FORM

Validation and error-trapping

- If statements are used to ensure:
  - Role ID is positive
  - Role name is not empty and does not exceed 50 characters
  - Sector is not empty and does not exceed 50 characters
  - Authorisation level is zero or positive
- Try-catch statements will surround all database operations to handle errors and provide user feedback
- Try-catch statements executed during loading of the form will ensure that the form is locked down in the event of a database error
- If statements will be used to ensure that a valid role has been selected for manipulation

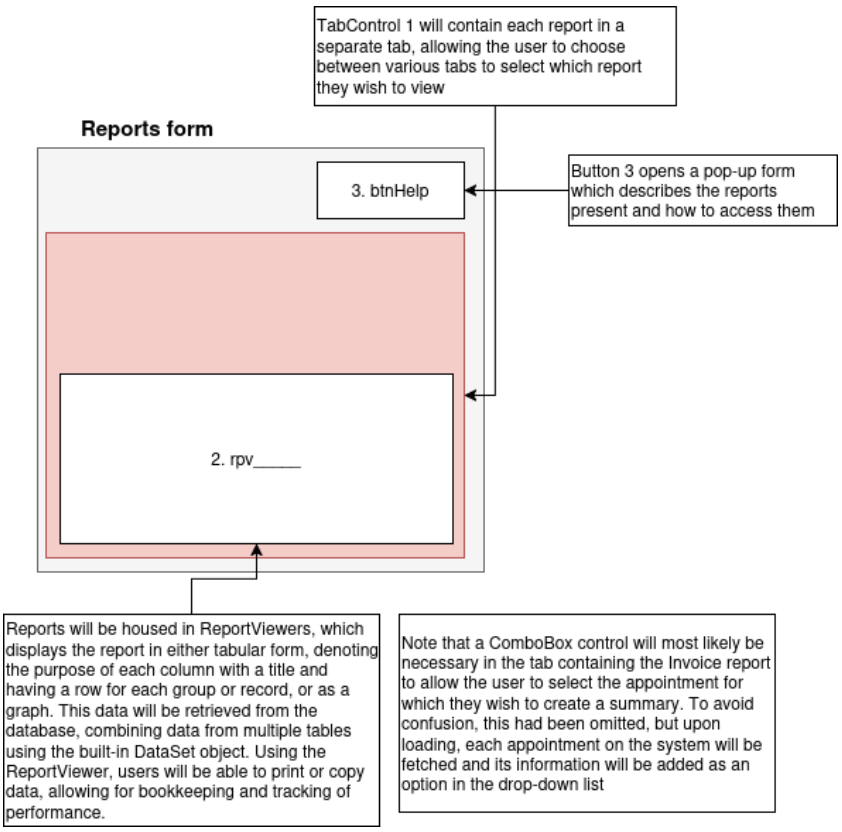
Input specifications

Field / Control	Data Type	Constraints / Validation Rules	Description / Purpose
txtRoleID	Integer (TextBox)	Required; >= 0; auto-populated when selecting a role	Unique identifier for the role
txtName	String (TextBox)	Required; non-empty; max 50 characters	Name of the role
comboAuthorisation Level	Integer (ComboBox)	0–2 (0 = Read-only, 1 = Restricted write, 2 = Unrestricted write)	Role permission level; determines access rights
comboSector	String (ComboBox)	Required; non-empty; max 50 characters	Sector the role belongs to

User requirements met: 6f, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16f, 23, 24, 26

frmReports

Drawing



Explanation

Accessible by customer service, administrative and finance staff, this form will display reports derived from the data stored within the database. In particular, the following reports are to be implemented: monthly profits due to service appointments grouped by mechanic (i.e. the summation of the price of those paid and completed, minus the cost of the parts); overall profits due to car sales, grouped by vehicle type (once again, sell price minus buy price summed); invoices for these appointments to allow staff to easily determine how much a customer is due to pay the business; customer interest by vehicle type (i.e., the total number of vehicles owned by customers grouped by vehicle type); missed appointments; and vehicle stock (the number of cars owned by dealership by vehicle type). With each of these reports, the ReportViewer control will allow for convenient saving or printing of the information produced, thus allowing for the keeping of records or external analysis. Note that the invoices tab may require a ComboBox element to select the specific appointment they wish to generate an overview for, and this will be handled by fetching all appointments from the database and parsing their appointment IDs, adding them as options.

Pseudocode and diagrams

As most functionality - from rendering reports and calculating values, to changing and displaying tabs - is handled by built-in methods, and only binding the reports is required on my end (which is rather mundane) no pseudocode will be provided. The aforementioned ComboBox would require parsing and population logic, but similar logic is handled in the CRUD forms and will not be included for the sake of brevity.

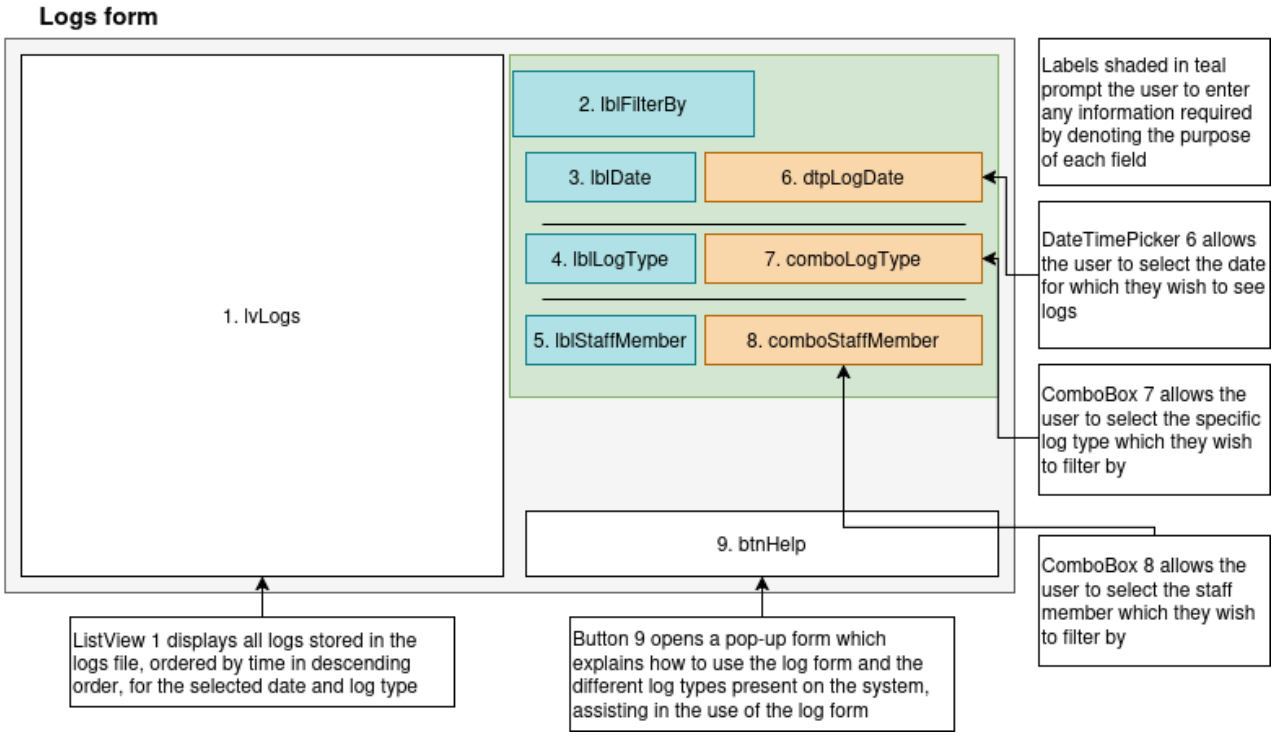
Validation and error-trapping

- If statements will be used when loading data from external sources to ensure that it was successfully retrieved before attempting to parse it

User requirements met: 15, 21

frmLogs

Drawing



Explanation

Accessible by HR and administrative staff, the logs form will display all logs accumulated in an external file, thus allowing these staff to know when each staff member performed what kind of action on which data, adding a form of accountability to the system. All relevant buttons on the system (CRUD, login and form access buttons) will be linked to the logging system, and clicking these will trigger an event which will save an appropriate message to an external file. These will be parsed by frmLogs on loading, and logs written on the chosen date (or the current date by default) will be displayed in the ListView control. It is important to note that no facilities are provided to modify the logs file, preventing any malicious users from covering up their tracks by editing or deleting logs. At the moment, 6 log types are to be considered - read, write, delete, access, successful login, unsuccessful login (i.e. login attempt).

Form event listener	Pseudocode

UpdateLogs	FETCH THE CHOSEN DATE, LOG TYPE AND STAFF MEMBER
	
	CLEAR ALL PREVIOUS LOADED LOGS
	
	LOAD THE LOG FILE INTO MEMORY
	
	FOREACH LOG IN THE FILE
	  CHECK THAT THE DATE SEGMENT IS PROPERLY FORMATTED
	
	  IF IT ISN'T
	  CONTINUE
	  END IF
	
	  PARSE THE DATE
	
	  CHECK THAT THE LOG TYPE SEGMENT IS PROPERLY FORMATTED
	
	  IF IT ISN'T
	  CONTINUE
	  END IF
	
	  MATCH THE LOG TYPE NUMBER TO THE APPROPRIATE LOG TYPE
	
	  REMOVE THE LOG TYPE NUMBER TO PRODUCE A CLEANED STRING FOR DISPLAY TO THE USER
	
	  CREATE A NEW ListViewItem OBJECT USING DATE, LOG TYPE AND STRING
	END FOREACH
	
	FILTER BY THE CRITERIA GIVEN
	ADD ALL VALID ListViewItems TO THE ListView CONTROL

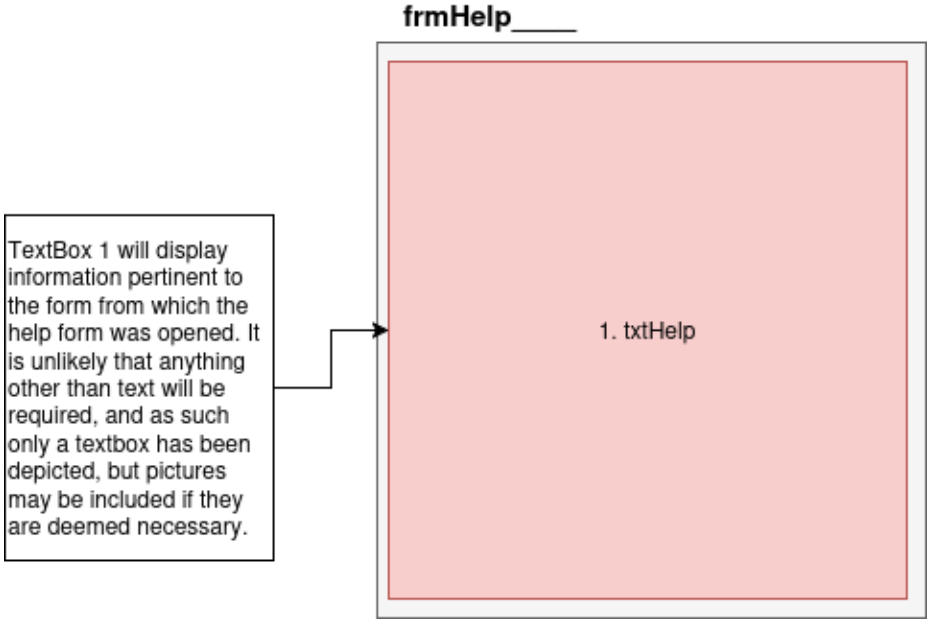
Validation and error-trapping

- If statements will ensure that if the log file does not exist in the expected location, a blank list is returned rather than attempting to load the nonexistent file
- Try-catch statements will be used during the parsing of logs to ensure that malformed logs are skipped
- If statements will be used during the parsing of logs to ensure that they are of the expected format
- Try-catch statements will be used while parsing log data such as dates and log types to ensure invalid values are ignored
- Default values will be used in switch statements to ensure that unexpected values are properly handled

User requirements met: 2, 15, 25

frmHelp____

Drawing



Explanation

Accessible from every form bar frmStartingMainMenuForm and frmLogin, the help forms will provide the user with information regarding the usage of the appropriate part of the system. A key focus will be any behaviours of the system that would not be intuitive. For example, the password field in frmStaffData being empty may not make sense to a first-time user, but the help form would explain this and provide valid reasoning as to why this is the case. Help will be predominantly textual, though images may be provided and are to be expected for the rather visual frmServiceAppointments.

Pseudocode and diagrams

There is no functionality for which high-level representations can be created

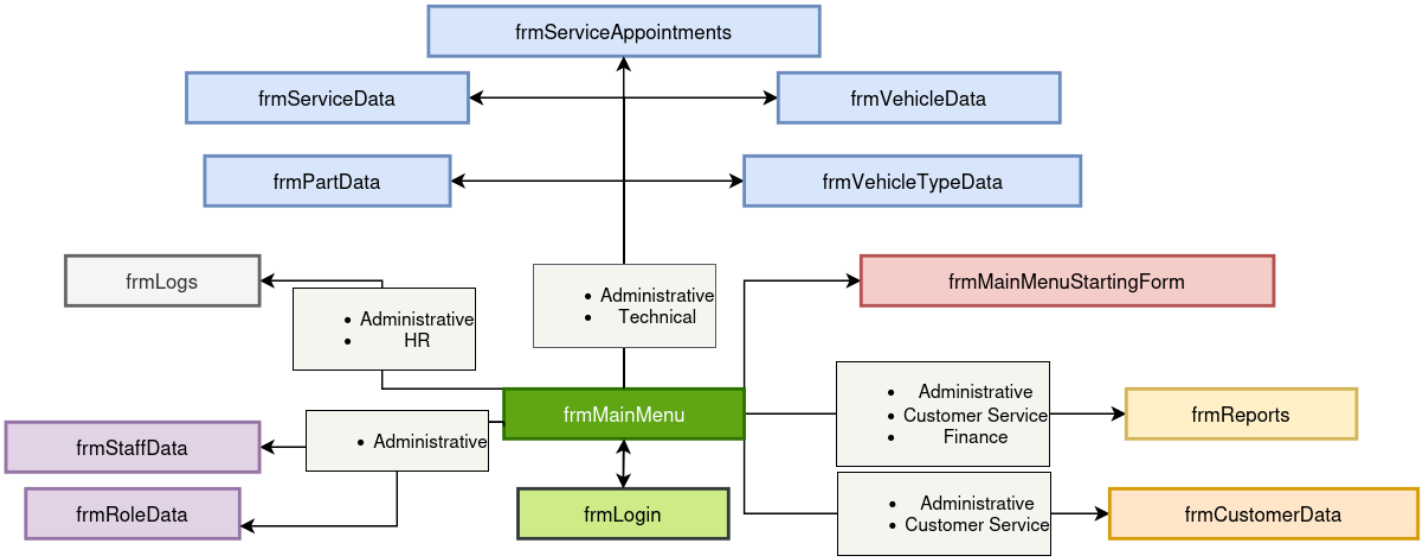
Validation and error-trapping

Forms of this type will not have dynamic data input or output, and thus require no validation.

User requirements met: 26

Program flow

The following is a diagram representation of the navigation flow within the program: the forms available for a user to access from other forms through click events. Attached to each flow arrow - bar the one connecting frmMainMenu to frmMainMenuStartingForm - is a list of role sectors which are able to access the respective forms.



All buttons on frmMainMenu will be disabled by default. Upon loading the form, the sector of the logged-in user's role will be fetched. Using this information, the appropriate navigation buttons will be enabled. As previously mentioned, forms will not be opened separately to frmMainMenu, but will instead be instantiated and inserted into a panel within frmMainMenu by clearing all of its controls, docking it and bringing it to the front. Ultimately, this reduces complexity when using the program and improves its continuity.

Omitted for clarity are the help forms present in every form except frmMainMenuStartingForm and frmLogin. Only one form can be open in the foreground at any time to prevent any unexpected behaviour occurring - with this usually being frmMainMenu - but help forms will open in conjunction with the form they belong to to assist the user with that particular form.

On logging out, all of the previous user's information is cleared from memory, ensuring that no states are carried over, thus allowing a new user to log in and use the system without any bugs.

Normalisation

As previously discussed, *Doherty Cars* already have their own (albeit ineffective) systems in place to record appointments, and as such they have their own representations of the entities pertaining to their business, forming the basis of the new database system. However, upon inspection of an old appointment sheet, it was identified that an appointment had nested repeating groups: an appointment has many services and a service has many parts. This structure introduces redundancy and complicates data management, making it unsuitable for a robust, scalable solution.

To address these issues, normalisation was applied to reorganise the data, minimising redundancy, improving data integrity, and simplifying the eventual implementation of the database system, ensuring a more elegant and maintainable design.

0NF

This represents the way the appointments were originally stored. The nested repeating groups enclosed by the curly brackets would compromise any relational database system attempting to directly implement this representation.

ServiceAppointments(**AppointmentID,** AppointmentStart, AppointmentEnd, Paid, Completed, Discounted, StaffID, StaffName, StaffUsername, StaffPassword, RoleID, RoleName, RoleAuthorisationLevel, RoleSector, VehicleID, VehicleTypeID, VehicleTypeName, VehicleComplexity, Model, Make, Year, VIN, LicensePlate, Condition, Available, Inspected, VehicleBuyPrice, VehicleSellPrice, VehicleStorageLocation, CustomerID, CustomerName, CustomerEmail, CustomerPhoneNumber, {ServiceID, ServiceName, ServiceDescription, ServiceDurationHours, ServicePrice {PartID, PartName, PartBuyPrice, PartSellPrice, PartQuantityInStock, PartQuantityUsed}})

1NF

With repeating groups removed through the primary-foreign key mechanism, the representation is now viable. Still, partial dependencies increase data redundancy.

ServiceAppointments(**AppointmentID,** AppointmentStart, AppointmentEnd, Paid, Completed, Discounted, StaffID, StaffFirstname, StaffSurname, StaffUsername, StaffPassword, RoleID, RoleName, RoleAuthorisationLevel, RoleSector, VehicleID, VehicleTypeID, VehicleTypeName, VehicleComplexity, Model, Make, Year, VIN, LicensePlate, Condition, Available, Inspected, VehicleBuyPrice, VehicleSellPrice, VehicleStorageLocation, CustomerID, CustomerFirstname, CustomerSurname, CustomerEmail, CustomerPhoneNumber)

ServiceAppointmentsServices(**AppointmentID*, ServiceID,** ServiceName, ServiceDescription, ServiceDurationHours, ServicePrice)

ServiceAppointmentsServicesParts(**AppointmentID*, ServiceID*, PartID,** PartName, PartBuyPrice, PartSellPrice, PartQuantityInStock, PartQuantityUsed)

2NF

With partial dependencies removed, the representation has less redundancy concerning appointments. However, transitive dependencies remain an issue.

ServiceAppointments(**AppointmentID,** AppointmentStart, AppointmentEnd, Paid, Completed, Discounted, StaffID, StaffFirstname, StaffSurname, StaffUsername, StaffPassword, RoleID, RoleName, RoleAuthorisationLevel, RoleSector, VehicleID, VehicleTypeID, VehicleTypeName, VehicleComplexity, Model, Year, VIN, LicensePlate, Condition, Available, Inspected, VehicleBuyPrice, VehicleSellPrice, VehicleStorageLocation, CustomerID, CustomerFirstname, CustomerSurname, CustomerEmail, CustomerPhoneNumber)

ServiceAppointmentsServices(**AppointmentID*, ServiceID***)

Services(**ServiceID**, Name, Description, DurationHours, Price)

ServiceAppointmentsServicesParts(**AppointmentID*, ServiceID*, PartID***)

Parts(**PartID,** Name, BuyPrice, SellPrice, Quantity)

3NF

Now in 3NF, the model is fit for implementation.

ServiceAppointments(**AppointmentID,** AppointmentStart, AppointmentEnd, Paid, Completed, Discounted, StaffID*, VehicleID*)

ServiceAppointmentsServices(**AppointmentID*, ServiceID***)

ServiceAppointmentsServicesParts(**AppointmentID*, ServiceID*, PartID*,** Quantity)

Parts(**PartID,** Name, BuyPrice, SellPrice, Quantity)

Staff(**StaffID,** RoleID*, Firstname, Surname, Username, Password)

Roles(**RoleID**, Name, AuthorisationLevel, Sector)

Services(**ServiceID,** Name, Description, DurationHours, Price)

Vehicles(**VehicleID**, CustomerID*, VehicleTypeID*, Model, Make, Year, VIN, LicensePlate, Condition, Available, Inspected, BuyPrice, SellPrice, StorageLocation)

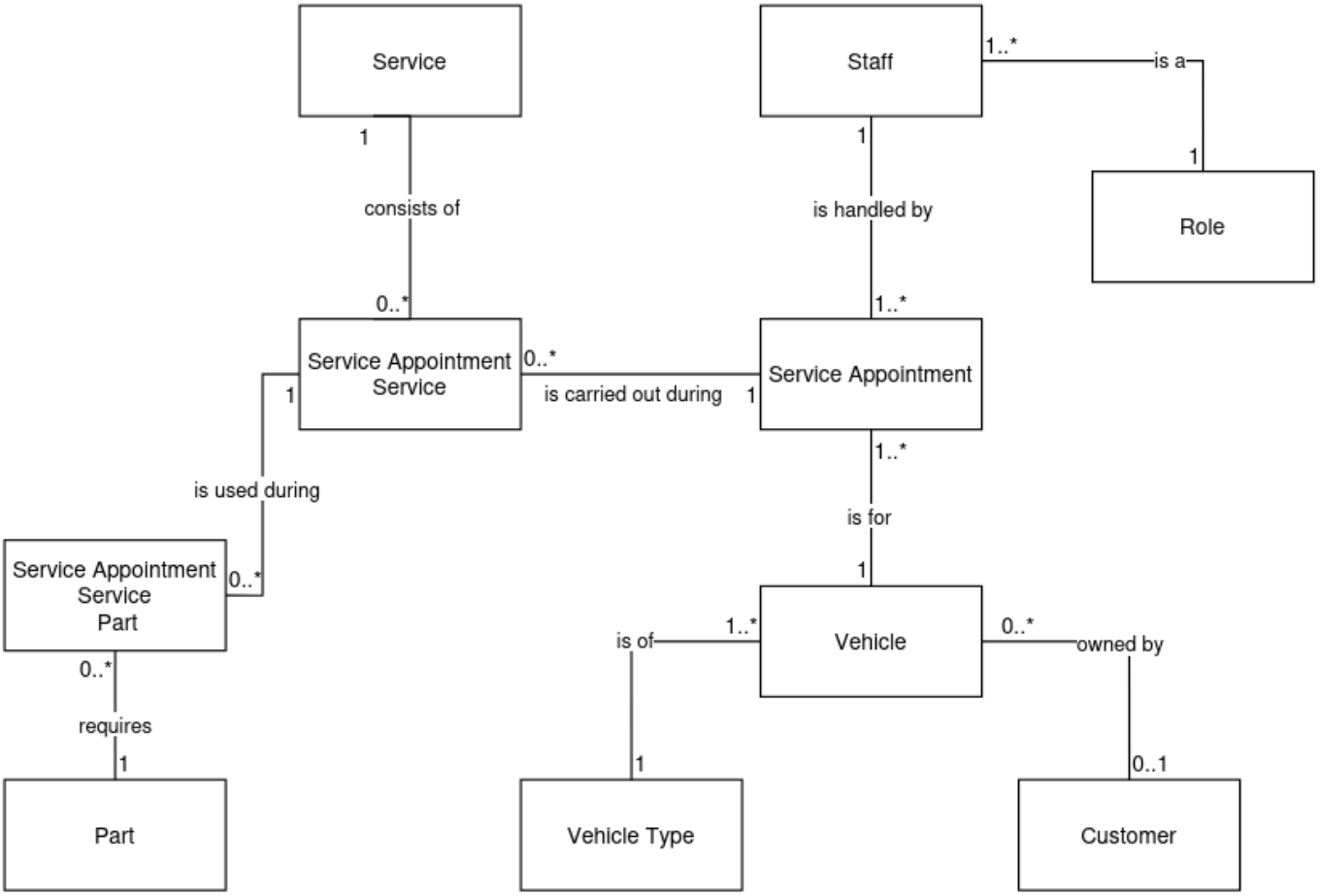
VehicleTypes(**VehicleTypeID**, Name, Complexity)

Customers(**CustomerID**, Firstname, Surname, Email, PhoneNumber)

Note that each table will have a DateDeleted field for soft deletes which has not been included in the normalisation

ER diagram

With the model now normalised, an ER diagram would both visually confirm that redundancy has been removed and directly guide the creation of the database.



Note that the optionality of certain foreign keys can be represented through cardinality in an ER diagram, unlike during the normalisation process. Particularly, a vehicle does not necessarily need to be owned by a customer (such as cars held in stock), and nor does a customer need to own a car (such as if they are a potential buyer). Furthermore, an appointment does not need a service (for example, in the event of an inspection), and nor does an appointment's service necessarily need parts.

Regardless, it is evident that the entities in the system and their relationships have been rigorously and correctly defined, and the model is now fit for implementation within a relational database.

Final database design

To comply with the user requirements and ensure soft deletes, a “DateDeleted” field has been added. Upon inserting a record, this will be set to NULL, indicating that the record exists on the system. However, once the record is deleted, this date will be set to the exact time of deletion, ensuring that the system can identify which records to ignore but that all information can be restored when required.

Customers

Represents clients of the business; any potential or former buyers. External validation will ensure that any emails inserted into the table follow the expected format “name@domain.com”

Field Name	Type of Key	Data Type	NULL/NOT NULL	Constraints
CustomerID	PK	INT	NOT NULL	IDENTITY(1,1)
Firstname		NVARCHAR(50)	NOT NULL	
Surname		NVARCHAR(50)	NOT NULL	
Email		NVARCHAR(50)	NULL	
DateDeleted		DATETIME	NULL	DEFAULT NULL
PhoneNumber		VARCHAR(15)	NULL	

Parts

Represents parts used when carrying out a service. No validation will be implemented to ensure that parts are sold at a profit in the event the business wants a loss leader.

Field Name	Type of Key	Data Type	NULL/NOT NULL	Constraints
PartID	PK	INT	NOT NULL	IDENTITY(1,1)
Name		NVARCHAR(50)	NOT NULL	
BuyPrice		MONEY	NOT NULL	
SellPrice		MONEY	NOT NULL	
Quantity		INT	NOT NULL	
DateDeleted		DATETIME	NULL	DEFAULT NULL

ServiceAppointments

Represents appointments that are to be carried out by a mechanic on a particular car.

Field Name	Type of Key	Data Type	NULL/NOT NULL	Constraints
AppointmentID	PK	INT	NOT NULL	IDENTITY(1,1)
VehicleID	FK	INT	NOT NULL	REFERENCES Vehicles(VehicleID)
StaffID	FK	INT	NOT NULL	REFERENCES Staff(StaffID)
AppointmentStart		DATETIME	NOT NULL	
AppointmentEnd		DATETIME	NOT NULL	
Paid		BIT	NOT NULL	DEFAULT 0
Completed		BIT	NOT NULL	DEFAULT 0
Discounted		BIT	NOT NULL	DEFAULT 0
DateDeleted		DATETIME	NULL	DEFAULT NULL

ServiceAppointmentsServices

Represents the services that are carried out during a particular appointment, accommodated using a composite key made up of foreign keys.

Field Name	Type of Key	Data Type	NULL/NOT NULL	Constraints
AppointmentID	PK, FK	INT	NOT NULL	REFERENCES ServiceAppointments(AppointmentID)
ServiceID	PK, FK	INT	NOT NULL	REFERENCES Services(ServiceID)
DateDeleted		DATETIME	NULL	DEFAULT NULL

ServiceAppointmentsServicesParts

Represents the parts that are to be used during a particular instance of a service during a particular appointment. Accommodated using a triple composite key of foreign keys.

Field Name	Type of Key	Data Type	NULL/NOT NULL	Constraints
AppointmentID	PK, FK	INT	NOT NULL	REFERENCES ServiceAppointmentsServices(AppointmentID, ServiceID)
ServiceID	PK, FK	INT	NOT NULL	REFERENCES ServiceAppointmentsServices(AppointmentID, ServiceID)
PartID	PK, FK	INT	NOT NULL	REFERENCES Parts(PartID)
Quantity		INT	NOT NULL	
DateDeleted		DATETIME	NULL	DEFAULT NULL

Services

Represents the services that can be carried out on vehicles during appointments. No limit is set on the duration, as it will change depending on the vehicle's complexity during appointment length calculation, and rigorous checks will be performed to ensure appointments are of a valid length.

Field Name	Type of Key	Data Type	NULL/NOT NULL	Constraints
ServiceID	PK	INT	NOT NULL	IDENTITY(1,1)
Name		NVARCHAR(50)	NOT NULL	
Description		NVARCHAR(500)	NOT NULL	
DurationHours		INT	NOT NULL	
Price		DECIMAL(10,2)	NOT NULL	
DateDeleted		DATETIME	NULL	DEFAULT NULL

Staff

Represents a member of staff who is able to interact with the system. As previously discussed, the password field will contain a hash as opposed to the plain text of a password.

Field Name	Type of Key	Data Type	NULL/NOT NULL	Constraints
StaffID	PK	INT	NOT NULL	IDENTITY(1,1)
Firstname		NVARCHAR(50)	NOT NULL	
Surname		NVARCHAR(50)	NOT NULL	
Username		NVARCHAR(50)	NOT NULL	
Password		NVARCHAR(255)	NOT NULL	
RoleID	FK	INT	NOT NULL	REFERENCES Roles(RoleID), DEFAULT 1
DateDeleted		DATETIME	NULL	DEFAULT NULL

Vehicles

Represents vehicles both stored by the dealership and sold to customers. In addition to the VehicleID, the VIN field will serve as a unique identifier of each vehicle which translates to the physical world, ensuring that the correct vehicles are modified in the database in the event of identical cars.

Field Name	Type of Key	Data Type	NULL/ NOT NULL	Constraints
VehicleID	PK	INT	NOT NULL	IDENTITY(1,1)
VehicleTypeID	FK	INT	NOT NULL	REFERENCES VehicleTypes(VehicleTypeID)
CustomerID	FK	INT	NULL	REFERENCES Customers(CustomerID)
Make		NVARCHAR(50)	NOT NULL	

Model		NVARCHAR(50)	NOT NULL	
Year		INT	NULL	
VIN		NVARCHAR(50)	NOT NULL	UNIQUE
LicensePlate		NVARCHAR(20)	NULL	
Condition		NVARCHAR(50)	NOT NULL	
Available		BIT	NOT NULL	
Inspected		BIT	NOT NULL	
BuyPrice		DECIMAL(10,2)	NULL	
SellPrice		DECIMAL(10,2)	NULL	
StorageLocation		NVARCHAR(50)	NULL	
DateDeleted		DATETIME	NULL	DEFAULT NULL

VehicleTypes

Represents the type or build of a vehicle, determining the time taken for it to be serviced. This is determined by the complexity field, which serves as a scalar multiplier for the base time taken for an appointment when considering its services.

Field Name	Type of Key	Data Type	NULL/NOT NULL	Constraints
VehicleTypeID	PK	INT	NOT NULL	IDENTITY(1,1)
Name		NVARCHAR(50)	NOT NULL	
Complexity		DECIMAL(2,1)	NOT NULL	DEFAULT 1
DateDeleted		DATETIME	NULL	DEFAULT NULL

Data requirements

For the system to properly interact with the database, entities must also be abstracted and implemented as classes and objects which can be instantiated during the loading of data. Seen below are high-level descriptions of the classes that will exist on the system.

ComboBoxItem (ComboBoxItem.cs)

Will be added to ComboBox controls as an Item when displaying data that requires a human-readable label to be understandable but an underlying identifier to be processable.

Property Name	Data Type	Description
DisplayValue	string	The text seen by the user in the ComboBox dropdown, returned when running the .ToString() method
HiddenValue	string	The underlying value associated with the item. Typically an ID or a key, this will be used internally for form logic or database operations, but not explicitly shown to the user.

CurrentUser (CurrentUser.cs)

Will be used to statically store the user who is currently logged in to ensure that their data can be easily retrieved, particularly when attempting actions that require authorisation or checking if they are authenticated for a certain form.

Property Name	Data Type	Description
DisplayValue	string	The text seen by the user in the ComboBox dropdown, returned when running the .ToString() method

Customer (Customer.cs)

Represents a record in the “Customers” table.

Property Name	Data Type	Description
CustomerID	int	The primary key field of the loaded record in the Customers table i.e. the record’s unique identifier
Firstname	string	The customer’s first name
Surname	string	The customer’s surname
Email	string	The customer’s email
PhoneNumber	string	The customer’s personal phone number

DBConnect (DBConnect.cs)

This class will not contain any properties, but rather will contain a static method to provide a connection string to the database, abstracting the interaction with the database and consequently preventing the need for complex code.

Logging (Logging.cs)

Encapsulates logging behaviour so that forms can statically call it to write logs to the system. In addition to the property below, it will contain both a method to write logs to the external logs file and a method to parse those logs for display in frmLogs.

Property Name	Data Type	Description
LogType	enum	Contains each type of log in the system as an enum, allowing for easy comparison and assignment when parsing and viewing logs

MechanicSchedule (MechanicSchedule.cs)

Represents a particular mechanic's schedule, allowing for easy access of all the appointments associated with them, as well as keeping record of the particular way it is represented within frmServiceAppointments

Property Name	Data Type	Description
MechanicStaffID	int	The primary key of the mechanic whose schedule is represented by the object
MechanicName	string	The mechanic's name as displayed in frmServiceAppointments
Timeblocks	List<Timeblock>	A list of all appointments stored under the mechanic's ID in ServiceAppointments converted into Timeblock objects
Index	int	The zero-indexed row in which the current mechanic's timeblocks are drawn in frmserviceAppointments

Part (Part.cs)

Represents a record in the "Parts" table

Property Name	Data Type	Description
PartID	int	The primary key of the loaded part record
Name	string	The name of the part
BuyPrice	decimal	The price at which a part is bought from the supplier
SellPrice	decimal	The price at which a part is sold to customers
Quantity	int	The number of parts of the type loaded which are in stock at the dealership

Role (Role.cs)

Represents a staff role in the “Roles” table.

Property Name	Data Type	Description
RoleID	int	The primary key of the loaded role record
Name	string	The name of the role
AuthorisationLevel	int	The extent of permissions given to the role. As discussed previously, there will be level 0 (can only read records), level 1 (can perform CRUD actions of relatively little risk) and level 2 (can perform CRUD actions of a high risk i.e. that can compromise the security of the system if maliciously used)
Sector	string	The sector of the staff role. As discussed, currently there will be administrative, technical, HR, finance and customer service

Service (Service.cs)

Represents a service in the “Services” table.

Property Name	Data Type	Description
ServiceID	int	The primary key of the loaded service record
Name	string	The name of the service
Description	string	A longer, free-form description of the purpose and details of the service
DurationHours	int	The time taken for the service in whole hours. Minutes are not considered as *Doherty Cars* outlined that mechanics are only expected to complete a few appointments per day, and as such there is little need to provide facilities to start storing appointment lengths and - by extension - service lengths by the minute.
Price	decimal	The price charged to an appointment’s customer for carrying out a service

ServiceAppointments (ServiceAppointments.cs)

Represents a service appointment as represented in the “ServiceAppointments”, “ServiceAppointmentsServices” and “ServiceAppointmentsServicesParts” tables

Property Name	Data Type	Description
AppointmentID	int	The primary key of the loaded appointment record
VehicleID	int	The primary key of the vehicle being serviced
StaffID	int	The primary key of the staff member carrying out the services
AppointmentStart	DateTime	The starting time of the appointment
AppointmentEnd	DateTime	The ending time of the appointment
Paid	bool	Whether the customer has paid for the appointment yet
Completed	bool	Whether the services have been carried out on the vehicle yet
Discounted	bool	Whether a discount will be applied to the appointment
ServiceIDsAndRequiredPartIDs	Dictionary<int, Dictionary<int,int>>	Stores the service ID as a key, and the dictionary which maps part ID to quantity as a value.

Staff (Staff.cs)

Represents a record in the “Staff” table.

Property Name	Data Type	Description
StaffID	int	The primary key of the loaded staff record
Firstname	string	The first name of the staff member
Surname	string	The last name of the staff member
Username	string	The identifier used for authenticating the staff member during the login process
Password	string	The hash of the password used to authenticate the staff member during the login process
RoleID	int	The primary key of the role associated with the staff member

Timeblock (Timeblock.cs)

Used to represent the block of time occupied by a service appointment. Note that no additional fields or methods will be provided, but the class will serve as an abstraction of an actual appointment for faster and easier processing.

Property Name	Data Type	Description
AppointmentID	int	The primary key of the appointment which the Timeblock object represents
AppointmentStart	DateTime	The time at which the Timeblock starts
AppointmentEnd	DateTime	The time at which the Timeblock ends

Validation (Validation.cs)

This class will not contain any properties, but will contain public validation methods that are deemed useful globally as opposed to the unique methods that many forms require. Due to the early stage in development, it is difficult to determine what would be required in this class at this stage.

Vehicle (Vehicle.cs)

Represents a record in the “Vehicles” table.

Property Name	Data Type	Description
VehicleID	int	The primary key of the loaded vehicle record
VehicleTypeID	int	The primary key of the vehicle’s type
CustomerID	int	The primary key of the vehicle’s owner
Make	string	The name of the vehicle’s manufacturer
Model	string	The name of the particular model that the car is
VIN	string	The vehicle identification number, used to identify the individual vehicle in the event that those with identical details are stored
LicensePlate	string	The license plate number registered to the vehicle, if it has one
Condition	string	The description of the condition in which the vehicle is considered to be
StorageLocation	string	The name of the location in the dealership where the vehicle is currently stored, if it has not been sold yet
Year	int?	The year the vehicle was manufactured, if this is known
Available	bool	Whether the vehicle is available for sale as a result of being ordered and the necessary prerequisite processes being carried out
Inspected	bool	Whether the vehicle has been inspected as part of the process carried out before a vehicle is put out for sale
BuyPrice	double?	The price at which the vehicle was bought by the dealership
SellPrice	double?	The price at which the vehicle will be or has been sold to a customer

VehicleTypes (VehicleTypes.cs)

Represents a record in the “VehicleTypes” table.

Property Name	Data Type	Description
VehicleTypeID	int	The primary key of the loaded vehicle type record
Name	string	The name of the vehicle type
Complexity	decimal	The scalar multiplier used when calculating the duration of a service appointment carried out on vehicles of the particular type

Logs (logs.txt)

Stores each log as a line in the file. Parsing and editing will be defined by the Logging class, but the file is planned to be structured as

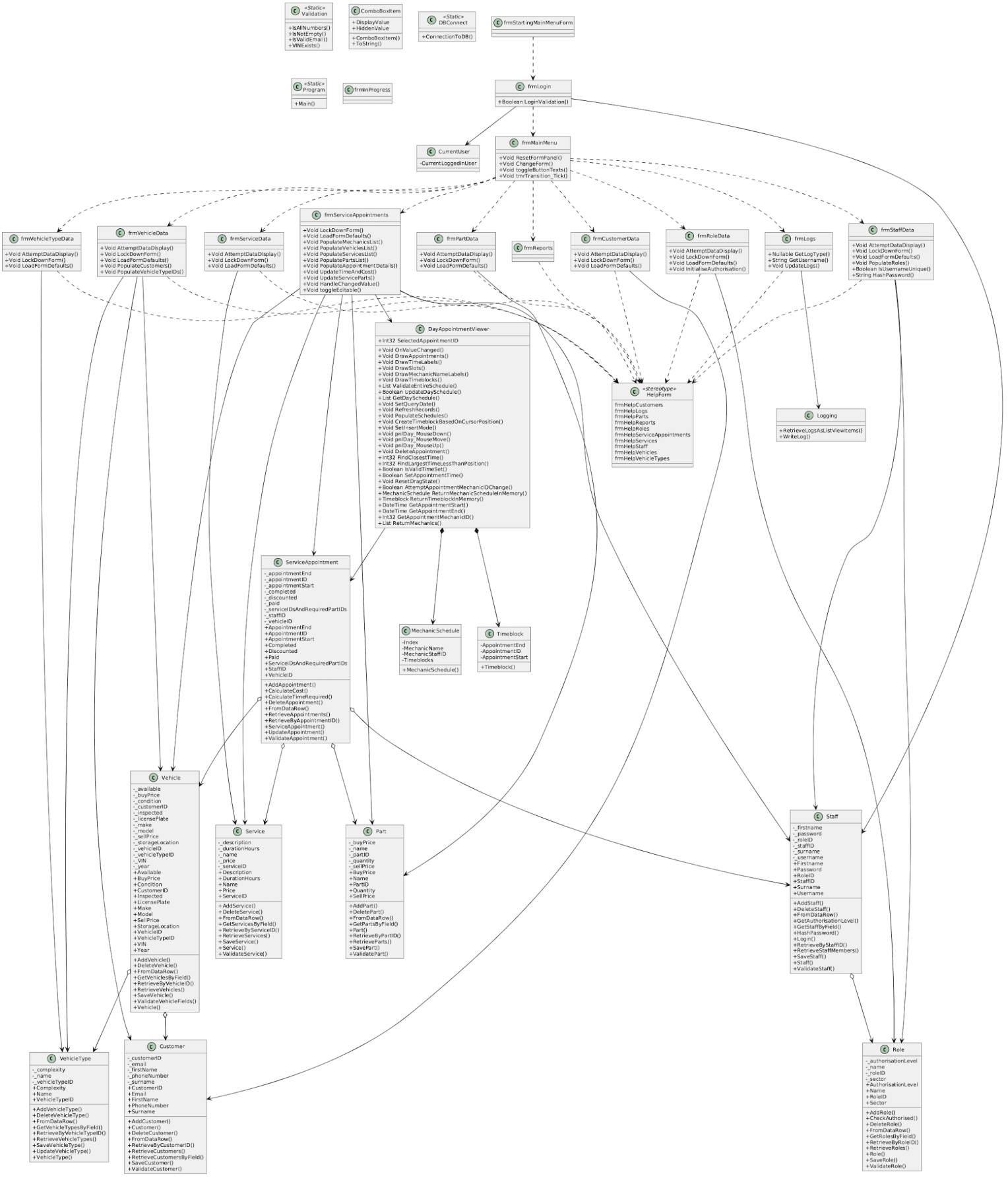
```
[DATE] - [USER] [ACTION] [LOGTYPE]
```

With:

- `[DATE]` as the date and time of the action down to seconds
- `[USER]` the username of the user on the system
- `[ACTION]` being a description of the action executed by the user, with further information depending on the type of action (e.g. update actions will contain the record ID and old name)
- `[LOGTYPE]` being the number of the type of action as defined in the Logging enum order

When parsing, each line will be parsed individually, with the date parsed using the DateTime class, the appropriate LogType value allocated to the log object depending on the number provided, and then the display text stored as the entire line minus the number at the end

Class diagram



Report specifications

rptAppointmentMonthlyProfitsByMechanic

Purpose

The report will calculate the monthly profit generated per mechanic by combining the service revenue and parts revenue of their appointments, applying discounts (which, at the moment, are at a fixed rate of 10%), and subtracting parts costs. Ultimately, the aim is to help management track the performance of the mechanics, as well as the profitability of certain executive decisions such as the implementation of discounts.

Data Sources

As the report will require mechanic, service, appointment etc. data all combined, the following tables will be combined using "LEFT JOIN" statements as some appointments may not have parts or services:

- **ServiceAppointments (sa)** - for appointments, completion status, payments, and discounts. Only completed and paid appointments will be considered.
- **ServiceAppointmentsServices (sas)** - link the services available to appointments
- **Services (s)** - for service pricing
- **ServiceAppointmentsServicesParts (sasp)** - link parts used in each service for each appointment
- **Parts (p)** - contains parts buy and sell prices
- **Staff (s)** - contains mechanic details (firstname, surname).

Content/Fields

Using the tables previously mentioned, the following fields will be generated:

- **StaffID** - unique staff identifier
- **AppointmentMonth** - the month of the appointment (formatted as YYYY-MM-01 for homogeneity)
- **Firstname** - the mechanic's first name
- **Surname** - the mechanic's surname.
- **MonthlyProfit** - the profit for that staff member in that month, calculated as (services + partsrevenue) * 0.9 if discounted - partscost

Design

The report will be in the form of a clustered bar graph. As this is directly provided through the ReportViewer package in Visual Studio, no sketch must be provided.

rptCustomerInterest

Purpose

The report will track the number of vehicles sold per vehicle type by the dealership, thus helping management understand sales distribution by vehicle type and identify which types are performing best.

Data Sources

The following tables will be inner joined when querying the database:

- **Vehicles (v)** - contains individual vehicle records
- **VehicleTypes (vt)** - contains vehicle type details (e.g., Sedan, SUV, Truck).

Content/Fields

- **VehiclesSold** - number of vehicles sold of a particular type (i.e. count of the number of vehicles of a particular type which have a customer associated with them)
- **VehicleTypeName** - name of the vehicle type

Design

The report will be in the form of a bar graph. As this is directly provided through the ReportViewer package in Visual Studio, no sketch must be provided.

rptInvoices

Purpose

This report provides a detailed summary for a single service appointment, showing services, parts, discounts, and totals, allowing for the dealership to effectively execute billing, customer communication, and internal record-keeping all through a single report.

Data Sources

Multiple tables will have to be retrieved and combined using “LEFT JOIN” statements (as some appointments may not have associated customers, services or parts) and “WITH” statements to treat any calculations such as sums and counts as objects to be queried:

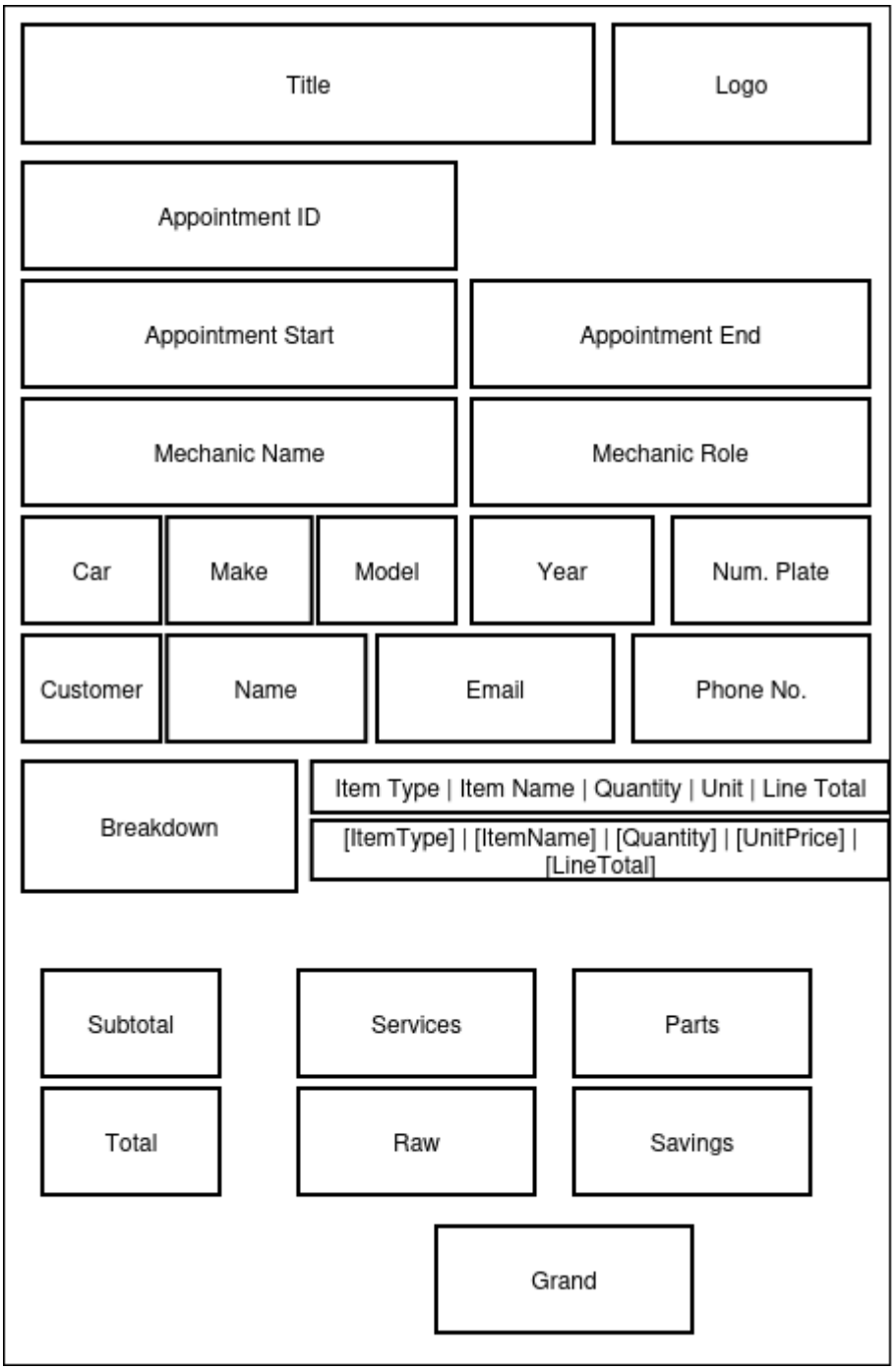
- **ServiceAppointments (sa)** - appointment details including date, staff, vehicle, and discount status
- **ServiceAppointmentsServices (sas)** - links services to appointments
- **Services (s)** - service pricing
- **ServiceAppointmentsServicesParts (sasp)** - links parts to appointments
- **Parts (p)** - part prices
- **Staff (st)** - staff details
- **Roles (r)** - staff roles
- **Vehicles (v)** - vehicle details
- **Customers (c)** - customer details

Content

- **Appointment details:** AppointmentID, AppointmentStart, AppointmentEnd, Discounted
- **Staff details:** StaffFirstname, StaffSurname, StaffRole
- **Vehicle details:** Make, Model, Year, LicensePlate
- **Customer details:** CustomerFirstname, CustomerSurname, Email, PhoneNumber
- **Financials:**
    - SubtotalServices - total cost of services
    - SubtotalParts - total cost of parts
    - RawTotal - sum of services and parts before discount
    - TotalSavings - discount applied (if any)
    - GrandTotal - final amount after discount
- **Item details:** ItemType, ItemName, Quantity, UnitPrice, LineTotal

Design

The report will be structured like a regular invoice, with fields populated with details as necessary on selecting a specific appointment



rptMissedAppointments

Purpose

This report lists overdue or missed service appointments i.e. those that were not completed and whose end time has passed, thus helping management track staff performance and prioritise follow-ups with customers.

Data Sources

The following tables must be inner joined:

- **ServiceAppointments (sa)** - appointment details including completion status and start/end times
- **Staff (s)** - staff details (firstname, surname)
- **Vehicles (v)** - vehicle make and model

Content

- **AppointmentID**
- **Mechanic** - full name of the assigned staff member (concatenated Firstname + Surname)
- **Car** - vehicle make and model (concatenated Make + Model)
- **AppointmentStart** - start time/date of the appointment

Design

This report will be implemented as a basic table, so no additional design is necessary.

rptVehicleStock

Purpose

This report shows the number of vehicles currently in stock (unsold) for each vehicle type, allowing management to monitor inventory and identify which vehicle types may need promotion.

Data Sources

The following tables will be inner joined:

- **Vehicles (v)** - records of all vehicles
- **VehicleTypes (vt)** - vehicle type details (e.g., Sedan, SUV, Truck).

Content

- **VehicleCount** - number of unsold vehicles per type
- **VehicleTypeName** - name of the vehicle type

Design

This will be implemented as a bar chart, so no further design is necessary.

---

rptVehicleTypeProfits

Purpose

This report calculates total profit earned per vehicle type by subtracting the purchase price from the selling price for sold vehicles, helping management identify which vehicle types generate the most profit.

Data Sources

The following tables will be inner joined:

- **Vehicles (v)** - vehicle records
- **VehicleTypes (vt)** - vehicle type details (e.g., Sedan, SUV, Truck)

Content

- **TotalProfits** - total profit per vehicle type
- **VehicleType** - name of the vehicle type

Design

This report will be implemented as a pie chart, so no further design is necessary.

Testing

Testing

Purpose of testing

Thorough testing is essential in improving the system's alignment with user requirements by identifying missing functionality and eliminating errors in the source code. A well-tested program ensures robustness, efficiency and effectiveness, ultimately fostering a more seamless and reliable user experience. This means easy access to precise and trustworthy information for all staff members, enhancing decision-making and efficiency.

Types of testing undergone

1. **White box testing** – analysing the internal logic of the program to identify flaws in program flow
2. **Unit testing** – verifying that individual modules function correctly in isolation, catching defects early in development
3. **Integration testing** – ensuring that groups of modules interact correctly, communicating effectively
4. **System testing** – evaluating the overall performance and reliability of the software under real-world conditions, ensuring it meets user requirements

Types of test data used

1. **Null** - this will ensure that the system can cope when no data is entered or selected
2. **Valid data** - this will ensure that the system operates as expected with normal data
3. **Invalid data** - ensures that the system can process unexpected or unwanted data and does not crash when such is entered
4. **Extreme data -** used to test that the system can cope with values on the boundary of what is considered "valid", such as largest and smallest numbers in a validation check

Items to be tested

1. frmLogin 2. frmMainMenu 3. frmStartingMainMenuForm 4. frmCustomerData 5. frmStaffData 6. frmRoleData 7. frmVehicleData	8. frmVehicleTypeData 9. frmPartData 10. frmServiceData 11. frmServiceAppointments 12. frmLogs 13. Logging 14. frmReports

frmLogin

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
1.1.1	btnLogin	Ensures that, upon pressing the button, the input credentials are processed and validated correctly and the username and hashed password attempted to be matched to a valid staff record. If a match is found, the appropriate staff member is to be loaded into memory and the main menu opened, and if any errors occur an appropriate message is shown to the user.	Valid: patdoh Password1.	The user's details will be stored in memory, frmLogin will be closed, and frmMainMenu opened.	The user's details are stored in memory - as evidenced by frmStartingMainMenuForm updating - frmLogin is closed, and frmMainMenu is opened.	Pass	1, 2
1.1.2			Valid (enter key pressed): patdoh Password1.			Pass	1, 2
1.1.3			Valid (username randomly capitalised): pAtDoh Password1.			Pass	1, 2
1.1.4			Valid (username has leading and trailing spaces): patdoh Password1.			Pass	1, 2
1.1.5			Valid (password has leading and trailing spaces): patdoh Password1.			Pass	1, 2
1.1.6			Valid (logging out and logging back in - same account): patdoh Password1. patdoh Password1.			Pass	1, 2, 3

1.1.7			Valid (logging out and logging back in - different account): patdoh Password1. ethanh1 h3110w0r1d			Pass	1, 2, 3
1.1.8			Invalid (incorrect password): patdoh password	A message box will be displayed with the message "Incorrect username or password, please try again.", the fields will be cleared, and the username field focused	A message box is displayed with the message "Incorrect username or password, please try again.", fields cleared, and username field focused	Pass	1, 23
1.1.9			Invalid (password wrongly uppercased): patdoh PaSSword1.			Pass	1, 23
1.1.10			Invalid (mismatched credentials): ethanh1 Password1.			Pass	1, 23
1.1.11			Invalid (inputting encoded password hash): patdoh wwmgrLMQlq3GDymT1xDuBjL1ExxO+we/FdbH1X1hMh33eLKs			Pass	1, 23
1.1.12			Null: *empty* *empty*	A message box will be displayed with the text "Sorry, you cannot leave username and password blank." No further action will be taken.	A message box is displayed with the text "Sorry, you cannot leave username and password blank." No further action is taken.	Pass	1, 23

1.1.13			Null (no password): patdoh *empty*	A message box will be displayed with the text "Sorry, you cannot leave password blank." No further action will be taken.	A message box is displayed with the text "Password must be between 8 and 128 characters." No further action is taken.	Fail	1, 23
1.1.14			Null (no username): *empty* Password1.	A message box will be displayed with the text "Sorry, you cannot leave username blank." No further action will be taken.	A message box is displayed with the text "Username can only contain letters, numbers, underscores and full stops." No further action is taken.	Fail	1, 23
1.1.15			Invalid (username is less than 3 characters) pa Password1.	A message box will be displayed with the text "Username must be between 3 and 50 characters." No further action will be taken.	A message box is displayed with the text "Username must be between 3 and 50 characters." No further action is taken.	Pass	1, 23
1.1.16			Invalid (username is more than 50 characters): patdohpatdohpatdohpatdohpatdohpatdohpatdohpatdohpatdohpatdohpatdohpatdoh Password1.			Pass	1, 23
1.1.17			Extreme (username is 3 characters): eth password	The user's details will be stored in memory, frmLogin will be closed, and frmMainMenu opened.	The user's details are stored in memory - as evidenced by frmStartingMainMenuForm updating - frmLogin is closed, and frmMainMenu is opened.	Pass	1, 2
1.1.18			Extreme (username is 50 characters): ethethethethethethethethethethethethethethethethethe			Pass	1, 2

			ethethethethet password				
1.1.19			Invalid (password less than 8 characters): patdoh pass	A message box will be displayed with the text "Password must be between 8 and 128 characters." No further action will be taken.	A message box is displayed with the text "Password must be between 8 and 128 characters." No further action is taken.	Pass	1, 23
1.1.20			Invalid (password more than 128 characters): patdoh passwordpassword passwordpassword passwordpassword passwordpassword passwordpassword passwordpassword passwordpassword passwordpassword password			Pass	1, 23
1.1.21			Extreme (password is 8 characters): eth password	The user's details will be stored in memory, frmLogin will be closed, and frmMainMenu opened.	The user's details are stored in memory - as evidenced by frmStartingMainMenuForm updating - frmLogin is closed, and frmMainMenu is opened.	Pass	1, 2

1.1.22			Extreme (password is 128 characters): eth passwordpassword passwordpassword passwordpassword passwordpassword passwordpassword passwordpassword passwordpassword passwordpassword			Pass	1, 2
1.1.23			Invalid (username does not contain only alphanumeric characters, underscores or fullstops) p£tdoh Password1.	A message box will be displayed with the text "Username can only contain letters, numbers, underscores and full stops." No further action will be taken.	A message box is displayed with the text "Username can only contain letters, numbers, underscores and full stops." No further action is taken.	Pass	1, 23
1.1.24			Null (Staff table missing): patdoh Password1.	An error message will be displayed as a result of being unable to load user details, the fields in the form will be cleared, and the username field focused	A message box is displayed with the message "Incorrect username or password, please try again.", fields cleared, and username field focused	Fail	1, 23
1.2.1	chkToggleHide	Ensures that the checkbox toggles obfuscation of the password, showing it as plain text or replacing it with asterisks as appropriate.	Click event (while password obfuscated)	txtPassword's text will be revealed as plain text	txtPassword's text is revealed as plain text	Pass	1
1.2.2			Click event (while password unobfuscated)	txtPassword's text will be hidden, with each character replaced with an asterisk	txtPassword's text is hidden, with each character replaced with an asterisk	Pass	1

frmMainMenu

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
2.1.1	frmMainMenu	Ensures that, upon loading, the form ensures that the user has accessed the form through a valid login action. Then, it ensures that the user is only able to access forms that they have been allocated.	Load event with valid user patdoh	The form will successfully load, with buttons dynamically updating based on the user's sector	The form successfully loads, with buttons updating based on the user's sector	Pass	2, 4, 5
2.1.2			Load event with null user *No user loaded into memory*	A message box will be shown with the message "Session expired. Please log in again." before the main menu is closed and the login form opened.	A message box is shown with the message "Session expired. Please log in again." before the main menu is closed and the login form opened.	Pass	5, 24
2.1.3			Load event with valid user but unexpected sector Administrated	A message box will be shown with the message "Role configuration error.", with no buttons authorised. No further action will be taken.	A message box is shown with the message "Role configuration error.", with no buttons authorised. No further action is taken.	Pass	2, 4, 5, 23
2.1.4			Load event with valid user with sector None	Form will load with no buttons authorised. No further action will be taken.	Form loads with no buttons authorised. No further action is taken.	Pass	2, 4, 5
2.1.5			Load event with valid user with sector Administrative	Form will load with all buttons authorised. No further action will be taken.	Form loads with all buttons authorised. No further action is taken.	Pass	2, 4, 5
2.1.6			Load event with valid user with sector Customer Service	Form will load with customer data, reports and enquiries buttons authorised. No further action will be taken.	Form loads with customer data, reports and enquiries buttons authorised. No further action is taken.	Pass	2, 4, 5

2.1.7			Load event with valid user with sector Technical	Form will load with service appointments, services, vehicles, parts, vehicle types, test drive appointments and point of sale buttons authorised. No further action will be taken.	Form loads with service appointments, services, vehicles, parts, vehicle types, test drive appointments and point of sale buttons authorised. No further action is taken.	Pass	2, 4, 5
2.1.8			Load event with valid user with sector Finance	Form will load with reports and point of sale buttons authorised. No further action will be taken.	Form loads with reports and point of sale buttons authorised. No further action is taken.	Pass	2, 4, 5
2.1.9			Load event with valid user with sector HR	Form will load with logs and enquiries buttons authorised. No further action will be taken.	Form loads with logs and enquiries buttons authorised. No further action is taken.	Pass	2, 4, 5
2.2.1	pnlButtons	Ensures that, upon a hover event, the panel is animated as expected, with it expanding when hovered over and collapsing when the mouse is moved away.	Mouse hover event over panel while panel in initial state	The panel and buttons contained within the panel will horizontally expand to the right until they reach a maximum width, at which point the descriptory text of the buttons within the panel will appear	Panel and buttons contained within panel horizontally expand to the right until they reach a maximum width, at which point the descriptory text of the buttons within the panel appears	Pass	
2.2.2			Mouse hover event over buttons while panel in initial state			Pass	
2.2.3			Mouse moved out of panel while panel fully expanded	The descriptory text of the buttons in the panel will disappear, and the panel and buttons contained within the panel will horizontally collapse to the left until they reach a minimum width	The descriptory text of the buttons in the panel disappears, and the panel and buttons contained within the panel horizontally collapse to the left until they reach a minimum width	Pass	
2.2.4			Mouse moved out of buttons while panel fully expanded			Pass	

2.2.5			Mouse moved out of panel/buttons while panel is expanding	The panel and buttons contained within the panel will stop expanding and immediately begin collapsing horizontally to the left until they reach a minimum width, with button text being hidden if necessary	The panel and buttons contained within the panel stop expanding and immediately begin collapsing horizontally to the left until they reach a minimum width, with button text being hidden if necessary	Pass	
2.2.6			Mouse hover event over panel/buttons while panel is collapsing	The panel and buttons contained within the panel will stop collapsing and immediately begin expanding horizontally to the right until they reach a maximum width, at which point the buttons' descriptory text will appear	The panel and buttons contained within the panel stop collapsing and immediately begin expanding horizontally to the right until they reach a maximum width, at which point the buttons' descriptory text will appear	Pass	
2.3.1	btnStaffData	Ensures that the button allows the user to access frmStaffData if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmStaffData opened in its place.	The form in the viewing panel is closed, and frmStaffData opened in its place.	Pass	2, 4, 5, 8
2.3.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 8
2.4.1	btnCustomer Data	Ensures that the button allows the user to access frmCustomerData if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmCustomerData opened in its place.	The form in the viewing panel is closed, and frmCustomerData opened in its place.	Pass	2, 4, 5, 8
2.4.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 8
2.5.1	btnVehicleData	Ensures that the button allows the user to access frmVehicleData if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmVehicleData opened in its place.	The form in the viewing panel is closed, and frmVehicleData opened in its place.	Pass	2, 4, 5, 8

2.5.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 8
2.6.1	btnVehicleTypeData	Ensures that the button allows the user to access frmVehicleTypeData if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmVehicleTypeData opened in its place.	The form in the viewing panel is closed, and frmVehicleTypeData opened in its place.	Pass	2, 4, 5, 8
2.6.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 8
2.7.1	btnServiceAppointments	Ensures that the button allows the user to access frmServiceAppointments if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmServiceAppointments opened in its place.	The form in the viewing panel is closed, and frmServiceAppointments opened in its place.	Pass	2, 4, 5, 8
2.7.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 8
2.8.1	btnServiceData	Ensures that the button allows the user to access frmServiceData if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmServiceData opened in its place.	The form in the viewing panel is closed, and frmServiceData opened in its place.	Pass	2, 4, 5, 8
2.8.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 8
2.9.1	btnPartData	Ensures that the button allows the user to access frmPartData if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmPartData opened in its place.	The form in the viewing panel is closed, and frmPartData opened in its place.	Pass	2, 4, 5, 8
2.9.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 8
2.10.1	btnRoleData	Ensures that the button allows the user to access frmRoleData if	Click event while authorised	The form in the viewing panel will be closed, and frmRoleData opened in its place.	The form in the viewing panel is closed, and frmRoleData opened in its place.	Pass	2, 4, 5, 8

2.10.2		permitted	Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 8
2.11.1	btnReports	Ensures that the button allows the user to access frmReports if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmReports opened in its place.	The form in the viewing panel is closed, and frmReports opened in its place.	Pass	2, 4, 5, 21
2.11.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 21
2.12.1	btnLogs	Ensures that the button allows the user to access frmLogs if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmLogs opened in its place.	The form in the viewing panel is closed, and frmLogs opened in its place.	Pass	2, 4, 5, 25
2.12.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5, 25
2.13.1	btnEnquiries	Ensures that the button allows the user to access frmInProgress if permitted	Click event while authorised	The form in the viewing panel will be closed, and frmInProgress opened in its place.	The form in the viewing panel is closed, and frmInProgress opened in its place.	Pass	2, 4, 5
2.13.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5
2.14.1	btnTestDrive Appointments		Click event while authorised	The form in the viewing panel will be closed, and frmInProgress opened in its place.	The form in the viewing panel is closed, and frmInProgress opened in its place.	Pass	2, 4, 5
2.14.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5
2.15.1	btnPointOfSale		Click event while authorised	The form in the viewing panel will be closed, and frmInProgress opened in its place.	The form in the viewing panel is closed, and frmInProgress opened in its place.	Pass	2, 4, 5
2.15.2			Click event while unauthorised	No further action will be taken.	No further action is taken.	Pass	2, 4, 5

2.16.1	btnMainMenu	Ensures that the button allows the user to access frmStartingMainMenu Form	Click event	The form in the viewing panel will be closed, and frmStartingMainMenuForm opened in its place.	The form in the viewing panel is closed, and frmStartingMainMenuForm opened in its place.	Pass	5
2.17.1	btnLogOut	Ensures that the button allows the user to access frmLogin, clearing the user's details from memory in doing so	Click event	frmMainMenu will be closed, and frmLogin displayed, with the current user's details removed from memory	frmMainMenu is closed, and frmLogin displayed, with the current user's details removed from memory	Pass	3
2.18.1	pnlFormDisplay	Ensures that the panel serves as a working interface for the rest of the forms present within the system	Interaction with panel while a form has been loaded: frmStaffData	Form will behave as if it were opened regularly, accepting key and mouse inputs	Form behaves as if it were opened regularly, accepting key and mouse inputs	Pass	5
2.18.2			Returning to a form after closing it: frmStaffData *Staff record modified* frmVehicleData frmStaffData	Reopened form will display the latest version of the data stored in the database after a change was made initially	Form displays the latest version of the data stored in the database after being reopened, reflecting the change made initially	Pass	5

frmStartingMainMenuForm

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
3.1.1	frmStartingMainMenuForm	Ensures that the form properly greets the user on loading, handling missing details gracefully	Load event with valid user: Patrick Administrator	lblWelcomeMessage will display the text “Welcome, Patrick” lblDate will display the text “It is “ followed by the current date written out in long form lblRole will display the text “You are logged in as a(n) Administrator”	lblWelcomeMessage displays the text “Welcome, Patrick” lblDate displays the text “It is “ followed by the current date written out in long form lblRole displays the text “You are logged in as a(n) Administrator”	Pass	2
3.1.2			Load event with no valid user: *No user present*	A message box will be displayed with the text "Your name could not be retrieved. Please ensure you have logged in correctly.” No further action will be taken.	A message box is displayed with the text "Your name could not be retrieved. Please ensure you have logged in correctly.” The system crashes.	Fail	2

frmCustomerData

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
4.1.1	frmCustomerData	Ensures that customer details are loaded and displayed properly on access, handling missing data gracefully	Load event	dgvCustomers will be populated with the customer records stored on the database. No further action will be taken.	dgvCustomers is populated with the customer records stored on the database. No further action is taken.	Pass	6a, 8
4.1.2			Load event with no Customers table present	A message box will be displayed with the text "Error retrieving customers: " followed by an appropriate error message, followed by another with the text "There was an error when attempting to load customer data. Please amend erroneous data in the database before proceeding." before all CRUD and search controls are disabled.	A message box is displayed with the text "Error retrieving customers: " followed by an appropriate error message, followed by another with the text "There was an error when attempting to load customer data. Please amend erroneous data in the database before proceeding." before all CRUD and search controls are disabled. The program crashes.	Fail	15, 23
4.2.1	dgvCustomers	Ensures that the user is able to select customer records for editing or deleting	Click event on a record	The customer's details are populated in the correct field, with their first name, surname, email, and phone number inserted into the correct textboxes.	The customer's details are populated in the correct field, with their first name, surname, email, and phone number inserted into the correct textboxes.	Pass	6a, 8, 11
4.2.2			Click event on a record after reset			Pass	6a, 8, 11
4.2.3			Click event on a newly added record			Pass	6a, 8, 11

4.2.4			Click updated record	The new customer's details are populated in the correct field, with their first name, surname, email, and phone number inserted into the correct textboxes.	The new customer's details are populated in the correct field, with their first name, surname, email, and phone number inserted into the correct textboxes.	Pass	6a, 8, 11
4.3.1	btnSave	Ensures that the user is able to save updated customer details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text "This action requires "Restricted write" permission, but you have "Read-only." No further action will be taken.	A message box is displayed with the text "This action requires "Restricted write" permission, but you have "Read-only." No further action is taken.	Pass	2, 7, 12, 23
4.3.2			Extreme: Click event - just permitted (user has authorisation level 1)	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" - the user is allowed to update the customer	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" - the user is allowed to update the customer	Pass	2, 7, 12, 24
4.3.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 12, 24
4.3.4			Click event - no customer chosen 1 Dennis Kleveckas dklev@gmail.com 1	A message box will be displayed with the text "Choose a customer to edit." No further action will be taken	A message box is displayed with the text "Choose a customer to edit." No further action is taken	Pass	9, 12, 23
4.3.5			Click event after a reset			Pass	9, 12, 14, 23

4.3.6			Valid (“Yes” clicked at dialog): 1 Johnn Yossarian catch22@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
4.3.7			Valid (“No” clicked at dialog): 1 Johnn Yossarian catch22@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action will be taken.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action is taken.	Pass	9, 12, 24
4.3.8			Valid (Same customer updated again): 1 Johnnn Yossarian catch22@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the latest details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the latest details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

4.3.9			Null (first name blank): 1 *empty* Yossarian catch22@pianosa.com 07221984421	A message box of errors will be displayed with the message “First name cannot be empty.”	A message box of errors is displayed with the message “First name cannot be empty.”	Pass	9, 12, 14, 23
4.3.10			Null (first name is whitespace): 1 Yossarian catch22@pianosa.com 07221984421			Pass	9, 12, 14, 23
4.3.11			Valid (first name has leading spaces): 1 John Yossarian catch22@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. The first name will have been trimmed of its whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. The first name has been trimmed of its whitespace.	Pass	9, 12, 14, 24
4.3.12			Valid (first name has trailing spaces): 1 John Yossarian catch22@pianosa.com 07221984421			Pass	9, 12, 14, 24

4.3.13			Invalid (first name longer than 50 characters): 1 JohnJohnJohn JohnJohnJohn JohnJohnJohn JohnJohnJohn John Yossarian catch22@pianosa.com 07221984421	A message box of errors will be displayed with the message "First name cannot be longer than 50 characters"	A message box of errors is displayed with the message "First name cannot be longer than 50 characters"	Pass	9, 12, 14, 23
4.3.14			Extreme (first name is 50 characters): 1 JohnJohnJohn JohnJohnJohn JohnJohnJohn JohnJohnJohn Jo Yossarian catch22@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
4.3.15			Null (surname blank): 1 John *empty* catch22@pianosa.com 07221984421	A message box of errors will be displayed with the message “Surname cannot be empty.”	A message box of errors is displayed with the message “Surname cannot be empty.”	Pass	9, 12, 14, 23

4.3.16			Null (surname is whitespace): 1 John catch22@pianosa.com 07221984421			Pass	9, 12, 14, 23
4.3.17			Invalid (surname longer than 50 characters): 1 John JohnJohnJohn JohnJohnJohn JohnJohnJohn JohnJohnJohn John catch22@pianosa.com 07221984421	A message box of errors will be displayed with the message "Surname cannot be longer than 50 characters"	A message box of errors is displayed with the message "Surname cannot be longer than 50 characters"	Pass	9, 12, 14, 23
4.3.18			Extreme (surname is 50 characters): 1 John JohnJohnJohn JohnJohnJohn JohnJohnJohn JohnJohnJohn Jo catch22@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

4.3.19			Null (email blank): 1 John Yossarian *empty* 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. The email field for the selected customer will now be blank.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. The email field for the selected customer is now blank.	Pass	9, 12, 14, 24
4.3.20			Invalid (email not properly formatted): 1 John Yossarian catch22pianosa.com 07221984421	A message box of errors will be displayed with the message “Email must be properly formatted.”	A message box of errors is displayed with the message “Email must be properly formatted.”	Pass	9, 12, 14, 23
4.3.21			Valid (email has special characters): 1 John Yossarian catch+22@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

4.3.22			Invalid (email longer than 50 characters): 1 John Yossarian catchcatchcatchcatchcatchcatchcatchcatchcatchcatchcatch@pianosa.com 07221984421	A message box of errors will be displayed with the message "Email cannot be longer than 50 characters."	A message box of errors is displayed with the message "Email cannot be longer than 50 characters."	Pass	9, 12, 14, 23
4.3.23			Extreme (email is 50 characters): 1 John Yossarian aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
4.3.24			Null (email and phone number only whitespace): 1 John Yossarian	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. Both email and phone number will be stored as null.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. Both email and phone number are stored as null.	Pass	9, 12, 14, 24

4.3.25			Null (phone number empty): 1 John Yossarian catch22@pianosa.com *empty*	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. Phone number will be stored as null.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. Phone number is stored as null.	Pass	9, 12, 14, 24
4.3.26			Invalid (phone number greater than 15 characters): 1 John Yossarian catch22@pianosa.com 11111111111111	A message box of errors will be displayed with the message “Phone number cannot be longer than 15 characters"	A message box of errors is displayed with the message “Phone number cannot be longer than 15 characters"	Pass	9, 12, 14, 23
4.3.27			Extreme (phone number is 15 characters): 1 John Yossarian catch22@pianosa.com 111111111111111	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

4.3.28			Valid (SQL injection attempt): 1 ; DROP TABLE Customers; Yossarian catch22@pianosa.com 07221984421	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. The database will not be affected.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated customer" displayed, and the form refreshed with the new data. The database is not affected.	Pass	9, 12, 14, 24
4.4.1	btnAdd	Ensures that the user is able to add new customer details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 10, 23
4.4.2			Extreme: Click event - just permitted (user has authorisation level 1)	The adding logic will be executed as expected.	The adding logic is executed as expected.	Pass	2, 7, 10, 24
4.4.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 10, 24

4.4.4			Click event after a reset *empty* *empty* *empty* *empty* *empty*	A message box of errors will be displayed with the text “ - First name cannot be empty - Surname cannot be empty “	A message box of errors is displayed with the text “ - First name cannot be empty - Surname cannot be empty “	Pass	9, 10, 14, 23
4.4.5			Valid: 1 Alex Cooper ac@proton.me 07223424421	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	Pass	9, 10, 14, 24
4.4.6			Valid (Same details added again): 1 Alex Cooper ac@proton.me 07223424421			Pass	9, 10, 14, 24
4.4.7			Null (first name blank): 1 *empty* Yossarian catch22@pianosa.com 07221984421	A message box of errors will be displayed with the message “First name cannot be empty.”	A message box of errors is displayed with the message “First name cannot be empty.”	Pass	9, 10, 14, 23

4.4.8			Null (first name is whitespace): 1 Yossarian catch22@pianosa.com 07221984421			Pass	9, 10, 14, 23
4.4.9			Valid (first name has leading spaces): 1 John Yossarian catch22@pianosa.com 07221984421	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record. The first name will have the whitespace trimmed off.	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record. The first name has the whitespace trimmed off.	Pass	9, 10, 14, 24
4.4.10			Valid (first name has trailing spaces): 1 John Yossarian catch22@pianosa.com 07221984421			Pass	9, 10, 14, 24
4.4.11			Invalid (first name longer than 50 characters): 1 JohnJohnJohn JohnJohnJohn JohnJohnJohn JohnJohnJohn	A message box of errors will be displayed with the message "First name cannot be longer than 50 characters"	A message box of errors is displayed with the message "First name cannot be longer than 50 characters"	Pass	9, 10, 14, 23

			John Yossarian catch22@pianosa.com 07221984421				
4.4.12			Extreme (first name is 50 characters): 1 JohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJohnJo Yossarian catch22@pianosa.com 07221984421	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	Pass	9, 10, 14, 24
4.4.13			Null (surname blank): 1 John *empty* catch22@pianosa.com 07221984421	A message box of errors will be displayed with the message “Surname cannot be empty.”	A message box of errors is displayed with the message “Surname cannot be empty.”	Pass	9, 10, 14, 23
4.4.14			Null (surname is whitespace): 1 John catch22@pianosa.com 07221984421			Pass	9, 10, 14, 23

4.4.15			Invalid (surname longer than 50 characters): 1 John JohnJohnJohn JohnJohnJohn JohnJohnJohn JohnJohnJohn John catch22@pianosa.com 07221984421	A message box of errors will be displayed with the message "Surname cannot be longer than 50 characters"	A message box of errors is displayed with the message "Surname cannot be longer than 50 characters"	Pass	9, 10, 14, 23
4.4.16			Extreme (surname is 50 characters): 1 John JohnJohnJohn JohnJohnJohn JohnJohnJohn JohnJohnJohn Jo catch22@pianosa.com 07221984421	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	Pass	9, 10, 14, 24
4.4.17			Null (email blank): 1 John Yossarian *empty* 07221984421	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record. The email field will be left null.	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record. The email field is left null.	Pass	9, 10, 14, 24

4.4.18			Invalid (email not properly formatted): 1 John Yossarian catch22pianosa.com 07221984421	A message box of errors will be displayed with the message “Email must be properly formatted.”	A message box of errors is displayed with the message “Email must be properly formatted.”	Pass	9, 10, 14, 23
4.4.19			Valid (email has special characters): 1 John Yossarian catch+22@pianosa.com 07221984421	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	Pass	9, 10, 14, 24
4.4.20			Invalid (email longer than 50 characters): 1 John Yossarian catchcatchcatchcatchcatchcatchcatchcatchcatchcatch@pianosa.com 07221984421	A message box of errors will be displayed with the message "Email cannot be longer than 50 characters."	A message box of errors is displayed with the message "Email cannot be longer than 50 characters."	Pass	9, 10, 14, 23

4.4.21			Extreme (email is 50 characters): 1 John Yossarian aaaa

			11				
4.4.25			Extreme (phone number is 15 characters): 1 John Yossarian catch22@pianosa.com 111111111111111	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record.	Pass	9, 10, 14, 24
4.4.26			Valid (SQL injection attempt): 1 ; DROP TABLE Customers; Yossarian catch22@pianosa.com 07221984421	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record. The database will not be affected	A message box will be displayed with the text “Customer has been added successfully”, and the table will be refreshed with the data containing the new record. The database has not been affected	Pass	9, 10, 14, 24
4.5.1	btnDelete	Ensures that the user is allowed to mark records as deleted in the database, preventing future loading and consideration	Invalid: Click event - not permitted (user has authorisation level 1)	A message box will be displayed with the text “This action requires “Unrestricted write” permission, but you have “Restricted write.” No further action will be taken.	A message box is displayed with the text “This action requires “Unrestricted write” permission, but you have “Restricted write.” No further action is taken.	Pass	2, 7, 13, 23
4.5.2			Extreme: Click event - just permitted (user has authorisation level 2)	The delete logic will be executed as expected.	The delete logic executes as expected.	Pass	2, 7, 13, 24

4.5.3			Click event - (“No” clicked at dialog) 1	No further action will be taken.	No further action is taken.	Pass	13, 24
4.5.4			Click event - (“Yes” clicked at dialog) 1	The selected user will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record.	The selected user is marked as deleted from the database, and this is indicated with the absence of the record in the table.	Pass	13, 14, 24
4.5.5			Click event - no customer selected	A message box will be displayed with the text “Select a customer to delete.” No further action will be taken.	A message box is displayed with the text “Select a customer to delete.” No further action is taken.	Pass	9, 13, 23
4.5.6			Click event - after resetting			Pass	9, 13, 23
4.5.7			Click event - customer has a vehicle associated with them Customer ID: 1 Vehicle ID: 29	The selected customer will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record. The same applies for the associated vehicle in the vehicle data form.	The selected customer is marked as deleted from the database, and this is indicated with the absence of the record in the table. The vehicle has also been removed from the vehicle data form.	Pass	13, 14, 16a, 24
4.5.8			Click event - customer has a vehicle associated with them, which itself has a service appointment	The selected customer will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record. The same applies for the	The selected customer is marked as deleted from the database, and this is indicated with the absence of the record in the table. The vehicle has also been removed from the vehicle data form, and the	Pass	13, 14, 16a, 24

			associated with it Customer ID: 6 Vehicle ID: 27 Appointment ID: 18	associated vehicle in the vehicle data form, and its associated appointment in the service appointment data form.	appointment from the service appointment data form.		
4.6.1	comboSearchField	Ensures that the user can select which field they wish to search the database against	Selected item changed from “First name” to “Surname”	Any searches on the table will be with respect to the “Surname” field	All searches are now with respect to the “Surname” field	Pass	8
4.6.2			Selected item changed back to “First name” from “Surname”	Any searches on the table will be with respect to the “First name” field	All searches are now with respect to the “First name” field	Pass	8
4.7.1	btnReset	Ensures that the user is able to clear all fields in the form and deselect the currently selected customer, preparing the form for new data entry	Click event - values had been typed in	All fields will be cleared, and any customers will be unselected	All fields have been cleared, and the selected customer has been unselected	Pass	14
4.7.2			Click event - values had been populated after selecting a customer			Pass	14
4.8.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpCustomers will be opened on top of frmCustomerData	frmHelpCustomers is opened on top of frmCustomerData	Pass	26
4.8.2			Click event away from frmHelp Customers	frmHelpCustomers will be closed	frmHelpCustomers is closed	Pass	26

4.9.1	txtSearch	Ensures that the user is able to search the database using the selected search type, search field and provided search substring	Text input after form initially loads. “Contains” and “Customer ID” appear selected 6	The table will be updated to display only records which contain the digit “6” in their ID	The table is updated to display only records which contain the digit “6” in their ID	Pass	8
4.9.2			“Exact” and “Customer ID” are selected 6	The table will be updated to display only the record with “6” as its ID	The table is updated to display only the record with “6” as its ID	Pass	8
4.9.3			Click event - btnReset is clicked while a search has been input	The table will display all records on the system, ignoring the search until it is updated again	The table displays all records on the system, ignoring the search. When the search is updated again, the table is filtered once more.	Pass	8
4.9.4			Invalid: “Exact” and “Customer ID” are selected a	The table will be empty	The table is empty	Pass	8, 9
4.9.5			Null: “Contains” and “Customer ID” are selected *empty*	The table will display all records present in the database	The table displays all records present in the database	Pass	8

frmStaffData

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
5.1.1	frmStaffData	Ensures that staff details are loaded and displayed properly on access, handling missing data gracefully	Load event	dgvStaff will be populated with the staff records stored on the database. No further action will be taken.	dgvStaff is populated with the staff records stored on the database. No further action is taken.	Pass	6b, 8
5.1.2			Load event with no Staff table present	A message box will be displayed with the text "Error retrieving staff records: " followed by an appropriate error message, followed by another with the text "There was an error attempting to load staff data. Please amend erroneous data in the database before proceeding." All controls associated with searching and data entry will be disabled. No further action will be taken.	A message box is displayed with the text "Error retrieving staff records: " followed by an appropriate error message, followed by another with the text "There was an error attempting to load staff data. Please amend erroneous data in the database before proceeding." All controls associated with searching and data entry are disabled. No further action is taken.	Pass	15, 23
5.1.3			Load event with no Roles table present	A message box will be displayed with the text "Error retrieving role records: " followed by an appropriate error message, followed by another with the text "Error loading roles." All controls associated with searching and data entry will be disabled. The staff table will be loaded for reading only. No further action will be taken.	A message box is displayed with the text "Error retrieving role records: " followed by an appropriate error message, followed by another with the text "Error loading roles." All controls associated with searching and data entry are disabled. The staff table is loaded for reading only. No further action is taken.	Pass	15, 23

5.2.1	dgvStaff	Ensures that the user is able to select staff records for editing or deleting	Click event on a record	The details for that record will be populated in the appropriate fields, with the exception of password which will remain blank	The details for that record are populated in the appropriate fields, with the password remaining blank	Pass	6b, 8, 11
5.2.2			Click event on a record after reset			Pass	6b, 8, 11
5.2.3			Click event on a newly added record			Pass	6b, 8, 11
5.2.4			Click event on an updated record	The most recent details for that record will be populated in the appropriate fields, with the exception of password which will remain blank	The most recent details for that record are populated in the appropriate fields, with the password remaining blank	Pass	6b, 8, 11
5.3.1	btnSave	Ensures that the user is able to save updated staff details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 12, 23
5.3.2			Extreme: Click event - just permitted (user has authorisation level 1)	The save logic will be executed as expected.	The save logic executes as expected.	Pass	2, 7, 12, 24
5.3.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 12, 24

5.3.4			Click event - no staff member selected 1 Patrick Doherty patdoh password 2	A message box will be displayed with the text "Choose a staff member to edit." No further action will be taken	A message box is displayed with the text "Choose a staff member to edit." No further action will be taken	Pass	9, 12, 23
5.3.5			Click event after reset			Pass	9, 12, 23
5.3.6			Valid ("Yes" clicked at dialog): 1 Patri Doherty patdoh password 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking "Yes", the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking "Yes", the new details are stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data.	Pass	9, 12, 24, 27
5.3.7			Valid (same staff updated): 1 Patric Doherty patdoh Password 3			Pass	9, 12, 24, 27

5.3.8			Valid (“No” clicked at dialog): 1 Patrick Doherty patdoh password 5	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action will be taken.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action is taken.	Pass	9, 12, 23
5.3.9			Null (first name blank) 1 *empty* Doherty patdoh password 2	A message box of errors will be displayed with the message “First name cannot be empty.”	A message box of errors is displayed with the message “First name cannot be empty.”	Pass	9, 12, 23
5.3.10			Null (first name is whitespace) 1 Doherty patdoh password 2			Pass	9, 12, 23

5.3.11			Valid (first name has leading spaces): 1 Patrick Doherty patdoh. password 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data. The first name will be trimmed of any whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated staff” displayed, and the form refreshed with the new data. The first name is trimmed of any whitespace.	Pass	9, 12, 24, 27
5.3.12			Valid (first name has trailing spaces): 1 Patrick Doherty patdoh_ password 2			Pass	9, 12, 24, 27
5.3.13			Invalid (first name longer than 50 characters): 1 PatrickPatric kPatrickPatri ckPatrickPatr ickPatrickPa Doherty patdoh password 2	A message box of errors will be displayed with the message “First name cannot be longer than 50 characters.”	A message box of errors is displayed with the message “First name cannot be longer than 50 characters.”	Pass	9, 12, 23

5.3.14			Extreme (first name is 50 characters): 1 PatrickPatrickPatrickPatrickPatrickPatrickPatrickPatrickPatrickP Doherty patdoh password 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated staff” displayed, and the form refreshed with the new data.	Pass	9, 12, 24, 27
5.3.15			Null (surname blank) 1 Patrick *empty* patdoh password 2	A message box of errors will be displayed with the message “Surname cannot be empty.”	A message box of errors is displayed with the message “Surname cannot be empty.”	Pass	9, 12, 23
5.3.16			Null (surname is whitespace) 1 Patrick patdoh password 2			Pass	9, 12, 23

5.3.17			Invalid (surname longer than 50 characters): 1 Patrick DohertyDoherty DohertyDohertyDohertyDohertyDohertyDohertyDohertyDohertyDo patdoh password 2	A message box of errors will be displayed with the message “Surname cannot be longer than 50 characters.”	A message box of errors is displayed with the message “Surname cannot be longer than 50 characters.”	Pass	9, 12, 23
5.3.18			Extreme (surname is 50 characters): 1 Patrick DohertyDoherty DohertyDohertyDohertyDohertyDohertyDohertyDohertyDohertyD patdoh password 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated staff” displayed, and the form refreshed with the new data.	Pass	9, 12, 24, 27
5.3.19			Click event - all fields blank *empty* *empty* *empty* *empty* 2	A message box of errors will be displayed with the message “ - First name cannot be empty - Surname cannot be empty - Username cannot be empty ”	A message box of errors is displayed with the message “ - First name cannot be empty - Surname cannot be empty - Username cannot be empty ”	Pass	9, 12, 23

5.3.20			Invalid (username is not purely alphanumeric, with underscores or periods) 1 Patrick Doherty pat£doh password 2	A message box of errors will be displayed with the message "Username can only contain letters, numbers, underscores, or full stops."	A message box of errors is displayed with the message "Username can only contain letters, numbers, underscores, or full stops."	Pass	9, 12, 23
5.3.21			Invalid (username is less than 3 characters) 1 Patrick Doherty pa password 2	A message box of errors will be displayed with the message "Username must be at least 3 characters."	A message box of errors is displayed with the message "Username must be at least 3 characters."	Pass	9, 12, 23
5.3.22			Extreme (username is 3 characters) 1 Patrick Doherty pat password 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated staff” displayed, and the form refreshed with the new data.	Pass	9, 12, 24, 27

5.3.23			Extreme (username is 50 characters) 1 Patrick Doherty patpatpatpat patpatpatpat patpatpatpat patpatpatpat pa password 2			Pass	9, 12, 24, 27
5.3.24			Invalid (username more than 50 characters) 1 Patrick Doherty patpatpatpat patpatpatpat patpatpatpat patpatpatpat patpat password 2	A message box of errors will be displayed with the message "Username cannot be longer than 50 characters."	A message box of errors is displayed with the message "Username cannot be longer than 50 characters."	Pass	9, 12, 23

5.3.25			Invalid (username already exists on the system and does not already belong to the selected user) 1 Patrick Doherty ethanh1 password 2	A message box will be displayed with the text "Username already exists. Choose a different username.”	A message box is displayed with the text "Username already exists. Choose a different username.”	Pass	9, 12, 23
5.3.26			Invalid (password shorter than 8 characters): 1 Patrick Doherty patdoh passwor 2	A message box of errors will be displayed with the message "Password must be at least 8 characters."	A message box of errors is displayed with the message "Password must be at least 8 characters."	Pass	9, 12, 23
5.3.27			Extreme (password is 8 characters): 1 Patrick Doherty patdoh password 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated staff” displayed, and the form refreshed with the new data.	Pass	9, 12, 24, 27

5.3.28			Extreme (password is 128 characters): 1 Patrick Doherty patdoh passwordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpassword 2			Pass	9, 12, 24, 27
5.3.29			Invalid (password more than 128 characters): 1 Patrick Doherty patdoh passwordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpassword 2	A message box of errors will be displayed with the message "Password cannot be longer than 128 characters."	A message box of errors is displayed with the message "Password cannot be longer than 128 characters."	Pass	9, 12, 23

5.3.30			Valid (password kept the same): 1 Patrick Doherty patdoh *empty* 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data. The password hash will remain the same in the table	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated staff” displayed, and the form refreshed with the new data. The password hash remains the same in the table.	Pass	9, 12, 24, 27
5.3.31			Valid (currently logged-in user modified) 1 Patric Doherty patdoh Password 3 *Logged in as patdoh*	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data. The user details in memory will have also been updated as evidenced by the change in permission.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated staff” displayed, and the form refreshed with the new data. The user details in memory have been updated as evidenced by the change in permission when attempting to interact with the form.	Pass	9, 12, 24, 27
5.3.32			Valid (SQL injection attempt) 1 ; DROP TABLE Staff ; Doherty patdoh Password 3	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data. The table will not be affected.	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated staff" displayed, and the form refreshed with the new data. The table is not affected.	Pass	9, 12, 24, 27

5.4.1	btnAdd	Ensures that the user is able to add new staff details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 1)	A message box will be displayed with the text “This action requires “Unrestricted write” permission, but you have “Restricted write.” No further action will be taken.	A message box is displayed with the text “This action requires “Unrestricted write” permission, but you have “Restricted write.” No further action will be taken.	Pass	2, 7, 10, 23
5.4.2			Extreme: Click event - just permitted (user has authorisation level 2)	The add logic will execute as expected.	The add logic executes as expected.	Pass	2, 7, 10, 24
5.4.3			Click event after reset *empty* *empty* *empty* *empty* 1	A message box of errors will be displayed with the text “ - First name cannot be empty - Surname cannot be empty - Username cannot be empty - Username must be at least 3 characters - Password cannot be empty “	A message box of errors is displayed with the text “ - First name cannot be empty - Surname cannot be empty - Username cannot be empty - Username must be at least 3 characters - Password cannot be empty “	Pass	9, 10, 14, 23
5.4.4			Valid: *empty* Patri Doherty patdoh.. password 2	A message box will be displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 17, 24, 27

5.4.5			Invalid (same details added again): *empty* Patri Doherty patdoh.. password 2	A message box will be displayed with the text “Username already exists. Choose a different username.” No further action will be taken.	A message box is displayed with the text “Username already exists. Choose a different username.” No further action is taken.	Pass	9, 10, 14, 23
5.4.6			Null (first name blank) 1 *empty* Doherty patdoh password 2	A message box of errors will be displayed with the message “First name cannot be empty.”	A message box of errors is displayed with the message “First name cannot be empty.”	Pass	9, 10, 14, 23
5.4.7			Null (first name is whitespace) 1 Doherty patdoh password 2			Pass	9, 10, 14, 23

5.4.8			Valid (first name has leading spaces): 1 Patrick Doherty patdoh7 password 2	A message box will be displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record. First name will have whitespace trimmed off.	A message box is displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record. First name has whitespace trimmed off.	Pass	9, 10, 14, 17, 24, 27
5.4.9			Valid (first name has trailing spaces): 1 Patrick Doherty patdoh8 password 2			Pass	9, 10, 14, 17, 24, 27
5.4.10			Invalid (first name longer than 50 characters): 1 PatrickPatrickPatrickPatrickPatrickPatrickPatrickPatrickPa Doherty patdoh password 2	A message box of errors will be displayed with the message “First name cannot be longer than 50 characters.”	A message box of errors is displayed with the message “First name cannot be longer than 50 characters.”	Pass	9, 10, 14, 23

5.4.11			Extreme (first name is 50 characters): 1 PatrickPatrickPatrickPatrickPatrickPatrickPatrickPatrickP Doherty patdoh9 password 2	A message box will be displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 17, 24, 27
5.4.12			Null (surname blank) 1 Patrick *empty* patdoh password 2	A message box of errors will be displayed with the message “Surname cannot be empty.”	A message box of errors is displayed with the message “Surname cannot be empty.”	Pass	9, 10, 14, 23
5.4.13			Null (surname is whitespace) 1 Patrick patdoh password 2			Pass	9, 10, 14, 23
5.4.14			Invalid (surname longer than 50 characters): 1 Patrick DohertyDohe	A message box of errors will be displayed with the message “Surname cannot be longer than 50 characters.”	A message box of errors is displayed with the message “Surname cannot be longer than 50 characters.”	Pass	9, 10, 14, 23

			rtyDohertyDohertyDohertyDohertyDohertyDoherty DohertyDoherty patdoh password 2				
5.4.15			Extreme (surname is 50 characters): 1 Patrick DohertyDoherty DohertyDoherty DohertyDoherty DohertyDoherty DohertyD patdoh11 password 2	A message box will be displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 17, 24, 27
5.4.16			Click event - all fields blank *empty* *empty* *empty* *empty* 2	A message box of errors will be displayed with the message “ - First name cannot be empty - Surname cannot be empty - Username cannot be empty - Password cannot be empty ”	A message box of errors is displayed with the message “ - First name cannot be empty - Surname cannot be empty - Username cannot be empty - Username must be at least 3 characters - Password cannot be empty - ”	Pass	9, 10, 14, 23

5.4.17			Invalid (username is not purely alphanumeric, with underscores or periods) 1 Patrick Doherty pat£doh password 2	A message box of errors will be displayed with the message "Username can only contain letters, numbers, underscores, or full stops."	A message box of errors is displayed with the message "Username can only contain letters, numbers, underscores, or full stops."	Pass	9, 10, 14, 23
5.4.18			Invalid (username is less than 3 characters) 1 Patrick Doherty pa password 2	A message box of errors will be displayed with the message "Username must be at least 3 characters."	A message box of errors is displayed with the message "Username must be at least 3 characters."	Pass	9, 10, 14, 23
5.4.19			Extreme (username is 3 characters) 1 Patrick Doherty pat password 2	A message box will be displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 17, 24, 27

5.4.20			Extreme (username is 50 characters) 1 Patrick Doherty patpatpatpat patpatpatpat patpatpatpat patpatpatpat pa password 2			Pass	9, 10, 14, 17, 24, 27
5.4.21			Invalid (username more than 50 characters) 1 Patrick Doherty patpatpatpat patpatpatpat patpatpatpat patpatpatpat patpat password 2	A message box of errors will be displayed with the message "Username cannot be longer than 50 characters."	A message box of errors is displayed with the message "Username cannot be longer than 50 characters."	Pass	9, 10, 14, 23

5.4.22			Invalid (username already exists on the system) 1 Patrick Doherty patdoh password 2	A message box will be displayed with the text "Username already exists. Choose a different username."	A message box is displayed with the text "Username already exists. Choose a different username."	Pass	9, 10, 14, 23
5.4.23			Invalid (password shorter than 8 characters): 1 Patrick Doherty patdoh passwor 2	A message box of errors will be displayed with the message "Password must be at least 8 characters."	A message box of errors is displayed with the message "Password must be at least 8 characters."	Pass	9, 10, 14, 23
5.4.24			Extreme (password is 8 characters): 1 Patrick Doherty patdoh17 password 2	A message box will be displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 17, 24, 27

5.4.25			Extreme (password is 128 characters): 1 Patrick Doherty patdoh passwordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpassword 2			Pass	9, 10, 14, 17, 24, 27
5.4.26			Invalid (password more than 128 characters): 1 Patrick Doherty patdoh passwordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpasswordpassword 2	A message box of errors will be displayed with the message "Password cannot be longer than 128 characters."	A message box of errors is displayed with the message "Password cannot be longer than 128 characters."	Pass	9, 10, 14, 23

5.4.27			Null (password left blank): 1 Patrick Doherty patdoh *empty* 2	A message box will be displayed with the message “Password cannot be empty.” No further action will be taken.	A message box is displayed with the message “Password cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23
5.4.28			Valid (SQL injection attempt) 1 ; DROP TABLE Staff ; Doherty patdoh1121 Password 3	A message box will be displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record. The table will not be affected.	A message box is displayed with the text “Staff member has been added successfully” before the table is updated with the latest version of the data including the new record. The table is not affected.	Pass	9, 10, 14, 17, 24, 27
5.5.1	btnDelete	Ensures that the user is allowed to mark records as deleted in the database, preventing future loading and consideration	Invalid: Click event - not permitted (user has authorisation level 1)	A message box will be displayed with the text “This action requires “Unrestricted write” permission, but you have “Restricted write.” No further action will be taken.	A message box is displayed with the text “This action requires “Unrestricted write” permission, but you have “Restricted write.” No further action is taken.	Pass	2, 7, 13, 23
5.5.2			Extreme: Click event - just permitted (user has authorisation level 2)	The delete logic will be executed as expected.	The delete logic executes as expected.	Pass	2, 7, 13, 24
5.5.3			Click event - (“No” clicked at dialog) 7	No further action will be taken.	No further action is taken.	Pass	13, 24

5.5.4			Click event - (“Yes” clicked at dialog) 7	The selected staff member will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record.	The selected staff member is marked as deleted from the database, and this is indicated with the absence of the record in the table.	Pass	13, 14, 24
5.5.5			Click event - no staff member selected	A message box will be displayed with the text “Select a user to delete.” No further action will be taken.	A message box is displayed with the text “Select a user to delete.” No further action is taken.	Pass	9, 13, 23
5.5.6			Click event - after resetting			Pass	9, 13, 23
5.5.7			Click event - while the selected user has appointments associated with them	A message box will appear with the text “Staff delete unsuccessful: - Staff member is assigned to service appointments and cannot be deleted.” No further action will be taken.	A message box appears with the text “Staff delete unsuccessful: - Staff member is assigned to service appointments and cannot be deleted.” No further action is taken.	Pass	13, 14, 16c, 23
5.5.8			Click event - with the currently logged-in user selected	A message box will appear with the text “You cannot delete the user that is currently logged in.” No further action will be taken.	A message box appears with the text “You cannot delete the user that is currently logged in.” No further action is taken.	Pass	9, 13, 23
5.6.1	comboSearch Field	Ensures that the user can select which field they wish to search the database against	Selected item changed from “First name” to “Surname”	Any searches on the table will be with respect to the “Surname” field	All searches are now with respect to the “Surname” field	Pass	8
5.6.2			Selected item changed back to “First name” from “Surname”	Any searches on the table will be with respect to the “First name” field	All searches are now with respect to the “First name” field	Pass	8

5.7.1	btnReset	Ensures that the user is able to clear all fields in the form and deselect the currently selected staff member, preparing the form for new data entry	Click event - values had been typed in	All fields will be cleared, and any staff members will be unselected	All fields have been cleared, and the selected staff member has been unselected	Pass	14
5.7.2			Click event - values had been populated after selecting a staff member			Pass	14
5.8.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpStaff will be opened on top of frmStaffData	frmHelpStaff is opened on top of frmStaffData	Pass	26
5.8.2			Click event away from frmHelpStaff	frmHelpStaff will be closed	frmHelpStaff is closed	Pass	26
5.9.1	txtSearch	Ensures that the user is able to search the database using the selected search type, search field and provided search substring	Text input after form initially loads. “Contains” and “Staff ID” appear selected 6	The table will be updated to display only records which contain the digit “6” in their ID	The table is updated to display only records which contain the digit “6” in their ID	Pass	8
5.9.2			“Exact” and “Staff ID” are selected 6	The table will be updated to display only the record with “6” as its ID	The table is updated to display only the record with “6” as its ID	Pass	8
5.9.3			Click event - btnReset is clicked while a search has been input	The table will display all records on the system, ignoring the search until it is updated again	The table displays all records on the system, ignoring the search. When the search is updated again, the table is filtered once more.	Pass	8

5.9.4			Invalid: “Exact” and “Staff ID” are selected a	The table will be empty	The table is empty	Pass	8, 9
5.9.5			Null: “Contains” and “Staff ID” are selected *empty*	The table will display all records present in the database	The table displays all records present in the database	Pass	8

frmRoleData

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
6.1.1	frmRoleData	Ensures that role details are loaded and displayed properly on access, handling missing data gracefully	Load event	dgvRoles will be populated with the role records stored on the database. No further action will be taken.	dgvCustomers is populated with the role records stored on the database. No further action is taken.	Pass	6f, 8
6.1.2			Load event - no Roles table present	A message box will be displayed with the text "Error retrieving role records: " followed by an appropriate error message, followed by another with the text "There was an error attempting to load roles. Please amend erroneous data in the database before proceeding." before all CRUD and search controls are disabled.	A message box is displayed with the text "Error retrieving role records: " followed by an appropriate error message, followed by another with the text "There was an error attempting to load roles. Please amend erroneous data in the database before proceeding." before all CRUD and search controls are disabled.	Pass	15, 23
6.2.1	dgvRoles	Ensures that the user is able to select role records for editing or deleting	Click event on a record	The details for that record will be populated in the appropriate fields	The details for that record are populated in the appropriate fields	Pass	6f, 8, 11
6.2.2			Click event on a record after a reset			Pass	6f, 8, 11
6.2.3			Click event on a recently added record			Pass	6f, 8, 11
6.2.4			Click event on a record that has been updated	The most recent details for that record will be populated in the appropriate fields	The most recent details for that record are populated in the appropriate fields	Pass	6f, 8, 11

6.3.1	btnSave	Ensures that the user is able to save updated role details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 12, 23
6.3.2			Extreme: Click event - just permitted (user has authorisation level 1)	The save logic will be executed as expected.	The save logic executes as expected.	Pass	2, 7, 12, 24
6.3.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 12, 24
6.3.4			Click event - no role selected *empty* Flibulator Administrative 2	A message box will be displayed with the text "Select a role to edit." No further action will be taken.	A message box is displayed with the text "Select a role to edit." No further action is taken.	Pass	9, 12, 23
6.3.5			Click event after a reset			Pass	9, 12, 14, 23

6.3.6			Valid (“No” clicked at dialog): 1 Flibulator None 0	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action will be taken.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action is taken.	Pass	9, 12, 24
6.3.7			Valid (“Yes” clicked at dialog): 2 Mega Flibulator Administrative 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated role" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated role” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
6.3.8			Valid (same role updated again): 2 MEGA MEGA Flibulator Administrative 2			Pass	9, 12, 14, 24
6.3.9			Null (role name is empty) 1 *empty* None 1	A message box of errors will be displayed with the text “Role name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Role name cannot be empty.” No further action is taken.	Pass	9, 12, 14, 23

6.3.10			Null (role name is whitespace) 3 Customer Service 2			Pass	9, 12, 14, 23
6.3.11			Valid (role name has leading spaces) 3 Bloobus Administrative 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated role" displayed, and the form refreshed with the new data. The role name will be trimmed of its whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated role” displayed, and the form refreshed with the new data. The role name is trimmed of its whitespace.	Pass	9, 12, 14, 24
6.3.12			Valid (role name has leading spaces) 3 Bloobus2 Administrative 2			Pass	9, 12, 14, 24

6.3.13			Extreme (role name is 50 characters) 4 getoutofmyheadgetoutofmyheadgetoutofmyheadgetoutofTechnical 1	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated role" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated role” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
6.3.14			Invalid (role name is more than 50 characters) 4 getoutofmyheadgetoutofmyheadgetoutofmyheadgetoutofmyheadTechnical 1	A message box of errors will be displayed with the text “Role name cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Role name cannot be longer than 50 characters.” No further action is taken.	Pass	9, 12, 14, 23
6.3.15			Valid (authorisation level 0 selected): 1 Evil1 None 0	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated role" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated role” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

6.3.16			Valid (authorisation level 1 selected): 1 Evil2 None 1			Pass	9, 12, 14, 24
6.3.17			Valid (authorisation level 2 selected): 1 Evil3 None 2			Pass	9, 12, 14, 24
6.3.18			Valid (SQL injection attempt) 1 ; DROP TABLE Roles ; None 2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated role" displayed, and the form refreshed with the new data. The table will not be affected.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated role” displayed, and the form refreshed with the new data. The table is not affected.	Pass	9, 12, 14, 24
6.4.1	btnAdd	Ensures that the user is able to add new role details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 10, 23

6.4.2			Extreme: Click event - just permitted (user has authorisation level 1)	The add logic will be executed as expected.	The add logic executes as expected.	Pass	2, 7, 10, 24
6.4.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 10, 24
6.4.4			Click event after a reset *empty* *empty* None 0	A message box of errors will be displayed with the text “Role name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Role name cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23
6.4.5			Valid 2 Mega Flibulator Administrative 2	A message box will be displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24
6.4.6			Valid (same details added again): 2 Mega Flibulator Administrative 2			Pass	9, 10, 14, 24

6.4.7			Null (role name is empty) 1 *empty* None 1	A message box of errors will be displayed with the text “Role name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Role name cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23
6.4.8			Null (role name is whitespace) 3 Customer Service 2			Pass	9, 10, 14, 23
6.4.9			Valid (role name has leading spaces) 3 Bloobus Administrative 2	A message box will be displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record. The role name will have been stripped of its whitespace.	A message box is displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record. The role name has been stripped of its whitespace.	Pass	9, 10, 14, 24
6.4.10			Valid (role name has leading spaces) 3 Bloobus2 Administrative 2			Pass	9, 10, 14, 24

6.4.11			Extreme (role name is 50 characters) 4 getoutofmyheadgetoutofmyheadgetoutofmyheadgetoutof Technical 1	A message box will be displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24
6.4.12			Invalid (role name is more than 50 characters) 4 getoutofmyheadgetoutofmyheadgetoutofmyheadgetoutofmyhead Technical 1	A message box of errors will be displayed with the text “Role name cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Role name cannot be longer than 50 characters.” No further action is taken.	Pass	9, 10, 14, 23
6.4.13			Valid (authorisation level 0 selected): 1 Evil1 None 0	A message box will be displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24

6.4.14			Valid (authorisation level 1 selected): 1 Evil2 None 1			Pass	9, 10, 14, 24
6.4.15			Valid (authorisation level 2 selected): 1 Evil3 None 2			Pass	9, 10, 14, 24
6.4.16			Valid (SQL injection attempt) 1 ; DROP TABLE Roles ; None 2	A message box will be displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record. The table will not be affected.	A message box is displayed with the text “Role added successfully” before the table is updated with the latest version of the data including the new record. The table is not affected.	Pass	9, 10, 14, 24
6.5.1	btnDelete	Ensures that the user is allowed to mark records as deleted in the database, preventing future loading and consideration	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 13, 23

6.5.2			Extreme: Click event - just permitted (user has authorisation level 1)	The delete logic will be executed as expected.	The delete logic executes as expected.	Pass	2, 7, 13, 24
6.5.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 13, 24
6.5.4			Click event - (“No” clicked at dialog) 7	No further action will be taken.	No further action is taken.	Pass	13, 24
6.5.5			Valid: Click event (role selected) 7	The selected role will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record.	The selected role is marked as deleted from the database, and this is indicated with the absence of the record in the table.	Pass	13, 14, 24
6.5.6			Invalid: Click event (no role selected)	A message box will be displayed with the text “Select a user to delete.” No further action will be taken.	A message box is displayed with the text “Select a user to delete.” No further action is taken.	Pass	9, 13, 23
6.5.7			Invalid: Click event (after adding a new record)			Pass	9, 13, 23

6.5.8			Invalid: Click event (after saving a record)			Pass	9, 13, 23
6.5.9			Invalid: Click event (after resetting record)			Pass	9, 13, 23
6.5.10			Invalid: Click event (role selected is associated with staff members) 2	A message box will be displayed with the text “Role delete unsuccessful: Role is assigned to staff members and cannot be deleted.” No further action will be taken.	A message box is displayed with the text “Role delete unsuccessful: Role is assigned to staff members and cannot be deleted.” No further action is taken.	Pass	13, 14, 16f, 23
6.6.1	comboSearchField	Ensures that the user can select which field they wish to search the database against	Selected item changed from “Name” to “Sector”	Any searches on the table will be with respect to the “Sector” field	All searches are now with respect to the “Sector” field	Pass	8
6.6.2			Selected item changed back to “Sector” from “Name”	Any searches on the table will be with respect to the “Name” field	All searches are now with respect to the “Name” field	Pass	8
6.7.1	btnReset	Ensures that the user is able to clear all fields in the form and deselect the currently selected role, preparing the form for new data entry	Click event - values had been typed in	All fields will be cleared, and any roles will be unselected	All fields have been cleared, and the role has been unselected	Pass	14
6.7.2			Click event - values had been populated after selecting a staff member			Pass	14

6.8.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpRoles will be opened on top of frmRoleData	frmHelpRole is opened on top of frmRoleData	Pass	26
6.8.2			Click event away from frmHelpRoles	frmHelpRoles will be closed	frmHelpRoles is closed	Pass	26
6.9.1	txtSearch	Ensures that the user is able to search the database using the selected search type, search field and provided search substring	Text input after form initially loads. “Contains” and “Role ID” appear selected 6	The table will be updated to display only records which contain the digit “6” in their ID	The table is updated to display only records which contain the digit “6” in their ID	Pass	8
6.9.2			“Exact” and “Role ID” are selected 6	The table will be updated to display only the record with “6” as its ID	The table is updated to display only the record with “6” as its ID	Pass	8
6.9.3			Click event - btnReset is clicked while a search has been input	The table will display all records on the system, ignoring the search until it is updated again	The table displays all records on the system, ignoring the search. When the search is updated again, the table is filtered once more.	Pass	8
6.9.4			Invalid: “Exact” and “Role ID” are selected a	The table will be empty	The table is empty	Pass	8, 9

6.9.5			Null: “Contains” and “Role ID” are selected *empty*	The table will display all records present in the database	The table displays all records present in the database	Pass	8

frmVehicleData

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
7.1.1	frmVehicleData	Ensures that vehicle details are loaded and displayed properly on access, handling missing data gracefully	Load event	dgvVehicles will be populated with the vehicle records stored on the database. No further action will be taken.	dgvVehicles is populated with the vehicle records stored on the database. No further action is taken.	Pass	6c, 8
7.1.2			Load event - no Vehicles table present	A message box will be displayed with the text "An error occurred while retrieving vehicle data:" followed by an appropriate error message, followed by another with the text "There was an error when attempting to load vehicle data. Please amend erroneous data in the database before proceeding." before all CRUD and search controls are disabled.	A message box is displayed with the text "An error occurred while retrieving vehicle data:" followed by an appropriate error message, followed by another with the text "There was an error when attempting to load vehicle data. Please amend erroneous data in the database before proceeding." before all CRUD and search controls are disabled.	Pass	15, 23
7.1.3			Load event - no Customers table present	A message box will be displayed with the text "Error retrieving customers: " followed by an appropriate error message, followed by another with the text "No customers could be loaded." before all CRUD and search controls are disabled. Vehicles and vehicle types will be loaded, but will be read-only	A message box is displayed with the text "Error retrieving customers: " followed by an appropriate error message, followed by another with the text "No customers could be loaded." before all CRUD and search controls are disabled. Vehicles and vehicle types are loaded, but are read-only	Pass	15, 23

7.1.4			Load event - no VehicleTypes table present	A message box will be displayed with the text "Error retrieving vehicle types: " followed by an appropriate error message, followed by another with the text “No vehicle types could be loaded.” before all CRUD and search controls are disabled. Vehicles and customers will be loaded, but will be read-only.	A message box is displayed with the text "Error retrieving vehicle types: " followed by an appropriate error message, followed by another with the text “No vehicle types could be loaded.” before all CRUD and search controls are disabled. Vehicles and customers are loaded, but are read-only.	Pass	15, 23
7.2.1	dgvVehicles	Ensures that the user is able to select vehicle records for editing or deleting	Click event on a record	The details for that record will be populated in the appropriate fields	The details for that record are populated in the appropriate fields	Pass	6c, 8, 11
7.2.2			Click event on a record after a reset			Pass	6c, 8, 11
7.2.3			Click event on a recently added record			Pass	6c, 8, 11
7.2.4			Click event on a record that has been updated	The most recent details for that record will be populated in the appropriate fields	The most recent details for that record are populated in the appropriate fields	Pass	6c, 8, 11
7.3.1	btnSave	Ensures that the user is able to save updated vehicle details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 12, 23
7.3.2			Extreme: Click event - just permitted (user has authorisation level 1)	The save logic will be executed as expected.	The save logic executes as expected.	Pass	2, 7, 12, 24

7.3.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 12, 24
7.3.4			Click event - no vehicle selected *empty* 1 Honda Civic 2 2C3CCAAG2EH118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box will be displayed with the text "Choose a vehicle to edit." No further action will be taken.	A message box is displayed with the text "Choose a vehicle to edit." No further action is taken.	Pass	9, 12, 23
7.3.5			Click event after a reset			Pass	9, 12, 14, 23

7.3.6			Valid (“No” clicked at dialog): 4 1 Hondas Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action will be taken.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action is taken.	Pass	9, 12, 24
7.3.7			Valid (“Yes” clicked at dialog): 4 1 Hondas Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

7.3.8			Valid (same vehicle updated again): 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 12, 14, 24
7.3.9			Null (all fields left blank) 4 1 *empty* *empty* *empty* *empty* No customer None Unknown Fair *empty* *empty* False False	A message box of errors will be displayed with the text “ - Make cannot be empty - Model cannot be empty - VIN cannot be empty ” No further action will be taken.	A message box of errors has been displayed with the text “ - Make cannot be empty - Model cannot be empty - VIN cannot be empty ” No further action has been taken.	Pass	9, 12, 14, 23

7.3.10			Null (make is empty) 4 1 *empty* Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text “Make cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Make cannot be empty.” No further action is taken.	Pass	9, 12, 14, 23
7.3.11			Null (make is whitespace) 4 1 Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 12, 14, 23

7.3.12			Valid (make has leading spaces) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. Make will be stripped of its whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data. Make is stripped of its whitespace.	Pass	9, 12, 14, 24
7.3.13			Valid (make has trailing spaces) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 12, 14, 24

7.3.14			Extreme (make has 50 characters) 4 1 HondaHonda HondaHonda HondaHonda HondaHonda HondaHonda Civic 2C3CCAAG2EH118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

7.3.15			Invalid (make has more than 50 characters) 4 1 HondaHonda HondaHonda HondaHonda HondaHonda HondaHonda H Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text “Make cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Make cannot be longer than 50 characters.” No further action is taken.	Pass	9, 12, 14, 23
7.3.16			Null (model is empty) 4 1 Honda *empty* 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text “Model cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Model cannot be empty.” No further action is taken.	Pass	9, 12, 14, 23

7.3.17			Null (model is whitespace) 4 1 Honda 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 12, 14, 23
7.3.18			Valid (model has leading spaces) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. Model will be stripped of its whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data. Model is stripped of its whitespace.	Pass	9, 12, 14, 24

7.3.19			Valid (model has trailing spaces) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 12, 14, 24
7.3.20			Extreme (model is 50 characters) 4 1 Honda CivicCivicCivicCivicCivicCivicCivicCivicCivicCivic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

7.3.21			Invalid (model is more than 50 characters) 4 1 Honda CivicCivicCiv icCivicCivicCi vicCivicCivic CivicCivicC 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text “Model cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Model cannot be longer than 50 characters.” No further action is taken.	Pass	9, 12, 14, 23
7.3.22			Null (year is unknown) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None Unknown Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. Year will be represented with a null value.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data. Year is represented with a null value.	Pass	9, 12, 14, 24

7.3.23			Extreme (year is 1800) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 1800 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
7.3.24			Extreme (year is 2050) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2050 Fair 1200 2200 False True			Pass	9, 12, 14, 24

7.3.25			Invalid (VIN already exists and doesn't belong to the current car) 4 1 Honda Civic JH2KF3500LK 000012 VU02GFI 2 None 2020 Fair 1200 2200 False True	A message box of errors will be displayed with the text "VIN must be unique." No further action will be taken.	A message box of errors is displayed with the text "VIN must be unique." No further action is taken.	Pass	9, 12, 14, 23
7.3.26			Null (VIN is empty) 4 1 Honda Civic *empty* VU02GFI 2 None 2020 Fair 1200 2200 False True	A message box of errors will be displayed with the text "VIN cannot be empty." No further action will be taken.	A message box of errors is displayed with the text "VIN cannot be empty." No further action is taken.	Pass	9, 12, 14, 23

7.3.27			Null (VIN is whitespace) 4 1 Honda Civic VU02GFI 2 None 2020 Fair 1200 2200 False True			Pass	9, 12, 14, 23
7.3.28			Valid (VIN has leading spaces) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2020 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. VIN will be stripped of its whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data. VIN is stripped of its whitespace.	Pass	9, 12, 14, 24

7.3.29			Valid (VIN has trailing spaces) 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2020 Fair 1200 2200 False True			Pass	9, 12, 14, 24
7.3.30			Extreme (VIN is 50 characters long) 4 1 Honda Civic 7F3K9L2X8M4 ZQ1R6T5V8Y2 N4C7B9H3J6P 2D8S4A1E5G7 W9U3K6 VU02GFI 2 None 2020 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

7.3.31			Invalid (VIN is more than 50 characters long) 4 1 Honda Civic 7F3K9L2X8M4 ZQ1R6T5V8Y2 N4C7B9H3J6P 2D8S4A1E5G7 W9U3K6AB VU02GFIF 2 None 2020 Fair 1200 2200 False True	A message box of errors will be displayed with the text "VIN cannot be longer than 50 characters." No further action will be taken.	A message box of errors is displayed with the text "VIN cannot be longer than 50 characters." No further action is taken.	Pass	9, 12, 14, 23
7.3.32			Null (license plate is empty) 4 1 Honda Civic 2C3CCAAG2E H118886 *empty* 2 None 2020 Fair 1200	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. License plate will be stored as a null value.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data. License plate is stored as a null value.	Pass	9, 12, 14, 24

7.3.33			Extreme (license plate is 20 characters) 4 1 Honda Civic 2C3CCAAG2E H118886 BX23KLM9RT 56YQ81ZV4P 2 None 2006 Fair 1200 2200 False True	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
7.3.34			Invalid (license plate is more than 20 characters) 4 1 Honda Civic 2C3CCAAG2E H118886 BX23KLM9RT 56YQ81ZV4PP 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text "License plate cannot be longer than 20 characters." No further action will be taken.	A message box of errors is displayed with the text "License plate cannot be longer than 20 characters." No further action is taken.	Pass	9, 12, 14, 23

7.3.35			Valid (license plate has leading spaces) 4 1 Hondas Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. License plate will be stripped of its whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data. License plate is stripped of its whitespace.	Pass	9, 12, 14, 24
7.3.36			Valid (license plate has trailing spaces) 4 1 Hondas Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False			Pass	9, 12, 14, 24

7.3.37			Null (customer and location not chosen) 4 1 Hondas Civic 2C3CCAAG2E H118886 VU02GFI No customer None 2006 Fair 1200 2200 False	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. Both location and customer ID will be stored as null.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle” displayed, and the form refreshed with the new data. Both location and customer ID are stored as null.	Pass	9, 12, 14, 24
7.3.38			Valid (SQL injection attempt) 4 1 ; DROP TABLE Vehicles; Civic 2C3CCAAG2E H118886 VU02GFI No customer None 2006 Fair 1200 2200 False	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. The table will be unaffected.	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle" displayed, and the form refreshed with the new data. The table is unaffected.	Pass	9, 12, 14, 24
7.4.1	btnAdd	Ensures that the user is able to add new vehicle details to the	Invalid: Click event - not permitted	A message box will be displayed with the text “This action requires “Restricted	A message box is displayed with the text “This action requires “Restricted write”	Pass	2, 7, 10, 23

		database, preventing unauthorised or invalid data entry	(user has authorisation level 0)	write" permission, but you have "Read-only." No further action will be taken.	permission, but you have "Read-only." No further action is taken.		
7.4.2			Extreme: Click event - just permitted (user has authorisation level 1)	The add logic will be executed as expected.	The add logic executes as expected.	Pass	2, 7, 10, 24
7.4.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 10, 24
7.4.4			Click event after a reset *empty* 1 *empty* *empty* *empty* *empty* No customer 1800 Fair 0 0 False False	A message box of errors will be displayed with the text " - Make cannot be empty - Model cannot be empty - VIN cannot be empty " No further action will be taken.	A message box of errors has been displayed with the text " - Make cannot be empty - Model cannot be empty - VIN cannot be empty " No further action has been taken.	Pass	9, 10, 14, 23

7.4.5			Invalid (attempt to add already existing details): 4 1 Honda Civic 2C3CCAAG2E H118886 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text "VIN must be unique." No further action will be taken.	A message box of errors is displayed with the text "VIN must be unique." No further action is taken.	Pass	9, 10, 14, 23
7.4.6			Valid: 4 1 Honda Civic 1HGCM82633 A482719 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24

7.4.7			Null (make is empty) 4 1 *empty* Civic 3VWFE21C04 M000001 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text “Make cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Make cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23
7.4.8			Null (make is whitespace) 4 1 Civic 3VWFE21C04 M000001 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 10, 14, 23

7.4.9			Valid (make has leading spaces) 4 1 Honda Civic 3VWFE21C04 M000001 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record Make will be stripped of its whitespace.	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record Make is stripped of its whitespace.	Pass	9, 10, 14, 24
7.4.10			Valid (make has trailing spaces) 4 1 Honda Civic JH4KA9650M C012345 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 10, 14, 24

7.4.11			Extreme (make has 50 characters) 4 1 HondaHonda HondaHonda HondaHonda HondaHonda HondaHonda Civic 1N4AL11D75C109151 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24

7.4.12			Invalid (make has more than 50 characters) 4 1 HondaHonda HondaHonda HondaHonda HondaHonda HondaHonda H Civic KM8JM12D97 U123456 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text “Make cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Make cannot be longer than 50 characters.” No further action is taken.	Pass	9, 10, 14, 23
7.4.13			Null (model is empty) 4 1 Honda *empty* KM8JM12D97 U123456 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text “Model cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Model cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23

7.4.14			Null (model is whitespace) 4 1 Honda KM8JM12D97 U123456 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 10, 14, 23
7.4.15			Valid (model has leading spaces) 4 1 Honda Civic KM8JM12D97 U123456 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. Model will be stripped of its whitespace.	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. Model is stripped of its whitespace.	Pass	9, 10, 14, 24

7.4.16			Valid (model has trailing spaces) 4 1 Honda Civic SALTY124X8A 123456 VU02GFI 2 None 2006 Fair 1200 2200 False True			Pass	9, 10, 14, 24
7.4.17			Extreme (model is 50 characters) 4 1 Honda CivicCivicCivicCivicCivicCivicCivicCivicCivicCivic WAUZZZ8K9A A000001 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24

7.4.18			Invalid (model is more than 50 characters) 4 1 Honda CivicCivicCiv icCivicCivicCi vicCivicCivic CivicCivicC VF1BG0B0H12 345678 VU02GFI 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text “Model cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Model cannot be longer than 50 characters.” No further action is taken.	Pass	9, 10, 14, 23
7.4.19			Null (year is unknown) 4 1 Honda Civic 1C4RJFAG0FC 625478 VU02GFI 2 None Unknown Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. Year will be stored as null in the table	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. Year is stored as null in the table	Pass	9, 10, 14, 24

7.4.20			Extreme (year is 1800) 4 1 Honda Civic 2HGFB2F50D H512394 VU02GFI 2 None 1800 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24
7.4.21			Extreme (year is 2050) 4 1 Honda Civic 3FA6P0H73DR 256819 VU02GFI 2 None 2050 Fair 1200 2200 False True			Pass	9, 10, 14, 24

7.4.22			Null (VIN is empty) 4 1 Honda Civic *empty* VU02GFI 2 None 2020 Fair 1200 2200 False True	A message box of errors will be displayed with the text "VIN cannot be empty." No further action will be taken.	A message box of errors is displayed with the text "VIN cannot be empty." No further action is taken.	Pass	9, 10, 14, 23
7.4.23			Null (VIN is whitespace) 4 1 Honda Civic VU02GFI 2 None 2020 Fair 1200 2200 False True			Pass	9, 10, 14, 23

7.4.24			Valid (VIN has leading spaces) 4 1 Honda Civic JTDKN3DU0A 0123456 VU02GFI 2 None 2020 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. VIN will be stripped of its whitespace.	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. VIN is stripped of its whitespace.	Pass	9, 10, 14, 24
7.4.25			Valid (VIN has trailing spaces) 4 1 Honda Civic 5NPEB4AC4D H789123 VU02GFI 2 None 2020 Fair 1200 2200 False True			Pass	9, 10, 14, 24

7.4.26			Extreme (VIN is 50 characters long) 4 1 Honda Civic 7F3K9L2X8M4 ZQ1R6T5V8Y2 N4C7B9H3J6P 2D8S4A1E5G7 W9U3K6 VU02GFI 2 None 2020 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24

7.4.27			Invalid (VIN is more than 50 characters long) 4 1 Honda Civic 7F3K9L2X8M4 ZQ1R6T5V8Y2 N4C7B9H3J6P 2D8S4A1E5G7 W9U3K6AB VU02GFIF 2 None 2020 Fair 1200 2200 False True	A message box of errors will be displayed with the text "VIN cannot be longer than 50 characters." No further action will be taken.	A message box of errors is displayed with the text "VIN cannot be longer than 50 characters." No further action is taken.	Pass	9, 10, 14, 23
7.4.28			Null (license plate is empty) 4 1 Honda Civic 1FTFW1ET5E KE12345 *empty* 2 None 2020 Fair 1200	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. License plate will be stored as a null value.	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. License plate is stored as a null value.	Pass	9, 10, 14, 24

7.4.29			Extreme (license plate is 20 characters) 4 1 Honda Civic KMHD35LH6GU123789 BX23KLM9RT56YQ81ZV4P 2 None 2006 Fair 1200 2200 False True	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record	Pass	9, 10, 14, 24
7.4.30			Invalid (license plate is more than 20 characters) 4 1 Honda Civic WBA3A5C58CF123456 BX23KLM9RT56YQ81ZV4PP 2 None 2006 Fair 1200 2200 False True	A message box of errors will be displayed with the text "License plate cannot be longer than 20 characters." No further action will be taken.	A message box of errors is displayed with the text "License plate cannot be longer than 20 characters." No further action is taken.	Pass	9, 10, 14, 23

7.4.31			Valid (license plate has leading spaces) 4 1 Hondas Civic ZFF65LJA4E0 203045 VU02GFI 2 None 2006 Fair 1200 2200 False	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. License plate will be stripped of its whitespace	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. License plate is stripped of its whitespace	Pass	9, 10, 14, 24
7.4.32			Valid (license plate has trailing spaces) 4 1 Hondas Civic 1GNEK13ZX3 R298765 VU02GFI 2 None 2006 Fair 1200 2200 False			Pass	9, 10, 14, 24

7.4.33			Null (customer and location not chosen) 4 1 Hondas Civic 3N1AB7AP4FY 345678 VU02GFI No customer None 2006 Fair 1200 2200 False	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. Both location and customer ID will be stored as null.	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. Both location and customer ID are stored as null.	Pass	9, 10, 14, 24
7.4.34			Valid (SQL injection attempt) 4 1 ; DROP TABLE Vehicles; Civic JHMFA16586S 012345 VU02GFI No customer None 2006 Fair 1200 2200 False	A message box will be displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. The table will be unaffected.	A message box is displayed with the text “Vehicle has been added successfully” before the table is updated with the latest version of the data including the new record. The table is unaffected.	Pass	9, 10, 14, 24
7.5.1	btnDelete	Ensures that the user is allowed to mark records as deleted in	Invalid: Click event - not permitted	A message box will be displayed with the text “This action requires “Restricted	A message box is displayed with the text “This action requires “Restricted write”	Pass	2, 7, 13, 23

		the database, preventing future loading and consideration	(user has authorisation level 0)	write" permission, but you have "Read-only." No further action will be taken.	permission, but you have "Read-only." No further action is taken.		
7.5.2			Extreme: Click event - just permitted (user has authorisation level 1)	The delete logic will be executed as expected.	The delete logic executes as expected.	Pass	2, 7, 13, 24
7.5.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 13, 24
7.5.4			Click event - ("No" clicked at dialog) 29	No further action will be taken.	No further action is taken.	Pass	13, 24
7.5.5			Valid: Click event (vehicle selected) 29	The selected vehicle will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record.	The selected vehicle is marked as deleted from the database, and this is indicated with the absence of the record in the table.	Pass	13, 14, 24
7.5.6			Invalid: Click event (after adding a new record)	A message box will be displayed with the text "Select a vehicle to delete." No further action will be taken.	A message box is displayed with the text "Select a vehicle to delete." No further action is taken.	Pass	9, 13, 24
7.5.7			Invalid: Click event (after saving a record)			Pass	9, 13, 24

7.5.8			Invalid: Click event (after resetting record)			Pass	9, 13, 24
7.5.9			Valid: Click event (vehicle is associated with appointments) Vehicle ID: 4 Appointment ID: 17	The selected vehicle will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record. Furthermore, the associated appointment will no longer be visible in frmServiceAppointments	The selected vehicle will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record. The associated appointment is not visible in frmServiceAppointments either	Pass	13, 14, 16b, 24
7.6.1	comboSearchField	Ensures that the user can select which field they wish to search the database against	Selected item changed from “Make” to “License Plate”	Any searches on the table will be with respect to the “License Plate” field	All searches are now with respect to the “License Plate” field	Pass	8
7.6.2			Selected item changed back to “Make” from “License Plate”	Any searches on the table will be with respect to the “Make” field	All searches are now with respect to the “Make” field	Pass	8
7.7.1	btnReset	Ensures that the user is able to clear all fields in the form and deselect the currently selected vehicle, preparing the form for new data entry	Click event - values had been typed in	All fields will be cleared, and any vehicles will be unselected	All fields have been cleared, and the vehicle has been unselected	Pass	14
7.7.2			Click event - values had been populated after selecting a vehicle			Pass	14

7.8.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpVehicles will be opened on top of frmVehicleData	frmHelpVehicles is opened on top of frmVehicleData	Pass	26
7.8.2			Click event away from frmHelp Vehicles	frmHelpVehicles will be closed	frmHelpVehicles is closed	Pass	26
7.9.1	txtSearch	Ensures that the user is able to search the database using the selected search type, search field and provided search substring	Text input after form initially loads. “Contains” and “Vehicle ID” appear selected 6	The table will be updated to display only records which contain the digit “6” in their ID	The table is updated to display only records which contain the digit “6” in their ID	Pass	8
7.9.2			“Exact” and “Vehicle ID” are selected 26	The table will be updated to display only the record with “26” as its ID	The table is updated to display only the record with “26” as its ID	Pass	8
7.9.3			Click event - btnReset is clicked while a search has been input	The table will display all records on the system, ignoring the search until it is updated again	The table displays all records on the system, ignoring the search. When the search is updated again, the table is filtered once more.	Pass	8
7.9.4			Invalid: “Exact” and “Vehicle ID” are selected a	The table will be empty	The table is empty	Pass	8, 9

7.9.5			Null: “Contains” and “Vehicle ID” are selected *empty*	The table will display all records present in the database	The table displays all records present in the database	Pass	8
7.10.1	numYear	Ensures that the user can only input valid years of manufacture	Invalid (attempting to input a year below 1800): Attempt to click to and type 1799	When clicked, no further action will be taken. When typed in, the year will go to the minimum of 1800.	When clicked, no further action is taken. When typed in, the year goes to the minimum of 1800.	Pass	9
7.10.2			Invalid (attempting to input a year above 2050): Attempt to click to and type 2051	When clicked, no further action will be taken. When typed in, the year will go to the maximum of 2050.	When clicked, no further action is taken. When typed in, the year goes to the maximum of 2050.	Pass	9
7.10.3			Valid: 2020 Click event up then down	Year will go up to 2021, then back to 2020.	Year goes up to 2021, then back to 2020	Pass	9
7.10.4			Valid: 2020 Year typed in	Year will be set to 2020	Year is set to 2020	Pass	9

7.11.1	numBuyPrice	Ensures that the user can only input valid vehicle buy prices	Invalid (attempting to input a negative number): Attempting to click to and type -1	When clicked, no further action will be taken. When typed in, the price will go to the minimum of 0.	When clicked, no further action is taken. When typed in, the price goes to the minimum of 0.	Pass	9
7.11.2			Valid: 5 Click event up then down	Price will go up to 6, and then back to 5.	Price goes up to 6, and then back to 5.	Pass	9
7.11.3			Valid: 5 Price typed in	Price will be set to 5	Price is set to 5	Pass	9
7.11.4			Invalid: (attempting to overflow the integer value) Attempting to click to and type 2147483648	When clicked, no further action will be taken. When typed in, the price will go to the maximum of 2147483647.	When clicked, no further action is taken. When typed in, the price goes to the maximum of 2147483647.	Pass	9

7.12.1	numSellPrice	Ensures that the user can only input valid vehicle sell prices	Invalid (attempting to input a negative number): Attempting to click to and type -1	When clicked, no further action will be taken. When typed in, the price will go to the minimum of 0.	When clicked, no further action is taken. When typed in, the price goes to the minimum of 0.	Pass	9
7.12.2			Valid: 5 Click event up then down	Price will go up to 6, and then back to 5.	Price goes up to 6, and then back to 5.	Pass	9
7.12.3			Valid: 5 Price typed in	Price will be set to 5	Price is set to 5	Pass	9
7.12.4			Invalid: (attempting to overflow the integer value) Attempting to click to and type 2147483648	When clicked, no further action will be taken. When typed in, the price will go to the maximum of 2147483647.	When clicked, no further action is taken. When typed in, the price goes to the maximum of 2147483647.	Pass	9

frmVehicleTypeData

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
8.1.1	frmVehicleType Data	Ensures that vehicle type details are loaded and displayed properly on access, handling missing data gracefully	Load event	dgvVehicleTypes will be populated with the vehicle records stored on the database. No further action will be taken.	dgvVehicleTypes is populated with the vehicle records stored on the database. No further action is taken.	Pass	6d, 8
8.1.2			Load event - no VehicleTypes table present	A message box will be displayed with the text "Error retrieving vehicle types:" followed by an appropriate error message, followed by another with the text "There was an error attempting to load vehicle type data." before all CRUD and search controls are disabled.	A message box is displayed with the text "An error occurred while retrieving vehicle type data:" followed by an appropriate error message, followed by another with the text "There was an error attempting to load vehicle type data." before all CRUD and search controls are disabled.	Pass	15, 23
8.2.1	dgvVehicleTypes	Ensures that the user is able to select vehicle type records for editing or deleting	Click event on a record	The details for that record will be populated in the appropriate fields	The details for that record are populated in the appropriate fields	Pass	6d, 8, 11
8.2.2			Click event on a record after a reset			Pass	6d, 8, 11
8.2.3			Click event on a recently added record			Pass	6d, 8, 11
8.2.4			Click event on a record that has been updated	The most recent details for that record will be populated in the appropriate fields	The most recent details for that record are populated in the appropriate fields	Pass	6d, 8, 11

8.3.1	btnSave	Ensures that the user is able to save updated vehicle type details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 12, 23
8.3.2			Extreme: Click event - just permitted (user has authorisation level 1)	The save logic will be executed as expected.	The save logic executes as expected.	Pass	2, 7, 12, 24
8.3.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 12, 24
8.3.4			Click event - no vehicle type selected *empty* Sedan 1.0	A message box will be displayed with the text "Choose a vehicle type to edit." No further action will be taken.	A message box is displayed with the text "Choose a vehicle type to edit." No further action is taken.	Pass	9, 12, 23
8.3.5			Click event after a reset			Pass	9, 12, 14, 23
8.3.6			Valid (“No” clicked at dialog): 1 Sedan 1.2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action will be taken.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action is taken.	Pass	9, 12, 24

8.3.7			Valid (“Yes” clicked at dialog): 1 Sedan 1.2	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle type" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle type” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
8.3.8			Valid (same vehicle type updated again): 1 Sedans 1.2			Pass	9, 12, 14, 24
8.3.9			Null (all fields left blank) 1 *empty* *empty*	A message box of errors will be displayed with the text “Vehicle type name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Vehicle type name cannot be empty.” No further action is taken.	Pass	9, 12, 14, 23
8.3.10			Null (name is empty) 1 *empty* 1.0			Pass	9, 12, 14, 23
8.3.11			Null (name is whitespace) 1 1.0			Pass	9, 12, 14, 23

8.3.12			Valid (name has leading spaces) 1 Sedane 1.0	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle type" displayed, and the form refreshed with the new data. Vehicle type name will be stripped of whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle type” displayed, and the form refreshed with the new data. Vehicle type name is stripped of whitespace.	Pass	9, 12, 14, 24
8.3.13			Valid (name has trailing spaces) 1 Sedanv 1.0			Pass	9, 12, 14, 24
8.3.14			Extreme (name is 50 characters) 1 SedanSedan SedanSedan SedanSedan SedanSedan SedanSedan 1.0	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle type" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle type” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24
8.3.15			Invalid (name is longer than 50 characters) 1 SedanSedan SedanSedan SedanSedan SedanSedan SedanSedan Sedan 1.0	A message box of errors will be displayed with the text “Vehicle type name cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Vehicle type name cannot be longer than 50 characters.” No further action is taken.	Pass	9, 12, 14, 23

8.3.16			Valid (SQL injection attempt) 1 ; DROP TABLE VehicleTypes ; 1.0	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle type" displayed, and the form refreshed with the new data. The table will not be affected.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle type” displayed, and the form refreshed with the new data. The table is not affected.	Pass	9, 12, 14, 24
8.4.1	btnAdd	Ensures that the user is able to add new vehicle type details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 10, 23
8.4.2			Extreme: Click event - just permitted (user has authorisation level 1)	The add logic will be executed as expected.	The add logic executes as expected.	Pass	2, 7, 10, 24
8.4.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 10, 24
8.4.4			Click event after a reset *empty* *empty* 0.0	A message box of errors will be displayed with the text “Vehicle type name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Vehicle type name cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23

8.4.5			Valid: 1 Hatchie 1.2	A message box will be displayed with the text “Vehicle type has been added successfully” before the table is updated with the latest version of the data including the new record.	A message box is displayed with the text “Vehicle type has been added successfully” before the table is updated with the latest version of the data including the new record.	Pass	9, 10, 14, 24
8.4.6			Valid (existing details added): 1 Hatchie 1.2			Pass	9, 10, 14, 24
8.4.7			Null (name is empty) 1 *empty* 1.0	A message box of errors will be displayed with the text “Vehicle type name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Vehicle type name cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23
8.4.8			Null (name is whitespace) 1 1.0			Pass	9, 10, 14, 23
8.4.9			Valid (name has leading spaces) 1 Sedane 1.0	A message box will be displayed with the text “Vehicle type has been added successfully” before the table is updated with the latest version of the data including the new record. The vehicle type name will be stripped of its whitespace.	A message box is displayed with the text “Vehicle type has been added successfully” before the table is updated with the latest version of the data including the new record. The vehicle type name is stripped of its whitespace.	Pass	9, 10, 14, 24

8.4.10			Valid (name has trailing spaces) 1 Sedanv 1.0			Pass	9, 10, 14, 24
8.4.11			Extreme (name is 50 characters) 1 SedanSedan SedanSedan SedanSedan SedanSedan SedanSedan 1.0	A message box will be displayed with the text “Vehicle type has been added successfully” before the table is updated with the latest version of the data including the new record.	A message box is displayed with the text “Vehicle type has been added successfully” before the table is updated with the latest version of the data including the new record.	Pass	9, 10, 14, 24
8.4.12			Invalid (name is longer than 50 characters) 1 SedanSedan SedanSedan SedanSedan SedanSedan SedanSedan Sedan 1.0	A message box of errors will be displayed with the text “Vehicle type name cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Vehicle type name cannot be longer than 50 characters.” No further action is taken.	Pass	9, 10, 14, 23

8.4.13			Valid (SQL injection attempt) 1 ; DROP TABLE VehicleTypes ; 1.0	A message box will be displayed with the text “Vehicle type has been added successfully” before the table is updated with the latest version of the data including the new record. The table will not be affected.	A message box is displayed with the text “Vehicle type has been added successfully” before the table is updated with the latest version of the data including the new record. The table is not affected.	Pass	9, 10, 14, 24
8.5.1	btnDelete	Ensures that the user is allowed to mark records as deleted in the database, preventing future loading and consideration	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 13, 23
8.5.2			Extreme: Click event - just permitted (user has authorisation level 1)	The delete logic will be executed as expected.	The delete logic executes as expected.	Pass	2, 7, 13, 24
8.5.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 13, 24
8.5.4			Click event - (“No” clicked at dialog) 18	No further action will be taken.	No further action is taken.	Pass	13, 24

8.5.5			Valid: Click event (vehicle type selected) 18	The selected vehicle will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record.	The selected vehicle is marked as deleted from the database, and this is indicated with the absence of the record in the table.	Pass	13, 14, 24
8.5.6			Invalid: Click event (after adding a new record)	A message box will be displayed with the text “Select a vehicle to delete.” No further action will be taken.	A message box is displayed with the text “Select a vehicle to delete.” No further action is taken.	Pass	9, 13, 23
8.5.7			Invalid: Click event (after saving a record)			Pass	9, 13, 23
8.5.8			Invalid: Click event (after resetting record)			Pass	9, 13, 23
8.5.9			Invalid: Click event (vehicle type associated with existing vehicles) 1	A message box will be displayed with the text “Vehicle type delete unsuccessful: Vehicle type is in use and cannot be deleted” No further action will be taken.	A message box is displayed with the text “Vehicle type delete unsuccessful: Vehicle type is in use and cannot be deleted” No further action is taken.	Pass	13, 14, 16d, 23
8.6.1	comboSearch Field	Ensures that the user can select which field they wish to search the database against	Selected item changed from “Name” to “Complexity”	Any searches on the table will be with respect to the “Complexity” field	All searches are now with respect to the “Complexity” field	Pass	8

8.6.2			Selected item changed back to “Name” from “Complexity”	Any searches on the table will be with respect to the “Name” field	All searches are now with respect to the “Name” field	Pass	8
8.7.1	btnReset	Ensures that the user is able to clear all fields in the form and deselect the currently selected vehicle type, preparing the form for new data entry	Click event - values had been typed in	All fields will be cleared, and any vehicle types will be unselected	All fields have been cleared, and the vehicle type has been unselected	Pass	14
8.7.2			Click event - values had been populated after selecting a vehicle type			Pass	14
8.8.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpVehicleTypes will be opened on top of frmVehicleTypeData	frmHelpVehicleTypes is opened on top of frmVehicleTypeData	Pass	26
8.8.2			Click event away from frmHelp VehicleTypes	frmHelpVehicleTypes will be closed	frmHelpVehicleTypes is closed	Pass	26
8.9.1	txtSearch	Ensures that the user is able to search the database using the selected search type, search field and provided search substring	Text input after form initially loads. “Contains” and “Vehicle Type ID” appear selected	The table will be updated to display only records which contain the digit “6” in their ID	The table is updated to display only records which contain the digit “6” in their ID	Pass	8

8.9.2			“Exact” and “Vehicle Type ID” are selected 6	The table will be updated to display only the record with “6” as its ID	The table is updated to display only the record with “6” as its ID	Pass	8
8.9.3			Click event - btnReset is clicked while a search has been input	The table will display all records on the system, ignoring the search until it is updated again	The table displays all records on the system, ignoring the search. When the search is updated again, the table is filtered once more.	Pass	8
8.9.4			Invalid: “Exact” and “Vehicle Type ID” are selected a	The table will be empty	The table is empty	Pass	8, 9
8.9.5			Null: “Contains” and “Vehicle Type ID” are selected *empty*	The table will display all records present in the database	The table displays all records present in the database	Pass	8
8.10.1	numComplexity	Ensures that the user can only input valid vehicle type complexities	Invalid (attempting to input a complexity less than 0): Attempt to click to and type -0.1	When clicked, no further action will be taken. When typed in, the complexity will go to the minimum of 0.0.	When clicked, no further action is taken. When typed in, the complexity goes to the minimum of 0.0.	Pass	9

8.10.2			Invalid (attempting to input a complexity above 5): Attempt to click to and type 5.1	When clicked, no further action will be taken. When typed in, the complexity will go to the maximum of 5.0.	When clicked, no further action is taken. When typed in, the complexity goes to the maximum of 5.0.	Pass	9
8.10.3			Valid: 2.0 Click event up then down	Complexity will go up to 2.1, then back to 2.0.	Complexity goes up to 2.1, then back to 2.0	Pass	9
8.10.4			Valid: 3.1 Complexity typed in	Complexity will be set to 3.1	Complexity is set to 3.1	Pass	9

frmPartData

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
9.1.1	frmPartData	Ensures that part details are loaded and displayed properly on access, handling missing data gracefully	Load event	dgvParts will be populated with the part records stored on the database. No further action will be taken.	dgvParts is populated with the part records stored on the database. No further action is taken.	Pass	6e, 8
9.1.2			Load event - no Parts table present	A message box will be displayed with the text "Error retrieving parts: " followed by an appropriate error message, followed by another with the text "Error loading parts. Please contact the administrator." before all CRUD and search controls are disabled.	A message box is displayed with the text "Error retrieving parts:" followed by an appropriate error message, followed by another with the text "Error loading parts. Please contact the administrator." before all CRUD and search controls are disabled.	Pass	15, 23
9.2.1	dgvParts	Ensures that the user is able to select part records for editing or deleting	Click event on a record	The details for that record will be populated in the appropriate fields	The details for that record are populated in the appropriate fields	Pass	6e, 8, 11
9.2.2			Click event on a record after a reset			Pass	6e, 8, 11
9.2.3			Click event on a recently added record			Pass	6e, 8, 11
9.2.4			Click event on a record that has been updated	The most recent details for that record will be populated in the appropriate fields	The most recent details for that record are populated in the appropriate fields	Pass	6e, 8, 11
9.3.1	btnSave	Ensures that the user is able to save updated part details to the database, preventing	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 12, 23

9.3.2		unauthorised or invalid data entry	Extreme: Click event - just permitted (user has authorisation level 1)	The save logic will be executed as expected.	The save logic executes as expected.	Pass	2, 7, 12, 24
9.3.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 12, 24
9.3.4			Click event - no part selected *empty* Engine Oil (3L) 22.00 35.00 40	A message box will be displayed with the text "Select a part to edit." No further action will be taken.	A message box is displayed with the text "Select a part to edit." No further action is taken.	Pass	9, 12, 23
9.3.5			Click event after a reset			Pass	9, 12, 14, 23
9.3.6			Valid ("No" clicked at dialog): 1 Engine Oil (5L) 22.00 38.00 40	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking "No", no further action will be taken.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking "No", no further action is taken.	Pass	9, 12, 24
9.3.7			Valid ("Yes" clicked at dialog): 1 Engine Oil (5L) 22.00 38.00 40	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking "Yes", the new details will be stored to the database, a message box with the text "Successfully updated part" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking "Yes", the new details are stored to the database, a message box with the text "Successfully updated part" displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

9.3.8			Valid (same part updated again): 1 Engine Oil (10L) 22.00 38.00 40			Pass	9, 12, 14, 24
9.3.9			All fields blank 1 *empty* *empty* *empty* *empty*	A message box of errors will be displayed with the text “Part name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Part name cannot be empty.” No further action is taken.	Pass	9, 12, 14, 23
9.3.10			Null (name is empty) 1 *empty* 22.00 38.00 40			Pass	9, 12, 14, 23
9.3.11			Null (name is whitespace) 1 22.00 38.00 40			Pass	9, 12, 14, 23

9.3.12			Valid (name has leading spaces) 1 Engine Oil (30L) 22.00 38.00 40	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle type" displayed, and the form refreshed with the new data. Part name will be stripped of whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle type” displayed, and the form refreshed with the new data. Part name is stripped of whitespace.	Pass	9, 12, 14, 24
9.3.13			Valid (name has trailing spaces) 1 Engine Oil (40L) 22.00 38.00 40			Pass	9, 12, 14, 24
9.3.14			Extreme (name is 50 characters) 1 marcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarco 22.00 38.00 40	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle type" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated vehicle type” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

9.3.15			Invalid (name is longer than 50 characters) 1 marcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolo 22.00 38.00 40	A message box of errors will be displayed with the text “Part name cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Part name cannot be longer than 50 characters.” No further action is taken.	Pass	9, 12, 14, 23
9.3.16			Valid (SQL injection attempt) 1 ; DROP TABLE Parts; 22.00 38.00 40	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle type" displayed, and the form refreshed with the new data. The table will be unaffected.	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated vehicle type" displayed, and the form refreshed with the new data. The table is unaffected.	Pass	9, 12, 14, 24
9.4.1	btnAdd	Ensures that the user is able to add new part details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 10, 23
9.4.2			Extreme: Click event - just permitted (user has authorisation level 1)	The add logic will be executed as expected.	The add logic executes as expected.	Pass	2, 7, 10, 24

9.4.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 10, 24
9.4.4			Click event after a reset *empty* *empty* 0.0 0.0 0	A message box of errors will be displayed with the text “Part name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Part name cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23
9.4.5			Valid: 1 Engine Oi (5L) 22.00 38.00 40	A message box will be displayed with the text “Part added successfully” before the table is updated with the latest version of the data including the new record.	A message box is displayed with the text “Part added successfully” before the table is updated with the latest version of the data including the new record.	Pass	9, 10, 14, 24
9.4.6			Valid (existing details added): 1 Engine Oi (5L) 22.00 38.00 40			Pass	9, 10, 14, 24
9.4.7			Null (name is empty) 1 *empty* 22.00 38.00 40	A message box of errors will be displayed with the text “Part name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Part name cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23

9.4.8			Null (name is whitespace) 1 22.00 38.00 40			Pass	9, 10, 14, 23
9.4.9			Valid (name has leading spaces) 1 Engine Oil (30L) 22.00 38.00 40	A message box will be displayed with the text “Part added successfully” before the table is updated with the latest version of the data including the new record. Whitespace will be stripped off the name.	A message box is displayed with the text “Part added successfully” before the table is updated with the latest version of the data including the new record. Whitespace is stripped off the name.	Pass	9, 10, 14, 24
9.4.10			Valid (name has trailing spaces) 1 Engine Oil (40L) 22.00 38.00 40			Pass	9, 10, 14, 24
9.4.11			Extreme (name is 50 characters) 1 marcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarco 22.00 38.00 40	A message box will be displayed with the text “Part added successfully” before the table is updated with the latest version of the data including the new record.	A message box is displayed with the text “Part added successfully” before the table is updated with the latest version of the data including the new record.	Pass	9, 10, 14, 24

9.4.12			Invalid (name is longer than 50 characters) 1 marcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolomarcopolo 22.00 38.00 40	A message box of errors will be displayed with the text “Part name cannot be longer than 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Part name cannot be longer than 50 characters.” No further action is taken.	Pass	9, 10, 14, 23
9.4.13			Valid (SQL injection attempt) 1 ; DROP TABLE Parts; 22.00 38.00 40	A message box will be displayed with the text “Part added successfully” before the table is updated with the latest version of the data including the new record. The table will be unaffected.	A message box is displayed with the text “Part added successfully” before the table is updated with the latest version of the data including the new record. The table is unaffected.	Pass	9, 10, 14, 24
9.5.1	btnDelete	Ensures that the user is allowed to mark records as deleted in the database, preventing future loading and consideration	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 13, 23
9.5.2			Extreme: Click event - just permitted (user has authorisation level 1)	The delete logic will be executed as expected.	The delete logic executes as expected.	Pass	2, 7, 13, 24

9.5.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 13, 24
9.5.4			Click event - (“No” clicked at dialog) 17	No further action will be taken.	No further action is taken.	Pass	13, 24
9.5.5			Valid: Click event (part selected) 17	The selected part will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record.	The selected part is marked as deleted from the database, and this is indicated with the absence of the record in the table.	Pass	13, 14, 24
9.5.6			Invalid: Click event (after adding a new record)	A message box will be displayed with the text “Select a part to delete.” No further action will be taken.	A message box is displayed with the text “Select a part to delete.” No further action is taken.	Pass	9, 13, 23
9.5.7			Invalid: Click event (after saving a record)			Pass	9, 13, 23
9.5.8			Invalid: Click event (after resetting record)			Pass	9, 13, 23
9.5.9			Invalid: Click event (part associated with existing appointments) 1	A message box will be displayed with the text “Part delete unsuccessful: part is used in service appointments and cannot be deleted” No further action will be taken.	A message box is displayed with the text “Part delete unsuccessful: part is used in service appointments and cannot be deleted” No further action is taken.	Pass	13, 14, 16e, 23

9.6.1	comboSearch Field	Ensures that the user can select which field they wish to search the database against	Selected item changed from “Part Name” to “Quantity”	Any searches on the table will be with respect to the “Quantity” field	All searches are now with respect to the “Quantity” field	Pass	8
9.6.2			Selected item changed back to “Part Name” from “Quantity”	Any searches on the table will be with respect to the “Part Name” field	All searches are now with respect to the “Part Name” field	Pass	8
9.7.1	btnReset	Ensures that the user is able to clear all fields in the form and deselect the currently selected part, preparing the form for new data entry	Click event - values had been typed in	All fields will be cleared, and any parts will be unselected	All fields have been cleared, and the part has been unselected	Pass	14
9.7.2			Click event - values had been populated after selecting a part			Pass	14
9.8.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpParts will be opened on top of frmPartData	frmHelpParts is opened on top of frmPartData	Pass	26
9.8.2			Click event away from frmHelpParts	frmHelpParts will be closed	frmHelpParts is closed	Pass	26
9.9.1	txtSearch	Ensures that the user is able to search the database using the selected search type, search field and provided search substring	Text input after form initially loads. “Contains” and “Part ID” appear selected 6	The table will be updated to display only records which contain the digit “6” in their ID	The table is updated to display only records which contain the digit “6” in their ID	Pass	8
9.9.2			“Exact” and “Part ID” are selected 6	The table will be updated to display only the record with “6” as its ID	The table is updated to display only the record with “6” as its ID	Pass	8

9.9.3			Click event - btnReset is clicked while a search has been input	The table will display all records on the system, ignoring the search until it is updated again	The table displays all records on the system, ignoring the search. When the search is updated again, the table is filtered once more.	Pass	8
9.9.4			Invalid: “Exact” and “Part ID” are selected a	The table will be empty	The table is empty	Pass	8, 9
9.9.5			Null: “Contains” and “Part ID” are selected *empty*	The table will display all records present in the database	The table displays all records present in the database	Pass	8
9.10.1	numBuyPrice	Ensures that the user can only input valid part buy prices	Invalid (attempting to input a negative number): Attempting to click to and type -1	When clicked, no further action will be taken. When typed in, the price will go to the minimum of 0.	When clicked, no further action is taken. When typed in, the price goes to the minimum of 0.	Pass	9
9.10.2			Valid: 5.00 Click event up then down	Price will go up to 6, and then back to 5.	Price goes up to 6, and then back to 5.	Pass	9
9.10.3			Valid: 5.47 Price typed in	Price will be set to 5.47	Price is set to 5.47	Pass	9

9.10.4			Invalid: (attempting to overflow the integer value) Attempting to click to and type 2147483648	When clicked, no further action will be taken. When typed in, the price will go to the maximum of 2147483647.	When clicked, no further action is taken. When typed in, the price goes to the maximum of 2147483647.	Pass	9
9.11.1	numSellPrice	Ensures that the user can only input valid part sell prices	Invalid (attempting to input a negative number): Attempting to click to and type -1	When clicked, no further action will be taken. When typed in, the price will go to the minimum of 0.	When clicked, no further action is taken. When typed in, the price goes to the minimum of 0.	Pass	9
9.11.2			Valid: 5 Click event up then down	Price will go up to 6, and then back to 5.	Price goes up to 6, and then back to 5.	Pass	9
9.11.3			Valid: 5.48 Price typed in	Price will be set to 5.48	Price is set to 5.48	Pass	9

9.11.4			Invalid: (attempting to overflow the integer value) Attempting to click to and type 2147483648	When clicked, no further action will be taken. When typed in, the price will go to the maximum of 2147483647.	When clicked, no further action is taken. When typed in, the price goes to the maximum of 2147483647.	Pass	9
9.12.1	numQuantity	Ensures that the user can only input valid part quantities	Invalid (attempting to input a negative number): Attempting to click to and type -1	When clicked, no further action will be taken. When typed in, the quantity will go to the minimum of 0.	When clicked, no further action is taken. When typed in, the quantity goes to the minimum of 0.	Pass	9
9.12.2			Valid: 5 Click event up then down	Quantity will go up to 6, and then back to 5.	Price goes up to 6, and then back to 5.	Pass	9
9.12.3			Valid: 5 Price typed in	Price will be set to 5	Price is set to 5	Pass	9

9.12.4			Invalid: (attempting to overflow the integer value) Attempting to click to and type 2147483648	When clicked, no further action will be taken. When typed in, the price will go to the maximum of 2147483647.	When clicked, no further action is taken. When typed in, the price goes to the maximum of 2147483647.	Pass	9

frmServiceData

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
10.1.1	frmServiceData	Ensures that service details are loaded and displayed properly on access, handling missing data gracefully	Load event	dgvServices will be populated with the part records stored on the database. No further action will be taken.	dgvServices is populated with the part records stored on the database. No further action is taken.	Pass	6g, 8
10.1.2			Load event - no Services table present	A message box will be displayed with the text "There was an error retrieving services: " followed by an appropriate error message, followed by another with the text "There was an error attempting to load service data. Please amend erroneous data in the database before proceeding." before all CRUD and search controls are disabled.	A message box is displayed with the text "There was an error retrieving services: " followed by an appropriate error message, followed by another with the text "There was an error attempting to load service data. Please amend erroneous data in the database before proceeding." before all CRUD and search controls are disabled.	Pass	15, 23
10.2.1	dgvServices	Ensures that the user is able to select service records for editing or deleting	Click event on a record	The details for that record will be populated in the appropriate fields	The details for that record are populated in the appropriate fields	Pass	6g, 8, 11
10.2.2			Click event on a record after a reset			Pass	6g, 8, 11
10.2.3			Click event on a recently added record			Pass	6g, 8, 11
10.2.4			Click event on a record that has been updated	The most recent details for that record will be populated in the appropriate fields	The most recent details for that record are populated in the appropriate fields	Pass	6g, 8, 11
10.3.1	btnSave	Ensures that the user is able to save updated	Invalid: Click event - not permitted	A message box will be displayed with the text “This action requires “Restricted	A message box is displayed with the text “This action requires “Restricted write”	Pass	2, 7, 12, 23

		service details to the database, preventing unauthorised or invalid data entry	(user has authorisation level 0)	write" permission, but you have "Read-only." No further action will be taken.	permission, but you have "Read-only." No further action is taken.		
10.3.2			Extreme: Click event - just permitted (user has authorisation level 1)	The save logic will be executed as expected.	The save logic executes as expected.	Pass	2, 7, 12, 24
10.3.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 12, 24
10.3.4			Click event - no service selected *empty* Minor Service 120 2 "Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection."	A message box will be displayed with the text "Choose a service to edit." No further action will be taken.	A message box is displayed with the text "Choose a service to edit." No further action is taken.	Pass	9, 12, 23
10.3.5			Click event after a reset			Pass	9, 12, 14, 23

10.3.6			Valid (“No” clicked at dialog): 3 Minor Service 130 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action will be taken.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “No”, no further action is taken.	Pass	9, 12, 24
10.3.7			Valid (“Yes” clicked at dialog): 3 Minor Service 130 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated service" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated service” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

10.3.8			Valid (same service updated again): 3 Minor Service 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”			Pass	9, 12, 14, 24
10.3.9			Null (name is empty) 3 *empty* 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A message box of errors will be displayed with the text “Service name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Service name cannot be empty.” No further action is taken.	Pass	9, 12, 14, 23

10.3.10			Null (name is whitespace) 3 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”			Pass	9, 12, 14, 23
10.3.11			Valid (name has leading spaces) 3 Minor Services 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated service" displayed, and the form refreshed with the new data. The whitespace will be trimmed from the name.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated service” displayed, and the form refreshed with the new data. The whitespace is trimmed from the name.	Pass	9, 12, 14, 24

10.3.12			Valid (name has trailing spaces) 3 Minor Servic 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”			Pass	9, 12, 14, 24
10.3.13			Extreme (name is 50 characters) 3 blehblehble hblehblehbl ehblehblehb lehblehbleh blehbl 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated service" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated service” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

10.3.14			Invalid (name more than 50 characters) 3 blehblehble hblehblehbl ehblehblehb lehblehbleh blehbleh 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A message box of errors will be displayed with the text “Service name cannot exceed 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Service name cannot exceed 50 characters.” No further action is taken.	Pass	9, 12, 14, 23
10.3.15			Null (description empty) 3 Minor Service 150 2 *empty*	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated service" displayed, and the form refreshed with the new data. Description will be stored as an empty string.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated service” displayed, and the form refreshed with the new data. Description is stored as an empty string.	Pass	9, 12, 14, 24

10.3.16			Valid (description has leading spaces) 3 Minor Service 150 2 “ Basic maintenance ”	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated service" displayed, and the form refreshed with the new data. Description will be trimmed of its whitespace.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated service” displayed, and the form refreshed with the new data. Description is trimmed of its whitespace.	Pass	9, 12, 14, 24
10.3.17			Valid (description has trailing spaces) 3 Minor Service 150 2 “Basic maintenance ”			Pass	9, 12, 14, 24
10.3.18			Extreme (description is 500 characters) 3 Minor Service 150 2 “dummy dummy dummy dummy	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated service" displayed, and the form refreshed with the new data.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated service” displayed, and the form refreshed with the new data.	Pass	9, 12, 14, 24

			dummy dummy dummy

			dummy dummy dummy

10.3.19			Invalid (description is more than 500 characters) 3 Minor Service 150 2 “dummy dummy dummy

			dummy dummy dummy

			dummy dummy dummy dummy dummy dummy dummy dummy dummy”				
10.3.20			Valid (SQL injection attempt) 3 ; DROP TABLE Services; 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A dialog will be displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details will be stored to the database, a message box with the text "Successfully updated service" displayed, and the form refreshed with the new data. The table will not be affected.	A dialog is displayed with the text "Are you sure that you want to overwrite changes?" On clicking “Yes”, the new details are stored to the database, a message box with the text "Successfully updated service” displayed, and the form refreshed with the new data. The table is not affected.	Pass	9, 12, 14, 24
10.4.1	btnAdd	Ensures that the user is able to add new service details to the database, preventing unauthorised or invalid data entry	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 10, 23

10.4.2			Extreme: Click event - just permitted (user has authorisation level 1)	The add logic will be executed as expected.	The add logic executes as expected.	Pass	2, 7, 10, 24
10.4.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 10, 24
10.4.4			Click event after a reset *empty* *empty* 0 1 “”	A message box of errors will be displayed with the text “Service name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Service name cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23
10.4.5			Valid: 3 Minor Service 130 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety	A message box will be displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record.	A message box is displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record.	Pass	9, 10, 14, 24

			inspection.”				
10.4.6			Valid (existing details added): 3 Minor Service 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”			Pass	9, 10, 14, 24
10.4.7			Null (name is empty) 3 *empty* 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A message box of errors will be displayed with the text “Service name cannot be empty.” No further action will be taken.	A message box of errors is displayed with the text “Service name cannot be empty.” No further action is taken.	Pass	9, 10, 14, 23

10.4.8			Null (name is whitespace) 3 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”			Pass	9, 10, 14, 23
10.4.9			Valid (name has leading spaces) 3 Minor Services 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A message box will be displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record. The whitespace will be trimmed from the name.	A message box is displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record. The whitespace is trimmed from the name.	Pass	9, 10, 14, 24

10.4.10			Valid (name has trailing spaces) 3 Minor Servic 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”			Pass	9, 10, 14, 24
10.4.11			Extreme (name is 50 characters) 3 blehblehble hblehblehbl ehblehblehb lehblehbleh blehbl 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A message box will be displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record.	A message box is displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record.	Pass	9, 10, 14, 24

10.4.12			Invalid (name more than 50 characters) 3 blehblehble hblehblehbl ehblehblehb lehblehbleh blehbleh 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A message box of errors will be displayed with the text “Service name cannot exceed 50 characters.” No further action will be taken.	A message box of errors is displayed with the text “Service name cannot exceed 50 characters.” No further action is taken.	Pass	9, 10, 14, 23
10.4.13			Null (description empty) 3 Minor Service 150 2 *empty*	A message box will be displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record. Description will be stored as an empty string.	A message box is displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record. Description is stored as an empty string.	Pass	9, 10, 14, 24

10.4.14			Valid (description has leading spaces) 3 Minor Service 150 2 “ Basic maintenance ”	A message box will be displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record. Description will have its whitespace trimmed.	A message box is displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record. Description has its whitespace trimmed.	Pass	9, 10, 14, 24
10.4.15			Valid (description has trailing spaces) 3 Minor Service 150 2 “Basic maintenance ”			Pass	9, 10, 14, 24
10.4.16			Extreme (description is 500 characters) 3 Minor Service 150 2 “dummy dummy dummy dummy	A message box will be displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record.	A message box is displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record.	Pass	9, 10, 14, 24

			dummy dummy dummy

			dummy dummy dummy

10.4.17			Invalid (description is more than 500 characters) 3 Minor Service 150 2 “dummy dummy dummy

			dummy dummy dummy

			dummy dummy dummy dummy dummy dummy dummy dummy dummy”				
10.4.18			Valid (SQL injection attempt) 3 ; DROP TABLE Services; 150 2 “Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.”	A message box will be displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record. The table will not be affected.	A message box is displayed with the text “Service has added successfully” before the table is updated with the latest version of the data including the new record. The table is not affected.	Pass	9, 10, 14, 24
10.5.1	btnDelete	Ensures that the user is allowed to mark records as deleted in the database, preventing future loading and consideration	Invalid: Click event - not permitted (user has authorisation level 0)	A message box will be displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action will be taken.	A message box is displayed with the text “This action requires “Restricted write” permission, but you have “Read-only.” No further action is taken.	Pass	2, 7, 13, 23

10.5.2			Extreme: Click event - just permitted (user has authorisation level 1)	The delete logic will be executed as expected.	The delete logic executes as expected.	Pass	2, 7, 13, 24
10.5.3			Valid: Click event - just permitted (user has authorisation level 2)			Pass	2, 7, 13, 24
10.5.4			Click event - (“No” clicked at dialog) 9	No further action will be taken.	No further action is taken.	Pass	13, 24
10.5.5			Valid: Click event (part selected) 9	The selected service will be marked as deleted from the database, and this will be indicated with the table updating with the latest version of the data, excluding the deleted record.	The selected service is marked as deleted from the database, and this is indicated with the absence of the record in the table.	Pass	13, 14, 24
10.5.6			Invalid: Click event (after adding a new record)	A message box will be displayed with the text “Select a service to delete.” No further action will be taken.	A message box is displayed with the text “Select a service to delete.” No further action is taken.	Pass	9, 13, 23
10.5.7			Invalid: Click event (after saving a record)			Pass	9, 13, 23

			Invalid: Click event (after resetting record)			Pass	9, 13, 23
10.5.8			Invalid: Click event (service associated with existing appointments) 1	A message box will be displayed with the text “Service delete unsuccessful: Service is currently used in service appointments and cannot be deleted.” No further action will be taken.	A message box is displayed with the text “Service delete unsuccessful: Service is currently used in service appointments and cannot be deleted.” No further action is taken.	Pass	13, 14, 16g, 23
10.6.1	comboSearch Field	Ensures that the user can select which field they wish to search the database against	Selected item changed from “Name” to “Description”	Any searches on the table will be with respect to the “Description” field	All searches are now with respect to the “Description” field	Pass	8
10.6.2			Selected item changed back to “Name” from “Description”	Any searches on the table will be with respect to the “Name” field	All searches are now with respect to the “Name” field	Pass	8
10.7.1	btnReset	Ensures that the user is able to clear all fields in the form and deselect the currently selected service, preparing the form for new data entry	Click event - values had been typed in	All fields will be cleared, and any services will be unselected	All fields have been cleared, and the service has been unselected	Pass	14
10.7.2			Click event - values had been populated after selecting a service			Pass	14

10.8.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpServices will be opened on top of frmServiceData	frmHelpServices is opened on top of frmServiceData	Pass	26
10.8.2			Click event away from frmHelp Services	frmHelpServices will be closed	frmHelpServices is closed	Pass	26
10.9.1	txtSearch	Ensures that the user is able to search the database using the selected search type, search field and provided search substring	Text input after form initially loads. “Contains” and “Service ID” appear selected 6	The table will be updated to display only records which contain the digit “6” in their ID	The table is updated to display only records which contain the digit “6” in their ID	Pass	8
10.9.2			“Exact” and “Service ID” are selected 6	The table will be updated to display only the record with “6” as its ID	The table is updated to display only the record with “6” as its ID	Pass	8
10.9.3			Click event - btnReset is clicked while a search has been input	The table will display all records on the system, ignoring the search until it is updated again	The table displays all records on the system, ignoring the search. When the search is updated again, the table is filtered once more.	Pass	8
10.9.4			Invalid: “Exact” and “Service ID” are selected a	The table will be empty	The table is empty	Pass	8, 9

10.9.5			Null: “Contains” and “Service ID” are selected *empty*	The table will display all records present in the database	The table displays all records present in the database	Pass	8
10.10.1	numPrice	Ensures that the user can only input valid service prices	Invalid (attempting to input a negative number): Attempting to click to and type `-1`	When clicked, no further action will be taken. When typed in, the price will go to the minimum of 0.	When clicked, no further action is taken. When typed in, the price goes to the minimum of 0.	Pass	9
10.10.2			Valid: `5` Click event up then down	Price will go up to 6, and then back to 5.	Price goes up to 6, and then back to 5.	Pass	9
10.10.3			Valid: `5` Price typed in	Price will be set to 5	Price is set to 5	Pass	9

10.10.4			Invalid: (attempting to overflow the integer value) Attempting to click to and type 2147483648	When clicked, no further action will be taken. When typed in, the price will go to the maximum of 2147483647.	When clicked, no further action is taken. When typed in, the price goes to the maximum of 2147483647.	Pass	9
10.11.1	numDuration	Ensures that the user can only input valid service durations (between 1 and 10 hours inclusive)	Invalid (attempting to input a number less than 1): Attempting to click to and type 0	When clicked, no further action will be taken. When typed in, the duration will go to the minimum of 1.	When clicked, no further action is taken. When typed in, the duration goes to the minimum of 1.	Pass	9
10.11.2			Invalid (attempting to input a number greater than 10): Attempting to click to and type 11	When clicked, no further action will be taken. When typed in, the duration will go to the maximum of 10.	When clicked, no further action is taken. When typed in, the duration goes to the maximum of 10.	Pass	9

10.11.3			Valid: 5 Click event up then down	Duration will go up to 6, and then back to 5.	Duration goes up to 6, and then back to 5.	Pass	9
10.11.4			Valid: 5 Duration typed in	Duration will be set to 5	Duration is set to 5	Pass	9
10.11.5			Invalid: (attempting to overflow the integer value) Attempting to click to and type 2147483648	When clicked, no further action will be taken. When typed in, the duration will go to the maximum of 2147483647.	When clicked, no further action is taken. When typed in, the duration goes to the maximum of 2147483647.	Pass	9
10.12.1	lblRemainingCharacters	Ensures that the label dynamically updates as the user types text into the description textbox, indicating how many of the allocated 500 characters they have remaining	Valid (character typed into description)	The label text will become “Remaining characters: “ followed by one less than the previous value of the label text	The label text becomes“Remaining characters: “ followed by one less than the previous value of the label text	Pass	
10.12.2			Valid (character removed from description)	The label text will become “Remaining characters: “ followed by one more than the previous value of the label text	The label text becomes“Remaining characters: “ followed by one more than the previous value of the label text	Pass	

10.12.3			Extreme (description is empty)	The label text will become “Remaining characters: 500“	The label text becomes “Remaining characters: 500“	Pass	
10.12.4			Extreme (description is 501 characters)	The label text will become “Remaining characters: -1“	The label text becomes “Remaining characters: -1“	Pass	

frmServiceAppointments

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
11.1.1	frmServiceAppointments	Ensures that the form displays the most up-to-date version of the service appointments in the database, preventing any interaction in the event of missing data	Load event	dtpAppointmentDate will be set to the first valid date (whether it be the current day or the next workday), all CRUD controls will be disabled, and no appointment will be selected. No further action will be taken.	dtpAppointmentDate is set to the first valid date (in this case, the next workday), all CRUD controls are disabled, and no appointment is selected. No further action is taken.	Pass	6h, 8
11.1.2			Load event - no Service Appointments table present	A sequence of message boxes with the text “Error retrieving appointments: “ followed by an appropriate error message will be displayed. After these, a final message box will be displayed with the text "No appointments could be loaded for any mechanics. Please manually insert at least one appointment for a mechanic before attempting to load their appointments." All CRUD and search functionality will be disabled, and the appointment viewer will be removed.	A sequence of message boxes with the text “Error retrieving appointments: “ followed by an appropriate error message are displayed. After these, a final message box is displayed with the text "No appointments could be loaded for any mechanics. Please manually insert at least one appointment for a mechanic before attempting to load their appointments." All CRUD and search functionality is disabled, and the appointment viewer is removed.	Pass	15, 23

11.1.3			Load event - no Vehicles table present	A message box will be displayed with the text “An error occurred while retrieving vehicle data:” followed by an appropriate error message, followed by another with the text "No vehicles could be loaded. Please check the vehicles table before proceeding." All CRUD and search functionality will be disabled, and the appointment viewer will be removed.	A message box is displayed with the text “An error occurred while retrieving vehicle data:” followed by an appropriate error message, followed by another with the text "No vehicles could be loaded. Please check the vehicles table before proceeding." All CRUD and search functionality is disabled, and the appointment viewer is removed.	Pass	15, 23
11.1.4			Load event - no Services table present	A message box will be displayed with the text "There was an error retrieving services:” followed by an appropriate error message, followed by another with the text "No services could be loaded. Please check the services table before proceeding." All CRUD and search functionality will be disabled, and the appointment viewer will be removed.	A message box is displayed with the text "There was an error retrieving services:” followed by an appropriate error message, followed by another with the text "No services could be loaded. Please check the services table before proceeding." All CRUD and search functionality is disabled, and the appointment viewer is removed.	Pass	15, 23
11.1.5			Load event - no Parts table present	A sequence of message boxes will be displayed with the text "Error retrieving parts:” followed by an appropriate error message, followed by a single message box with the text "No parts could be loaded. Please check the parts table before proceeding." All CRUD and search functionality will be disabled, and the appointment viewer will be removed.	A sequence of message boxes is displayed with the text "Error retrieving parts:” followed by an appropriate error message, followed by a single message box with the text "No parts could be loaded. Please check the parts table before proceeding." All CRUD and search functionality is disabled, and the appointment viewer is removed.	Pass	15, 23

11.1.6			Load event - on a Saturday (21st of March)	dtpAppointmentDate will be set to the first valid date (the 23rd of March), all CRUD controls will be disabled, and no appointment will be selected. No further action will be taken.	dtpAppointmentDate is set to the first valid date (the 23rd of March), all CRUD controls are disabled, and no appointment is selected. No further action is taken.	Pass	6h, 8, 20
11.1.7			Load event - on a Saturday (22nd of March)			Pass	6h, 8, 20
11.1.8			Load event - reloading after modifying mechanic names in frmStaffData	The form will load as expected, with the new mechanic's name displayed on the left side of the timetable.	The form loads as expected, with the new mechanic's name displayed on the left side of the timetable.	Pass	6h, 8
11.2.1	pnlViewer	Ensures that the provided panel allows the user to effectively manipulate timeblocks - both existing and new - and select appointments for editing, preventing invalid operations	Valid: Click event on an appointment	The form's fields will be populated with the selected appointment's details as found on the database, including the parts used for each service	The form's fields are populated with the selected appointment's details as found on the database, including the parts used for each service	Pass	8, 11, 18
11.2.2			Valid: Click event on a different appointment			Pass	8, 11, 18
11.2.3			Valid: Click event on an appointment not belonging to the current user (mechanic has authorisation level 2)			Pass	2, 7, 8, 11, 18

11.2.4			Invalid: Click event on an unauthorised appointment (mechanic has authorisation level 1, and clicks on an appointment that's not theirs)	A message box will be displayed with the text "You are not permitted to edit your colleagues' appointments." No further action will be taken.	A message box is displayed with the text "You are not permitted to edit your colleagues' appointments." No further action is taken.	Pass	2, 7, 8, 11, 18, 23
11.2.5			Invalid: Click event on an unauthorised empty space (mechanic has authorisation level 1, and clicks on an empty space that's not theirs)			Pass	2, 7, 8, 11, 18, 23
11.2.6			Invalid: Click event on an authorised empty space after having selected an appointment	No further action will be taken.	No further action is taken.	Pass	8, 11, 18
11.2.7			Invalid: Click event on an empty space after a reset			Pass	8, 11, 14, 18

11.2.8			Invalid: Click event on where an appointment used to be (after changing times)			Pass	8, 11, 18
11.2.9			Invalid: Click event on where an appointment used to be (after deleting)			Pass	8, 11, 18
11.2.10			Invalid: Click event on where an appointment used to be (after changing its mechanic)			Pass	8, 11, 18
11.2.11			Invalid: Click event above the timetable (appointment present directly underneath)			Pass	8, 11, 18

11.2.12			Invalid: Click event to the left of the timetable (appointment present directly to the right)			Pass	8, 11, 18
11.2.13			Invalid: Click event to the right of the timetable (appointment present directly to the left)			Pass	8, 11, 18
11.2.14			Invalid: Click event underneath the timetable (appointment present directly above)	No further action will be taken.	The program crashes.	Fail	8, 11, 18
11.2.15			Valid: Clicking on an authorised empty space in insert mode (left half of empty space)	A new appointment will be created - a timeblock coloured orange - of length 1 hour contained entirely within the clicked cell. The appointment will be automatically selected, but no further action will be taken.	A new appointment is created - a timeblock coloured orange - of length 1 hour contained entirely within the clicked cell. The appointment is automatically selected, but no further action is taken.	Pass	8, 10, 18

11.2.16			Valid: Clicking on an authorised empty space in insert mode (right half of empty space)			Pass	8, 10, 18
11.2.17			Valid: Clicking on a new appointment in insert mode	No further action will be taken.	No further action is taken.	Pass	8, 10, 18
11.2.18			Invalid: Clicking on an existing appointment in insert mode			Pass	8, 10, 18
11.2.19			Invalid: Click event above the timetable in insert mode (appointment present directly underneath)			Pass	8, 10, 18
11.2.20			Invalid: Click event above the timetable in insert mode (appointment not present directly underneath)			Pass	8, 10, 18

11.2.21			Invalid: Click event to the left of the timetable in insert mode (appointment present directly to the right)	No further action will be taken.	A new appointment is created - a timeblock coloured orange - of length 1 hour contained entirely within the leftmost cell. The appointment is automatically selected, but no further action is taken.	Fail	8, 10, 18
11.2.22			Invalid: Click event to the left of the timetable in insert mode (appointment not present directly to the right)	A new appointment will be created - a timeblock coloured orange - of length 1 hour contained entirely within the leftmost cell. The appointment will be automatically selected, but no further action will be taken.	A new appointment is created - a timeblock coloured orange - of length 1 hour contained entirely within the leftmost cell. The appointment is automatically selected, but no further action is taken.	Pass	8, 10, 18
11.2.23			Invalid: Click event to the right of the timetable in insert mode (appointment present directly to the left)	No further action will be taken.	No further action is taken.	Pass	8, 10, 18
11.2.24			Invalid: Click event to the right of the timetable in insert mode (appointment not present directly to the left)			Pass	8, 10, 18

11.2.25			Invalid: Click event underneath the timetable in insert mode (appointment present directly above)			Pass	8, 10, 18
11.2.26			Invalid: Click event underneath the timetable in insert mode (appointment not present directly above)			Pass	8, 10, 18
11.2.27			Invalid: Click event on an unauthorised space in insert mode (mechanic clicks on a schedule that isn't theirs, and they have authorisation level 1)	A message box will be displayed with the text "You are not permitted to edit your colleagues' appointments." No further action will be taken.	A message box is displayed with the text "You are not permitted to edit your colleagues' appointments." No further action is taken.	Pass	2, 7, 8, 10, 18, 23
11.2.28			Invalid: Click event on an authorised space in insert mode when a	No further action will be taken.	No further action is taken.	Pass	8, 10, 18

			new appointment already exists				
11.2.29			Invalid: Click event on an existing appointment in insert mode			Pass	8, 10, 18
11.2.30			Valid: Dragging the left side of an appointment	When moving the cursor left, the left end of the appointment moves left, and when moving the cursor right, the left end of the appointment moves right	When moving the cursor left, the left end of the appointment moves left, and when moving the cursor right, the left end of the appointment moves right	Pass	8, 12, 18
11.2.31			Valid: Dragging the left side of a new appointment			Pass	8, 10, 18
11.2.32			Valid: Dragging the right side of an appointment	When moving the cursor left, the right end of the appointment moves left, and when moving the cursor right, the right end of the appointment moves right	When moving the cursor left, the right end of the appointment moves left, and when moving the cursor right, the right end of the appointment moves right	Pass	8, 12, 18
11.2.33			Valid: Dragging the right side of a new appointment			Pass	8, 10, 18
11.2.34			Valid: Dropping the right side of an appointment on the left half of an empty space as to	The right end of the appointment will snap down to the closest time (in this case, 10:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The right end of the appointment snaps down to the closest time (in this case, 10:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 12, 18

			snap down				
11.2.35			Valid: Dropping the right side of an appointment on the right half of an empty space as to snap up	The right end of the appointment will snap up to the closest time (in this case, 11:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The right end of the appointment snaps to the closest time (in this case, 11:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 12, 18
11.2.36			Valid: Dropping the left side of an appointment on the left half of an empty space as to snap down	The left end of the appointment will snap down to the closest time (in this case, 10:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The left end of the appointment snaps down to the closest time (in this case, 10:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 12, 18
11.2.37			Valid: Dropping the left side of an appointment on the right half of an empty space as to snap up	The left end of the appointment will snap up to the closest time (in this case, 11:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The left end of the appointment snaps to the closest time (in this case, 11:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 12, 18
11.2.38			Valid: Dropping the right side of a new appointment on the left half of an empty space as to snap down	The right end of the appointment will snap down to the closest time (in this case, 10:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The right end of the appointment snaps down to the closest time (in this case, 10:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 10, 18

11.2.39			Valid: Dropping the right side of a new appointment on the right half of an empty space as to snap up	The right end of the appointment will snap up to the closest time (in this case, 11:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The right end of the appointment snaps to the closest time (in this case, 11:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 10, 18
11.2.40			Valid: Dropping the left side of a new appointment on the left half of an empty space as to snap down	The left end of the appointment will snap down to the closest time (in this case, 10:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The left end of the appointment snaps down to the closest time (in this case, 10:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 10, 18
11.2.41			Valid: Dropping the left side of a new appointment on the right half of an empty space as to snap up	The left end of the appointment will snap up to the closest time (in this case, 11:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The left end of the appointment snaps to the closest time (in this case, 11:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 10, 18
11.2.42			Invalid: Dragging and dropping the right side of an appointment as to cause it to snap down to length 0	A message box will be displayed with the message “You cannot have an appointment of length 0!” before the state of the appointment viewer and fields is reverted back to how they were before dragging was initiated.	A message box is displayed with the message “You cannot have an appointment of length 0!” before the state of the appointment viewer and fields is reverted back to how they were before dragging was initiated.	Pass	8, 9, 12, 18, 20, 23

11.2.43			Invalid: Dragging and dropping the left side of an appointment as to cause it to snap up to length 0			Pass	8, 9, 12, 18, 20, 23
11.2.44			Invalid: Dragging and dropping the right side of a new appointment as to cause it to snap down to length 0			Pass	8, 9, 10, 18, 20, 23
11.2.45			Invalid: Dragging and dropping the left side of a new appointment as to cause it to snap up to length 0			Pass	8, 9, 10, 18, 20, 23
11.2.46			Invalid: attempting to drop the left side of an appointment onto an existing one	A message box will be displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments" before the state of the appointment viewer and fields is reverted back to how they were before dragging was initiated.	A message box is displayed with the message "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments" before the state of the appointment viewer and fields is reverted back to how they were before dragging was initiated.	Pass	8, 9, 12, 18, 20, 23
11.2.47			Invalid: attempting to drop the right side of an			Pass	8, 9, 12, 18, 20, 23

			appointment onto an existing one				
11.2.48			Invalid: attempting to drop the left side of a new appointment onto an existing one			Pass	8, 9, 10, 18, 20, 23
11.2.49			Invalid: attempting to drop the right side of a new appointment onto an existing one			Pass	8, 9, 10, 18, 20, 23
11.2.50			Valid: attempting to drop the left side of an appointment outside the timetable	The left end of the appointment will snap up to the earliest time (in this case, 8:00AM) and this change will be reflected in the date-time picker controls and the appointment viewer	The left end of the appointment snaps to the earliest time (in this case, 8:00AM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 9, 12, 18, 20
11.2.51			Valid: attempting to drop the right side of an appointment outside the timetable	The right end of the appointment will snap down to the latest time (in this case, 5:00PM) and this change will be reflected in the date-time picker controls and the appointment viewer	The right end of the appointment snaps to the latest time (in this case, 5:00PM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 9, 12, 18, 20
11.2.51			Valid: attempting to drop the left side of a new appointment	The left end of the appointment will snap up to the earliest time (in this case, 8:00AM) and this change will be reflected in the date-time	The left end of the appointment snaps to the earliest time (in this case, 8:00AM) and this change is reflected in the date-time picker controls and the appointment	Pass	8, 9, 10, 18, 20

			outside the timetable	picker controls and the appointment viewer	viewer		
11.2.53			Valid: attempting to drop the right side of a new appointment outside the timetable	The right end of the appointment will snap down to the latest time (in this case, 5:00PM) and this change will be reflected in the date-time picker controls and the appointment viewer	The right end of the appointment snaps to the latest time (in this case, 5:00PM) and this change is reflected in the date-time picker controls and the appointment viewer	Pass	8, 9, 10, 18, 20
11.2.54			Mouse move event after dropping an appointment	No further action will be taken.	No further action is taken.	Pass	18
11.3.1	dtpAppointmentDate	Ensures that the user is able to select the date for which they wish to see appointments, preventing the viewing of weekends and displaying the most up-to-date version of the data available	Valid: Change event to a workday with no appointments	An empty timetable will be shown, the last appointment unselected and all CRUD functionality disabled.	An empty timetable is shown, the last appointment unselected and all CRUD functionality disabled.	Pass	6h, 8, 14
11.3.2			Valid: Change event to a workday where one mechanic has appointments	The timetable will be populated with the appointments of the mechanic, maintaining the same mechanic order as before.	The timetable is populated with the appointments of the mechanic, maintaining the same mechanic order as before.	Pass	6h, 8, 14
11.3.3			Valid: Change event to a workday where two mechanics have appointments	The timetable will be populated with the appointments of the 2 mechanics, maintaining the same mechanic order as before.	The timetable is populated with the appointments of the 2 mechanics, maintaining the same mechanic order as before.	Pass	6h, 8, 14
11.3.4			Valid: Change event to a workday where three	The timetable will be populated with the appointments of all 3 mechanics, maintaining the	The timetable is populated with the appointments of all 3 mechanics, maintaining the same mechanic order as before.	Pass	6h, 8, 14

			mechanics have appointments	same mechanic order as before.			
11.3.5			Invalid: Change event to a Saturday	A message box will be displayed with the error “We’re closed during the weekend!” before the appointment date is set to the next workday i.e. the next Monday, with the appointments loaded for that day.	A message box is displayed with the error “We’re closed during the weekend!” before the appointment date is set to the next workday i.e. the next Monday, with the appointments loaded for that day.	Pass	6h, 8, 14, 20
11.3.6			Invalid: Change event to a Sunday			Pass	6h, 8, 14, 20
11.3.7			Valid: Change event with an appointment already selected	The appointments for the selected date will be loaded, the last appointment unselected and all CRUD functionality disabled.	The appointments for the selected date are loaded, the last appointment unselected and all CRUD functionality disabled.	Pass	6h, 8, 14
11.3.8			Valid: Change event in insert mode with a new appointment present	The appointments for the selected date will be loaded, the last appointment unselected, insert mode disabled and all CRUD functionality disabled.	The appointments for the selected date are loaded, the last appointment unselected, insert mode disabled and all CRUD functionality disabled.	Pass	6h, 8, 14
11.3.9			Valid: Change event to the same date in insert mode with a new appointment present	No further action will be taken.	No further action is taken.	Pass	6h, 8
11.3.10			Valid: Change event back to same date when	The appointments for the selected date will be loaded, the last appointment unselected and all CRUD	The appointments for the selected date are loaded, the last appointment unselected and all CRUD functionality disabled. The	Pass	6h, 8, 14

			some appointments have modified times	functionality disabled. The times in the database will be shown instead of the modified times.	times in the database are shown instead of the modified times.		
11.3.11			Valid: Change event back to the same date after deleting some appointments	The appointments for the selected date will be loaded, the last appointment unselected and all CRUD functionality disabled. The deleted appointments will not be visible.	The appointments for the selected date are loaded, the last appointment unselected and all CRUD functionality disabled. The deleted appointments are not visible.	Pass	6h, 8, 14
11.3.12			Invalid: Attempting to change the date by scrolling	No further action will be taken.	No further action is taken.	Pass	6h, 8
11.4.1	dtpAppointment Start	Ensures that the user can input valid appointment start times, preventing appointments from exceeding the provided time bounds and clashing with other appointments	Invalid: Attempting to type a time in with an appointment selected			Pass	20
11.4.2			Valid: Clicking up on the control with an appointment selected 10:00	The time will be set to 11:00, and this will be indicated by the left end of the appointment's timeblock moving up one cell	The time is set to 11:00, and this is indicated by the left end of the appointment's timeblock moving up one cell	Pass	12, 18

11.4.3			Valid: Clicking down on the control with an appointment selected 11:00	The time will be set to 10:00, and this will be indicated by the left end of the appointment's timeblock moving down one cell	The time is set to 10:00, and this is indicated by the left end of the appointment's timeblock moving down one cell	Pass	12, 18
11.4.4			Invalid: Attempting to click down when the time is already at the start of the day	A message box will be displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action will be taken.	A message box is displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action is taken.	Pass	9, 12, 18, 20
11.4.5			Invalid: Attempting to click up to cause an appointment to be of length 0			Pass	9, 12, 18, 20
11.4.6			Valid: Click event on an appointment in pnlViewer after modifying its starting time	The start time for the appointment will be displayed as the modified time.	The start time for the appointment is displayed as the modified time.	Pass	12, 18

11.4.7			Valid: Click event on a different appointment in pnlViewer	The start time for the appointment will be displayed as that appointment's starting time.	The start time for the appointment is displayed as that appointment's starting time.	Pass	12, 18
11.4.8			Valid: Clicking up on the control with a new appointment selected 10:00	The time will be set to 11:00, and this will be indicated by the left end of the appointment's timeblock moving up one cell	The time is set to 11:00, and this is indicated by the left end of the appointment's timeblock moving up one cell	Pass	10, 18
11.4.9			Valid: Clicking down on the control with a new appointment selected 11:00	The time will be set to 10:00, and this will be indicated by the left end of the appointment's timeblock moving down one cell	The time is set to 10:00, and this is indicated by the left end of the appointment's timeblock moving down one cell	Pass	10, 18
11.4.10			Invalid: Attempting to click down when the time is already at the start of the day with a new appointment selected	A message box will be displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action will be taken.	A message box is displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action is taken.	Pass	9, 10, 18, 20

11.4.11			Invalid: Attempting to click up to cause a new appointment to be of length 0			Pass	9, 10, 18, 20
11.4.12			Valid: Click event on an new appointment in pnlViewer after modifying its starting time	The start time for the appointment will be displayed as the modified time.	The start time for the appointment is displayed as the modified time.	Pass	10, 18
11.4.13			Invalid: Click event on a different appointment in pnlViewer in insert mode	No further action will be taken.	No further action is taken.	Pass	10, 18
11.4.14			Invalid: Click event down in order to attempt to cause the appointment to overlap with another	A message box will be displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action will be taken.	A message box is displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action is taken.	Pass	9, 12, 18, 20
11.4.15			Invalid: Click event down in order to attempt to			Pass	9, 10, 18, 20

			cause a new appointment to overlap with another				
11.5.1	dtpAppointmentEnd	Ensures that the user can input valid appointment end times, preventing appointments from exceeding the provided time bounds and clashing with other appointments	Invalid: Attempting to type a time in	No further action will be taken.	No further action is taken.	Pass	20
11.5.2			Valid: Clicking up on the control with an appointment selected 10:00	The time will be set to 11:00, and this will be indicated by the right end of the appointment's timeblock moving up one cell	The time is set to 11:00, and this is indicated by the right end of the appointment's timeblock moving up one cell	Pass	12, 18
11.5.3			Valid: Clicking down on the control with an appointment selected 11:00	The time will be set to 10:00, and this will be indicated by the right end of the appointment's timeblock moving down one cell	The time is set to 10:00, and this is indicated by the right end of the appointment's timeblock moving down one cell	Pass	12, 18
11.5.4			Invalid: Attempting to click up when the time is already at the end of the day	A message box will be displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action will be taken.	A message box is displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action is taken.	Pass	9, 12, 18, 20
11.5.5			Invalid: Attempting to click down to cause an appointment to be of length 0			Pass	9, 12, 18, 20

11.5.6			Valid: Click event on an appointment in pnlViewer after modifying its starting time	The end time for the appointment will be displayed as the modified time.	The end time for the appointment is displayed as the modified time.	Pass	12, 18
11.5.7			Valid: Click event on a different appointment in pnlViewer	The end time for the appointment will be displayed as that appointment's ending time.	The end time for the appointment is displayed as that appointment's ending time.	Pass	12, 18
11.5.8			Valid: Clicking up on the control with a new appointment selected 10:00	The time will be set to 11:00, and this will be indicated by the right end of the appointment's timeblock moving up one cell	The time is set to 11:00, and this is indicated by the right end of the appointment's timeblock moving up one cell	Pass	10, 18
11.5.9			Valid: Clicking down on the control with a new appointment selected 11:00	The time will be set to 10:00, and this will be indicated by the right end of the appointment's timeblock moving down one cell	The time is set to 10:00, and this is indicated by the right end of the appointment's timeblock moving down one cell	Pass	10, 18
11.5.10			Invalid: Attempting to click up when the time is already at the end of the day with a new appointment	A message box will be displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action will be taken.	A message box is displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action is taken.	Pass	9, 10, 18, 20

			selected				
11.5.11			Invalid: Attempting to click down to cause a new appointment to be of length 0			Pass	9, 10, 18, 20
11.5.12			Valid: Click event on an new appointment in pnlViewer after modifying its starting time	The start time for the appointment will be displayed as the modified time.	The start time for the appointment is displayed as the modified time.	Pass	10, 18
11.5.13			Invalid: Click event on a different appointment in pnlViewer in insert mode	No further action will be taken.	No further action is taken.	Pass	10, 18
11.5.14			Invalid: Click event up in order to attempt to cause the appointment to overlap with another	A message box will be displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action will be taken.	A message box is displayed with the text "New time must not cause the appointment to go beyond the working hours, nor overlap with other appointments." No further action is taken.	Pass	9, 12, 18, 20
11.5.15			Invalid: Click event up in order to attempt to			Pass	9, 10, 18, 20

			cause a new appointment to overlap with another				
11.6.1	comboMechanicID	Ensures that the user can appropriately allocate a mechanic to an appointment, preventing appointments from clashing with existing appointments for that mechanic and unauthorised allocation	Valid: Changing to a mechanic with a free slot in the valid position when authorised	The appointment's mechanic will change, as evidenced by the appointment having the same details - including time - but moving into the chosen mechanic's row	The appointment's mechanic changes, as evidenced by the appointment having the same details - including time - but moving into the chosen mechanic's row	Pass	12, 18
11.6.2			Invalid: Attempting to change mechanic when unauthorised (a slot that doesn't belong to the user when they are of authorisation level 1)	A message box will be displayed with the text "You can't give your colleague an appointment!" No further action will be taken.	A message box is displayed with the text "You can't give your colleague an appointment!" No further action is taken.	Pass	2, 7, 12, 18, 23

11.6.3			Invalid: Attempting to change mechanic when it would cause clashes with existing appointments	A message box will be displayed with the text “Sorry, this would overlap with an existing appointment belonging to this mechanic.” No further action will be taken.	A message box is displayed with the text “Sorry, this would overlap with an existing appointment belonging to this mechanic.” No further action is taken.	Pass	9, 12, 18, 20
11.6.4			Valid: Attempting to change to the same mechanic	No further action will be taken.	No further action is taken.	Pass	12, 18
11.7.1	comboVehicleID	Ensures that the user can allocate a vehicle to an appointment, with the displayed appointment time dynamically changing to represent the vehicle’s complexity	Valid: Changing to a different vehicle From 33 (complexity 1.3) to 34 (complexity 1.0) “Full Service” selected (4 hours base time)	The time required for the appointment as displayed in lblTimeRequiredValue updates to show the base time of the appointment (i.e. the sum of the service durations) multiplied by the new vehicle’s complexity, which will be “4 hours”	The label displays “4 hours”	Pass	18, 22
11.8.1	chkDiscounted	Ensures that the user can mark an appointment as discounted, toggling whether the appointment’s price is reduced by 10%	Check event £250.00	The cost will be multiplied by 0.9 and displayed in lblCostValue (text will be £225.00)	The label displays “£225.00”	Pass	18, 22
11.8.2			Uncheck event £225.00	The cost will be updated to its original value in lblCostValue (text will be £250.00)	The label displays “£250.00”	Pass	18, 22

11.9.1	chklbServices	Ensures that the user can allocate and deallocate services from an appointment, with their presence, cost and time taken all represented in the appropriate controls	Check event on a service that has not been selected "Full Service" 4 hours £250.00	The service will be added to the tree view as a new node with no parts associated. Furthermore, the service's time and cost will be added on to the running totals and this will be reflected in the updated label values.	The service is added to the tree view as a new node with no parts associated. The service's time and cost is added on and this is reflected in the updated label values.	Pass	18, 19, 22
11.9.2			Unchecking a service that is selected "Full Service" 4 hours £250.00	The service will be removed from the tree view and the service's time and cost will be removed from the running totals, with this reflected in the updated label values.	The service will be removed from the tree view and the service's time and cost will be removed from the running totals, with this reflected in the updated label values.	Pass	18, 19, 22
11.9.3			Rechecking a service that had parts "Full Service" 4 hours £250.00 Engine Oil (5L)	The service and its parts will be removed from memory initially, as indicated by the price and time going down, and on rechecking the service will be added to the tree view as a new node with no parts associated. Furthermore, the service's time and cost will be added on to the running totals and this will be reflected in the updated label values.	The service and its parts are removed from memory initially, as indicated by the price and time going down, and on rechecking the service is added to the tree view as a new node with no parts associated. Furthermore, the service's time and cost are added on to the running totals and this is reflected in the updated label values.	Pass	18, 19, 22
11.10.1	chkInsertMode	Ensures that the user can effectively switch between adding new appointments and editing existing ones	Check event when no appointment is selected	All CRUD controls will be disabled, the "Save" and "Save New Schedule" buttons hidden, and the "Add" button shown	All CRUD controls are disabled, the "Save" and "Save New Schedule" buttons hidden, and the "Add" button shown	Pass	10, 14, 18
11.10.2			Check event when an appointment is selected	The current appointment will be unselected, all CRUD controls will be disabled, the "Save" and "Save New Schedule" buttons hidden, and	The current appointment is unselected, all CRUD controls are disabled, the "Save" and "Save New Schedule" buttons hidden, and the "Add" button shown	Pass	10, 14, 18

				the “Add” button shown			
11.10.3			Unchecked event when no appointment has been created	CRUD controls will remain disabled, the “Add” button will be hidden, and the “Save” and “Save New Schedule” buttons shown	CRUD controls remain disabled, the “Add” button is hidden, and the “Save” and “Save New Schedule” buttons shown	Pass	10, 14, 18
11.10.4			Unchecked event when a temporary appointment is created	The appointment will be deleted, CRUD controls will become disabled, the “Add” button will be hidden, and the “Save” and “Save New Schedule” buttons shown.	The appointment is deleted, CRUD controls become disabled, the “Add” button is hidden, and the “Save” and “Save New Schedule” buttons shown	Pass	10, 14, 18
11.11.1	btnPartPlus	Ensures that the user can add parts to the selected service, with their cost represented in the appropriate controls	Invalid: Click event (no service has been chosen)	A message box will be displayed with the text “No service has been chosen to add a part to.” No further action will be taken.	A message box will be displayed with the text “No service has been chosen to add a part to.” No further action will be taken.	Pass	9, 18, 19, 22, 23
11.11.2			Invalid: Click event (no part has been chosen) “Full Service”	A message box will be displayed with the text “No part has been chosen to add to a service.” No further action will be taken.	A message box will be displayed with the text “No part has been chosen to add to a service.” No further action will be taken.	Pass	9, 18, 19, 22, 23
11.11.3			Valid: Click event (part already existed in service and is in stock) “Full Service” Engine Oil (5L) 1	The part node belonging to the “Full Service” node will be updated to have the text “Engine Oil (5L) - 2” to indicate another part has been added.	The part node now has the text “Engine Oil (5L) - 2”	Pass	18, 19, 22

11.11.4			Valid: Click event (part did not exist in service and is in stock) “Full Service” Engine Oil (5L)	A new node will be added to the “Full Service” node with the text “Engine Oil (5L) - 1” to indicate one part of that type is now associated with that service.	A new node is added to the “Full Service” node with the text “Engine Oil (5L) - 1”	Pass	18, 19, 22
11.11.5			Valid: Click event (part already existed in service and is not in stock as a result of that appointment) “Full Service” Engine Oil (5L) 40	A message box will be displayed with the text “You don’t have enough of this part in stock!” No further action is taken.	A message box is displayed with the text “You don’t have enough of this part in stock!” No further action is taken.	Pass	18, 19, 22

11.11.6			Valid: Click event (part already existed in service and is not in stock as a result of multiple appointments) "Full Service" Engine Oil (5L) 22 "Minor Service" Engine Oil (5L) 18			Pass	18, 19, 22
11.11.7			Valid: Click event (part did not exist in service and is not in stock)			Pass	18, 19, 22
11.12.1	btnPartMinus	Ensures that the user can remove parts from the selected service, with their cost deduction represented in the appropriate controls	Invalid: Click event (no service has been chosen)	A message box will be displayed with the text "No service has been chosen to remove a part from." No further action will be taken.	A message box will be displayed with the text "No service has been chosen to remove a part from." No further action will be taken.	Pass	9, 18, 19, 22, 23
11.12.2			Invalid: Click event (no part has been chosen) "Full Service"	A message box will be displayed with the text "No part has been chosen to remove from the service." No further action will be taken.	A message box will be displayed with the text "No part has been chosen to remove from the service." No further action will be taken.	Pass	9, 18, 19, 22, 23

11.12.3			Valid: Click event (part already existed in service and had more than 1 present) "Full Service" Engine Oil (5L) 2	The part node belonging to the "Full Service" node will be updated to have the text "Engine Oil (5L) - 1" to indicate a part has been removed.	The part node now has the text "Engine Oil (5L) - 1"	Pass	18, 19, 22
11.12.4			Valid: Click event (part already existed in service and had 1 present) "Full Service" Engine Oil (5L) 1	The part node belonging to the "Full Service" node will be removed to indicate no part of that kind remains.	The part node is removed.	Pass	18, 19, 22
11.12.5			Invalid: Click event (part was not present in service) "Full Service" Engine Oil (5L)	A message box will be displayed with the text "The current service doesn't have any of that item!" No further action is taken.	A message box is displayed with the text "The current service doesn't have any of that item!" No further action is taken.	Pass	9, 18, 19, 22, 23
11.13.1	btnDelete	Ensures that the user can delete the selected appointment from the database, preventing further	Valid: Click event ("No" selected at dialog) 51	No further action will be taken.	No further action is taken.	Pass	13, 18, 24

11.13.2		loading into the timetable	Valid: Click event (“Yes” selected at dialog) 51	All CRUD functionality will be disabled, and the appointment and all of its satellite data regarding services and parts marked as deleted from the database as indicated by the updated timetable no longer displaying the selected appointment.	All CRUD functionality is disabled, and the appointment and all of its satellite data regarding services and parts marked as deleted from the database as indicated by the updated timetable no longer displaying the selected appointment.	Pass	13, 18, 24
11.13.3			Valid: Click event in insert mode 0	The new appointment will be deleted from memory as indicated by its disappearance from the timetable, all CRUD functionality will be disabled, and insert mode will be disabled.	The new appointment is deleted from memory as indicated by its disappearance from the timetable, all CRUD functionality is disabled, and insert mode is disabled.	Pass	13, 18, 24
11.14.1	btnAdd	Ensures that the user can add a new appointment to the database with valid details, becoming a permanent addition to the timetable	Valid: Monday 21st March, 2026 11:00 16:00 False True False 3 36 Full Service • Engine Oil (5L) x 2 • Battery Wheel Alignment • Wheel Alignment Bolt	A message box will be displayed with the message “Appointment has been added successfully.” The new appointment will be added to the database with all of the input details, as indicated by the new appointment becoming blue and the input details being populated into the forms’ fields when clicking the appointment.	A message box is displayed with the message “Appointment has been added successfully.” The appointment is added to the database, the temporary timeblock becomes blue to indicate its addition to the timetable, and clicking on the appointment displays the same details input	Pass	10, 14, 18, 24

11.14.2			Invalid (time allocated less than required time) Monday 2nd February, 2026 9:00 10:00 False False False 3 4 Full Service	A message box will be displayed with the text "It is impossible (or at least very unlikely) the appointment will be completed in the allotted time. Either remove some services, or service a less complex car." No further action will be taken.	A message box is displayed with the text "It is impossible (or at least very unlikely) the appointment will be completed in the allotted time. Either remove some services, or service a less complex car." No further action is taken.	Pass	9, 10, 18, 20, 23
11.14.3			Valid (no services allocated) Monday 2nd February, 2026 9:00 10:00 False False False 3 4	A message box will be displayed with the message “Appointment has been added successfully.” The new appointment will be added to the database with all of the input details, as indicated by the new appointment becoming blue and the input details being populated into the forms’ fields when clicking the appointment.	A message box is displayed with the message “Appointment has been added successfully.” The appointment is added to the database, the temporary timeblock becomes blue to indicate its addition to the timetable, and clicking on the appointment displays the same details input	Pass	10, 14, 18, 24

11.14.4			Valid (multiple services require the same parts) Wednesday 4th February, 2026 8:00 16:00 True False True 6 33 Full Service • Engine Oil (5L) x 2 Minor Service • Engine Oil (5L) x 2			Pass	10, 14, 18, 24
11.15.1	btnSave	Ensures that the user can add a save new details to an existing appointment, preventing invalid details from being input	Valid (“No” selected at dialog) Wednesday 4th February, 2026 8:00 17:00 True True True 6 33 Full Service • Engine Oil (5L) x 3 Minor Service	No further action will be taken.	No further action is taken.	Pass	12, 24

11.15.2			Valid (“Yes” selected at dialog) Wednesday 4th February, 2026 8:00 17:00 True True True 6 33 Full Service • Engine Oil (5L) x 3 Minor Service	A message box will be displayed with the message “Save successful.” The new details will be stored under the same appointment, as indicated by them populating the forms’ fields when the appointment is selected	A message box is displayed with the message “Save successful.” The new details is stored under the same appointment, as indicated by them populating the forms’ fields when the appointment is selected	Pass	9, 12, 18, 20, 24

11.15.3			Invalid (time allocated less than required time) Wednesday 4th February, 2026 8:00 12:00 True True True 6 33 Full Service • Engine Oil (5L) x 3 Minor Service	A message box will be displayed with the text "It is impossible (or at least very unlikely) the appointment will be completed in the allotted time. Either remove some services, or service a less complex car." No further action will be taken.	A message box is displayed with the text "It is impossible (or at least very unlikely) the appointment will be completed in the allotted time. Either remove some services, or service a less complex car." No further action is taken.	Pass	9, 12, 18, 20, 23
11.15.4			Valid (no services allocated) Wednesday 4th February, 2026 8:00 12:00 True True True 6 33	A message box will be displayed with the message “Save successful.” The new details will be stored under the same appointment, as indicated by them populating the forms’ fields when the appointment is selected	A message box is displayed with the message “Save successful.” The new details is stored under the same appointment, as indicated by them populating the forms’ fields when the appointment is selected	Pass	9, 12, 18, 20, 24
11.16.1	btnSaveScheduleOnly	Ensures that the user can save only the new times of existing appointments,	Valid (“No” selected at dialog)	No further action will be taken.	No further action is taken.	Pass	12, 24

11.16.2		seeing which appointments need additional time in the event of invalid time allocated	Valid (“Yes” selected at dialog)	All the modified appointment times will be saved to the database for the appropriate appointments, as indicated by the times remaining when the same date is reselected. No further details will be saved.	All the modified appointment times are saved to the database for the appropriate appointments, as indicated by the times remaining when the same date is reselected. No further details are saved.	Pass	9, 12, 18, 20, 24
11.16.3			Valid (times not affected)	Although times are saved to the database, no change will be visible.	No change is visible.	Pass	9, 12, 18, 20, 24
11.16.4			Invalid (multiple appointments not allocated enough time)	A message box will be displayed with the text “The following timeblocks need amended” followed by each appointment’s ID, mechanic and required additional time. No further action will be taken.	A message box is displayed with the text “The following timeblocks need amended” followed by each appointment’s ID, mechanic and required additional time. No further action is taken.	Pass	9, 12, 18, 20, 23
11.17.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpServiceAppointments will be displayed over frmServiceAppointments	frmHelpServiceAppointments is displayed over frmServiceAppointments	Pass	26
11.17.2			Click event away from frmHelp Service Appointments	frmHelpServiceAppointments will be closed	frmHelpServiceAppointments is closed	Pass	26
11.18.1	btnSearch	Ensures that the user is able to query the database with the selected vehicle or mechanic, outputting each appointment’s details in the event of a match	Click event (neither field has been selected)	A message box will be displayed with the text “You must select something to search” No further action will be taken.	A message box is displayed with the text “You must select something to search” No further action is taken.	Pass	8

11.18.2			Click event (a mechanic has been chosen) 3	A message box will be displayed with the appointment ID, date and completion status of each appointment matching the search criteria	A message box is displayed with the appointment ID, date and completion status of each appointment matching the search criteria	Pass	8
11.18.3			Click event (a vehicle has been chosen) 27			Pass	8
11.18.4			Click event (both a mechanic and vehicle have been chosen) 6 27			Pass	8
11.18.5			Valid (no appointments match) 3 27	A message box will be displayed with the text “No appointments were found with the given criteria.” No further action will be taken.	A message box is displayed with the text “No appointments were found with the given criteria.” No further action is taken.	Pass	8
11.18.6			Click event (setting mechanic back to “None”) 27	A message box will be displayed with the appointment ID, date and completion status of each appointment matching the search criteria. The criteria will no longer include the mechanic.	A message box is displayed with the appointment ID, date and completion status of each appointment matching the search criteria. The criteria no longer includes the mechanic.	Pass	8

11.18.7			Click event (setting vehicle back to “None”) 3	A message box will be displayed with the appointment ID, date and completion status of each appointment matching the search criteria. The criteria will no longer include the vehicle.	A message box is displayed with the appointment ID, date and completion status of each appointment matching the search criteria. The criteria no longer includes the vehicle.	Pass	8

frmLogs

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
12.1.1	frmLogs	Ensures that logs for the current day are loaded and displayed properly on access, handling missing data gracefully	Load event	Each staff member will be added as a filtering option to the combo box, and the logs for the current date will be displayed.	Each staff member is added as a filtering option to the combo box, and the logs for the current date are displayed.	Pass	6b, 25
12.1.2			Load event - no Staff table present	A message box will be displayed with the text “Error retrieving staff records: ", followed by an error message, then another with the text "No staff members could be loaded.” and the logs for the current date will be displayed.	A message box is displayed with the text “Error retrieving staff records: ", followed by an error message, then another with the text "No staff members could be loaded.” and the logs for the current date are displayed.	Pass	23
12.2.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpLogs is displayed above frmLogs	frmHelpLogs is displayed above frmLogs	Pass	26
12.2.2			Click event away from frmHelpLogs	frmHelpLogs is closed	frmHelpLogs is closed	Pass	26
12.3.1	dtpLogData	Ensures that the user is able to change the date for which they wish to view logs, displaying appropriate data when changed	Date changed to one with logs	All the logs for that date will be displayed in descending time order, with the log types removed from the end	All the logs for that date are displayed in descending time order, with the log types removed from the end	Pass	25
12.3.2			Date changed to one without logs	Nothing will be displayed	Nothing is displayed	Pass	25

12.3.3			Date changed while filtering by staff and log type	All the logs for that date will be displayed in descending time order, with the filter still active and with the log types removed from the end	All the logs for that date are displayed in descending time order, with the filter still active and with the log types removed from the end	Pass	25
12.4.1	comboLogType	Ensures that the user is able to filter the type of log which they wish to see for the selected day, updating the log list as appropriate	“None” selected	All logs for the selected date, irrespective of log type, will be displayed	All logs for the selected date, irrespective of log type, are displayed	Pass	25
12.4.2			“LoginAttempt” selected	All logs for the selected date marked as a login attempt will be displayed	All logs for the selected date marked as a login attempt are displayed	Pass	25
12.4.3			“Login” selected	All logs for the selected date marked as a successful login will be displayed	All logs for the selected date marked as a successful login are displayed	Pass	25
12.4.4			“Access” selected	All logs for the selected date marked as a form access event will be displayed	All logs for the selected date marked as a form access event are displayed	Pass	25
12.4.5			“Create” selected	All logs for the selected date marked as a record creation will be displayed	All logs for the selected date marked as a record creation are displayed	Pass	25
12.4.6			“Update” selected	All logs for the selected date marked as a record update action will be displayed	All logs for the selected date marked as a record update action are displayed	Pass	25
12.4.7			“Delete” selected	All logs for the selected date marked as a record deletion will be displayed	All logs for the selected date marked as a record deletion are displayed	Pass	25

12.4.8			“Delete” selected - no “Delete” logs for that day	Nothing will be displayed	Nothing is displayed	Pass	25
12.4.9			“Delete” selected with user “patdoh”	All logs for the selected date marked as a record deletion carried out by “patdoh” will be displayed	All logs for the selected date marked as a record deletion carried out by “patdoh” are displayed	Pass	25
12.5.1	comboStaffMember	Ensures that the user is able to filter the staff member which they wish to see for the selected day, updating the log list as appropriate	“None” selected	All logs for the selected date, irrespective of staff member, will be displayed	All logs for the selected date, irrespective of staff member, are displayed	Pass	25
12.5.2			Staff member “patdoh” selected	All logs for the selected date attributed to “patdoh” will be displayed	All logs for the selected date attributed to “patdoh” are displayed	Pass	25
12.5.3			Staff member changed to “ethanh1” from “patdoh”	All logs for the selected date attributed to “ethanh1” will be displayed	All logs for the selected date attributed to “ethanh1” are displayed	Pass	25
12.5.4			“None” reselected from “ethanh1”	All logs for the selected date, irrespective of staff member, will be displayed	All logs for the selected date, irrespective of staff member, are displayed	Pass	25
12.6.1	lvLogs	Ensures that the log list displays only valid logs, handling improper data gracefully	No logs present for given filters	Nothing will be displayed	Nothing is displayed	Pass	25
12.6.2			Log malformed - date unparsable (contains character ‘a’)	The log will not be displayed	The log is not displayed	Pass	25

12.6.3			Log malformed - invalid type (34)	The log will not be displayed	The log is not displayed	Pass	25
12.6.4			Log malformed - brackets not in correct positions (around name instead of date)	The log will not be displayed	The log is not displayed	Pass	25

Logging

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
13.1.1	WriteLog	Ensures that the system is able to write logs externally for relevant user actions, dynamically saving appropriate details depending on the log type	Login attempt: patdoh password	A log will be written to logs.txt with the following message: [*current date/time*] - User patdoh had an unsuccessful login attempt [1]	A log is written to logs.txt with the following message: [*current date/time*] - User patdoh had an unsuccessful login attempt [1]	Pass	2, 25
13.1.2			Login: patdoh Password1.	A log will be written to logs.txt with the following message: [*current date/time*] - User patdoh logged in [2]	A log is written to logs.txt with the following message: [*current date/time*] - User patdoh logged in [2]	Pass	2, 25
13.1.3			Access to form frmStaffData by patdoh	A log will be written to logs.txt with the following message: [*current date/time*] - User patdoh accessed the Staff Data form [3]	A log is written to logs.txt with the following message: [*current date/time*] - User patdoh accessed the Staff Data form [3]	Pass	2, 25
13.1.4			New vehicle type record created by patdoh: Minibus 1.2	A log will be written to logs.txt with the following message: [*current date/time*] - User patdoh added a new vehicle type (Minibus) [4]	A log is written to logs.txt with the following message: [*current date/time*] - User patdoh added a new vehicle type (Minibus) [4]	Pass	2, 25

13.1.5			Vehicle type record updated by patdoh: 18 Minibus 1.2 → 18 Megabus 2.0	A log will be written to logs.txt with the following message: [*current date/time*] - User patdoh modified vehicle type 18 (Megabus) [5]	A log is written to logs.txt with the following message: [*current date/time*] - User patdoh modified vehicle type 18 (Megabus) [5]	Pass	2, 25
13.1.6			Vehicle type record deleted by patdoh: 18 Megabus 2.0	A log will be written to logs.txt with the following message: [*current date/time*] - User patdoh deleted vehicle type 18 (Megabus) [6]	A log is written to logs.txt with the following message: [*current date/time*] - User patdoh deleted vehicle type 18 (Megabus) [6]	Pass	2, 25
13.1.7			Login attempt: patdoh password logs.txt not present	A new logs.txt file will be created, and the following message appended: [*current date/time*] - User patdoh had an unsuccessful login attempt [1]	A new logs.txt file is created, and the following message appended: [*current date/time*] - User patdoh had an unsuccessful login attempt [1]	Pass	2, 25

frmReports

Test ID	Test item	Description	Test data	Expected outcome	Actual (observed) outcome	Pass/Fail	User requirement
14.1.1	frmReports	Ensures that, on loading, the reports in the form are appropriately initialised, with each report successfully fetching and displaying the appropriate data	Load event	The combobox in the “Invoices” tab will be populated with a list of all service appointments in the database, and all report viewer have their respective report loaded and displayed	The combobox in the “Invoices” tab is populated with all the service appointments in the database, and the report viewer display their respective reports as expected	Pass	6h, 21
14.2.1	btnHelp	Ensures that the user is able to access the help form for details regarding the usage of the current form	Click event	frmHelpReports will open and display over frmReports	frmHelpReports opens and displays over frmReports	Pass	26
14.2.2			Click event away from frmHelpReports	frmHelpReports will close	frmHelpReports closes	Pass	26
14.3.1	tabReports	Ensures that the user is able to effectively switch between reports in the form through the use of tabs	Click events Invoices → Mechanic Monthly Profits → Profits by Vehicle Types → Vehicle Stock → Customer Interest → Missed Appointments	After each click event, the displayed tab will change to the appropriate tab, rendering the report contained within the tab	After each click event, the appropriate tab is displayed, with the report within the tab rendered properly	Pass	21
14.4.1	tabInvoices	Ensures that the user is able to generate an invoice displaying appropriate information for an appointment when selecting a specific appointment	Load event	No data will be initially displayed.	No data is initially displayed.	Pass	6a, 6b, 6c, 6e, 6f, 6g, 6h, 21

14.4.2			Appointment selected after loading	The details for that appointment are displayed in a report, including the appointment's ID, start and end; mechanic's name and role; vehicle's make, model, year and license plate number; customer's name, email and phone number; a list of services and parts used; and a breakdown of individual costs, raw total cost, savings and grand total.	The details for that appointment are displayed, with all of the aforementioned information included.	Pass	6a, 6b, 6c, 6e, 6f, 6g, 6h, 21
14.4.3			Different appointment selected from current appointment	The details for the previous appointment are cleared, and replaced with the details for the new appointment	The details for the previous appointment are cleared, and replaced with the details for the new appointment	Pass	6a, 6b, 6c, 6e, 6f, 6g, 6h, 21
14.4.4			Appointment selected with no customer associated with the vehicle	All of the aforementioned details will be displayed, with the exception of customer data, which will be left blank	All of the aforementioned details are displayed, with customer data left blank	Pass	6a, 6b, 6c, 6e, 6f, 6g, 6h, 21
14.4.5			Appointment selected again after making a change to it	The details of the appointment in the invoice will have updated to reflect the latest version in the database	The details in the invoice have updated to reflect the latest version in the database	Pass	6a, 6b, 6c, 6e, 6f, 6g, 6h, 21

14.5.1	tabAppointmentMonthlyProfitsByMechanic	Ensures that the user is able to view the profits generated from appointments, grouped by mechanic and then by month	Load event	The total profits generated by each mechanic during each month as a result of their appointments are displayed in a clustered bar chart	The total profits generated by each mechanic during each month as a result of their appointments are displayed in a clustered bar chart	Pass	6b, 6e, 6g, 6h, 21
14.5.2			Load event after the profits generated in appointments are modified	The new total profits generated by each mechanic during each month as a result of their appointments are displayed	The new total profits generated by each mechanic during each month as a result of their appointments are displayed	Pass	6b, 6e, 6g, 6h, 21
14.6.1	tabVehicleTypeProfits	Ensures that the user is able to view the profits generated from each type of vehicle	Load event	The total profits generated by the vehicles sold of each vehicle type are displayed	The total profits generated by the vehicles sold of each vehicle type are displayed	Pass	6c, 6d, 21
14.6.2			Load event after the profits generated from each vehicle type are modified	The new total profits generated by the vehicles sold of each vehicle type are displayed	The new total profits generated by the vehicles sold of each vehicle type are displayed	Pass	6c, 6d, 21
14.7.1	tabVehicleStock	Ensures that the user is able to view the vehicles that have not been sold to a customer yet	Load event	The total number of unsold vehicles grouped by vehicle type is displayed	The total number of unsold vehicles grouped by vehicle type is displayed	Pass	6c, 6d, 21
14.7.2			Load event after the number of vehicles in stock is changed	The new total number of unsold vehicles grouped by vehicle type is displayed	The new total number of unsold vehicles grouped by vehicle type is displayed	Pass	6c, 6d, 21
14.8.1	tabCustomerInterest	Ensures that the user is able to view the most popular vehicle types based on the number	Load event	The total number of vehicles sold grouped by vehicle type is displayed	The total number of vehicles sold grouped by vehicle type is displayed	Pass	6c, 6d, 21

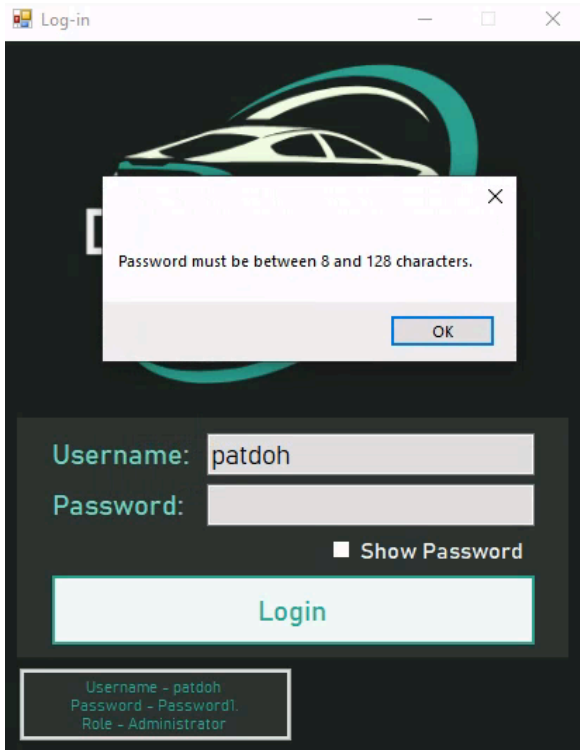
14.8.2		of customers associated with each vehicle type	Load event after the number of vehicles sold to customers is changed	The new total number of vehicles sold grouped by vehicle type is displayed	The new total number of vehicles sold grouped by vehicle type is displayed	Pass	6c, 6d, 21
14.9.1	tabMissedAppointments	Ensures that the user is able to view the details of appointments for which the scheduled time has passed but are still not marked as completed	Load event	A list of the details for each missed appointment, including appointment ID, mechanic name, car details and start of appointment, will be displayed	A list of the details for each missed appointment, including appointment ID, mechanic name, car details and start of appointment, will be displayed	Pass	6b, 6h, 21
14.9.2			Load event after the number of appointments missed is changed	An updated list of the details for each missed appointment, including appointment ID, mechanic name, car details and start of appointment, will be displayed	An updated list of the details for each missed appointment, including appointment ID, mechanic name, car details and start of appointment, will be displayed	Pass	6b, 6h, 21

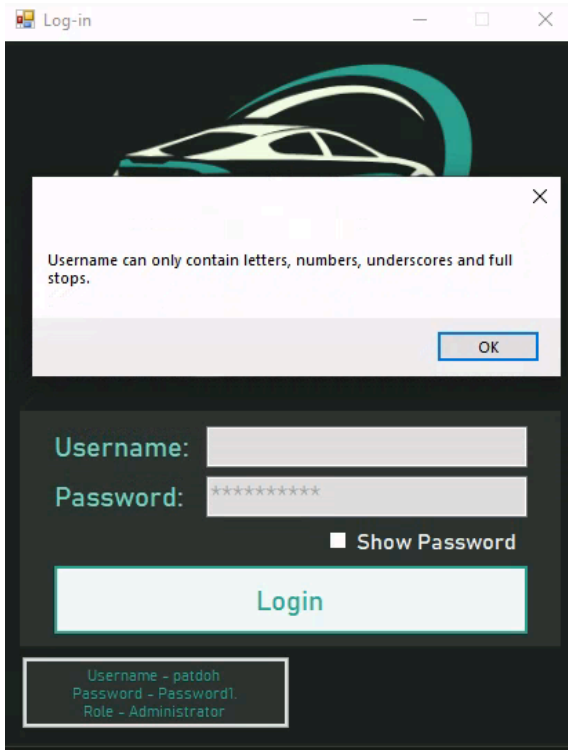
Corrective action

frmLogin

1.1.13-1.1.14

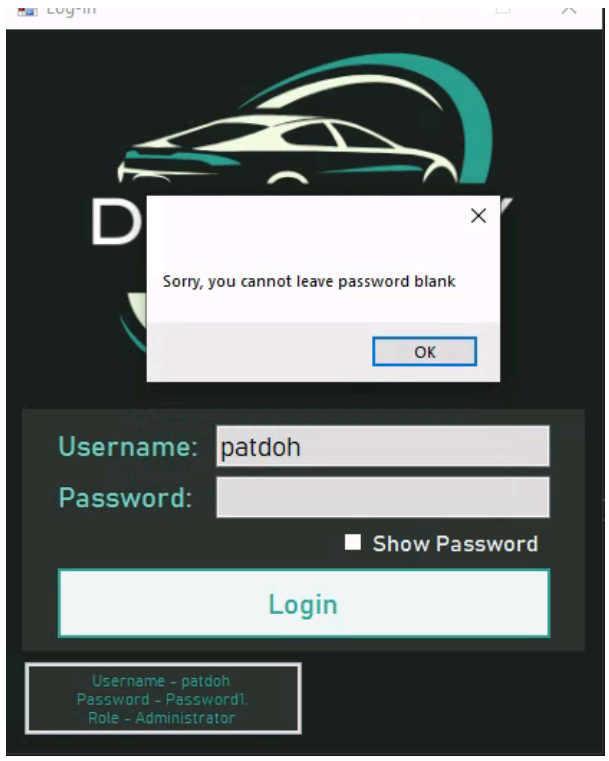
Before:

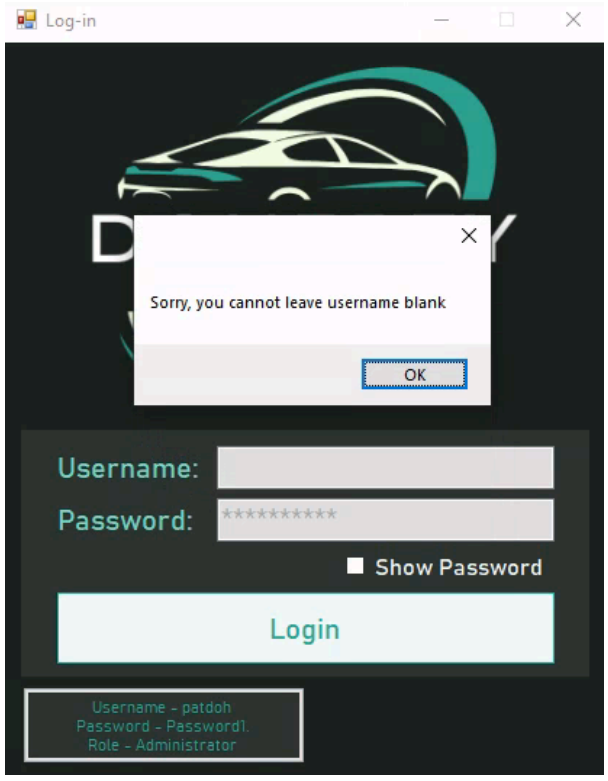




```
// ensure the user inputs valid data in the fields
1 reference
public bool LoginValidation(string username, string password)
{
    // if both fields are blank
    if (string.IsNullOrWhiteSpace(username) && string.IsNullOrWhiteSpace(password))
    {
        MessageBox.Show("Sorry, you cannot leave username and password blank");
        return false;
    }

    // if the username is not alphanumeric or with underscores
    if (!Regex.IsMatch(username, @"^[a-z0-9_.]+$"))
    {
        MessageBox.Show("Username can only contain letters, numbers, underscores and full stops.");
        return false;
    }

    // if the username is too short/long
    if (username.Length < 3 || username.Length > 50)
    {
        MessageBox.Show("Username must be between 3 and 50 characters.");
        return false;
    }

    // if the password is too short/long
    if (password.Length < 8 || password.Length > 128)
    {
        MessageBox.Show("Password must be between 8 and 128 characters.");
        return false;
    }

    // if only one field is blank
    if (string.IsNullOrWhiteSpace(username))
    {
        MessageBox.Show("Sorry, you cannot leave username blank");
        return false;
    }

    if (string.IsNullOrWhiteSpace(password))
    {
        MessageBox.Show("Sorry, you cannot leave password blank");
        return false;
    }
    return true;
}
```

After:






```
// ensure the user inputs valid data in the fields
1 reference
public bool LoginValidation(string username, string password)
{
    // if both fields are blank
    if (string.IsNullOrWhiteSpace(username) && string.IsNullOrWhiteSpace(password))
    {
        MessageBox.Show("Sorry, you cannot leave username and password blank");
        return false;
    }


    // if only one field is blank
    if (string.IsNullOrWhiteSpace(username))
    {
        MessageBox.Show("Sorry, you cannot leave username blank");
        return false;
    }

    if (string.IsNullOrWhiteSpace(password))
    {
        MessageBox.Show("Sorry, you cannot leave password blank");
        return false;
    }

    // if the username is not alphanumeric or with underscores
    if (!Regex.IsMatch(username, @"^[a-z0-9_.]+$"))
    {
        MessageBox.Show("Username can only contain letters, numbers, underscores and full stops.");
        return false;
    }

    // if the username is too short/long
    if (username.Length < 3 || username.Length > 50)
    {
        MessageBox.Show("Username must be between 3 and 50 characters.");
        return false;
    }

    // if the password is too short/long
    if (password.Length < 8 || password.Length > 128)
    {
        MessageBox.Show("Password must be between 8 and 128 characters.");
        return false;
    }

    return true;
}
```

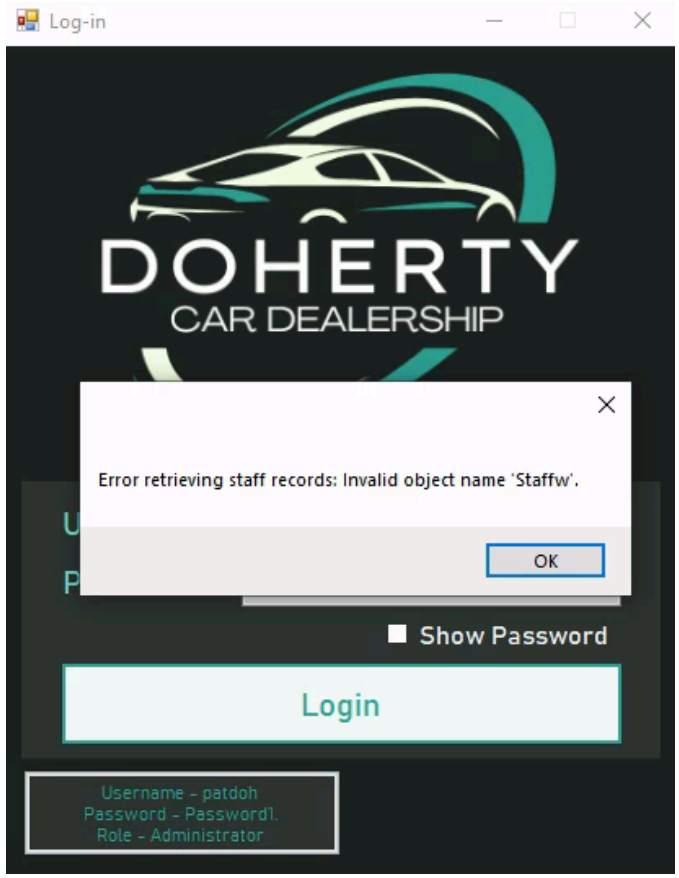
This error was a result of an improper sequence of checks. For the password, the length check initiated before the presence check which - as an empty password has length 0 - triggered the “Password must be between 8 and 128 characters” message before the “Sorry, you cannot leave password blank” message. Similarly, for the username’s regex check, an empty string will always return null, triggering the username’s format check before the presence check, once again displaying an inappropriate error message. The fix for this was to ensure that the presence check was the very first check made when validating input credentials.

1.1.24

Before:

```
else // else output an error message, log a failed attempt and clear fields
{

    MessageBox.Show("Incorrect username or password, please try again.");
```

After:



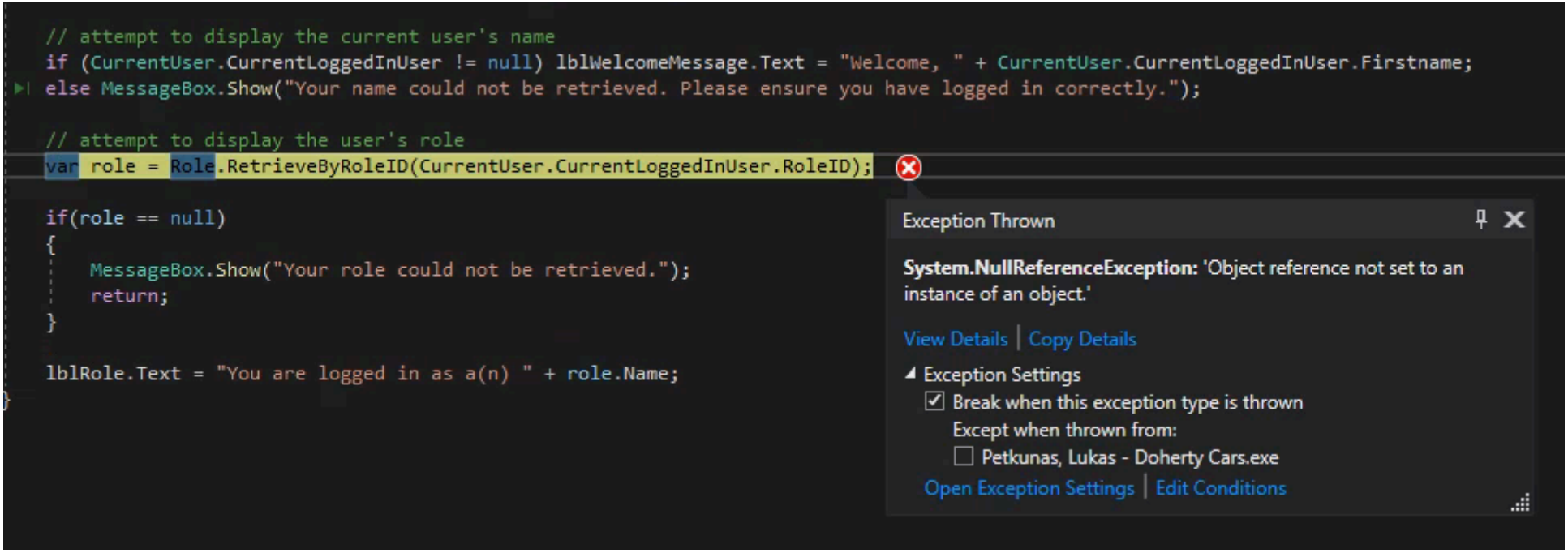

```
// check if no staff members were found because of a server error or just because of incorrect credentials
if(Staff.RetrieveStaffMembers() != null)
    MessageBox.Show("Incorrect username or password, please try again.");
```

This error was a result of a missing check, causing the program to be unable to differentiate between an empty result because of a lack of matches and an empty result because of a system error. Consequently, even if the program gracefully handled an error, it displayed an inappropriate message. The fix for this was to check if the retrieval method returned the output expected with an error (null) before displaying the incorrect username/password error.

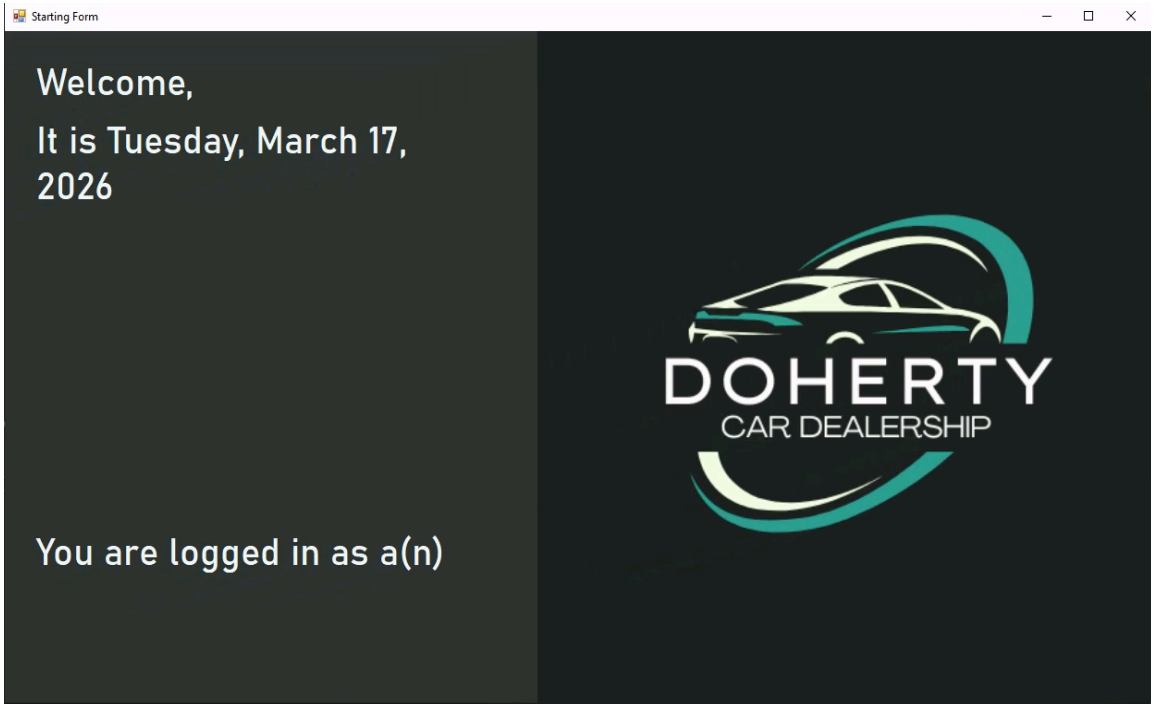
3.1.2

Before:

Program crashes



After:



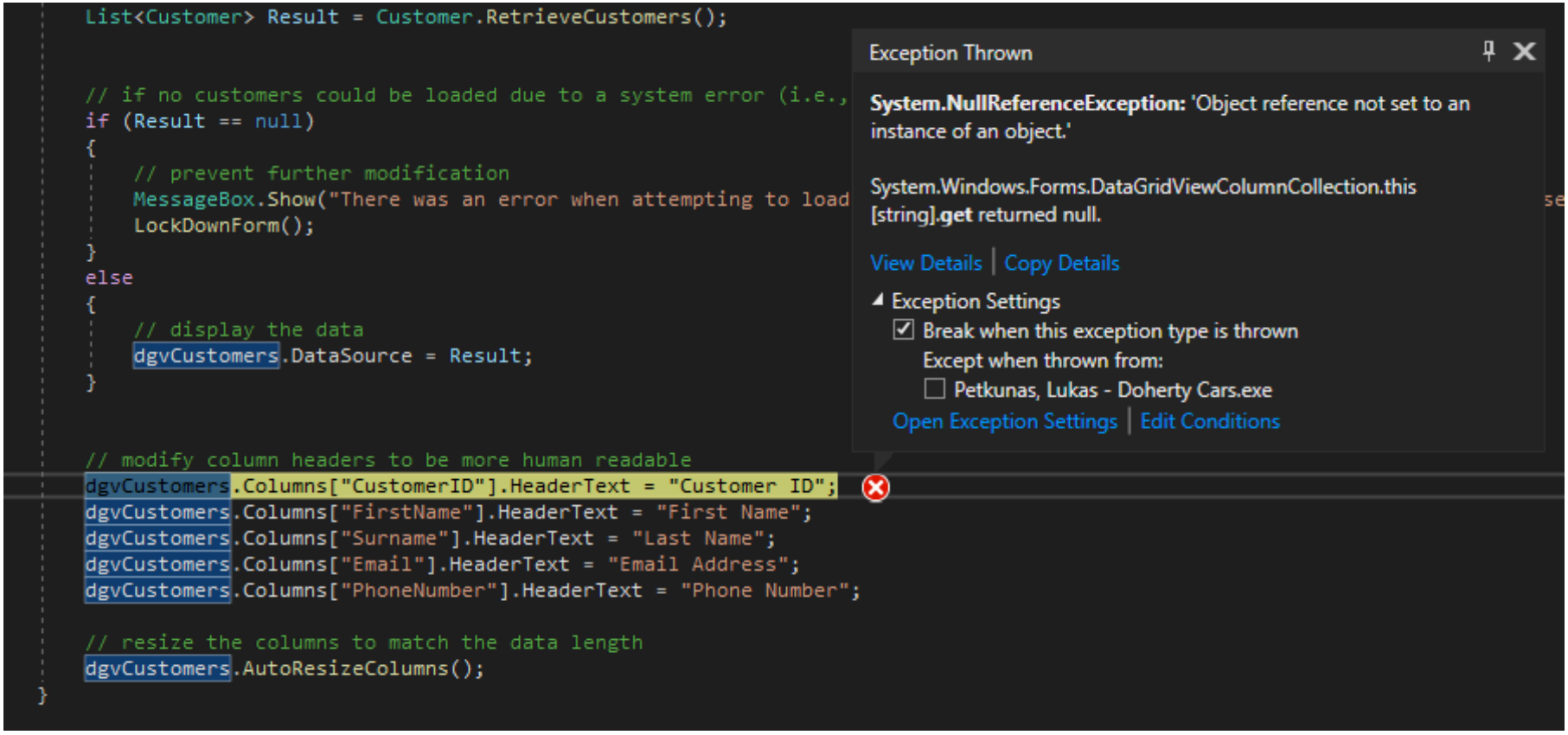

```
1 reference
private void frmStartingMainMenuForm_Load(object sender, EventArgs e)
{


    // display the current time
    lblDate.Text = "It is " + DateTime.Now.ToLongDateString();

    // attempt to display the current user's name
    if (CurrentUser.CurrentLoggedInUser != null) lblWelcomeMessage.Text = "Welcome, " + CurrentUser.CurrentLoggedInUser.Firstname;
    else
    {
        MessageBox.Show("Your name could not be retrieved. Please ensure you have logged in correctly.");
        return;
    }

    // attempt to display the user's role
    var role = Role.RetrieveByRoleID(CurrentUser.CurrentLoggedInUser.RoleID);
```

This error was a result of a lack of exiting in the event of a lack of a user object. Even though the check which ensured that a valid CurrentLoggedInUser exists was properly executed, the code still continued to execute and ultimately attempted to retrieve the RoleID property on a null object, throwing a Null Reference Exception. The fix for this was to return out of the method when an invalid setup was detected, preventing further processing.

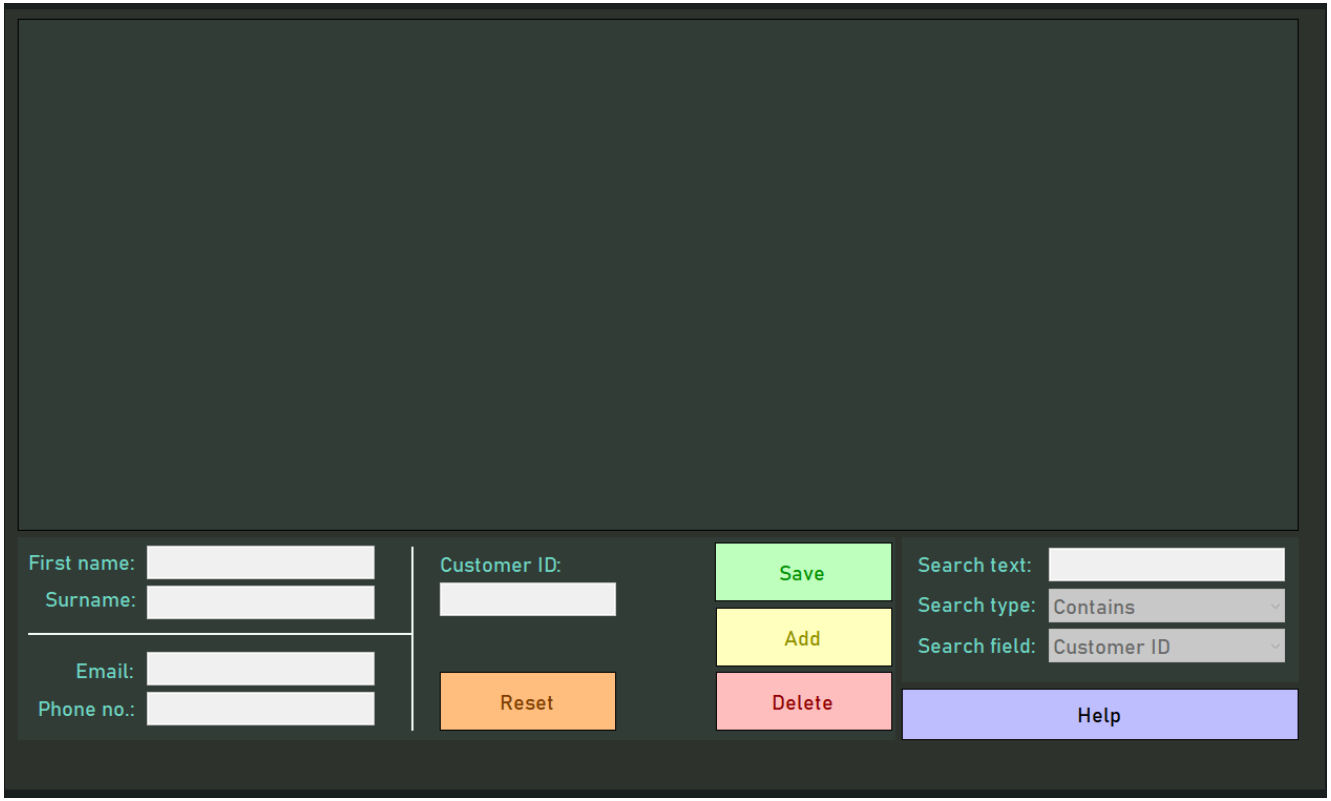
4.1.2

Before:

Program crashes



After:



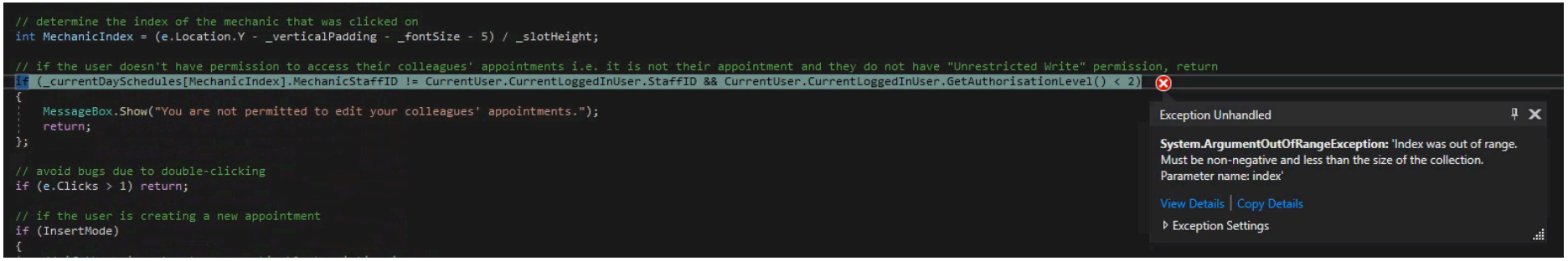

```
2 references
private void AttemptDataDisplay()
{
    // populate the data grid view with the latest version of the customers database stored when the form is loaded
    List<Customer> Result = Customer.RetrieveCustomers();


    // if no customers could be loaded due to a system error (i.e., null was returned instead of an empty list)
    if (Result == null)
    {
        // prevent further modification
        MessageBox.Show("There was an error when attempting to load customer data. Please amend erroneous data in the database before proceeding.");
        LockDownForm();
        return;
    }
    else
    {
        // display the data
        dgvCustomers.DataSource = Result;
    }
```

Similarly to 3.1.2, this error was a result of a lack of exiting when no data could be loaded. Even though the program could detect when no data could be retrieved as a result of a system error (denoted by the null check), no return statement was implemented and as a result the program still attempted to change the header text of the now non-existent fields, causing a Null Reference Exception. The fix for this was to implement a return statement when no data could be loaded. Note that most other CRUD forms had the same issue, but this was amended after this specific test case.

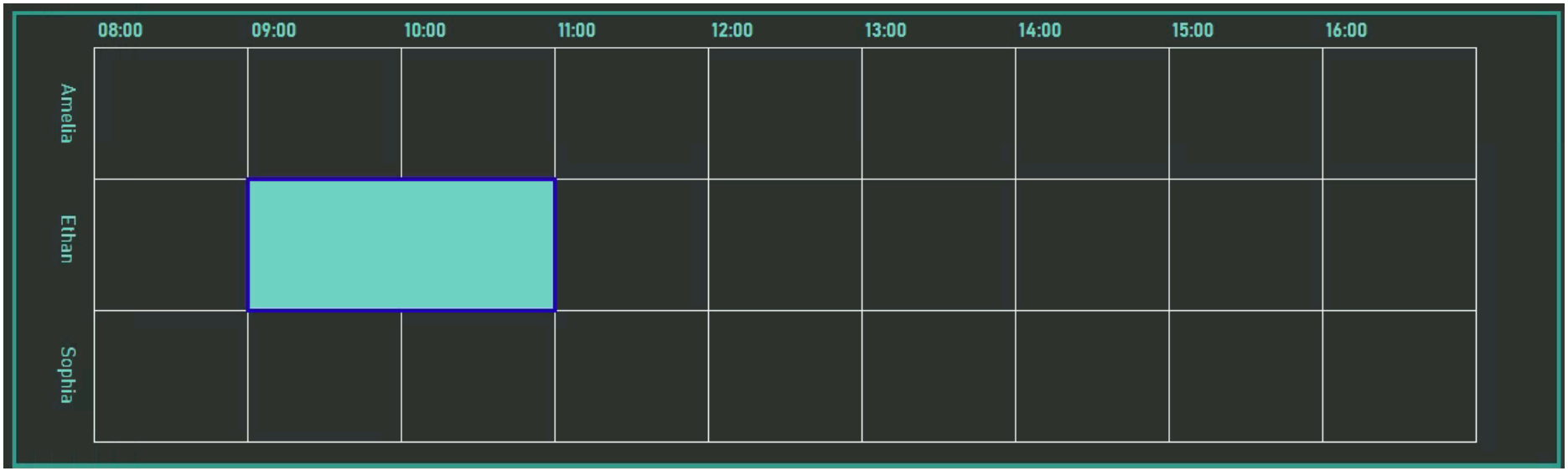
11.2.14

Before:

Program crashes



After:

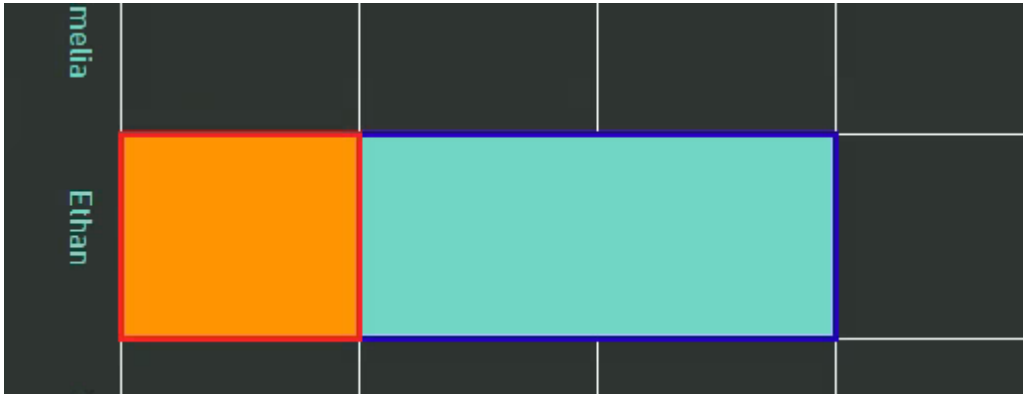



```
// determine the index of the mechanic that was clicked on
int MechanicIndex = (e.Location.Y - _verticalPadding - _fontSize - 5) / _slotHeight;

// if there is no such mechanic at that index, return
if (MechanicIndex >= _currentDaySchedules.Count) return;

// if the user doesn't have permission to access their colleagues' appointments i.e. it is not their appointment and they do not have "Unrestricted Write" permission, return
if (_currentDaySchedules[MechanicIndex].MechanicStaffID != CurrentUser.CurrentLoggedInUser.StaffID && CurrentUser.CurrentLoggedInUser.GetAuthorisationLevel() < 2)
{
```

This error was a result of improper validation regarding the calculation of the index which was clicked on in the panel viewer. Technically, the calculation returned the correct value - in this specific test case, the area below the timetable would qualify as the 4th position (index 3) if there was a 4th mechanic, but the lack of a mechanic in this index caused a Null Reference Exception when attempting to access the mechanic schedules using this calculated index. The fix for this was to implement a check which ensures that the calculated index falls within the bounds of the mechanic schedule array.

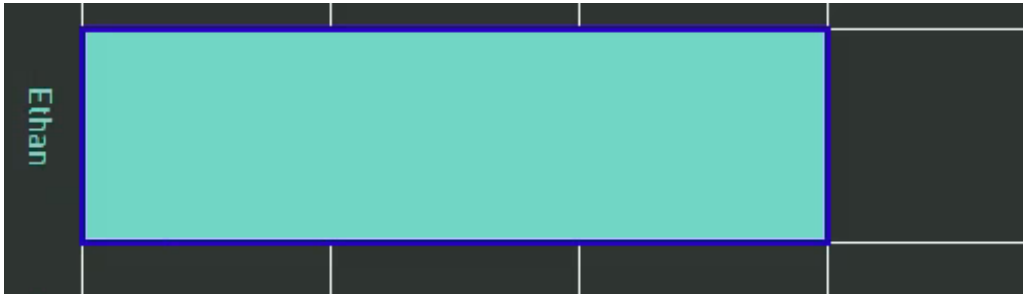
11.2.21

Before:




```
// if there is not a temporary timeblock existing in memory
if (!_visibleTimeblockCoordinates.ContainsKey(0))
{
    CreateTimeblockBasedOnCursorPosition(e.Location);

}
```

After:




```
// if there is not a temporary timeblock existing in memory
if (!_visibleTimeblockCoordinates.ContainsKey(0))
{
    CreateTimeblockBasedOnCursorPosition(e.Location);

    // if the time set clashes, just remove and skip
    if(!IsValidTimeSet(FindLargestTimeLessThanPosition(e.X), 0, false, true)){
        DeleteAppointment(0);
        return;
    }
}
```

A rather significant issue, this error was a result of lack of validation when attempting to create a new appointment. Even though the creation logic was intact, no checks were made to ensure that the new appointment would not conflict with any others when it was initially created, thus allowing it to intersect with another appointment, preventing further modification as any attempt at movement would cause a validation error, and allowing the user to add an appointment to the database which otherwise contradicts the validation set out in the design of the form. This would have also applied to just clicking on an existing appointment, but was fixed in this test case and as a result didn't show up later. As a result, a check is now made when the appointment is created to ensure that if it is in a valid position (i.e., if it were moved by 0 pixels would it intersect with another appointment.) If the position is invalid, the new appointment is immediately deleted, thus solving the problem.

Acceptance Testing

In order to ensure that the system is fit for purpose - i.e. it meets the user requirements, is approved for use and is ultimately complete - it is crucial that the end user is involved in the testing process. Thus, staff members from business sectors towards whom the initial requirements were targeted were invited to participate in an anonymous, unguided usage session and questionnaire to effectively evaluate their satisfaction with the final system. In particular, the administrative, technical and customer service business sectors were each invited to participate, and the results from this trial were compiled and analysed.

Questionnaire

To ensure that the system was fit for purpose, the questionnaire was designed with the user requirements in mind as seen on the next page.

Doherty Cars - Questionnaire for new system

Thank you for taking the time to test out the new system. Please take a few minutes to fill out the questionnaire below. If any of the questions are not applicable to you, please leave them unanswered.

1. Did you find it was easy to access the system?
   ☐ Yes ☐ No
2. Were you able to use the system for extended periods of time stably, including when logging in and out of the system?
   ☐ Yes ☐ No
3. Were you able to access only sections of the system that were relevant to your job title?
   ☐ Yes ☐ No
4. Did you find that the system prevented you from carrying out actions beyond your level of authorisation?
   ☐ Yes ☐ No
5. Were the records in the database displayed in a clear manner?
   ☐ Yes ☐ No
6. Were you able to create new records easily?
   ☐ Yes ☐ No
7. Were you able to edit records easily?
   ☐ Yes ☐ No
8. Were you able to delete records easily?
   ☐ Yes ☐ No
9. Did you find the system permitted you to carry out actions considered “invalid”?
   ☐ Yes ☐ No
10. Do you believe that the system is secure i.e. unauthorised personnel would find it difficult to access the system, and even harder to affect the data on the system?
    ☐ Yes ☐ No
11. Were you satisfied with the quality of the reports present on the system?
    ☐ Yes ☐ No
12. Did you find the system allowed you to create appointments effectively? ☐ Yes ☐ No
13. Did you find the system allowed you to view appointments effectively? ☐ Yes ☐ No
14. Did you find the system allowed you to delete appointments effectively? ☐ Yes ☐ No

15. Did you find the system allowed you to edit appointments effectively? ☐ Yes ☐ No
16. Were you satisfied with the feedback provided by the system in the event of an unsuccessful action?
    ☐ Yes ☐ No
17. Did you find the help forms provided useful?
    ☐ Yes ☐ No
18. Did you find the system visually appealing?
    ☐ Yes ☐ No
19. Did you find the system easy to navigate?
    ☐ Yes ☐ No
20. Was the system responsive (i.e. less than 2 seconds between input and output)?
    ☐ Yes ☐ No

Responses

The representative group consisted of 1 administrative staff (despite the anonymous nature of the test, it is unfortunate to say that this is guaranteed to be Patrick Doherty), 3 mechanics (technical) and 3 customer service staff. The responses gathered were as follows:

#	Question	A	M1	M2	M3	CS1	CS2	CS3	Total Y	Total N
1	Easy to access system?	Y	Y	Y	Y	Y	Y	Y	7	0
2	Use system stably (login/out)?	Y	Y	Y	Y	Y	Y	Y	7	0
3	Access only relevant sections?	Y	Y	Y	Y	Y	Y	Y	7	0
4	System prevented unauthorised actions?	Y	Y	Y	Y	Y	Y	Y	7	0
5	Records displayed clearly?	Y	Y	Y	Y	Y	N	Y	6	1
6	Create new records easily?	Y	Y	Y	Y	Y	Y	Y	7	0
7	Edit records easily?	Y	Y	Y	Y	Y	Y	N	6	1
8	Delete records easily?	N	Y	N	Y	Y	Y	Y	5	2
9	System permitted invalid actions?	N	N	N	N	N	N	N	0	7
10	System secure?	Y	Y	Y	Y	Y	Y	Y	7	0
11	Satisfied with reports?	Y	–	–	–	Y	Y	N	3	1
12	Create appointments effectively?	Y	Y	Y	Y	–	–	–	4	0
13	View appointments effectively?	Y	Y	Y	Y	–	–	–	4	0
14	Delete appointments effectively?	Y	Y	Y	N	–	–	–	3	1
15	Edit appointments effectively?	Y	Y	Y	Y	–	–	–	4	0
16	Feedback useful?	Y	Y	Y	Y	Y	Y	Y	7	0
17	Help forms useful?	Y	Y	Y	Y	Y	N	Y	6	1
18	System visually appealing?	Y	Y	N	Y	Y	Y	Y	6	1
19	Easy to navigate?	Y	Y	Y	Y	Y	Y	Y	7	0
20	System responsive (<2s)?	Y	Y	N	Y	Y	Y	Y	6	1

Analysis

As can be seen from the responses, users are overall satisfied with the quality of the system. Many “No” responses can be attributed to personal preferences. For example, question 8 has received the most negative feedback, and this is most likely due to the restrictions put in place that prevent certain records - such as vehicle types - from being deleted if they are associated with another record - in this case a vehicle. This forces the user to navigate another form to delete or modify the perpetrating child record, ultimately delaying the user for the sake of referential integrity.

Regarding other questions, many received all “Yes” responses (or in the case of question 9, all “No”s) so it may be assumed that those aspects of the system are complete and suitable. It is unknown why the customer service staff member in question 5 responded negatively, as the customer table will display all possible relevant data on the system (perhaps they were expecting additional information such as join date, address etc.) but as the owner is satisfied with viewing data it will be regarded as an anomaly. The same applies to question 7, as customer service members have it rather easy when it comes to the complexity of the data they edit, and the validation system is clear in its feedback.

Question 11’s negative response may have been justified, as although the information provided by the reports should be detailed enough, it could be presented in a clearer and more punctual manner, although this is more so a limitation of the library used than a fault of the system.

Question 14’s negative response is most likely an anomaly, as there is little reason to be unsatisfied with deleting when the system does not implement any restrictions on deleting selected appointments.

Question 17’s negative response may also be justified, as the help forms - though detailed - are rather verbose and could be made more effective through the use of images or more compartmentalised sections.

Questions 18 and 20 can be regarded as anomalies due to the subjective nature of the questions and the overall positive responses received.

Testing conclusion

Considering the rather thorough unit and integration testing phase and the success of the acceptance testing phase, it can be confidently said that no critical action must be taken at this point that would force certain aspects of the system to be redesigned or reimplemented, and the project can safely move onto the evaluation phase.

Doherty Cars

Program User Guide

User guide

Hardware and Software Requirements

Before you attempt to utilise the system, ensure that the following requirements are met on the workstation on which the system will be executed. These are considered the minimum specifications, and as such any problems caused by hardware or software issues cannot be attributed to the program if they occur.

If you are unsure of your system's specification, please contact the administrator or an IT specialist.

Software requirements:

- **Operating system:**
  - Windows 10 or later (64-bit)
  - OR Windows Server
- **Runtime/framework**
  - .NET 8 (Windows Desktop Runtime)
- **Dependencies**
  - Microsoft SQL Server
  - Visual Studio 2019
  - Relevant NuGet packages (refer to Visual Studio 2019's documentation)

Hardware requirements:

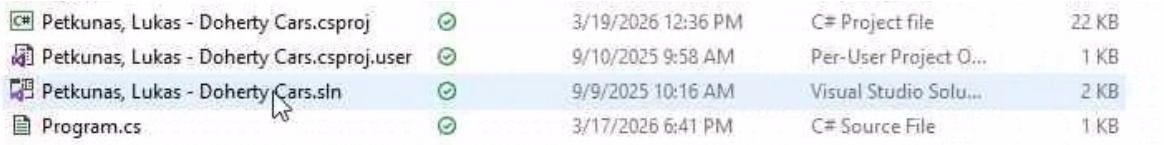
- **Minimum:**
  - Dual-core 1.6 GHz CPU
  - 4GB RAM
  - 500MB-2GB free disk space
  - 1366x768 resolution display
  - Stable LAN connection
- **Recommended:**
  - Intel i5/Ryzen 5 or equivalent
  - 8GB+ RAM
  - 2-5GB free SSD disk space
  - 1920x1080 resolution display

Installation

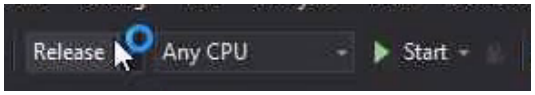
Before you begin utilising the system, it is crucial that you have the appropriate program files present on your workstation. As a prerequisite, ensure that you have the software system's project's files present in a readily accessible folder. If you do not, please consult an IT specialist or administrator regarding access to these files.

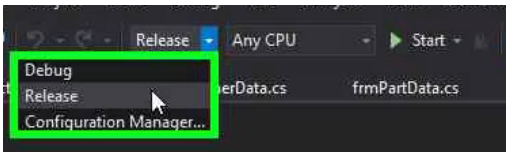
Compilation

1. With the project folder open, find the solution file and open it in Visual Studio 2019. This will open the project and attempt to load all of its files. Before attempting to compile, ensure that all necessary files have been loaded, especially if you rely on a cloud-based storage solution such as OneDrive.

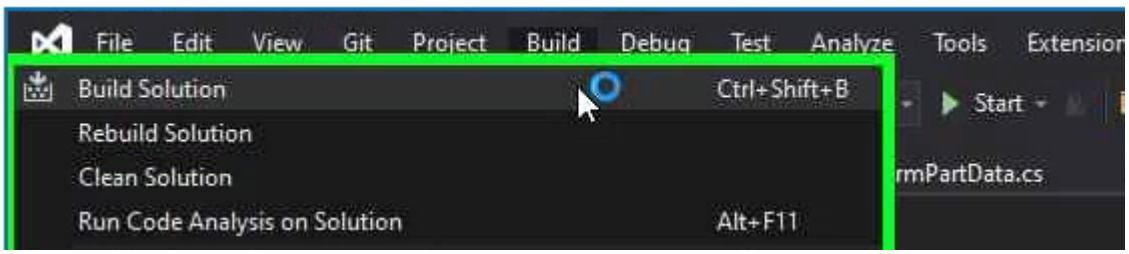


2. From here, locate the section in the toolbar with the button labelled "Start." Beside this should be a list from which you can select compilation methods. For standard usage, select "Release" before proceeding.





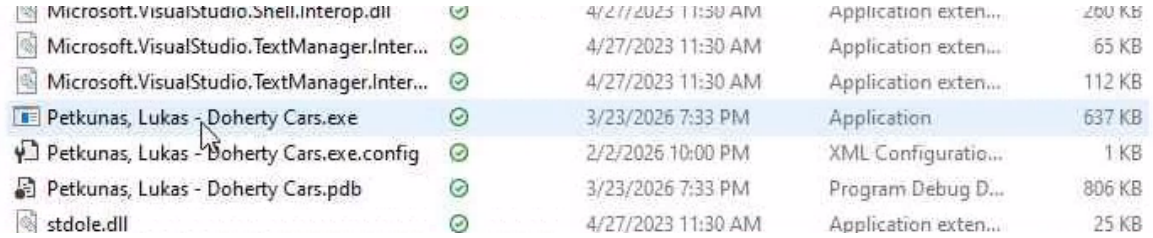
3. After selecting the desired compilation method, go to the "Build" section in the header and choose the first option.



4. After this, within the project folder, there will be a "bin" folder. In here, you will be able to enter the folder with the same name as the compilation method you chose (e.g. "Release") to access the compiled program files. For standard usage, the expected directory is "(project folder)/bin/Release/"

Execution

Within the appropriate folder as described in the “Compilation” section, you will find an executable file with the same name as the project. Double-clicking this will run the system locally.



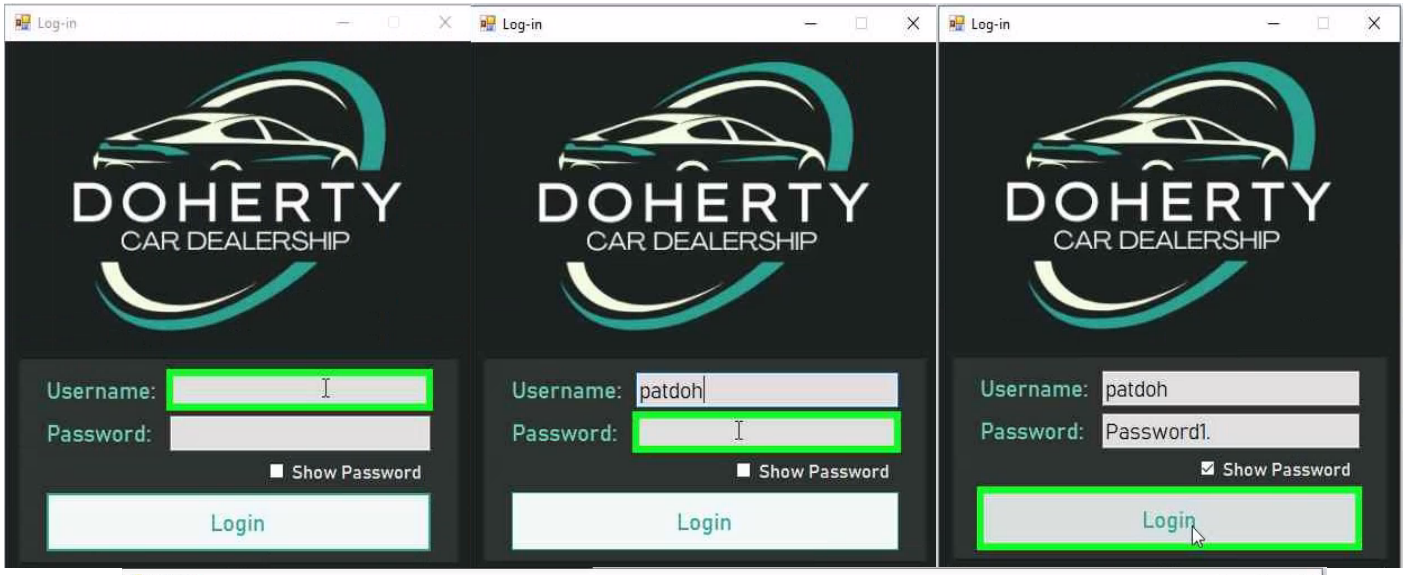
Note that while all the files within the folder are necessary for the program’s execution, the folder need not stay in the same directory. Therefore, once you are finished building, you may move the folder containing the executable to a different location, dispose of the project files and uninstall Visual Studio 2019 if it is unnecessary for you.

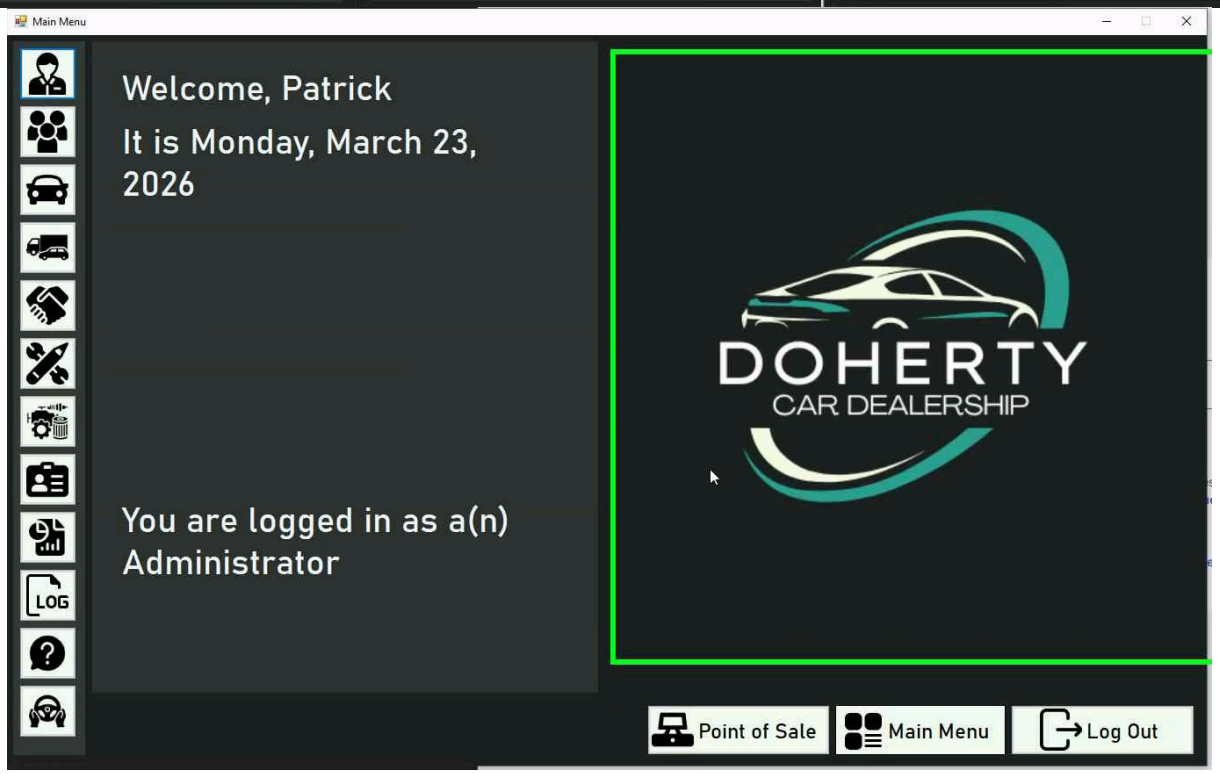
If you have any concerns regarding the setup of the program, do not hesitate to contact an administrator or IT specialist.

Operating the system

Logging-in

To log-in, simply input your credentials into the form that initially appears when opening the program, with your username and password in the appropriate fields, and hit “Log-in.” In the event of a successful log-in, you will be redirected to the main menu form. Otherwise, you will be given appropriate feedback. To ensure that you have entered your password correctly, you may avail of the provided “Show Password” checkbox

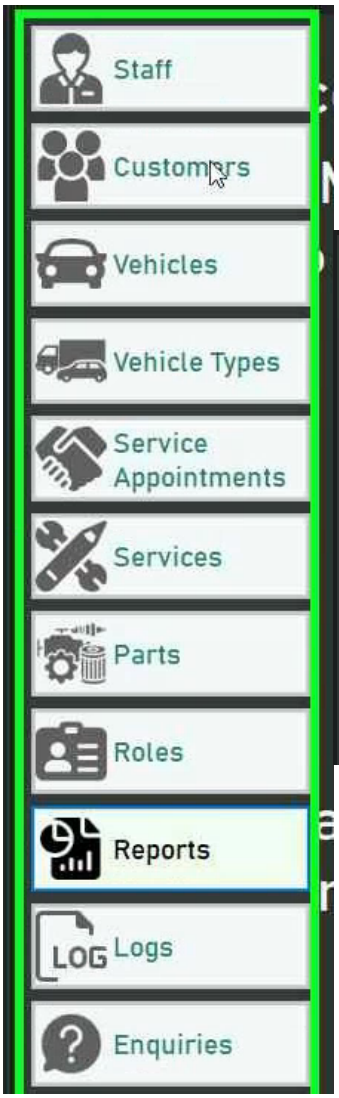


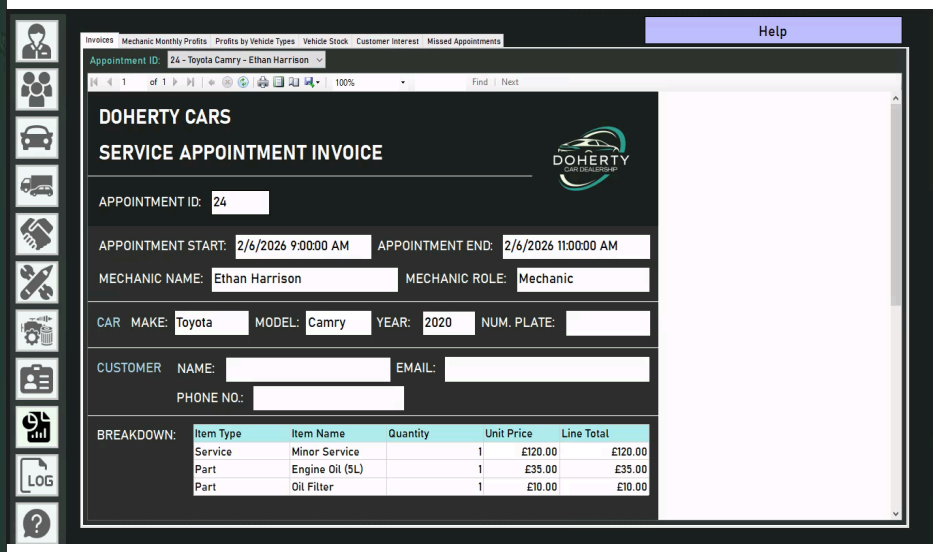


Navigation

Once you have logged in, you will notice unlabelled buttons to the left of the main menu. For more information on them, you may hover over the panel, which will expand the panel and buttons and label the buttons appropriately. If you are not authorised to access a form, the appropriate button will be disabled and made distinctly gray.

Clicking on any of the buttons will cause their respective form to open in the viewing panel, allowing you to interact with the data stored on the system. If at any point you wish to stop using the system, you may press the “Log Out” button.

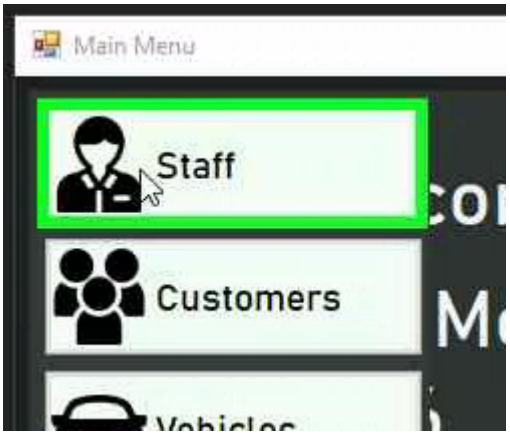




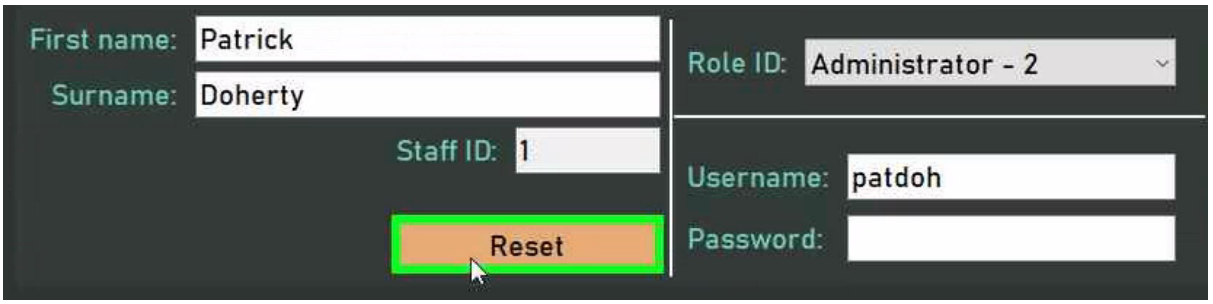
Creating a new user

For the sake of security, no registration form has been provided to prevent unauthorised access to the system. Instead, it is up to administrators (or similarly authorised users) to create user accounts for staff members. First, the administrator must access the system as demonstrated in the previous section.

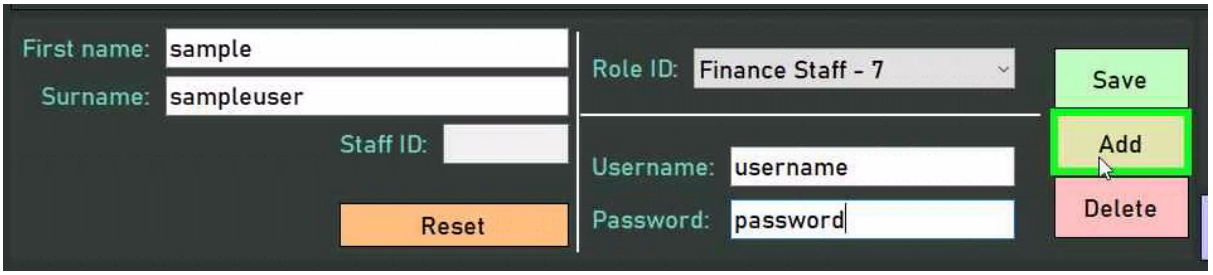
First, access the staff data form.



From here, you may find the fields automatically populated. Use the “Reset” button to clear them and make data input easier for you.



Input the new user’s details as appropriate. Make avail of the provided help form if any issues occur during this process. Once you are satisfied with the user details, you may press the “Add” button. This will create a new record with the provided details, and the user will be able to access the system using the credentials you have allocated to them.



CRUD operations (general)

As 7 of the provided forms use the same general CRUD system (these being the forms for staff, customers, vehicles, vehicle types, parts, services and roles,) much of the behaviour behind these forms carries over and thus each form will not be covered individually. Instead, the most complex of these forms - the vehicles form - will be covered in-depth to allow you to extrapolate the guidance provided onto the other forms.

If you find yourself stuck with any idiosyncrasies of certain forms, do not hesitate to consult the provided help form or - if needs be - an administrator.

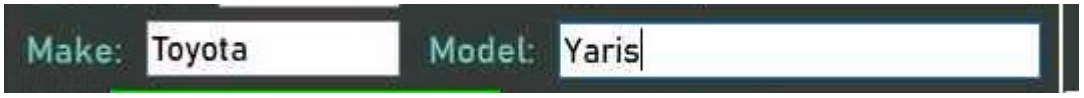
Creating

On accessing a form, you may find that details have been automatically populated. To clear the provided fields to prevent you from having to manually delete the populated data, hit the “Reset” button.

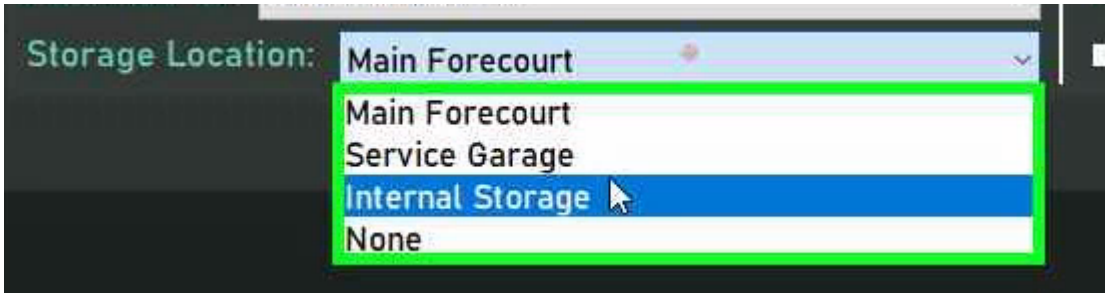
From here, you are able to input the desired data for the new record in the provided fields. An attempt has been made to denote the purpose of each field as clearly as possible using a text label to the left of it. Note that every form will have a read-only slot for a record’s ID. When creating a new record, this will be ignored, as the system will automatically handle allocating IDs to new records.

Text fields (such as “Make” and “Model” in this form) can simply be provided by typing in

the desired values upon selecting the appropriate TextBox.



Fields that must have restricted values are denoted by ComboBoxes. For fields that simply have a finite number of possible values, these will be determined before the form loads. However, for fields which depend on other tables - in this case, the customers table - all records from those tables will be retrieved, processed and provided as an option in the CheckBox, with pertinent information such as name denoting the record each option in the list represents.

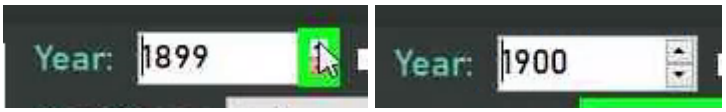


An example of predetermined values

An example of values fetched during runtime

Regardless, to select an option, click on the ComboBox to expand the list, and then click on the desired item in the list.

For numerical data, the NumericUpDown controls provided allow you to either manually go to the number which you desire (using the up and down arrows provided to increase and decrease the value respectively) or simply type in the desired number. Note that they will automatically reject numbers that are either invalid or do not fall within the predetermined ranges.



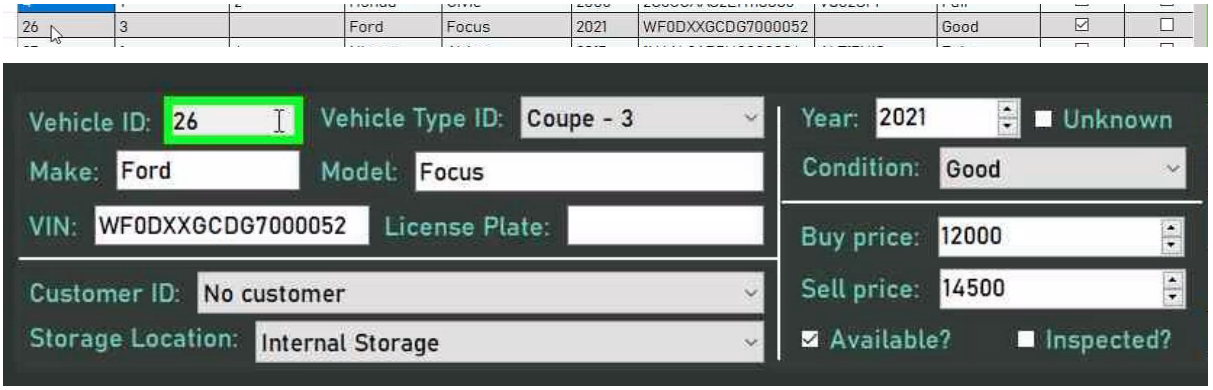
Lastly, for boolean data (that is, anything that is just true or false), CheckBoxes have been provided, which you can tick for “true” and untick for “false.”

Once you are satisfied with your details, click the “Add” button. This will perform validation on your provided data. If any of the data does not meet the rules enforced by the system, an error message will be displayed and you will be given the chance to amend your provided data. If it is deemed fine, the record will be added to the system, which should be evidenced by its appearance in the table.

Reading

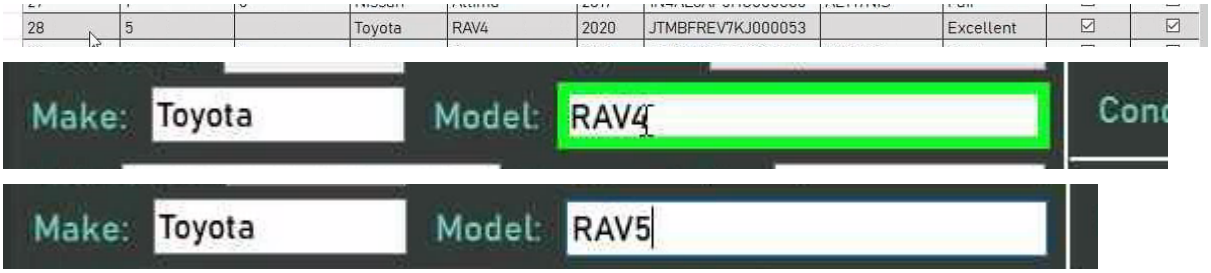
To select the record which you wish to edit, click on the appropriate row in the table provided. This will select that user for editing or deleting, populating their details into the fields provided.

Note that foreign keys (fields which come from other tables) are identified simply by their ID in the table, but upon selection the appropriate record will be selected in that ID's ComboBox, allowing you to see which foreign record the selected record is associated with.

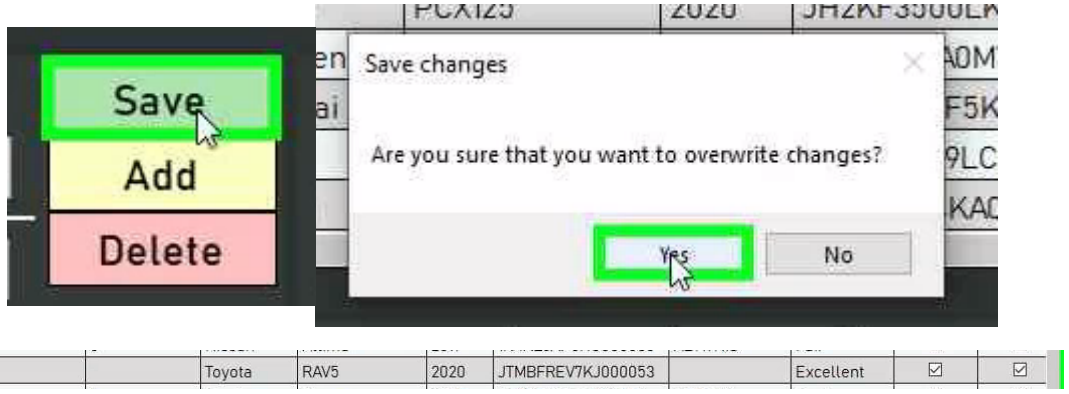


Updating

To update a record, first select the desired record as described in the previous section. Then, with the existing details populated into the appropriate fields, you are able to edit these details (or replace them with entirely new ones) with the same procedure described in the “Creating” section.

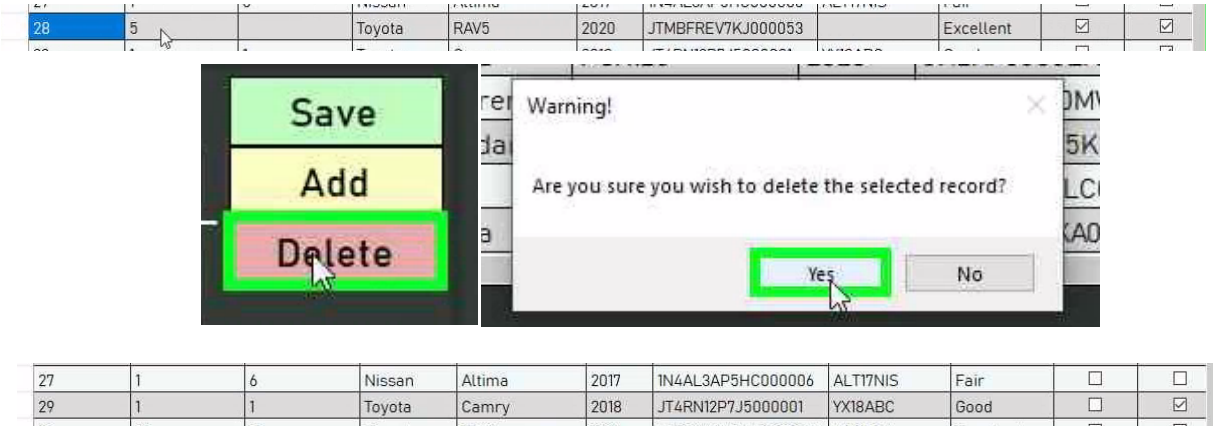


Once you are satisfied with your input details, click on the “Save” button. Your data will be validated, and if it passes the checks carried out by the system then the details input will be stored under the same ID as the selected record, as should be evidenced by the presence of new details in the table.



Deleting

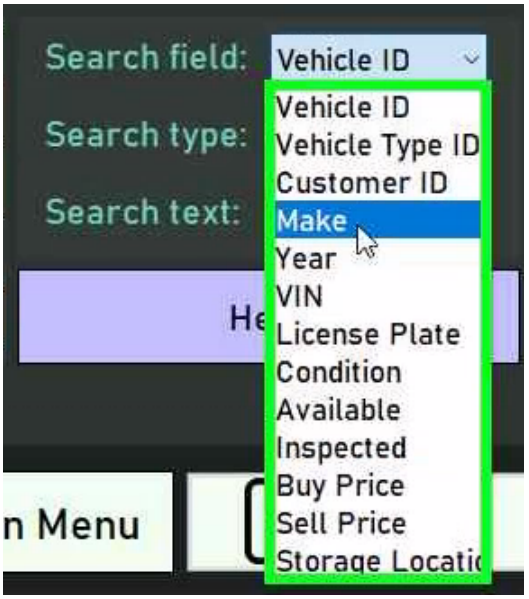
To delete a record, simply select it in the table and press the “Delete” button. Note that some forms may have certain rules implemented that could prevent deletion if the record is associated with records in another table, so consult the provided help form to see if these rules apply.



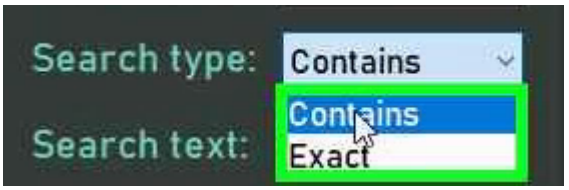
Searching

Searching is provided in each CRUD form through the use of 3 controls.

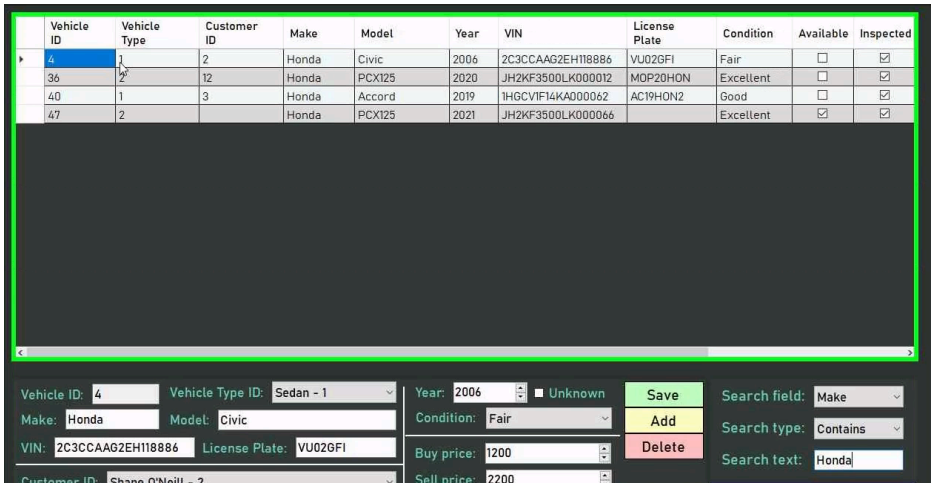
To select the field in the table you wish to search by, select it in the ComboBox.



Similarly, a ComboBox is provided for the type of search, of which there are presently “Contains” - where the provided text can make up part of or the entire field - or “Exact” - where the field must exactly match the provided text.



As you type into the search text field, you will find that the table automatically updates to match your query, thus allowing you to filter the database effectively.



CRUD operations (appointments)

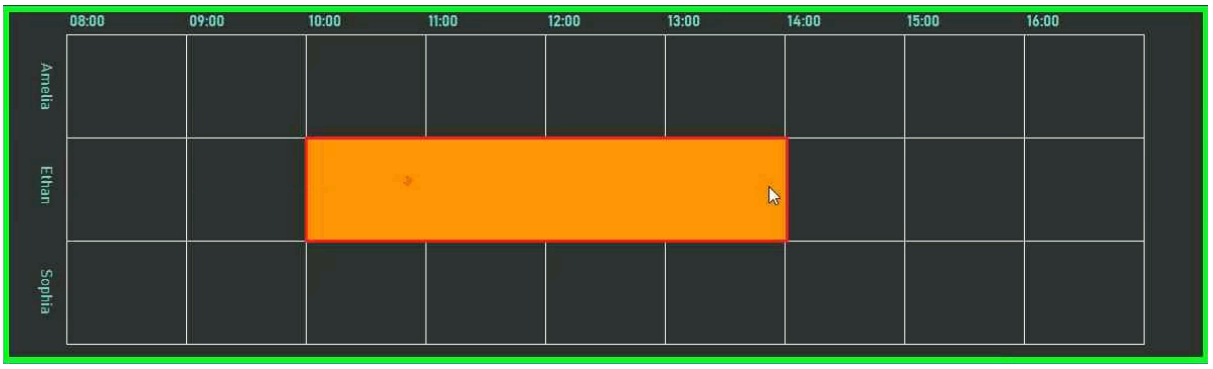
Due to the rather distinct nature of the service appointments form, an additional section has been provided to guide mechanics in performing CRUD operations in the form.

Creating

To create a timeblock, you must first activate “Insert Mode.” This will clear any changes you made to the timetable and deselect any appointments, so ensure you are ready to do so.

When you have activated “Insert Mode,” click on any free slot on the timetable. Note that you will be able to change the times later, so where you set it initially does not matter, but you must ensure the appointment is provided enough time to fit all of its services. Also note that you are not allowed to create appointments on your colleagues’ slots if you are not a lead mechanic.

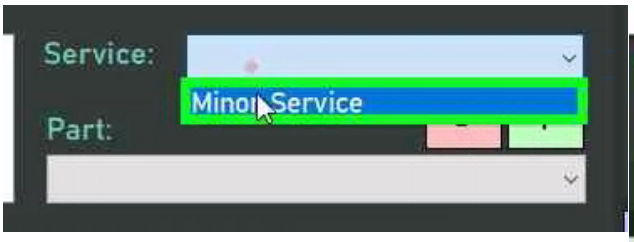
Once you have created the appointment, you may change the start and end times by either dragging and dropping the start or end of the appointment, or by using the provided NumericUpDown controls. Do note that these do not allow you to type times in.



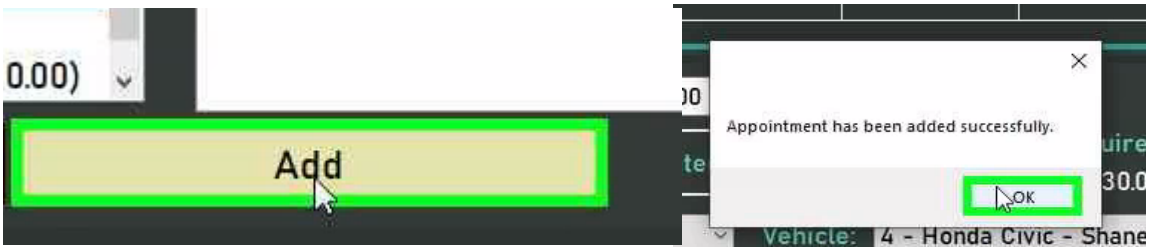
Once you are satisfied with the times, you will find that most of the fields are of a similar nature to the CRUD forms, so special focus will be made on the service and part selection system.

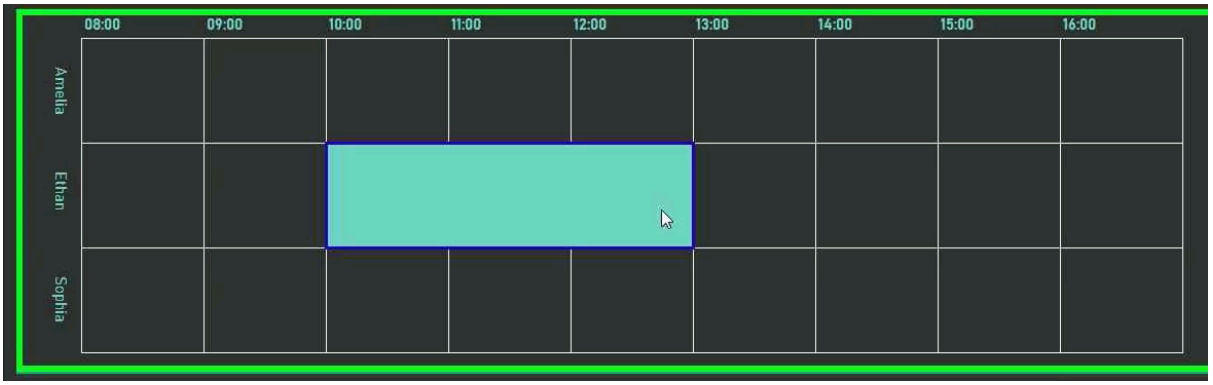
To select a service you wish to add on, simply double-click on it in the provided list. To deselect a service, simply click on it again. The service's price and time will be added on to the running total, as displayed in the appropriate fields.

Some services may require parts. For these, select the service you wish to add parts to, and the part you wish to add on, and then click the “Plus” button to add and “Minus” button to subtract parts. Note that checks will be made to ensure that you do not exceed the part quantity stored in the part table, and the parts added on will be subtracted from the parts table upon creating the appointment.

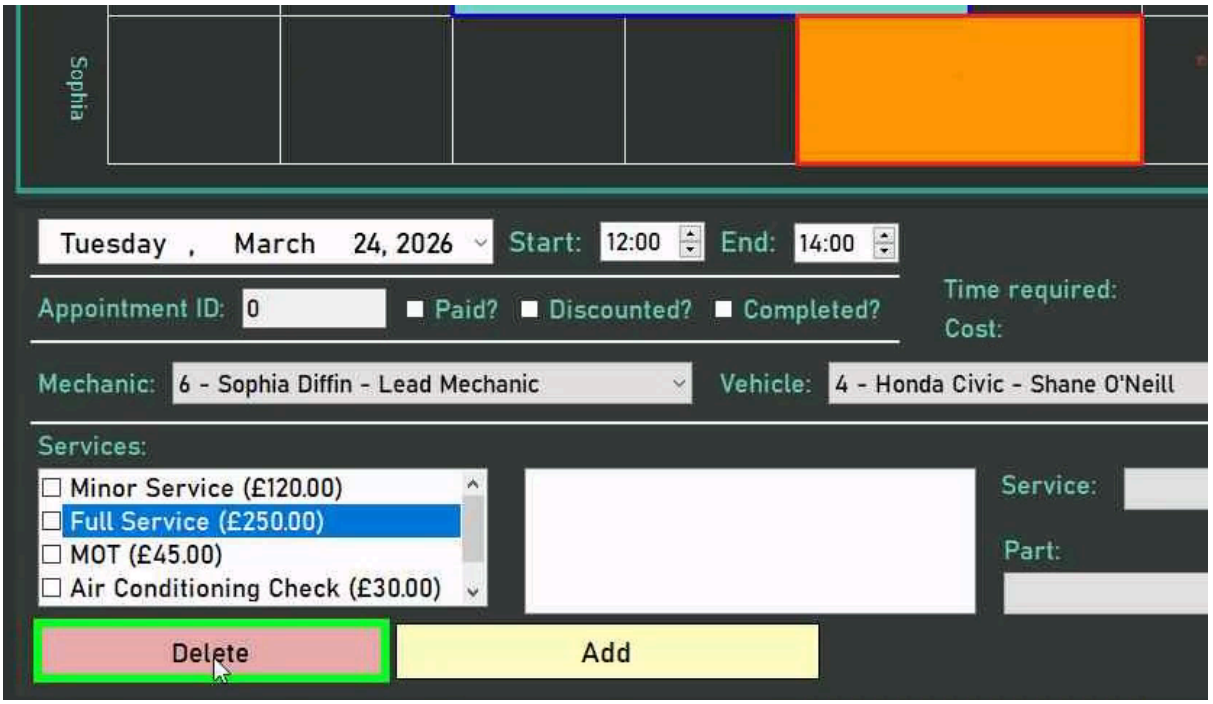


Once you are satisfied with the appointment details, click the “Add” button. If the appointment details are valid, the appointment will be added to the database as indicated by the orange timeblock turning blue.



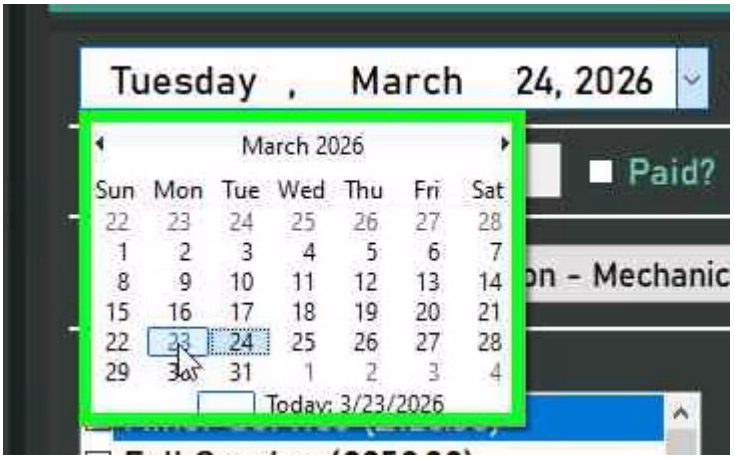


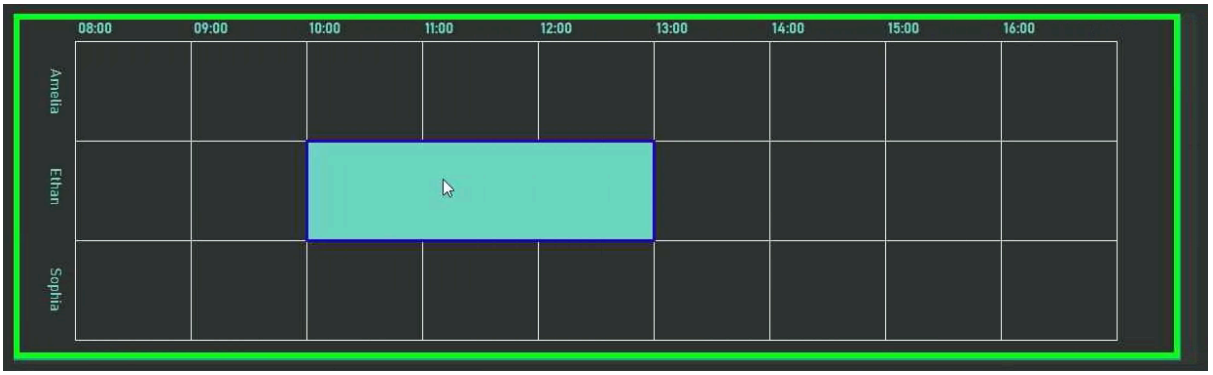
If at any point you wish to stop creating the timeblock, you may either disable “Insert Mode” or press the “Delete” button to delete the orange timeblock.

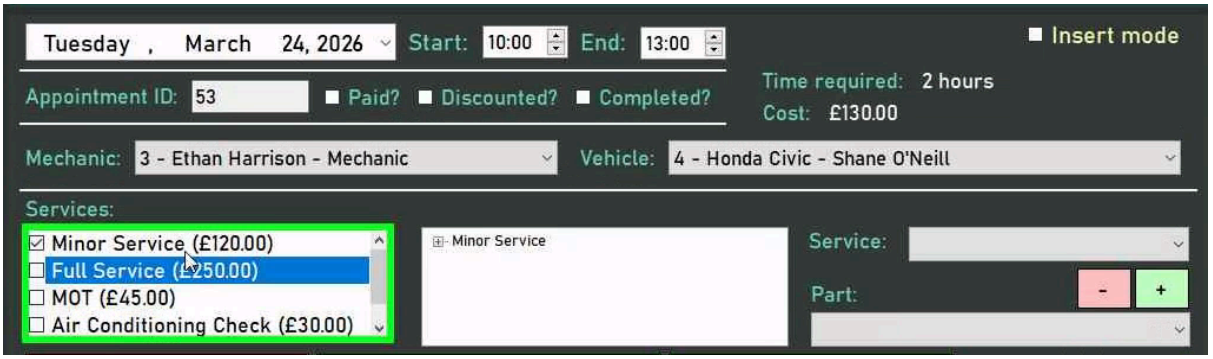


Reading

To access the timetable for a specific day, choose that day in the DateTimePicker provided. Once in the desired timetable, to view the details for a desired appointment, click on its timeblock representation. This will automatically populate its details into the appropriate fields.

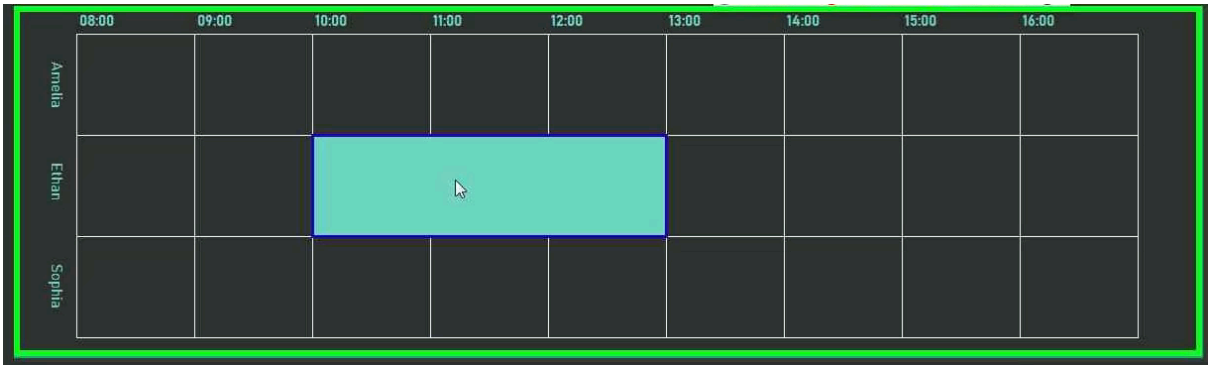


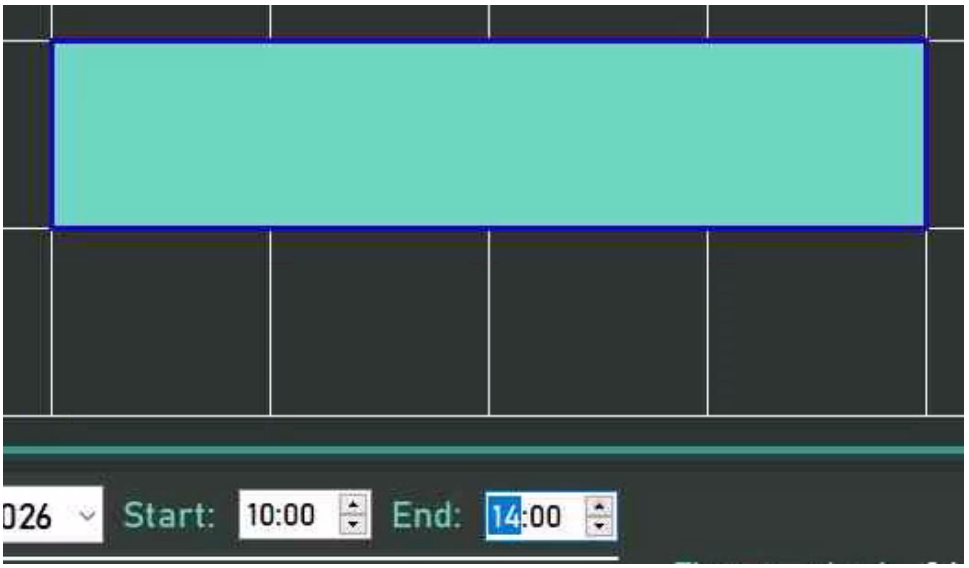


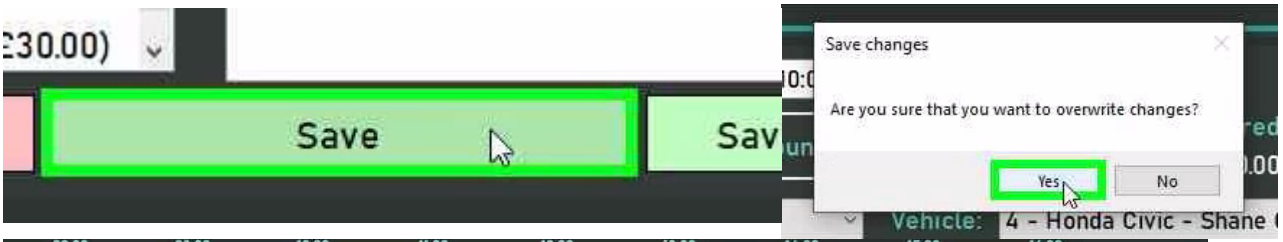


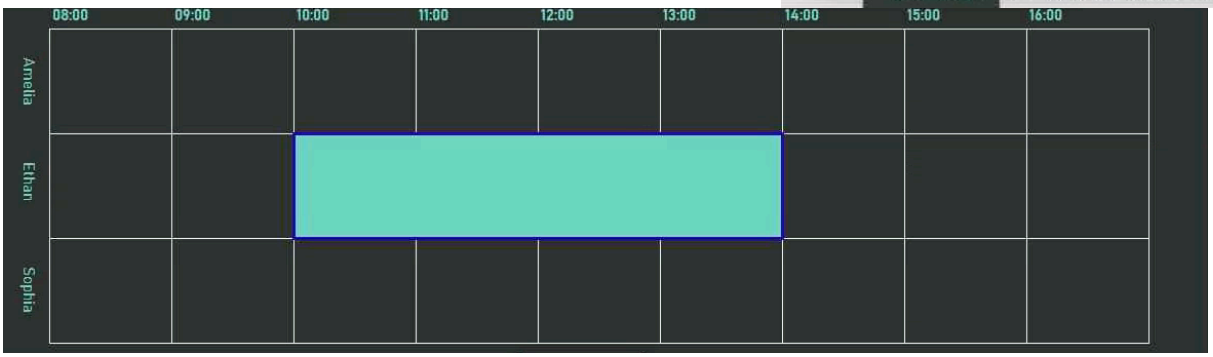
Updating

To update an appointment, select it by clicking on the timeblock representation, then modify the appointment's details much in the same way as in the "Creating" section. Once you are satisfied with the appointment details, simply click the "Save" button. This will validate your appointment, and if all is deemed correct the new details will be stored in the database.



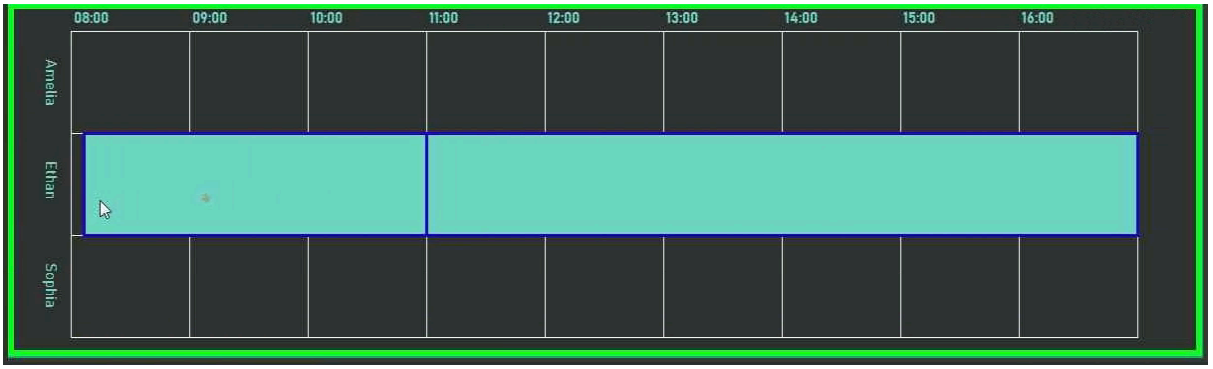


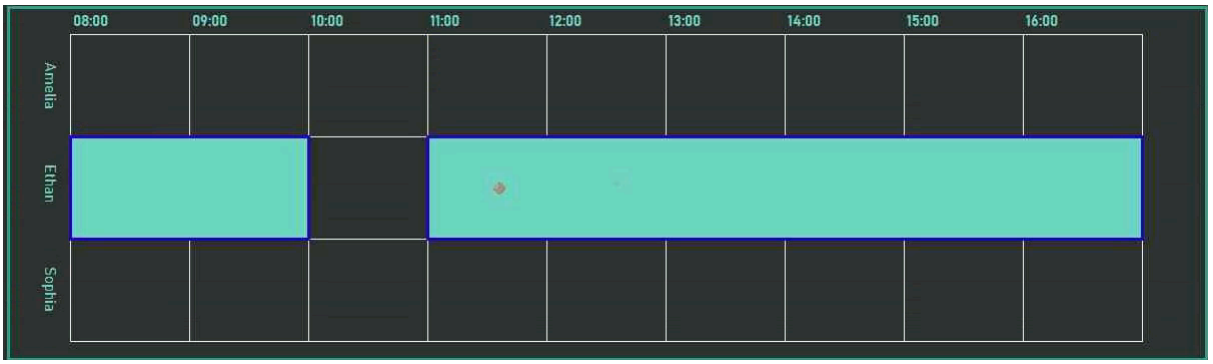


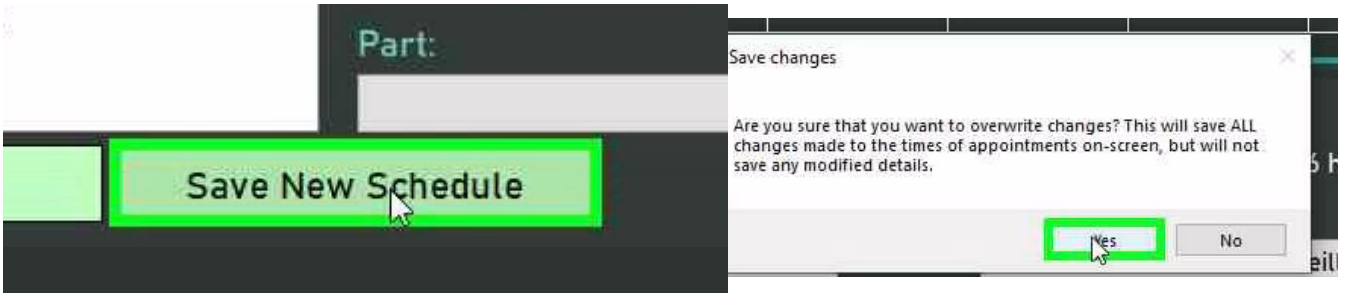


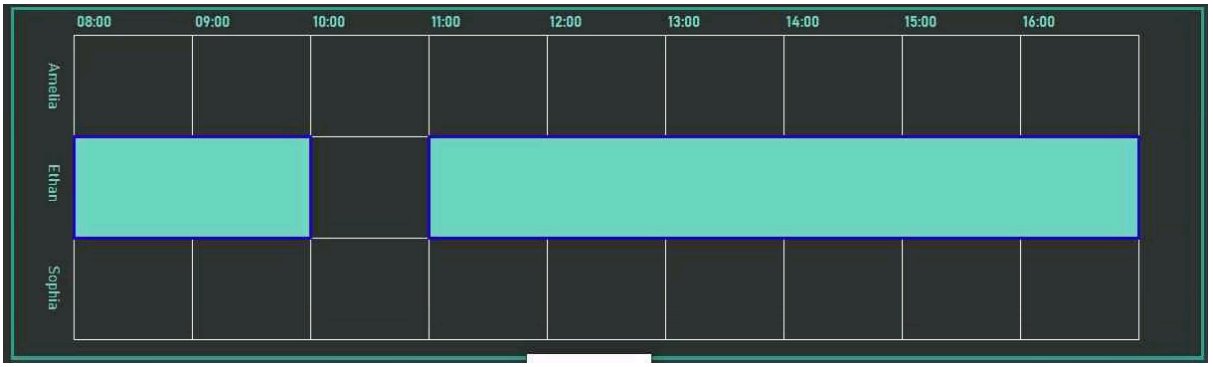
Updating Schedule

Updating a schedule undergoes much the same process as regular updating, except that only times will be saved for multiple appointments. To do this, first manipulate the relevant appointments to have the desired times, and then click the “Save New Schedule” button. If the times are valid, they will be saved to the database. Otherwise, you will be told which appointments need allocated more time.



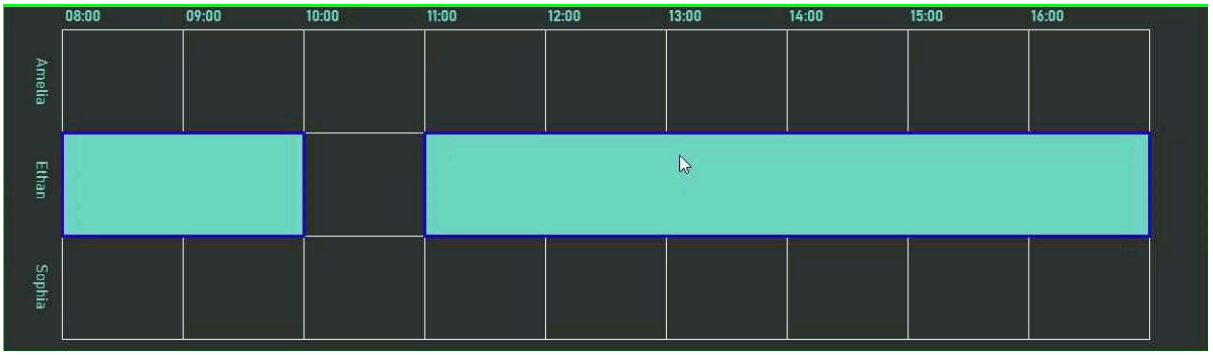


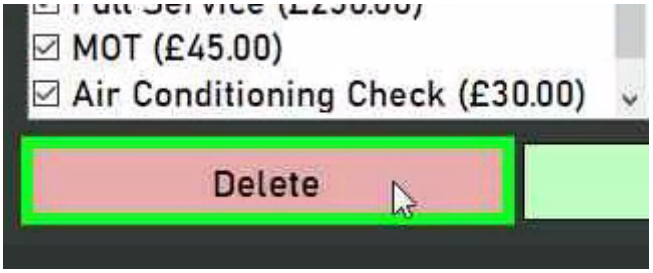




Deleting

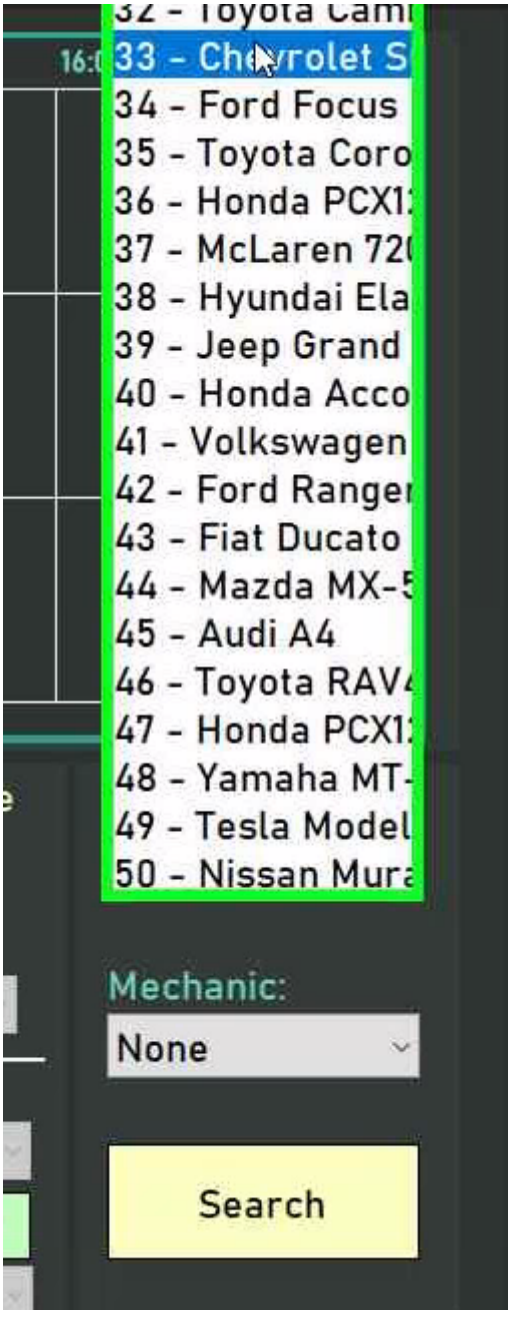
To update an appointment, select it by clicking on the timeblock representation, then click the “Delete” button.

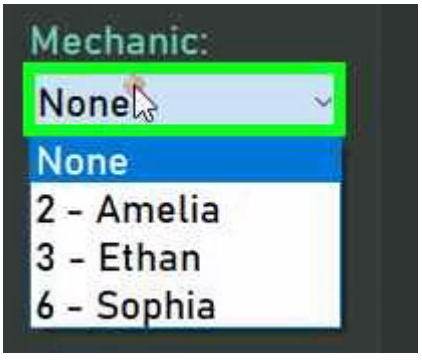




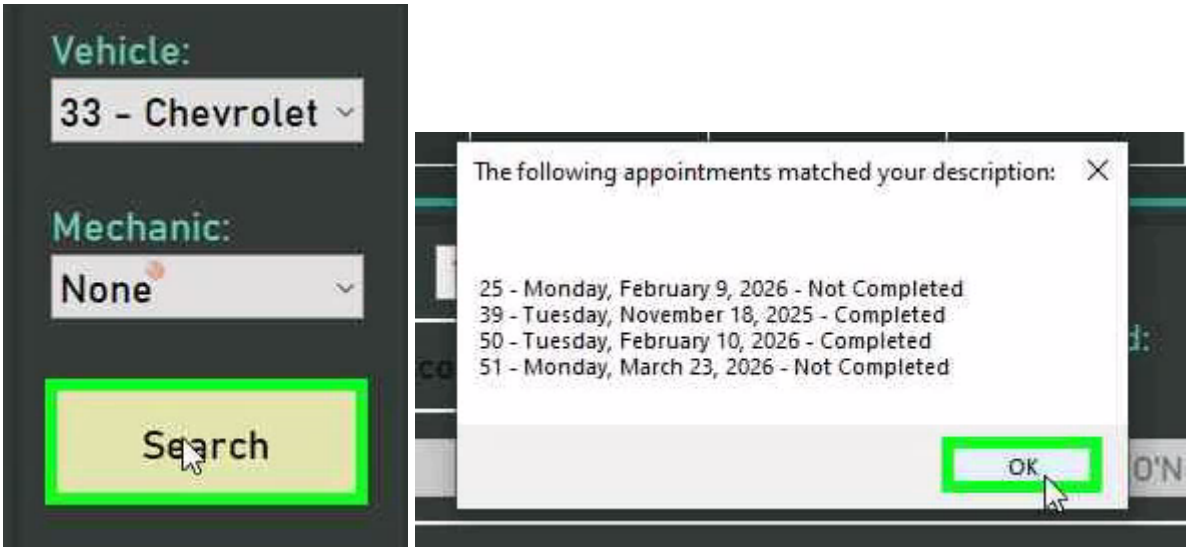
Searching

You may search for specific appointments using the allocated section. Here, 2 ComboBoxes have been provided: one for mechanics, and one for vehicles. If you wish to make a search more specific, you may select one of the available options in the ComboBoxes, otherwise you may leave them in their default state of “None.”





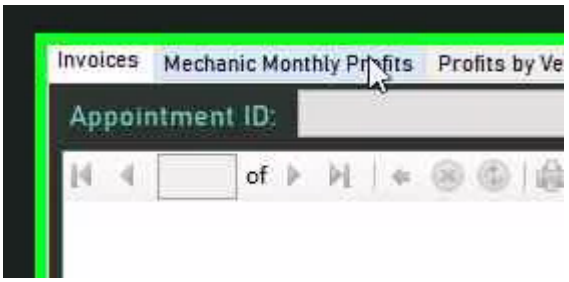
Once you are satisfied with the criteria, you may press the “Search” button to query the database with the provided filter. This will output a MessageBox displaying the details of all the appointments that match your search criteria if any were found.

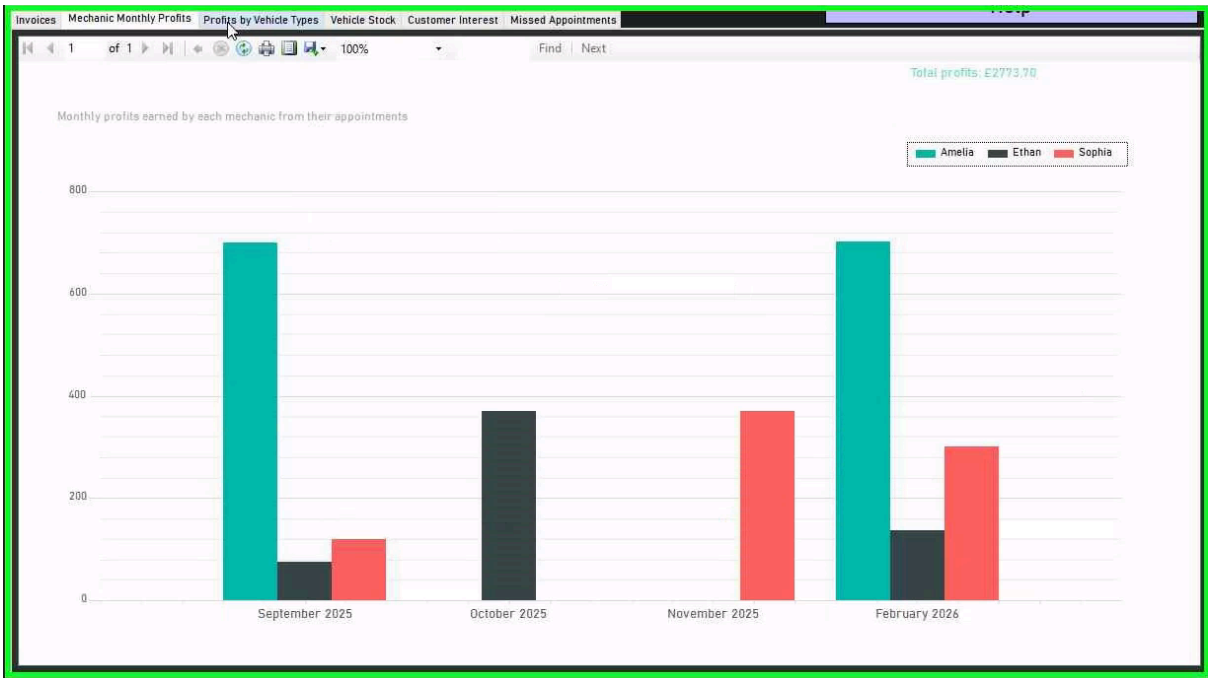

Vehicle:
33 - Chevrolet
Mechanic:
None
Search
The following appointments matched your description:
25 - Monday, February 9, 2026 - Not Completed
39 - Tuesday, November 18, 2025 - Completed
50 - Tuesday, February 10, 2026 - Completed
51 - Monday, March 23, 2026 - Not Completed
OK

Reports

Changing reports

To change to a different report in the reports menu, simply click the tab with the name of the report you wish to access. This will automatically display the desired report with the most up-to-date information.

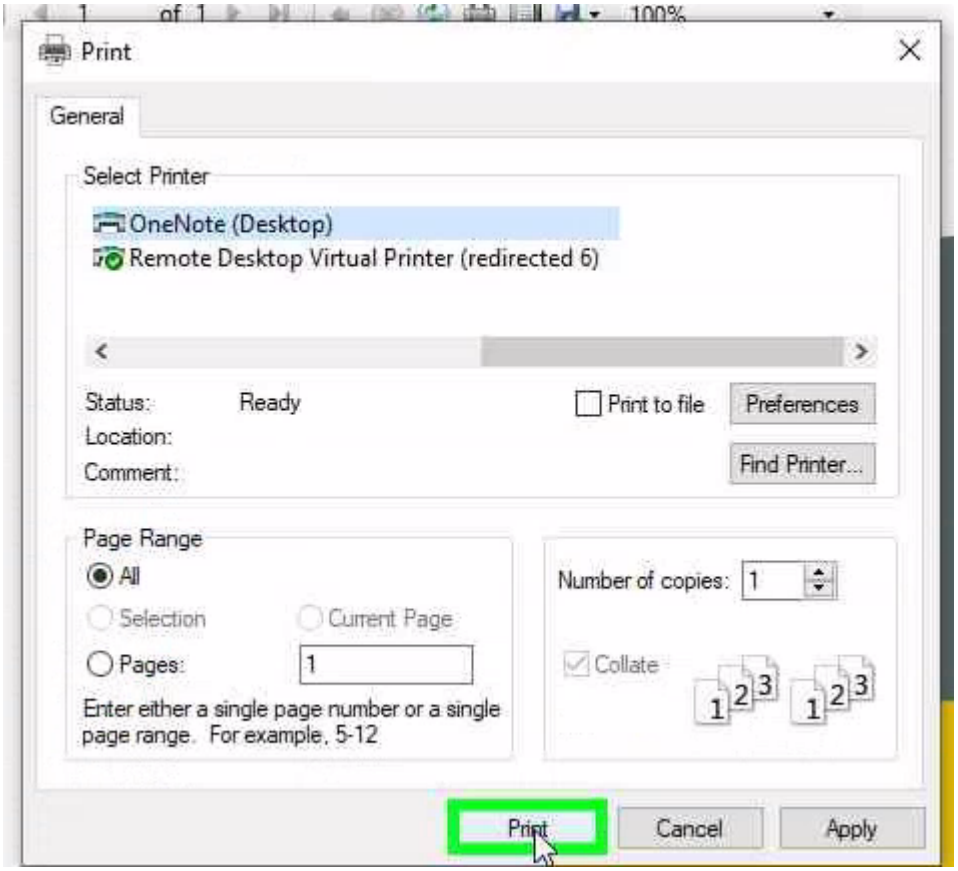




Printing reports

To print a report, click the printer icon seen in the toolbar at the top of the report.

This will open the default system print dialog, from which you can select the printer with which you wish to print the report and any other settings. The specific details of this dialog depend on the system you are using, and thus will not be discussed further.

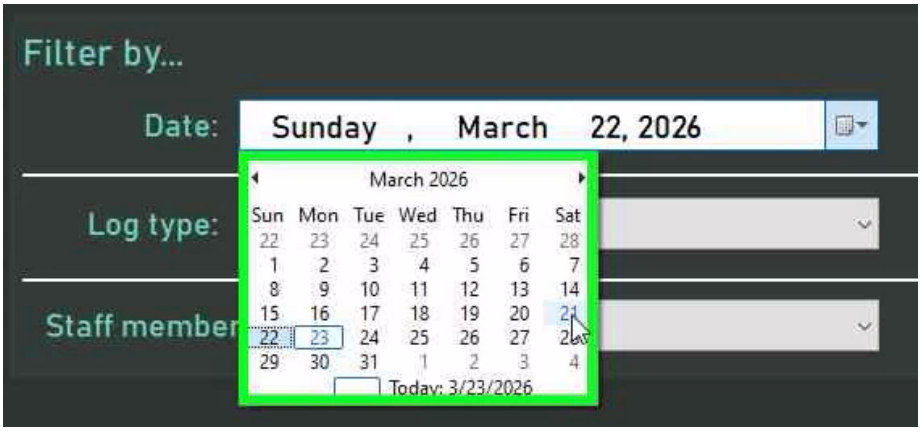


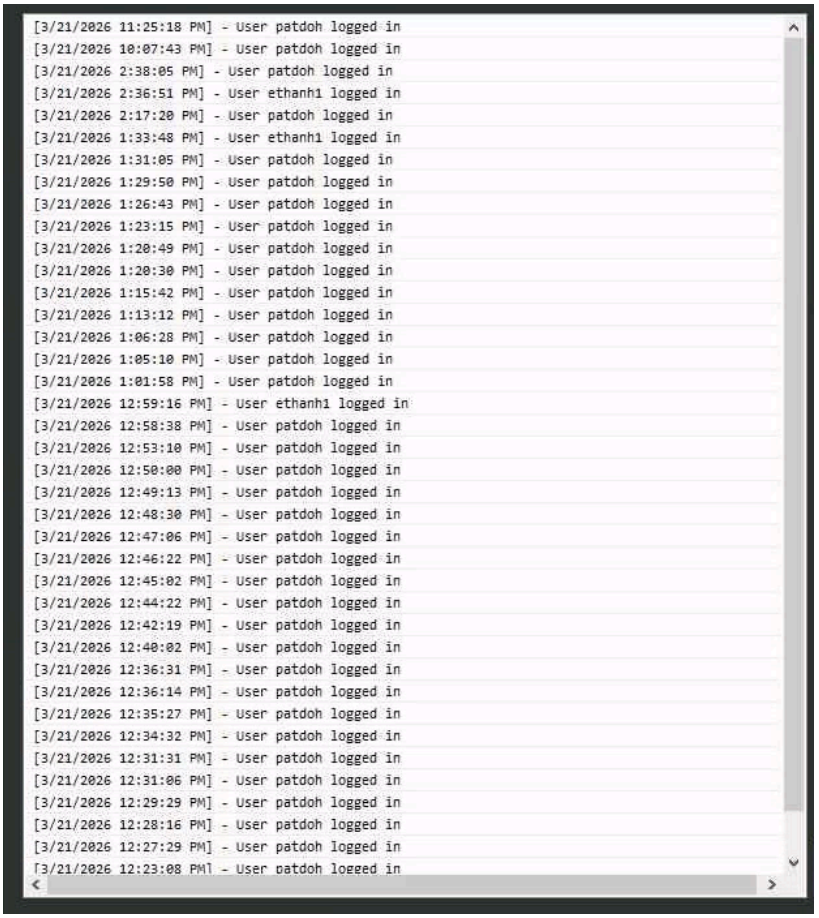
Logs

Filtering

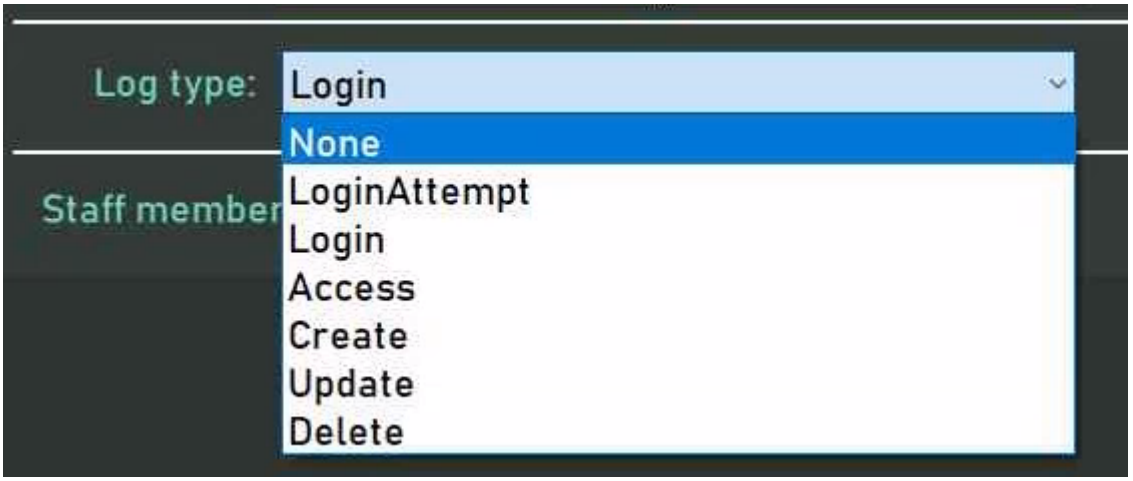
The logs form provides 3 filtering options.

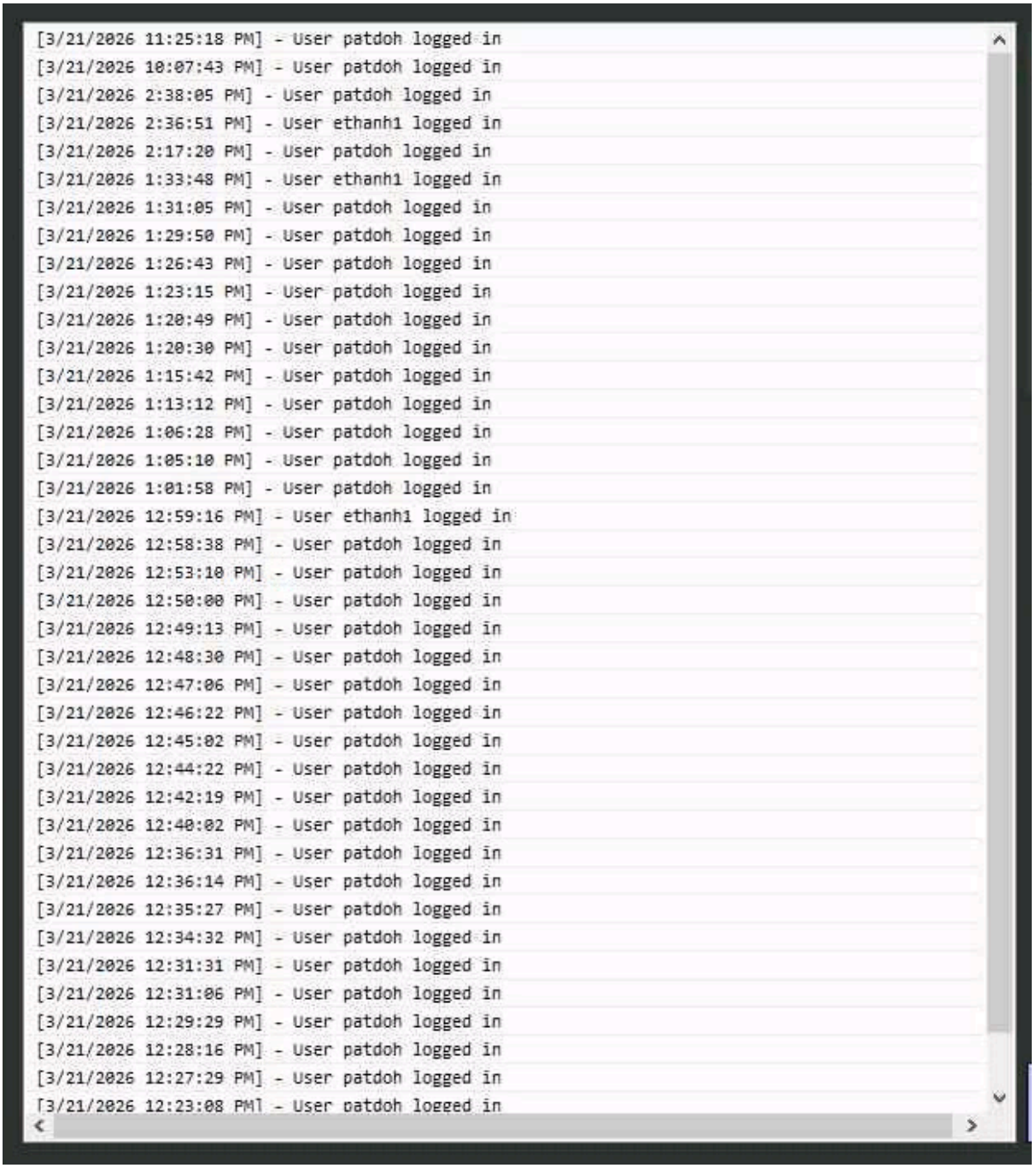
By default, the form displays the logs for the current day, but the log date can be selected using the DateTimePicker.





To further filter logs, the user can select the staff member associated with a log or the log type, both accommodated using ComboBoxes.





Logs can only be read to ensure integrity and accountability.

FAQ

1. **What does the password "hash" stored in the staff table mean?**
   A password hash is a one-way transformation of a password that lets the system verify it without ever storing the actual password, ensuring the security of the system
2. **What is a vehicle type's "complexity"?**
   A vehicle type's complexity is the scalar factor by which the duration of a service appointment is multiplied when a vehicle of that type is being serviced, aiming to ensure that more complex cars are given ample time to be serviced thoroughly.
3. **What is a role's "authorisation level"?**
   A role's authorisation level denotes the level of access which a user of that role has in the system. It does not denote which forms they can access - that is the job of the sector - and presently has 3 values: 0, representing read-only, where a user can only read and not modify record; 1, representing restricted write, where a user has permission to make changes to less crucial data on the system, but is restricted from performing more critical tasks; and 2, representing unrestricted write, where the user has unrestricted access to operations on the data present in the database.
4. **Why has _____ not been implemented yet?**
   Time.
5. **What do I do in the event of an error?**
   A number of help forms have been provided which aim to guide you if the event is a result of any behaviour that is unique to that form. If the help form does not help you, contact the administrator or a member of the IT department.
6. **How do I add on more mechanics / how do I modify the working hours for mechanics?**
   It is currently assumed that the number of 3 mechanics will remain the same in the foreseeable future, and likewise for working hours, and as such only editing the source code will allow you to modify the parameters of the appointment viewer. Note that the appointment viewer was designed to be dynamic, and so should respond to such change appropriately, but technical knowhow is still required to make these changes.
7. **Why are my changes not saving to the database?**
   Changes may not be saved to a database either because of the use of the "Debug" compilation option, or system error preventing data from being saved. Either way, more thorough investigation on your end will be required.
8. **What are the log types on the system?**
   At the moment, to reflect the current operations available to the user on the system, the log types available are unsuccessful login attempts (LoginAttempt), successful logins (Login), accessing forms (Access), creating new data (Create), updating existing data (Update), and deleting existing data (Delete.) These are not final, and may be amended in the future as more functionality is added.

Evaluation

Purpose of evaluation

By comprehensively evaluating my performance and the effectiveness of my decisions throughout the design, development, and testing phases, crucial feedback can be generated which can feed into future development cycles, thus optimising my performance in the future. Furthermore, I can explicitly demonstrate the suitability of the program for *Doherty Car's*, thus helping prove I met the requirements set out initially. Additionally, the quality of the final solution can be effectively assessed, highlighting key sections of the system that may require future modification or expansion in future development cycles. This evaluation will aim not only to identify my successes, but also my shortcomings in both my technical capabilities and personal effectiveness.

By critically analysing the design phase, my efficiency in identifying the problems faced by *Doherty Cars* and the effectiveness of my proposed solutions to each subproblem can be evaluated, allowing for informed determination as to the practicality and effectiveness of my initial approaches. Ultimately, this helps me identify weaknesses in my problem-solving processes. Similarly, determining whether the selected development approach and estimated timescale were effective and accurate allows for an informed determination of changes to planning that may have to be implemented in the future, ensuring I adhere to the most structured and efficient approach possible.

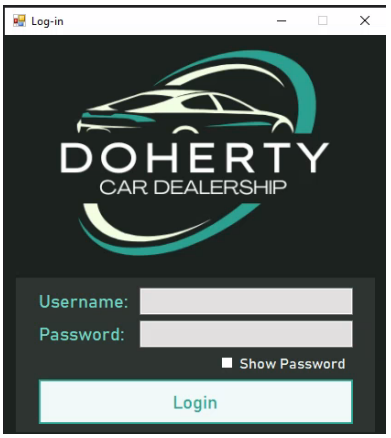
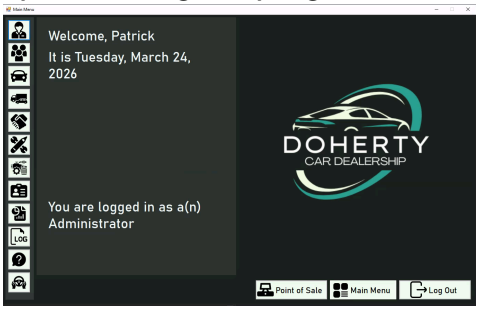
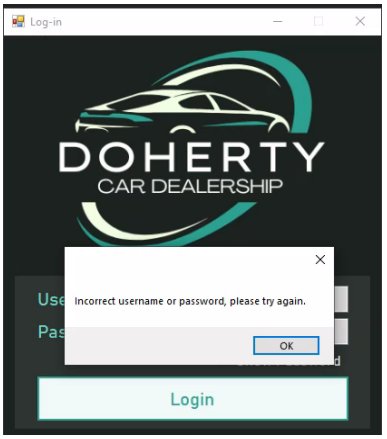
Similarly, evaluating the development process enables the identification of stages where my performance fell short or where user requirements were not fully met. This ensures that future iterations are executed with greater precision, efficiency, and adherence to user requirements.

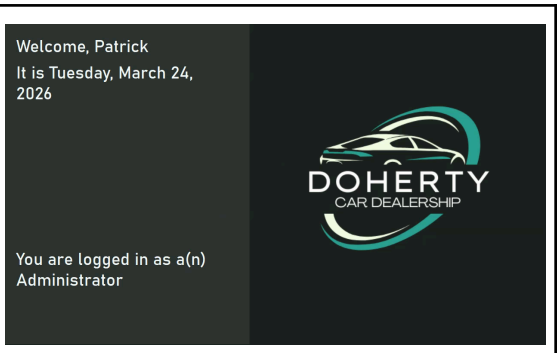
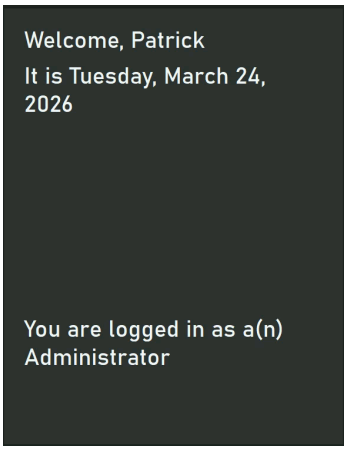
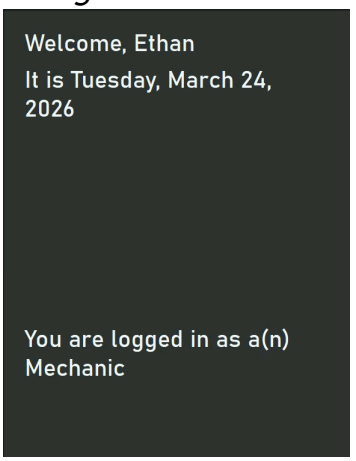
Lastly, a thorough review of the testing phase ensures that test cases are comprehensive, verifying that all potential execution paths were considered. This mitigates the risk of system crashes and inconsistencies, reinforcing the robustness of the final system. Furthermore, it ensures that my fact-finding techniques were effective in gathering feedback from the end users, preventing me from making invalid or incomplete conclusions in the future.

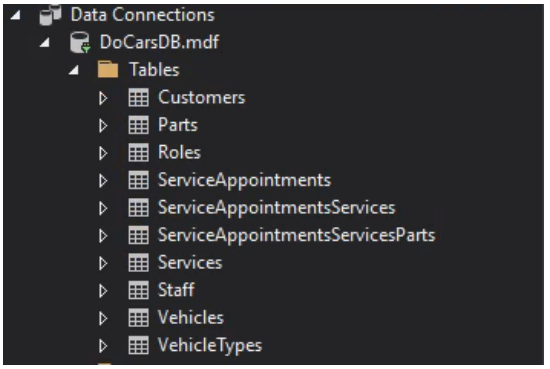
Ultimately, this iterative evaluation process not only strengthens the integrity of the program but also enhances my ability to deliver high-quality solutions in future projects.

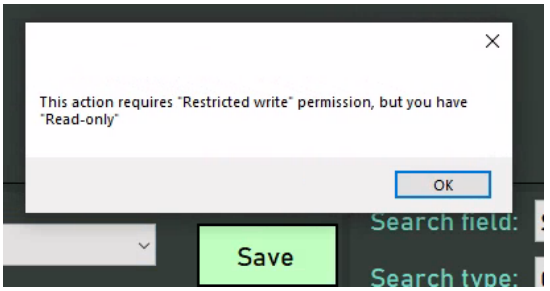
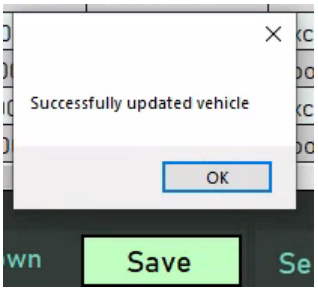
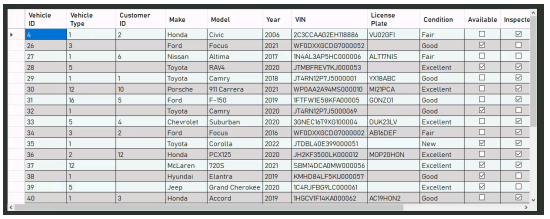
Meeting user requirements

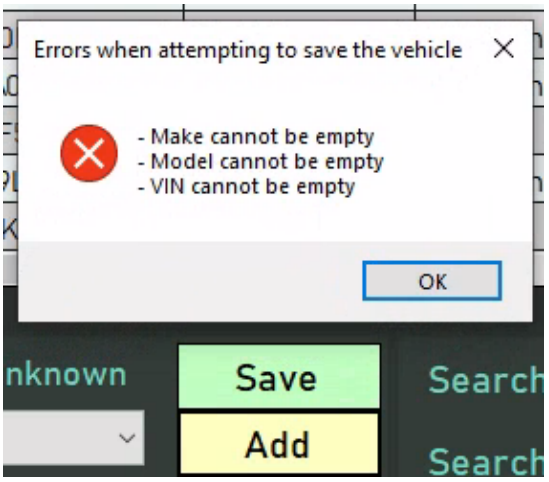
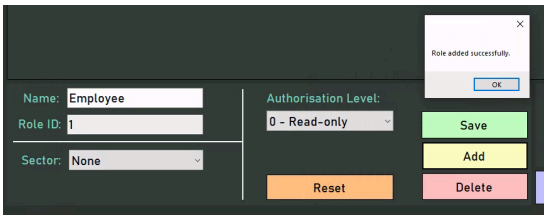
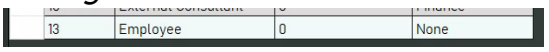
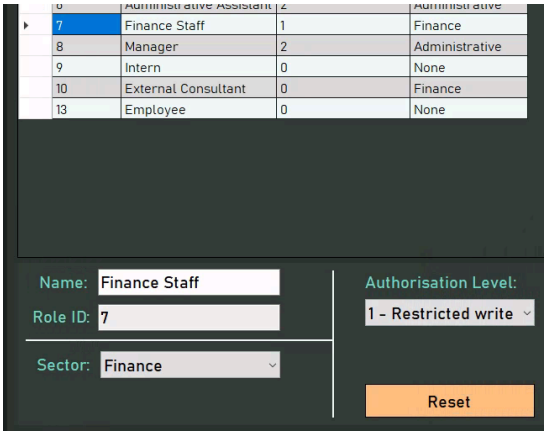
The following table evaluates the extent to which the finished system succeeds in fully meeting the original user requirements produced through analysis of the problems faced by Doherty Cars, thus ensuring that the final product is suitable for their needs and does not deviate from their initial expectations. Screenshots will be provided as evidence for each requirement.

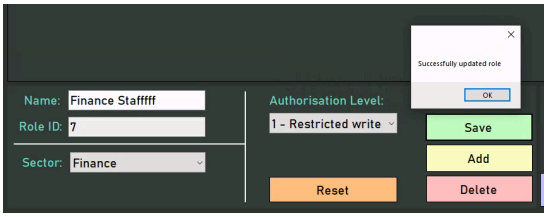
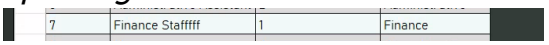
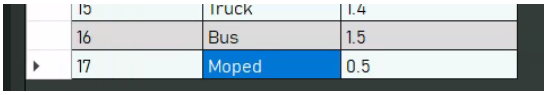
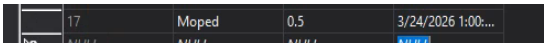
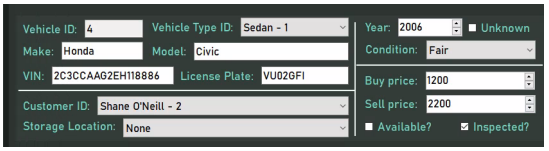
User requirement	Description	Screenshot evidence
1. …implement a login system that authenticates staff members using their username and password, preventing unauthorised system access	This requirement has been fully fulfilled through the implementation of frmLogin, which appears upon the program starting up. Staff members can only access the system if they input a valid pair of credentials - an existing username and a password that hashes to form the same hash associated with that username - and hit “Login”, otherwise they are left barred from the rest of the system. This ultimately ensures unauthorised access is prevented, as without knowing a valid set of credentials - and due to the inability to create another on startup - the only way access can be provided is if a user is registered onto the system by an administrator, as discussed in requirement 17.	 *Upon starting the program*  *On logging in with valid credentials*  *On an invalid login attempt*

2. …keep a record of the logged-in user in memory, enabling dynamic modification of the system and its functionality using their information	This requirement has been fully fulfilled through the implementation of the static CurrentUser class, which is used to store the object representation of the staff record containing the credentials that were used to log in. From here, these details are publicly accessible throughout the entire program. This is most evident in the frmStartingMainMenu form, where the class is used to display the current user's name, but it is also the driving force behind the logging, sector and authorisation logic of the program, as the logging functionality constantly retrieves the current user's username when writing a new log, and the authorisation and sector functionality both require access to the current user's role ID to determine permissions during runtime.	 *The current user's name and role displayed in frmStartingMainMenu*
3. …allow users to log out and log back in with different credentials	This requirement has been fully fulfilled through the provision of a "Log Out" button in frmMainMenu which is constantly visible to any logged-in user. Upon clicking it, the user is redirected back to the login screen, whereupon the details of the previous user are erased from memory to prevent any clashes with the new user about to log in. Thus, the usage of different credentials during the same session has been accommodated for.	 *1st login*  *2nd login*

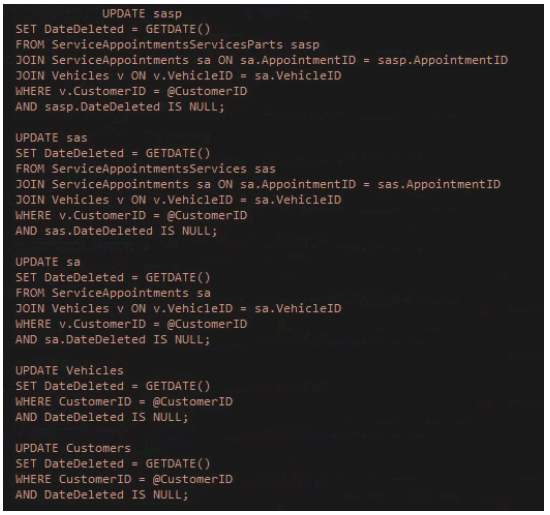
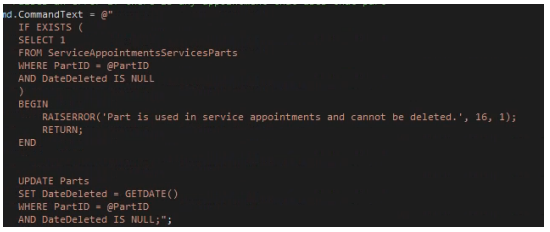
4. …assign access to certain forms based on the user's role, ensuring they only modify data that is relevant to them	This requirement has been fully implemented through the combination of the "Sector" property in the Role class and custom logic in frmMainMenu. On loading the form, the current user's role is fetched, and using this role their sector is fetched. By default, all buttons in the form are automatically disabled to prevent unauthorised access, and only through explicit association with a sector in a dictionary is a button re-enabled. These sector-to-button associations were established during the design phase, and ensure that each sector's accessible data is directly relevant to the operation of that sector.	 *The buttons seen in the main menu form, allowing for navigation through the system. Note the grayed out buttons, indicating the current user is not authorised to access them.*
5. …allow users to easily access relevant forms through the use of a centralised main menu form	This requirement has been fulfilled through the frmMainMenu form. With the exception of frmLogin, all forms within the system can be loaded into memory through the comprehensive selection of buttons provided. The form's centralised nature stems from the fact that these forms are displayed in the viewing panel provided directly within the main menu, with forms being disposed and instantiated dynamically as the user cycles through the buttons. Consequently, the user never has to leave the form to carry out any actions required, providing ease of use and ensuring that their train of thought is not interrupted when having to change between forms rapidly.	
6. …maintain records relevant to the business externally in a database, including tables for: a. Customers b. Staff c. Vehicles d. Vehicle types e. Parts f. Roles g. Services h. Service appointments	This requirement has been fully implemented through the use of a comprehensive relational database solution. Each of the entities relevant to the business has its own table (with the exception of service appointments, which has two additional tables to handle parts and services logic) which themselves contain fields representing all the crucial characteristics identified. The integrity of the data is maintained through the use of constraints applied to certain fields, most notably primary-foreign key relationships in staff, vehicles, and service appointments, and composite keys in all 3 service appointment tables.	 *The tables viewed in SQL server*

7. …allocate certain permissions - namely read, restricted write and unrestricted write - based on the user's role's authorisation level, determining the actions they can take on the system	This requirement has been fully implemented through the use of the authorisation level property within the Role class, with 0 representing read-only, 1 representing restricted write and 2 representing unrestricted write. Within each CRUD form, a static method is executed during each CRUD action which compares the authorisation level of the current user to the authorisation level required for that action. If the user is below the required level, they are informed of the level they require and are not permitted to execute that action. Otherwise, the form behaves as expected. Ultimately, this aims to reduce the absence of accountability present in the old systems that were implemented by Doherty Cars, as only trusted users are able to carry out the actions that plagued Mr Doherty in the past.	 *Attempting to save (which requires authorisation level 1 in this form) while the user has authorisation level 0*  *Otherwise*
8. …enable users to access pertinent records in the database by providing a form for each table and displaying it in a tabular, interactive format	This requirement has been implemented through the use of the frm_____Data forms (and to an extent, the frmServiceAppointments form with its appointment viewing table) which each contain a DataGridView control. On the loading of a data form, all the records present on the database in the table which the form is concerned with are fetched and processed to instantiate objects of the representative entity class. This list of objects is then set to be the data source of the DataGridView, which is automatically populated with the fields and records present in the list. Furthermore, the DataGridView dynamically updates upon the execution of a CRUD action, displaying the most recent version of the data present on the system, and allows for the selection of records by clicking on their respective row, feeding into the "interactive" aspect of the requirement.	 *Sample table seen in frmVehicleData. The rest of the frm____Data forms look similar to this.* RoleID | Name | Authorisation Level | Sector 1 | Employee | 0 | None 2 | Administrator | 2 | Administrative 3 | Sales Advisor | 2 | Customer Service 4 | Mechanic | 1 | Technical 5 | Lead Mechanic | 2 | Technical 6 | Administrative Assistant | 2 | Administrative 7 | Finance Staff | 1 | Finance 8 | Manager | 2 | Administrative 9 | Intern | 0 | None 10 | External Consultant | 0 | Finance *Same for frmRoleData.*

9. …validate any data input to the system before making changes to the database to ensure data is consistent, accurate and correct	This requirement has been implemented extensively throughout the program, with each entity class having its own validation method which ensures that the provided fields follow business logic as well as checks in its properties, preventing invalid objects from being created. Furthermore, many user controls have hard-coded limitations (such as NumericUpDown controls not taking values outside of the provided range.) Lastly, a separate “Validation” class has been implemented which holds commonly required validation methods, thus aiding in the validation process.	 *Sample validation error message in frmVehicleData*
10. …enable users to create new records in the database by inputting the data into the respective fields provided within the table’s form	This requirement has been implemented through the use of the frm_____Data forms’ and frmServiceAppointments’ “Add” buttons. Users are permitted to input the details they desire into the fields provided in the form, and - once they are satisfied - are able to press the “Add” button to attempt to add a record to the system. Validation will be carried out initially, but if the data passes this then a new entity object will be created in memory and used to add a new record to the appropriate table.	 *Adding a new record*  *The new record*
11. …allow users to easily select records that they wish to modify or delete, either through the use of controls or mouse interaction	This requirement has been implemented through the use of the frm_____Data forms’ DataGridView and frmServiceAppointments’ appointment viewer. In the data forms, selecting a record is simply a matter of clicking on the row for that specific record, which will automatically populate that record’s details in the provided fields. In frmServiceAppointments, the user must simply click on the timeblock representing the relevant appointment. In both cases, it is simply a matter of clicking on the desired record.	 *Clicking the record (in blue) has populated its details in the appropriate fields*

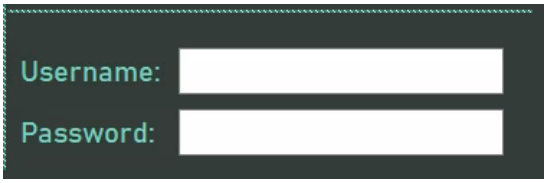
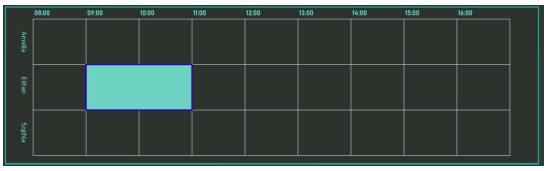
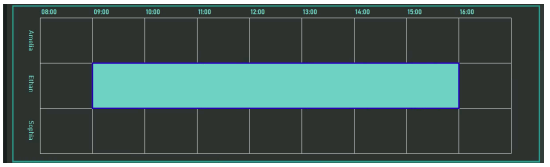
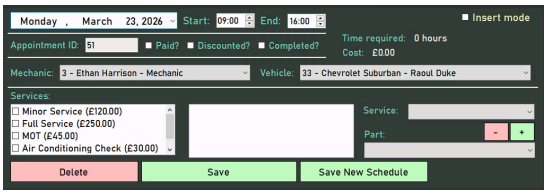
12. …enable users to update a selected record in the database by making changes to the data automatically populated into the form	This requirement has been implemented through the use of the frm_____Data forms' and frmServiceAppointments' "Save" buttons. Users are permitted to input the details they desire into the fields provided in the form, and - once they are satisfied - are able to press the "Save" button to attempt to save new details to the system. Validation will be carried out initially, but if the data passes this then the details are saved under the same ID as the selected record	 *Updating a record*  *The updated record*
13. …allow users to soft delete a selected record, ensuring that it is ignored by the system during data display and processing, but still exists on the database	This requirement has been implemented through the use of the frm_____Data forms' and frmServiceAppointments' "Delete" buttons. Upon selecting a record through the functionality provided in requirement 11, the user is simply able to press the "Delete" button which - if the action is valid - will set the DateDeleted flag for the selected record to the current date, thus ensuring that the data for that record still exists, but has been visibly labelled as redundant to the system.	 *Before clicking delete* *After clicking delete*  *The record still exists*
14. …clear a form's fields after a CRUD action to accommodate new data entry	This requirement has been implemented through the use of the frm_____Data forms' "Reset" buttons, and the frm_____Data forms' and frmServiceAppointments' LoadFormDefaults() methods. The user can deliberately clear the fields in a data form by pressing the "Reset" button, additionally unselecting any selected records, which aids in data entry by ensuring there is no conflict with populated data. Similarly, upon creating, updating or deleting a record in either the data or service appointments forms, the fields will be cleared and any records unselected to prevent any issues or conflicts with the previous state of the form when inputting new data.	 *Before updating the vehicle* *After updating the vehicle*

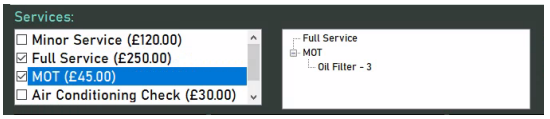
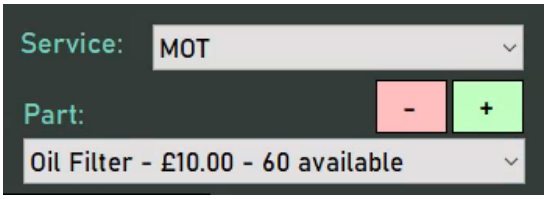
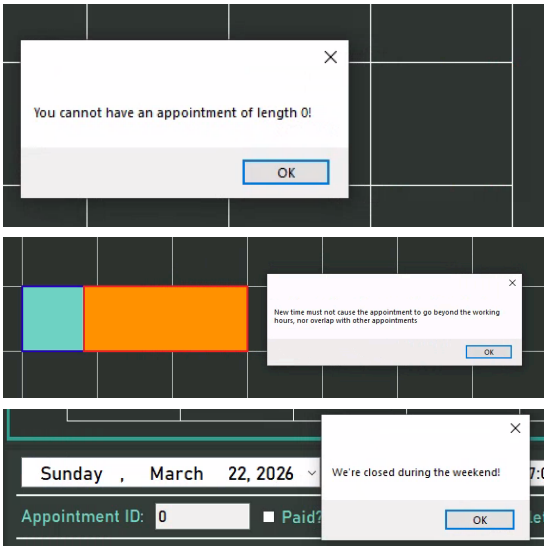
15. …prevent data entry upon a system error, ensuring data cannot become corrupted or lost due to a non-existent or broken database	This requirement has been implemented in virtually every form, as most forms which retrieve data from the database make explicit checks to ensure that valid data - or at the very least, no data for a valid reason - was received, thus preventing crashes as a result of Null Reference or SQL exceptions. This is most evident in the data forms, where checks are made when loading all records from a table to ensure that no errors occurred. If they did, then the entire form is locked down, with all search and CRUD functionality disabled, ultimately preventing further potential damage to the system through attempting to push new data to an already broken setup.	 *The code checking to see if a valid result was returned*

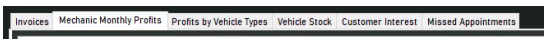
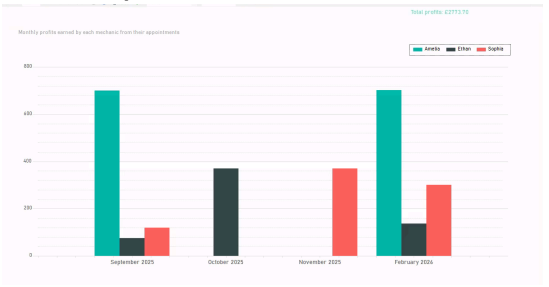
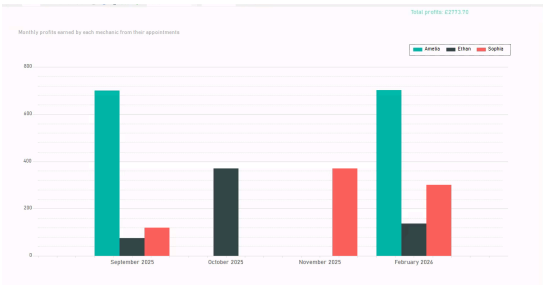
16. …properly handle entity relationships, ensuring referential integrity through foreign keys. In particular, when soft-deleting, the following rules will apply: a. Deleting a customer will delete any of their vehicles and service appointments related to those vehicles b. Deleting a vehicle will delete any of its service appointments c. Deleting a staff member will not be permitted while they have existing service appointments d. Deleting a vehicle type will not be permitted while its being used by existing vehicles e. Deleting a part will not be permitted while its being used in an existing service appointment f. Deleting a role will not be permitted while it applies to existing staff g. Deleting a service will not be permitted while its being used in an existing service appointment	This requirement has been fulfilled within each relevant entity's representative class. In the customer and vehicle classes, statements are combined in their respective delete methods which attempt to delete any associated records in other tables as well, thus ensuring that changes cascade through the system and that no references are maintained to non-existent data. In the rest of the classes, a check is made before attempting to delete a record to see if the relevant record has any associated records in other tables. If it does, an exception is thrown - which is handled by the method - with the error message that such an action would be invalid, and no further action is taken. Ultimately, referential integrity is adamantly maintained throughout the system to prevent any mishaps due to careless deletion.	 *The SQL for deleting a customer. Notice that vehicles, service appointments and satellite data are all processed when a customer is deleted*  *The SQL for deleting a part. Notice that an error is thrown if a part used for an appointment is attempted to be deleted.*

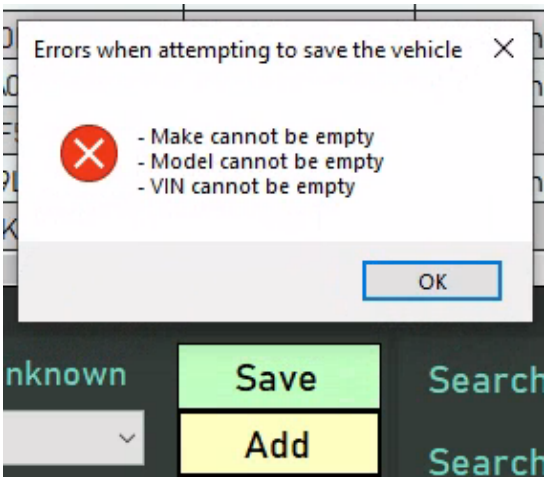
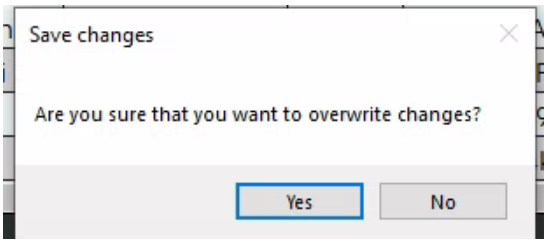
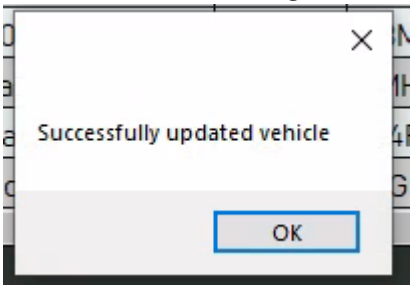
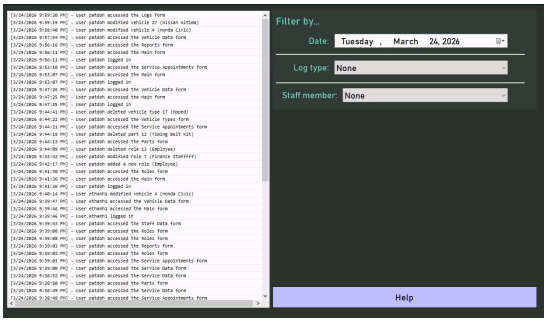
```
            UPDATE sasp
SET DateDeleted = GETDATE()
FROM ServiceAppointmentsServicesParts sasp
JOIN ServiceAppointments sa ON sa.AppointmentID = sasp.AppointmentID
JOIN Vehicles v ON v.VehicleID = sa.VehicleID
WHERE v.CustomerID = @CustomerID
AND sasp.DateDeleted IS NULL;

UPDATE sas
SET DateDeleted = GETDATE()
FROM ServiceAppointmentsServices sas
JOIN ServiceAppointments sa ON sa.AppointmentID = sas.AppointmentID
JOIN Vehicles v ON v.VehicleID = sa.VehicleID
WHERE v.CustomerID = @CustomerID
AND sas.DateDeleted IS NULL;

UPDATE sa
SET DateDeleted = GETDATE()
FROM ServiceAppointments sa
JOIN Vehicles v ON v.VehicleID = sa.VehicleID
WHERE v.CustomerID = @CustomerID
AND sa.DateDeleted IS NULL;

UPDATE Vehicles
SET DateDeleted = GETDATE()
WHERE CustomerID = @CustomerID
AND DateDeleted IS NULL;

UPDATE Customers
SET DateDeleted = GETDATE()
WHERE CustomerID = @CustomerID
AND DateDeleted IS NULL;
```

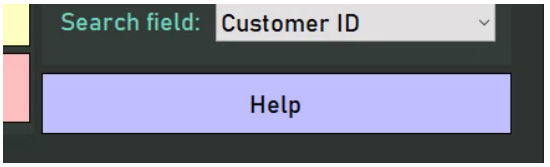
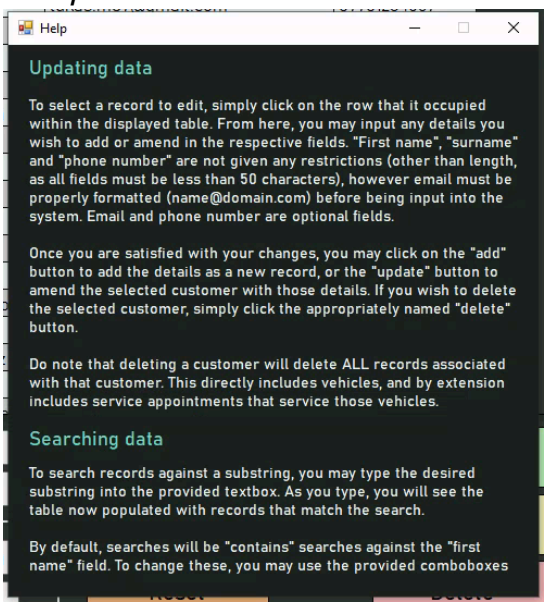
```
cmd.CommandText = @"
  IF EXISTS (
  SELECT 1
  FROM ServiceAppointmentsServicesParts
  WHERE PartID = @PartID
  AND DateDeleted IS NULL
  )
  BEGIN
      RAISERROR('Part is used in service appointments and cannot be deleted.', 16, 1);
      RETURN;
  END


  UPDATE Parts
  SET DateDeleted = GETDATE()
  WHERE PartID = @PartID
  AND DateDeleted IS NULL;";
```

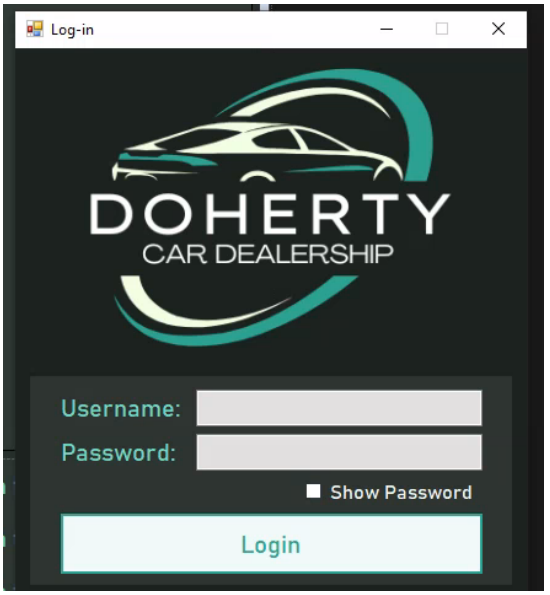
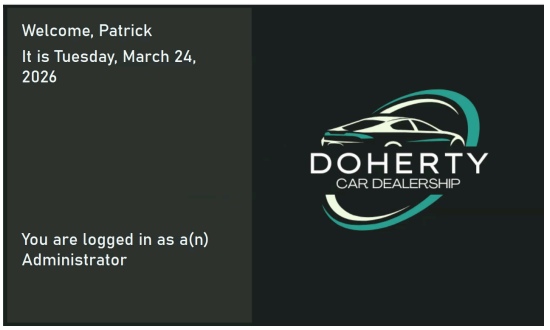
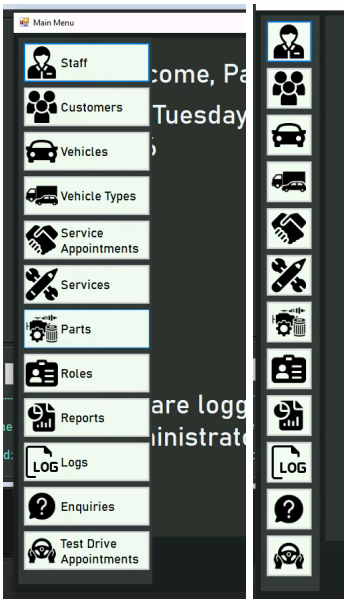
17. …ensure that new users added to the system by administrators can log in using the administrators' determined username and password	This has been implemented through the frmLogin form, Staff table and frmStaffData form. When a new user is added to the system, they are required to be provided a valid username and password, which is then pushed to the system alongside the new record to the Staff table. When the user attempts to login, the Staff table is referenced, thus forming a connection between the CRUD and login behaviour of the system. Important to note is the "administrator" part of the requirement, as sufficiently high authorisation level - and a relatively unrestricted sector - is required before adding new users	 *The fields provided*
18. …allow mechanics to easily view, modify, create and delete appointments using a drag and drop system, displaying each day's available and taken timeslots	This requirement has been fully fulfilled through the use of the custom control provided in frmServiceAppointments. Within it, each day is presented as a timetable, with a row for each mechanic, allowing for easy viewing of the appointments scheduled for each day. From here, appointments can be easily selected by clicking on their representative timeblocks, populating the provided fields with the appointment data. The user is then permitted to edit the fields as required, and use the "Save" button to save the new appointment details to the database. If they wish to delete the selected appointment instead, they may use the "Delete" button. Similarly, a user may use the provided timetable in "Insert Mode" and click on an empty space to create a new appointment which can then be populated with details. The drag-and-drop aspect of the requirement is fulfilled through the control's ability to dynamically modify times when a timeblock is dragged on, simplifying the appointment modification process by abstracting appointment times. This all culminates in a suitable solution to the appointment booking problems initially laid out by Doherty Cars, providing a sleek, intuitive solution to an otherwise convoluted scheduling problem.	 *The viewer*  *After dragging the appointment*  *The fields provided*

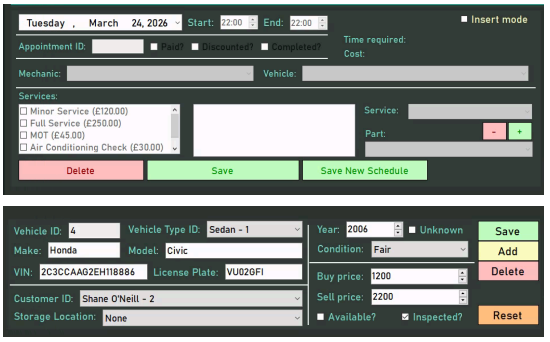
19. …enable mechanics to select the services and parts used for each appointment	This requirement has been fully fulfilled through the CheckedListBox and ComboBox provided in frmServiceAppointments. Each service present on the system is provided as an option in the CheckedListBox. To add a service on to a selected appointment, the mechanic may double click the desired appointment, and the same applies for removing services. Once a service has been added, it is added as an option in the ComboBox provided for adding parts, wherein the mechanic can choose a service they wish to add parts to and the part they wish to add, and use the provided “Plus” and “Minus” buttons to change quantities. When the appointment is saved, these service and service-part relationships are stored in the appropriate tables, helping handle an otherwise complex process.	 *Service selection*  *Part selection*
20. …ensure that all appointments created are valid i.e. don’t conflict with other appointments, fit within one work day, and don’t occur on a weekend	This requirement has been fully fulfilled through extensive validation in the frmServiceAppointments provided control. Many checks are implemented to ensure that dragging and manually setting times does not allow the user to cause an appointment to overlap with another, cause it to go out of the bounds of the day or cause it to become 0 hours long. Similar checks are implemented when changing the mechanic that an appointment belongs to. Additionally, it is impossible for the user to set the appointment date to be a weekend. All of these checks culminate into a robust scheduling solution for the mechanics, ensuring none of the confusion of the previous paper-based system remains.	 *A range of errors*

21. …generate reports and invoices using data stored on the database to analyse the business' performance and assist with customer service	This requirement has been fully fulfilled in the frmReports form, where each report has been provided a separate tab. Each report fetches data from the database and combines and processes it differently, providing unique insights into the business' operation. Each report is dynamic, as updating the data on the database and reopening the report in the same session results in an altered report. For the sake of brevity (as the report design has already been specified in the design phase), specific details of reports will be omitted, but many reports are provided for the outlined purposes, with vehicle and appointment profit reports for business performance, and customer vehicle interest and missed appointments for customer service.	 *All the reports available*  *A sample report*
22. …automatically determine an appointment's length and cost based on the services carried out, parts used and the type of vehicle serviced	This requirement has been fully fulfilled through the use of the provided labels in frmServiceAppointments and the reports provided. In frmServiceAppointments, as a user adds on services and parts, their time and cost are added on in memory and this is then reflected in the provided labels, ensuring that mechanics can stick within a certain time or budget. Similarly, the reports provided automatically calculate the cost of an appointment using its parts and services, allowing management to effectively determine the productivity of mechanics and the most profitable services. Ultimately, both implementations of this requirement contribute to the efficient operation of the business.	 *As seen in the service appointments form*  *As seen in the reports form*

23. …capture any error messages resulting from invalid data or system errors and display them to the user so they can respond accordingly	This requirement has been fully fulfilled, with all CRUD forms and the login form all containing validation checks to ensure that the data input follows the rules set beforehand - outputting an appropriate message if not - and try-catch statement outputting exception messages in the event that any action executes unsuccessfully. This allows the user to amend any erroneous data or pinpoint an issue in the system, contributing to the usability of the system.	 *Sample validation error message in frmVehicleData*
24. …provide relevant feedback through confirmation and success messages to provide certainty	This requirement has been fully fulfilled, most evident in CRUD forms where confirmation messages are displayed for update and delete actions, and success messages are displayed for create, update and delete actions, ultimately ensuring that the user is certain that their action was executed successfully and contributing to the ease of use of the system.	 *Confirmation message*  *Success message*
25. …log user actions externally with timestamps, activity type and user, ensuring that no critical actions can be taken on the system without accountability	This requirement has been fulfilled within each form through the use of a Logging class which provides a method to write logs to an external logs.txt file. Particularly, LoginAttempt and Login logs are captured in frmLogin, Access logs in frmMainMenu, and Create, Update and Delete logs in all data forms and frmServiceAppointments. Each log also contains the timestamp of when it was recorded and the username of the user who carried out the action, preventing actions on the system from being carried out without accountability.	 *The logs form. Note the list of logs on the left, and the filter options on the right*

26. …offer help forms to ensure that new users of the system are able to navigate and make use of it effectively, mitigating any risk of error and thus data loss	This requirement has been fulfilled within each form (except frmLogin, frmMainMenu and frmStartingMainMenuForm) through the use of a dedicated button. Upon clicking the button, a help form appears containing text pertaining to the usage of the form from which it originated, with a focus on the intricacies of the CRUD functionalities in data forms, thus ensuring that anyone unfamiliar with the system can quickly understand how to use the system and any limitations set in place, preventing data loss through invalid CRUD attempts.	 *A help button*  *A help form*
27. …hash passwords before storing them to ensure the security of the system	This requirement has been fulfilled in frmStaffData, as the plain text passwords provided are run through a hashing algorithm before they are stored in the “Password” field of the database. Consequently, users are able to freely log in, as their password is hashed using the same process and compared to the same hash to authenticate them, but any malicious actors are unable to do anything with the hash, thus ensuring the security of the system.	 *Passwords as stored on the system*

28. …implement an intuitive and visually appealing interface to maximise staff efficiency	This requirement has been fulfilled, as clearly labelled fields and buttons are present throughout the system, helping even a technologically illiterate audience navigate the system, and many important functionalities are colour-coded (e.g. red for delete, green for save, yellow for add.) This ultimately contributes to the aesthetic appeal of the program by making it seem less bland and the efficiency aspect of the requirement by ensuring staff members quickly locate their desired functionality. Furthermore, the presence of the company's logo in the login form and starting main menu form (as well as the dynamic nature of the starting form) both serve to "bulk out" the program and make it seem more sophisticated.	 *Login form*  *Starting main menu form*
29. …ensure users can easily navigate between their required forms, and that navigation is fluid	The requirement is fulfilled through the use of the expanding panel in frmMainMenu, which uses the sine function to fluidly expand to reveal the purpose of each button, making navigation significantly more pleasant. Furthermore, the purpose of each button is explicitly denoted, making locating the desired form easier and ultimately contributing to the fluid nature of the system's navigation.	 *Expanded and unexpanded buttons*

30. …implement icons where applicable to graphically denote the purpose of each section and aid in navigation	The requirement is fulfilled through the use of icons in frmMainMenu to represent the function of each button, allowing users to easily locate the button which they require by looking at the abstract representation of each form's functionality, and consequently making the system easier to navigate and less visually basic.	 *The icons seen*
31. …have designs consistent throughout the program. In particular, all menus will: follow a similar colour scheme implement the same fonts and and similar font sizes throughout	This requirement has been fulfilled throughout the program through the use of a consistent teal colour scheme - a colour which is both aesthetically pleasing and calming - with all elements of the system adhering to it, and the use of Bahnschrift (size 11) font throughout most of the program, with both aspects only having slight variations where required for clarity. Ultimately, this serves to create a consistent and easily navigable system for the user, aiding in its use.	 *As can be seen, controls are consistently coloured and have identical fonts throughout*
32. …respond to user input within acceptable time frames (i.e less than 2 seconds)	This requirement has been fulfilled throughout the program through the use of efficient methods and logic, as only frmReports regularly takes longer than half a second to load fully - most due to the inefficacies of the Microsoft Report Viewer - but still falls under the two second limit, meaning the system is sufficiently fast for a regular user.	*No screenshots can be provided for this*

Future development

Although it has been evidenced that I met the requirements set out during the project initiation phase - and thus the scope and operation of the program is as anticipated - this is not to say that the project can be considered to be in a fully completed state, with many problem areas initially identified by Doherty Cars still to be addressed. With this in mind, the program was developed to be extensible in the future, particularly with the presence of buttons for features not yet present and the modularity of the appointment viewer. Given more time (and with a potential development cycle in the future) many further issues faced by the business can be mitigated.

One of the most obvious sections of the system that can be properly addressed in future iterations of development is vehicle stock management. Although the present system is satisfactory in storing the details of vehicles as outlined in the problem specification, much of the data regarding the lifecycles of these vehicles relies on manual input and is not sustainable as the number of vehicles in storage increases with the business' growth. As a result, rather than a tabular form, a more abstract representation of vehicle stock could have been created, with the business able to create visual top-down models of garages and storage spaces within which they are able to identify parking spaces. Each of these parking spaces could have been colour-coded depending on the status of the space or the vehicle on the space, such as "empty," "checking," "logging" or "preparing," consequently allowing staff members to briefly glance at a vehicle and quickly comprehend the stage of its lifecycle it is at. This would also mitigate the issue of sold vehicles (i.e., those no longer in the company's storage) being displayed alongside unsold vehicles as in the current system.

From here, each vehicle could be edited by clicking on its parking space, and vehicles could be searched by supplying the desired vehicle's details, with the matching vehicle automatically highlighted or - if the vehicle is in a different storage location - that location chosen. Furthermore, supplier information could have been stored in the database in a separate table, allowing for management to easily assess the quality and reliability of cars sourced from certain suppliers. Similarly, important documentation such as warranty and delivery information and supplier invoices can be embedded alongside every vehicle - with viewers for these documents embedded directly into the system - allowing for easy bookkeeping, which prevents the confusion caused by the scattered maintenance of these documents with the current system implemented.

This proposed system would work for both used and new cars - perhaps with identifiers for each such as a different parking space shape - with additional statuses for used cars (e.g. “valuating.”) To address Mr Doherty’s concern with vehicles remaining in storage for longer than commercially viable, prompts could be implemented which force users accessing the system to decide the course of action regarding these vehicles once there is a certain amount of time between the purchase of a vehicle and the current date. This proposed solution would prevent the administrative issues currently involved in vehicle management by Doherty Cars, ultimately contributing to the efficiency of the business’ operations. Furthermore, it would slot into the point of sale system also to be developed, ensuring that sales are only carried out with valid vehicles and at the most appropriate price.

Another section of the system that could be derived from existing implementation is test drive bookings. Much like service appointments, test drive bookings are not allowed to intersect with each other, nor are vehicles allowed to be booked if they are currently under maintenance. As such, the custom appointment viewer control can be modified for test drive appointments, allowing the user to select the vehicle for which they wish to view and edit appointments. Using existing service appointments and test drive bookings, vehicle availability could be determined by finding times in which neither occur for the selected vehicles. The fields provided for this implementation would be similar to those for service appointments, allowing the user to select the customer record, staff record and the validation status for the booking. Dynamic calculation of booking costs can be executed using the duration of the bookings, and bookings can be linked across days to ensure extended bookings - such as those across weekends - are accurately calculated. All of these features would culminate in increased user satisfaction and reduced staff confusion, as validation and cross-table checks would prevent clashes caused by the current paper calendar system.

Lastly, an enquiries system could be implemented, drawing from the existing functionalities of the customer, vehicle and service appointment tables. An “inbox” system could be created, where - upon receiving communications from an existing or potential customer - users would be able to categorise messages by customer or enquiry type. This would allow users to filter by enquiries relevant to their business sector and - once an enquiry has been followed through with a sale or appointment depending on the enquiry type - associate them with records from other tables. Ultimately, by creating a centralised inbox system, the disorganised management system would be alleviated, ensuring that customers are treated to a prompt and professional customer service experience, which would drive sales. To further ensure that enquiries are followed up, a “reply” system could be implemented which ensures that staff members can respond to enquiries from within the system, automatically marking the enquiry as “in progress” or “completed” and thus preventing multiple staff members from responding to the same enquiry, increasing the business’ appearance of professionalism. Similarly, the integration of the enquiries system with the existing customer table would ensure that returning customers are appropriately marked, allowing these valued clients to be treated to appropriate discounts and thus aiding the business’ reputation for reliable customer service.

As evidenced by these proposed solutions, many of the solutions would simply integrate into or be derived from existing functionality in the system, thus ensuring that the issues of the current system are addressed in a uniform and homogenous manner, aiding in the business' effective operation. Given more time, the current hindering paper-based system could be fully phased out, but at the present only sections of the system most affected by it have been addressed.

Strengths and weaknesses of the system

Feature / Functionality	Strengths	Weaknesses	Potential improvements
Login system	➢ By accessing user details stored in the database in a reliable and robust manner, it provides a mechanism through which users access the system, allowing them to utilise the specific sections of the program authorised for their sector ➢ By only allowing users with unique usernames to access the system, it introduces a way to for the logging system to identify the users behind specific actions, accommodating the accountability desired by Mr Doherty ➢ Implements extensive validation and error-trapping, ensuring that the login processes operate smoothly, and that valid user data is saved	➢ The passwords in textboxes are stored as plain text in memory ➢ Users are not given a way to graphically represent themselves, such as through a profile picture ➢ The password hashing method uses PBKDF2 with a static iteration count of 75623. This is somewhat arbitrary and not updated dynamically, thus leading to potential weakness in regards to brute force attacks	➢ Use a more computationally expensive hashing protocol, such as Argon2, to prevent brute-force attacks even with specialised equipment such as GPUs ➢ Store passwords using C#'s built-in SecureString class, ensuring passwords are removed from memory as soon as possible ➢ Allow users to upload a custom image to represent themselves, serving as an avatar throughout the system
Navigation	➢ Every button within the system is distinctly labelled, indicating its purpose and thus, in the event of a button that opens another form, its destination. This ensures that users can easily traverse the system, able to trivially find their desired form ➢ Every form is optimised to ensure that loading occurs promptly upon opening, ensuring the	➢ The loading of frmReports is often slower than the rest of the forms on the system due to the inefficacies of the ReportViewer package, and as a result the user may be unsatisfied with the response times when attempting to view these reports	➢ Load logic could be implemented in frmReports which ensures that rather than loading all reports simultaneously, only those that have their tab opened are to have their data fetched and displayed. This would ultimately prevent the unnecessary loading of data not relevant to

	system is pleasant to use by minimising wait times. ➢ The hover-to-expand logic in the main menu ensures that the users can easily access buttons when they wish to change forms, but that the buttons do not take up unnecessary space in the form when they are not being used, thus providing the user with a minimalistic experience		the current user and optimise the form's performance
Service appointments	➢ Exhaustive validation concerning the movement and manipulation of timeblocks ensures that all changes made to the appointments result in a valid state, consequently preventing the confusion and clashes of the previous system in place. ➢ The drag-and-drop nature of the form ensures that even less technically proficient users are able to make changes to appointment times, as much of the time logic is now abstracted behind the timeblock representations provided ➢ The modular design of the appointment viewer control - with much of the logic (particularly regarding rendering, business hours and mechanics) depending on constant values that can be changed within the source code to accommodate changes to the circumstances of the business. As a result, minimal time will be spent on maintenance, as changes to the	➢ Currently, although the timeblocks make it easy to understand the timings of an appointment, they leave a lot to be desired with regards to the details of the appointment such as vehicle, services, completion status etc, requiring these details to be populated into the fields provided before the user can view them. ➢ Currently, not as much feedback is provided to the user as would be desired. For example, when adding enough services to go over the time allocated to the appointment, the user only becomes aware of this when they attempt to add or save the appointment. Similarly, no visual feedback is provided when a user clicks on an unauthorised timeblock.	➢ Rather than a constant teal colour scheme, the appointments could be colour-coded to indicate their status. For example, a paid appointment could be golden, completed green, and missed red, allowing the user to quickly gauge which appointments are important ➢ In a similar vein to the above, appointments could dynamically change colour in response to invalid actions. For example, clicking on a colleagues' appointment could trigger it to flash red if unauthorised, or an appointment could go through a gradient depending on whether the time allocated to an appointment is comfortable, just on the verge or insufficient.

	control must be made only in one place.		
Logging	➢ By storing the username associated with an interaction with the system, it is relatively easy to identify who the perpetrating user is when a malicious action is executed ➢ By storing the time of a log and allowing filtering by date, the precise log can be easily located by reverse engineering the times between which a malicious action occurred and checking between those	➢ Logs are stored in a plain text file externally to the system, meaning even if the system does not provide the means for editing logs, a sufficiently persistent user could delete logs from the text file and cover up their tracks	➢ Store logs in an encrypted file, thus ensuring that even if a user opens the log file externally, they are not able to identify the required log and delete it. Their only resort would be to delete all logs on the system, a relatively easy action to trace on a workstation.
Help forms	➢ Due to the ubiquitous nature of the help forms, it is easy for users to find the information concerning the specific form they require assistance with. This minimises user frustration by ensuring that they are able to quickly learn the intricacies of the system	➢ Although much comprehensive information is provided concerning the various aspects of the system, the help forms are notably lacking in images. As a result, the help functionality appears rather bland and verbose, making it difficult for the user to effectively find the information they desire	➢ Images could be embedded for the CRUD form or, for the service appointments form, animated GIFs to show the usage of the system, thus ensuring that users can gain a quick overview of the system rather than having to parse through large chunks of text
Reports	➢ The invoices report is interactive through the use of a ComboBox of service appointments, simplifying the billing and bookkeeping process by displaying all pertinent details for an appointment in one centralised document ➢ Most reports provide a graph/chart which allows for a comprehensive overview of the operations of the business, allowing management to easily assess the performance of the business and the profitability of certain executive decisions	➢ Most reports are rather brief in nature, providing a general overview of the entire operations of the business.	➢ Provide search criteria for the reports, allowing users to select the specific time periods, records etc. they wish to include in the report, thus allowing for more precise analysis

Database	➢ Through the extensive use of constraints, the integrity of the data is ensured. For example, foreign key relationships are enforced within most tables provided, ensuring that referential integrity is maintained and that the system does not break when attempting to refer to a non-existent record. Similarly, many of the field length guidelines are hard-coded into the table, ensuring that all data provided is reasonable and appropriate. ➢ As users are not able to access the CRUD forms - and thus the interface to their related tables - without proper authorisation or sector, the security of the data on the database is maintained, ensuring that malicious or naive actors are not able to view or modify restricted data. ➢ As the database is in 3NF, data redundancy is minimised, thus reducing the storage space occupied by the database and preventing an invalid setup through the presence of repeating groups, particularly with services and service parts in appointments	➢ Though measures have been implemented to ensure that unintentional or malicious manipulation of the database is impossible through the provided system, the database is still present locally on the workstation, meaning the records could easily be edited and deleted when opened in an application such as SQL Server ➢ Similarly, the data on the database is not encrypted. If someone were to open the database externally, all the data would be in plain text, compromising the system's security ➢ Some of the fields in the database have predetermined values (such as the "Sector" field in the Roles table,) and these would most likely be better off as foreign keys from a separate table as to ensure that errors when inputting the hard-coded values do not compromise the robustness of the system	➢ Use a cloud-based service to store the database online, thus not only ensuring that users on a workstation could access the database just through the file, but also that staff members could access the system remotely, aiding in productivity. ➢ Encrypt the data input into the database using a suitable encryption method such as Argon2 to prevent unauthorised access ➢ Implement additional tables, such as a new "Sectors" table, to ensure that there is no reliance on strings as keys
CRUD logic	➢ A plethora of try-catch statements and checks are implemented when interacting with the database to ensure that any system errors are gracefully handled and that the source of any errors is promptly located.	➢ When reading data from tables which contains foreign keys, the keys are displayed as they are found on the table, making it difficult to visualise the data associated with a particular record	➢ A custom class implementation for each entity specifically used for displaying in a DataGridView could be created, containing additional fields for information about foreign records not

	➢ Extensive validation is provided when attempting to create or update records to ensure it adheres both to the rules set out by the database and the business, ensuring the integrity of the data provided ➢ Where necessary, custom delete logic has been implemented which ensures that where deletes cascade, the appropriate records are manipulated using the same foreign key, and where they are forbidden, checks are made using the foreign key and errors are thrown in the event of an invalid action ➢ When reading records from the database, a ComboBox has been provided in each form which allows the user to view information regarding the foreign record(s) associated with the selected record and change the record associated if necessary, thus maintaining referential integrity in a simple, easy-to-use way.	➢ The search functionality leaves a lot to be desired, with only 2 search modes provided and only search by substring available.	found in the entity's table, which would then be instantiated when fetching records from a table and would consequently be displayed, helping the user visualise the data associated with each record ➢ Additional search modes could be introduced, such as "Starts with", "Ends with" and "Does not contain." Furthermore, searching by text length could be implemented
Design	➢ The system is visually consistent; the same font and similar font colours are used throughout, and the same colour scheme of teal, with accent colours of white, is implemented in every form. This ensures that a professional atmosphere is maintained throughout, and that using the system is as pleasant for the user as possible, improving retention	➢ Little is implemented beyond the essentials for the system to operate, thus making the system seem rather rudimentary and dull	➢ Tastefully add picture boxes in appropriate places. This can include a picture box for the user's avatar in the main menu form, a logo in the login and main menu forms, etc.

Design process

During the design process, many of the tasks executed served as crucial to the overall completeness and usability of the system. During the initial client meeting and discussion of problems, the precise and encapsulated selection of issues to address ultimately helped ensure that the scope of the program was kept manageable, preventing me from experiencing burnout from “crunching” to attempt to deliver features too grandiose for the provided time, or from leaving the client unsatisfied through the project not meeting the outlined requirements. From these selected issues, the 32 user requirements generated served to effectively steer the direction of development and ultimately allow me to generate a tentative timeline and scope, which ensured effective communication as to the progress of the project to *Doherty Cars*. From these issues, a comprehensive list of user requirements was generated, which provided a general notion of what was expected to be delivered.

Though the project management tools were novel to me - particularly the use of PERT charts - they helped me outline the exact tasks required for the project to be fully completed, as well as task dependencies, which assisted in the prioritisation of critical project sections. This helped me ensure that the project adhered to the project timeline, consequently reducing the pressure involved in developing the system by providing a sense of direction. This indirectly helped ensure the quality of the project by allowing me to focus solely on the details of implementation rather than the more general structure of the system. If not for these tools, I do not believe I would have been able to adhere to the time constraints provided by *Doherty Cars*.

The use of both the initial storyboards and the final, computerised storyboards proved essential in translating the requirements set out by the business into a working system. The hand-drawn sketches allowed for my interpretation of the system to be depicted before attempting to start building the system, and thus our ideas to be consolidated by discussing these sketches, allowing both parties to pinpoint potential challenges or opportunities that might not have been apparent from the outset. Consequently, the storyboarding process facilitated a clearer mutual understanding, reducing the risk of miscommunication and saving time that might otherwise have been spent revising the system due to misunderstandings.

Using the user's feedback on the initial, hand-drawn sketches, more accurate computerised sketches of the forms were created with the feedback's criticisms in mind. These were detailed enough to be fit for implementation, identifying the final controls which were to be present in each form, and a general annotation regarding their purpose. This made the start of the building process substantially more convenient, as it was a mere case of translating the design into Windows forms. Furthermore, it made programming the system more tangible, as the general purpose and functionality of each control was already identified, and thus the most optimal way to fit each method together could be created prospectively, reducing the need for retrospective modification of the system. This effect is further emphasised by the validation identified and pseudocode written for each control in each form: not only was a general notion of the form's functionality fostered, but the fundamental structure of each method was predetermined, allowing for the main concern when building the system to be writing the code, as opposed to developing the solution. Additionally, identifying the structure of the database - including specifics such as constraints and data types - before building the solution ensured that the solution was accurately designed to retrieve and process the data required to effectively manipulate the entities relevant to the business.

One part of the design process, which may be deemed excessive, is the use of UML diagrams such as sequence and activity diagrams, which outline the behaviour and structure of the system in an abstract, top-down manner. Once again, these assisted in translating the design into a working solution, but a significant amount of time was spent constructing diagrams which could have otherwise been spent building the solution. If another methodology such as Agile had been chosen, these may have been crucial in receiving user feedback between iterations, helping guide the direction of the system, but the Waterfall methodology used meant that these diagrams were left ineffective when implementing the system at a low level. Therefore, it has been identified that textual descriptions of the behaviour of the system would have served equally effectively, meaning development time could have been reduced.

Despite this criticism, time management was not a major concern during the design process, especially considering the benefits of executing it; more time was saved programming with a concrete idea in mind rather than blindly throwing code together in an attempt to create a solution than was lost creating the initial design. Therefore, I believe that the design process was ultimately successful.

Implementation process

Given the novelty of the development area when compared to my previous projects, the initial building process may have not occurred as smoothly as I had anticipated, with much of the CRUD functionality's intricacies having to be identified during development. However, as mentioned in the evaluation of the design process, the sketches produced were easily translatable into Windows forms, helping me focus on grappling with aspects I was less comfortable with. Much of the CRUD functionality was rather simple to implement - most notably the creation and update logic - but the reading logic required for more complex tables (i.e. those with primary-foreign key mechanisms involved) and reports involving joins, as well as the delete logic (preventing invalid deletes or ensuring deletes cascaded) require more extensive planning and implementation due to the less logical and more descriptive nature of SQL. As for the design of the system, the colour scheme and structure of the system were identified preemptively in the design process, meaning little thought had to be put into laying out and appropriately executing the teal colour scheme seen in the final product.

Furthermore, the final quality of my code was acceptable. During the process in which I identified the specific implementation for each aspect of the forms, what would be deemed "spaghetti" code was created in an attempt to design an initial working solution upon which I could build. This may have caused some time loss, as additional time was spent refactoring the code in order to make it more modular, and this could've been prevented by reversing my programming process, rather than writing comments to explain code to the user, I should've written code to explain comments to the computer, thus ensuring that structure of the solution was identified before programming it. In a similar vein, the names of some variables and the explanations of the purpose of certain lines of code in the solution leave a lot to be desired, and could've been improved by planning before developing the solution. An attempt was made to ensure that all methods and public properties were written in PascalCase, and all private properties in _camelCase, but there may be sections of the program where this convention was abandoned when focusing on the implementation of the system.

However, I am satisfied with the robustness of the code; various try-catch and if statements are included throughout the program - even where they may be deemed edge cases - to ensure that the program does not crash as a result of an invalid operation. Consequently, users may feel comfortable using the system with the knowledge that their actions are unlikely to cause any unintended interruptions, especially to the database system. Furthermore, the modular, encapsulated nature of the code - with each form calling the CRUD and validation logic of the representative entity classes - ensures that maintenance is easy, as any desired changes must be made only in one place to be put in action throughout the form, and test cases are easily identified, as test data can be developed with one specific aspect of implementation in mind. Additionally, the code was separated into regions, helping with the readability and navigation of large files.

Once again, the novel nature of SQL to me may have slightly stalled me during the implementation process, but I am ultimately satisfied with the reliability and robustness of the queries produced, particularly in the effective handling of database errors, ensuring that the system does not crash as a result of an invalid operation. However, one aspect notably missing from the final implementation is the use of transactions when executing multiple statements at once - for example, in the saving and adding of new service appointments. Though preliminary checks ensure the database is operational before the CRUD form for service appointments is ever loaded, errors can still occur during runtime and - if occurring during these SQL statements - may result in partially complete records in the database, reducing the integrity of the data. However, this will have to be fixed in the next development iteration.

One notable absence in my solution is the absence of any images, islands or additional “decorative” functionality throughout the program, which may cause the program to appear bland and uninteresting. However, I believe that the business-oriented purpose of the program does not call for this, and this criticism may be overlooked considering the program operates efficiently and effectively.

```
// stores the id of the appointment that was inserted into the ServiceAppointments table
int LastInsertedId;
string AppointmentQuery = @"
    INSERT INTO ServiceAppointments
    (VehicleID, StaffID, AppointmentStart, AppointmentEnd, Paid, Completed, Discounted)
    VALUES
    (@VehicleID, @StaffID, @AppointmentStart, @AppointmentEnd, @Paid, @Completed, @Discounted);

    SELECT SCOPE_IDENTITY();";

string AppointmentServicesQuery = @"
    INSERT INTO ServiceAppointmentsServices
    (AppointmentID, ServiceID)
    VALUES
    (@AppointmentID, @ServiceID);";

string AppointmentServicesPartsQuery = @"
    INSERT INTO ServiceAppointmentsServicesParts
    (AppointmentID, ServiceID, PartID, Quantity)
    VALUES
    (@AppointmentID, @ServiceID, @PartID, @Quantity);"; ;
```

An example of a method that would require transactions. Each statement is executed sequentially, so if one does not execute correctly then the data from the previous statements will still remain.

Evaluation of chosen methodology

As outlined in the planning stage, the Waterfall model was followed throughout development. Though seen as rudimentary in the professional development world, this sequential and structured approach offered by it proved it well-suited to the project's constraints when compared to other methodologies, ensuring that time was managed effectively and that scope was controlled to prevent excessive workload or delays.

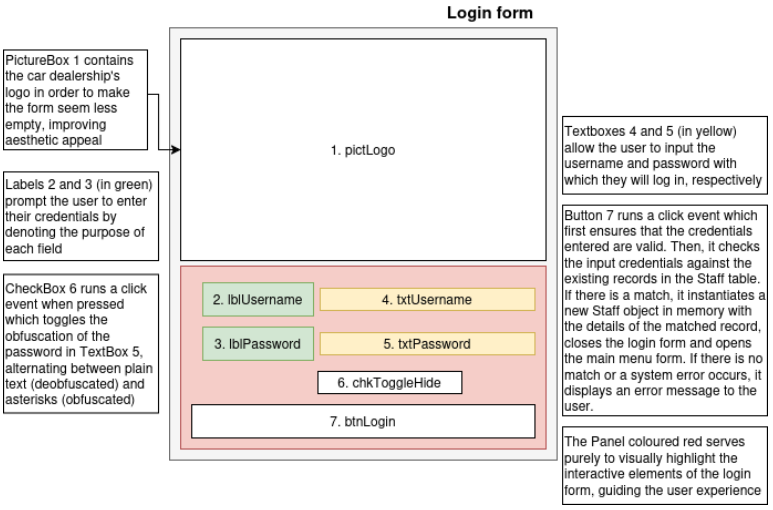
A key advantage of the Waterfall model is its linear progression: upon completing a section of the project, there was no need to return to it, allowing full focus on a single task. This contrasts with RAD (specifically with iterative prototyping), where multiple sections of the system can be developed, tested and evaluated simultaneously to provide feedback. While this may have helped with defining user requirements if they were changing, it would have greatly increased complexity due to the need to schedule client meetings, plan the next course of action and accommodate any changes to the project plan. Specifically, the final nature of the "requirements" phase in Waterfall prevented this - once the requirements were initially declared, they were fixed for the project's duration, ensuring that scope creep was avoided and that the requirements to be implemented were reasonable for the provided time and resources.

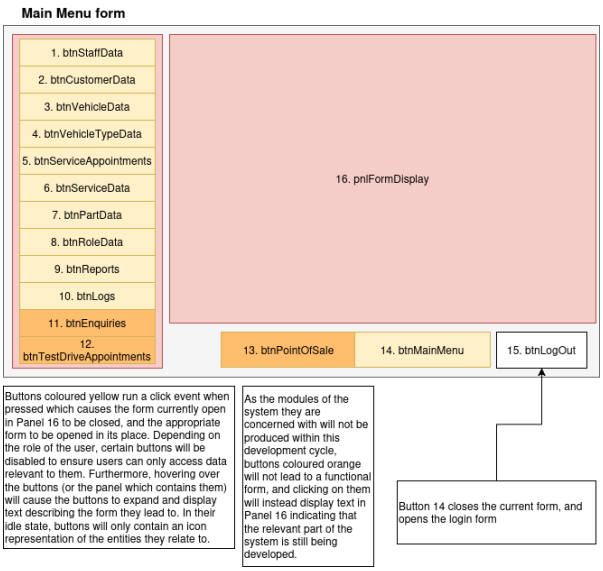
Another benefit to the Waterfall model was its approach to implementation: while the system's development was split into phases (as evident on the Gantt chart provided), once one of these batches was completed, there was no need to return to it, leaving a working subset of the system that could be integrated with future batches. This contrasts starkly with Agile (specifically its subsets of XP, DSDM and Scrum) where the focus is on features as opposed to sections of the system, taking a more quantity-over-quality approach. Although a functional system covering all bases could have been developed early on, there would have been a requirement to constantly return to parts of the system already partially developed - which would have required more documentation to ensure that the progress and objectives in each part were clear - and ensure that any additional functionality did not cause any conflicts with the systems already in place. All of this would have culminated in a hindrance in an already sufficiently complex project.

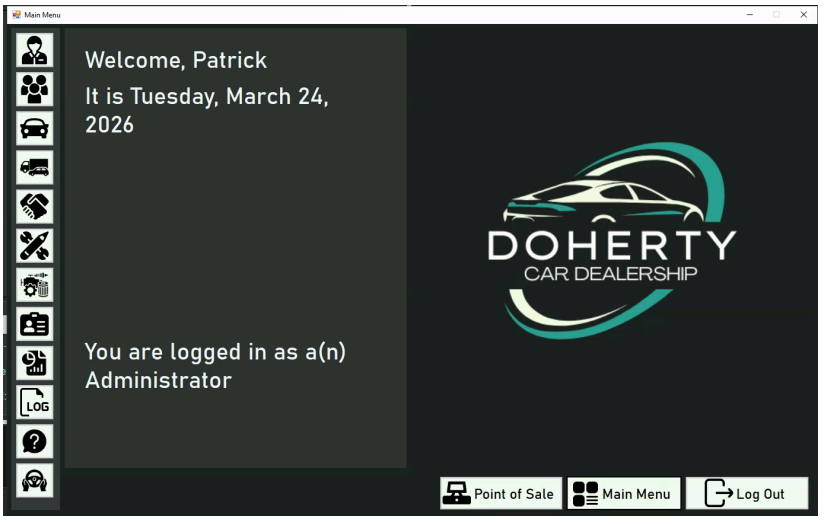
However, one thing that I found to be unsatisfactory regarding the Waterfall model was its approach to testing. Unlike RAD and Agile, where testing is carried out throughout the program's development, Waterfall executes all of its testing on the supposedly completed system at the end. Consequently, if any severe problems - particularly structural - were to be found at the end of the development lifecycle, this would have required a complete redesign and rework, causing a significant delay in the project's timescale. Luckily, this did not occur, but must be kept in mind in future projects.

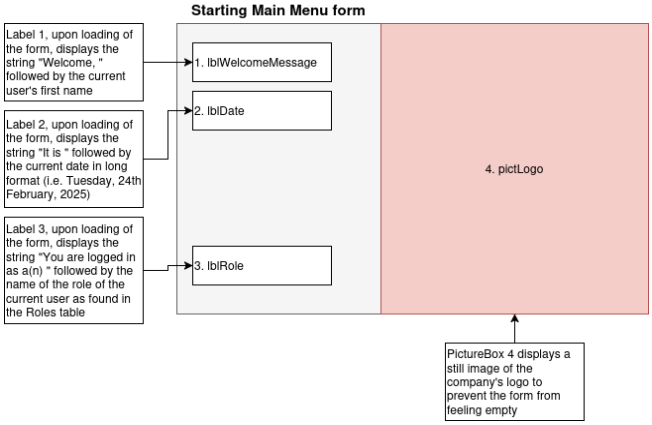
Comparison between final design and final system

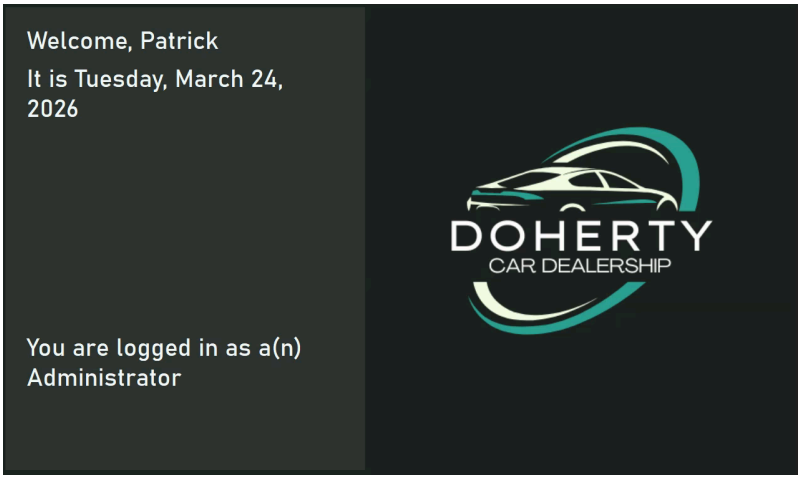
As evidenced by the screenshots below, the final system only minorly deviated from the final designs produced during the design phase of the project, highlighting their success in driving the development of the project and their effectiveness in providing a base from which the developer can work off of. Where forms deviated, it is only in minor features such as button placement or text.

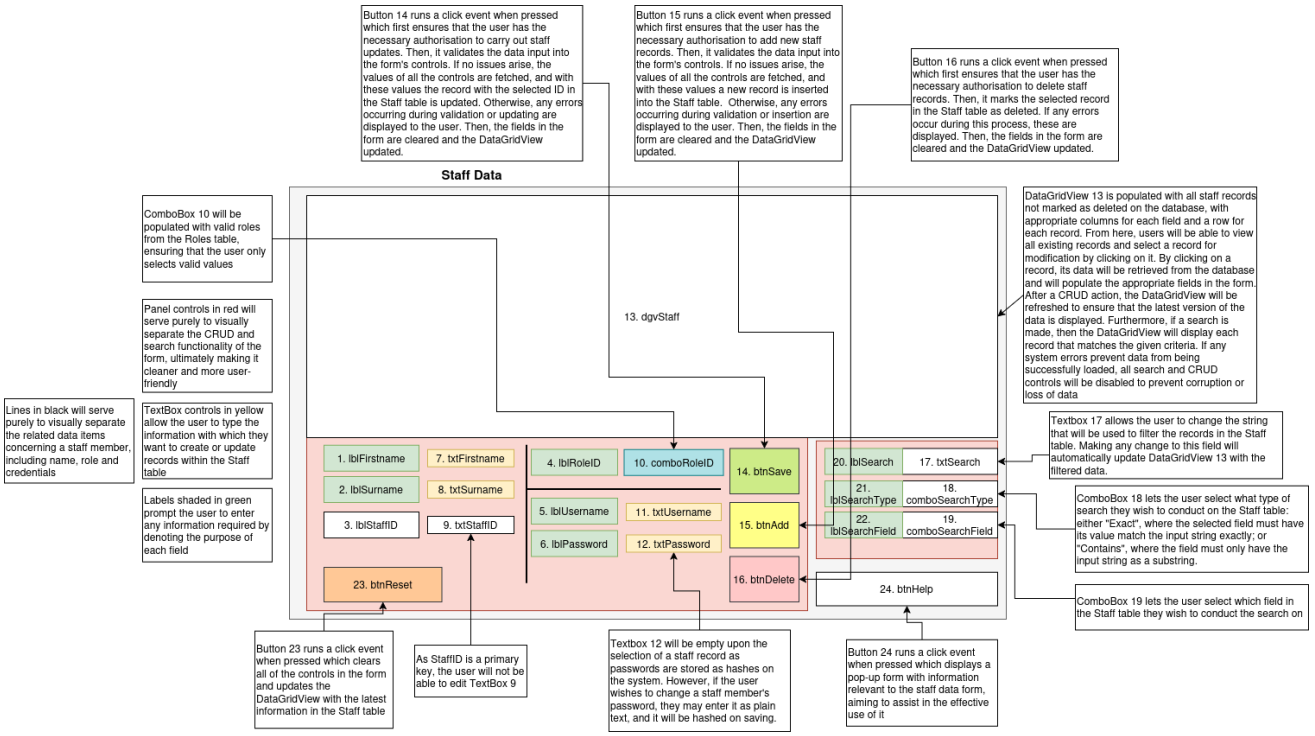


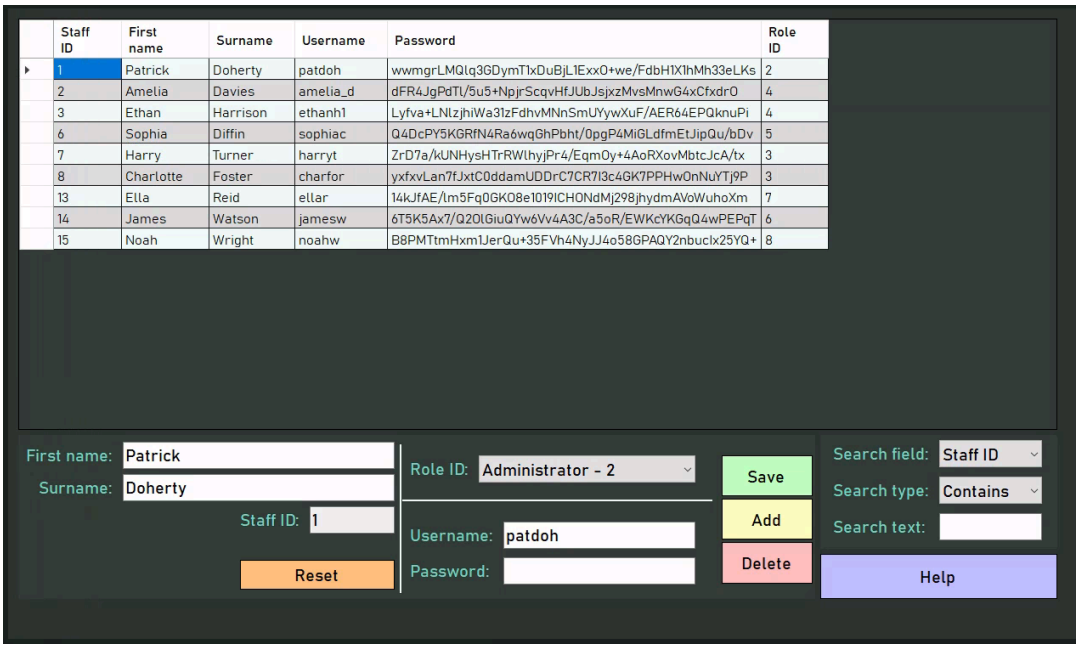


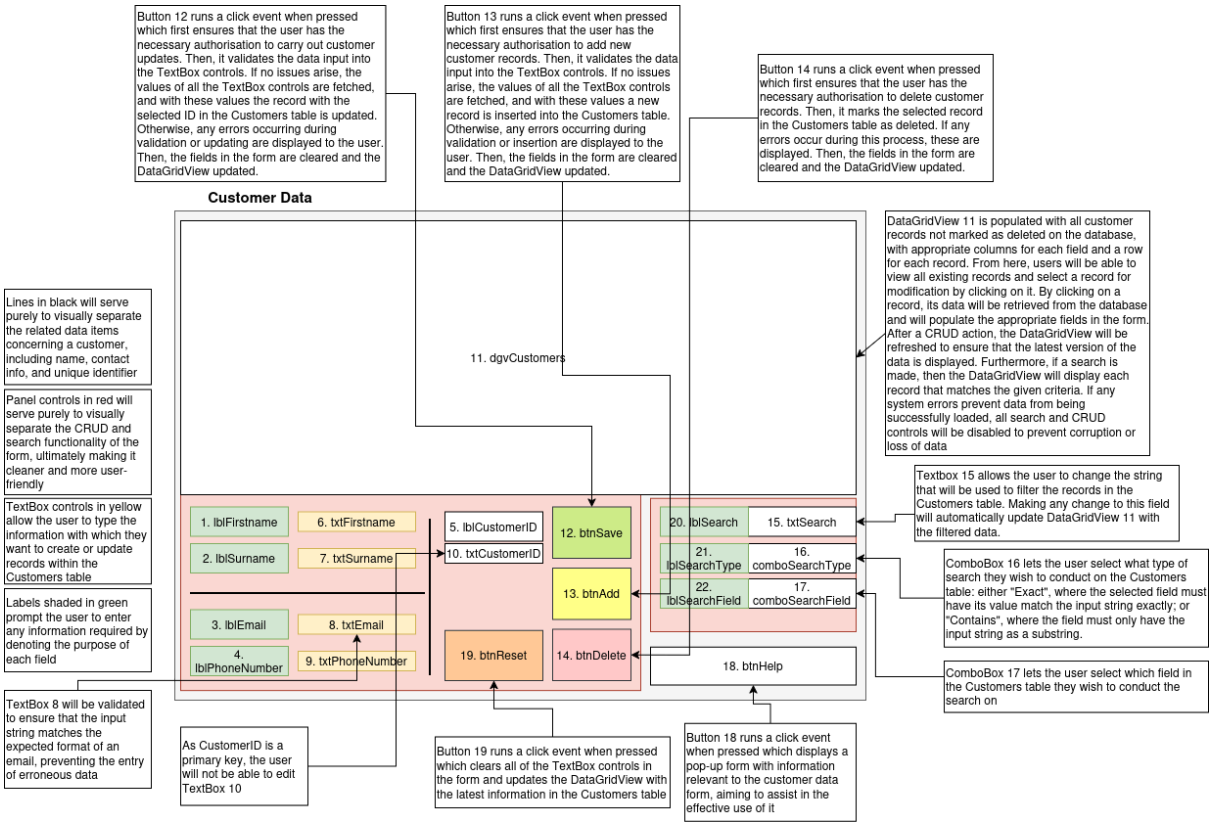



Starting Main Menu form
Label 1, upon loading of the form, displays the string "Welcome, " followed by the current user's first name
Label 2, upon loading of the form, displays the string "It is " followed by the current date in long format (i.e. Tuesday, 24th February, 2025)
Label 3, upon loading of the form, displays the string "You are logged in as a(n) " followed by the name of the role of the current user as found in the Roles table
1. lblWelcomeMessage
2. lblDate
3. lblRole
4. pictLogo
PictureBox 4 displays a still image of the company's logo to prevent the form from feeling empty


Welcome, Patrick
It is Tuesday, March 24, 2026
You are logged in as a(n) Administrator
DOHERTY
CAR DEALERSHIP

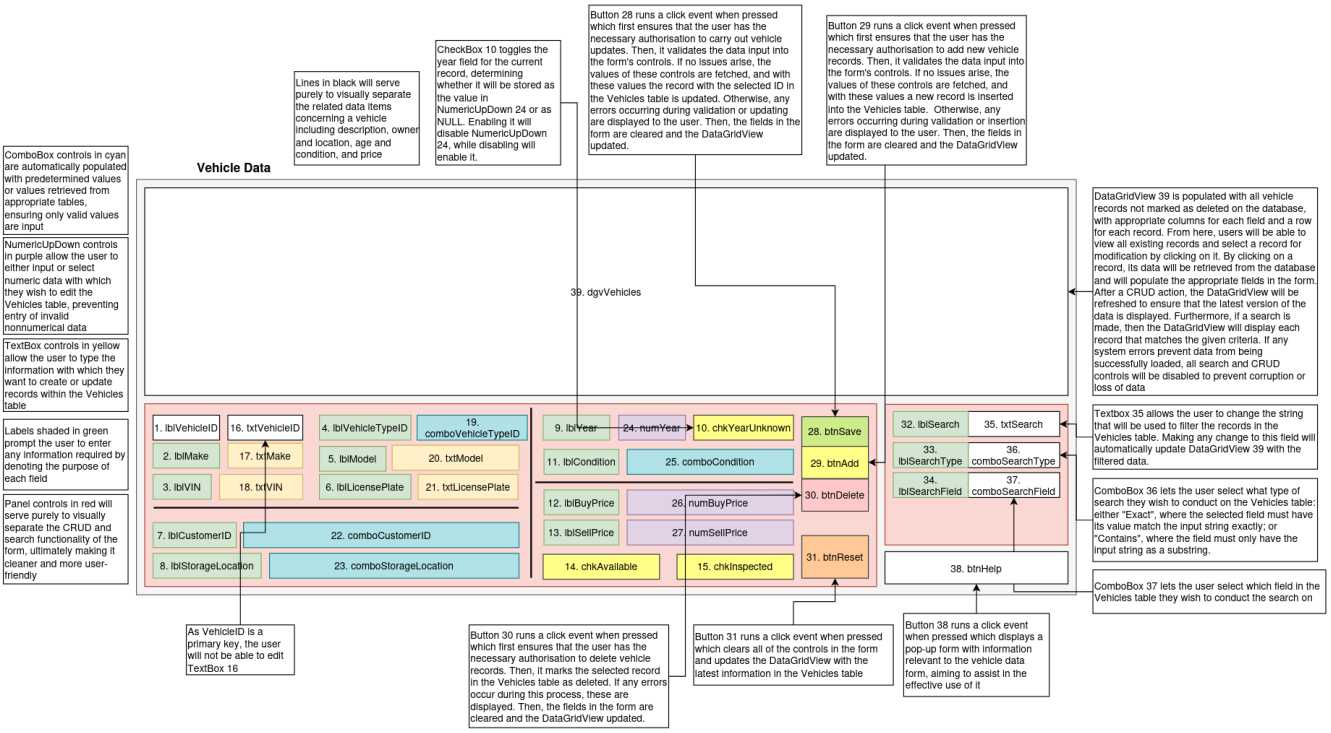

Button 14 runs a click event when pressed which first ensures that the user has the necessary authorisation to carry out staff updates. Then, it validates the data input into the form's controls. If no issues arise, the values of all the controls are fetched, and with these values the record with the selected ID in the Staff table is updated. Otherwise, any errors occurring during validation or updating are displayed to the user. Then, the fields in the form are cleared and the DataGridView updated.
Button 15 runs a click event when pressed which first ensures that the user has the necessary authorisation to add new staff records. Then, it validates the data input into the form's controls. If no issues arise, the values of all the controls are fetched, and with these values a new record is inserted into the Staff table. Otherwise, any errors occurring during validation or insertion are displayed to the user. Then, the fields in the form are cleared and the DataGridView updated.
Button 16 runs a click event when pressed which first ensures that the user has the necessary authorisation to delete staff records. Then, it marks the selected record in the Staff table as deleted. If any errors occur during this process, these are displayed. Then, the fields in the form are cleared and the DataGridView updated.
Staff Data
ComboBox 10 will be populated with valid roles from the Roles table, ensuring that the user only selects valid values
Panel controls in red will serve purely to visually separate the CRUD and search functionality of the form, ultimately making it cleaner and more user-friendly
Lines in black will serve purely to visually separate the related data items concerning a staff member, including name, role and credentials
TextBox controls in yellow allow the user to type the information with which they want to create or update records within the Staff table
Labels shaded in green prompt the user to enter any information required by denoting the purpose of each field
13. dgvStaff
DataGridView 13 is populated with all staff records not marked as deleted on the database, with appropriate columns for each field and a row for each record. From here, users will be able to view all existing records and select a record for modification by clicking on it. By clicking on a record, its data will be retrieved from the database and will populate the appropriate fields in the form. After a CRUD action, the DataGridView will be refreshed to ensure that the latest version of the data is displayed. Furthermore, if a search is made, then the DataGridView will display each record that matches the given criteria. If any system errors prevent data from being successfully loaded, all search and CRUD controls will be disabled to prevent corruption or loss of data
1. lblFirstname
7. txtFirstname
2. lblSurname
8. txtSurname
3. lblStaffID
9. txtStaffID
4. lblRoleID
10. comboRoleID
5. lblUsername
11. txtUsername
6. lblPassword
12. txtPassword
14. btnSave
15. btnAdd
16. btnDelete
23. btnReset
20. lblSearch
17. txtSearch
21. lblSearchType
18. comboSearchType
22. lblSearchField
19. comboSearchField
24. btnHelp
Textbox 17 allows the user to change the string that will be used to filter the records in the Staff table. Making any change to this field will automatically update DataGridView 13 with the filtered data.
ComboBox 18 lets the user select what type of search they wish to conduct on the Staff table: either "Exact", where the selected field must have its value match the input string exactly; or "Contains", where the field must only have the input string as a substring.
ComboBox 19 lets the user select which field in the Staff table they wish to conduct the search on
Button 23 runs a click event when pressed which clears all of the controls in the form and updates the DataGridView with the latest information in the Staff table
As StaffID is a primary key, the user will not be able to edit TextBox 9
Textbox 12 will be empty upon the selection of a staff record as passwords are stored as hashes on the system. However, if the user wishes to change a staff member's password, they may enter it as plain text, and it will be hashed on saving.
Button 24 runs a click event when pressed which displays a pop-up form with information relevant to the staff data form, aiming to assist in the effective use of it


Staff ID
First name
Surname
Username
Password
Role ID
1 Patrick Doherty patdoh wwmgrLMQlq3GDymT1xDuBjL1Exx0+we/FdbH1X1hMh33eLKs 2
2 Amelia Davies amelia_d dFR4JgPdTl/5u5+NpjrScqvHfJUbJsjxzMvsMnwG4xCfxdr0 4
3 Ethan Harrison ethanh1 Lyfva+LNlzjhiWa31zFdhvMNnSmUYywXuF/AER64EPQknuPi 4
6 Sophia Diffin sophiac Q4DcPY5KGRfN4Ra6wqGhPbht/0pgP4MiGLdfmEtJipQu/bDv 5
7 Harry Turner harryt ZrD7a/kUNHysHTrRWlhyjPr4/EqmOy+4AoRXovMbtcJcA/tx 3
8 Charlotte Foster charfor yxfxvLan7fJxtC0ddamUDDrC7CR7l3c4GK7PPHw0nNuYTj9P 3
13 Ella Reid ellar 14kJfAE/lm5Fq0GKO8e1019ICHONdMj298jhydmAVoWuhoXm 7
14 James Watson jamesw 6T5K5Ax7/Q2OlGiuQYw6Vv4A3C/a5oR/EWKcYKGqQ4wPEPqT 6
15 Noah Wright noahw B8PMTtmHxm1JerQu+35FVh4NyJJ4o58GPAQY2nbuclx25YQ+ 8
First name: Patrick
Surname: Doherty
Staff ID: 1
Reset
Role ID: Administrator - 2
Username: patdoh
Password:
Save
Add
Delete
Search field: Staff ID
Search type: Contains
Search text:
Help



Customer ID	First Name	Last Name	Email Address	Phone Number
1	John	Yossarian	catch22@pianosa.com	07221984421
2	Shane	O'Neill	igbro@greennet.com	07435633115
3	Lukas	Miller	lukas.m87@gmail.com	07701234567
4	Raoul	Duke	fear.n.loathing@vegas.com	07802345678
5	Dr.	Gonzo	dr.gonzo@samoa.net	07903456789
6	Winston	Smith	winston.smith1984@bigbrother.com	
7	Noah	Johnson	noah.johnson01@mail.com	07404567890
8	Ava	Roberts		07305678901
9	Lucas	Evans	lucasevans77@outlook.com	07506789012
10	Mia	Davies	miah.davies@gmail.com	07707890123
11	Liam	White		
12	Isabella	Harris	bella.harris_3@yahoo.com	07808901234
13	James	Martin	james.martin88@gmail.com	07909012345
14	Charlotte	Thompson	charlotte.thompson@icloud.com	
15	Benjamin	Garcia	b.garcia23@hotmail.com	07410123456
16	Amelia	Martinez	amelia.m_92@gmail.com	07311234567
17	Alexander	Clark		07712345678
18	Ella	Rodriguez	ella.r.93@yahoo.co.uk	

First name: John
Surname: Yossarian
Email: catch22@pianosa.com
Phone no.: 07221984421
Customer ID: 1
Reset
Save
Add
Delete
Search text:
Search type: Contains
Search field: Customer ID
Help



Vehicle ID	Vehicle Type	Customer ID	Make	Model	Year	VIN	License Plate	Condition	Available	Inspecte
4	1	2	Honda	Civic	2006	2C3CCAAG2EH118886	VU02GFI	Fair	☐	☑
26	3		Ford	Focus	2021	WF0DXXGCDG7000052		Good	☑	☐
27	1	6	Nissan	Altima	2017	1N4AL3AP5HC000006	ALT17NIS	Fair	☐	☐
28	5		Toyota	RAV4	2020	JTMBFREV7KJ000053		Excellent	☑	☑
29	1	1	Toyota	Camry	2018	JT4RN12P7J5000001	YX18ABC	Good	☐	☑
30	12	10	Porsche	911 Carrera	2021	WP0AA2A94MS000010	MI21PCA	Excellent	☐	☑
31	16	5	Ford	F-150	2019	1FTFW1E58KFA00005	GONZO1	Good	☐	☑
32	1		Toyota	Camry	2020	JT4RN12P7J5000069		Good	☑	☐
33	5	4	Chevrolet	Suburban	2020	3GNEC16T9XG100004	DUK23LV	Excellent	☐	☑
34	3	2	Ford	Focus	2016	WF0DXXGCDG7000002	AB16DEF	Fair	☐	☑
35	1		Toyota	Corolla	2022	JTDBL40E399000051		New	☑	☑
36	2	12	Honda	PCX125	2020	JH2KF3500LK000012	MOP20HON	Excellent	☐	☑
37	12		McLaren	720S	2021	SBM14DCA0MW000056		Excellent	☑	☑
38	1		Hyundai	Elantra	2019	KMHD84LF5KU000057		Good	☑	☐
39	5		Jeep	Grand Cherokee	2020	1C4RJFBG9LC000061		Excellent	☑	☐
40	1	3	Honda	Accord	2019	1HGCV1F14KA000062	AC19HON2	Good	☐	☑

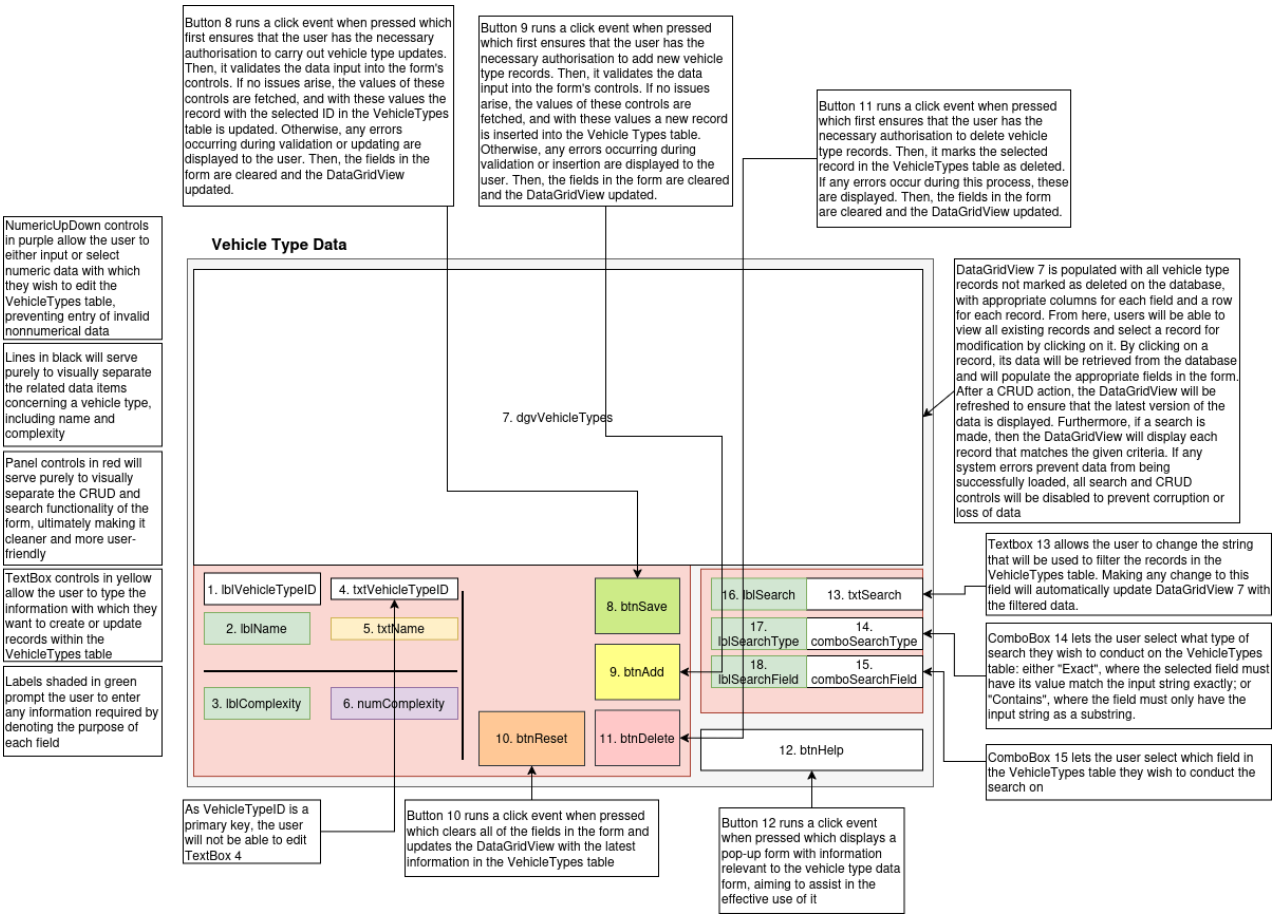
Vehicle ID: 4 | Vehicle Type ID: Sedan - 1 | Year: 2006 | Unknown | Save | Search field: Vehicle ID

Make: Honda | Model: Civic | Condition: Fair | Add | Search type: Contains

VIN: 2C3CCAAG2EH118886 | License Plate: VU02GFI | Buy price: 1200 | Delete | Search text:

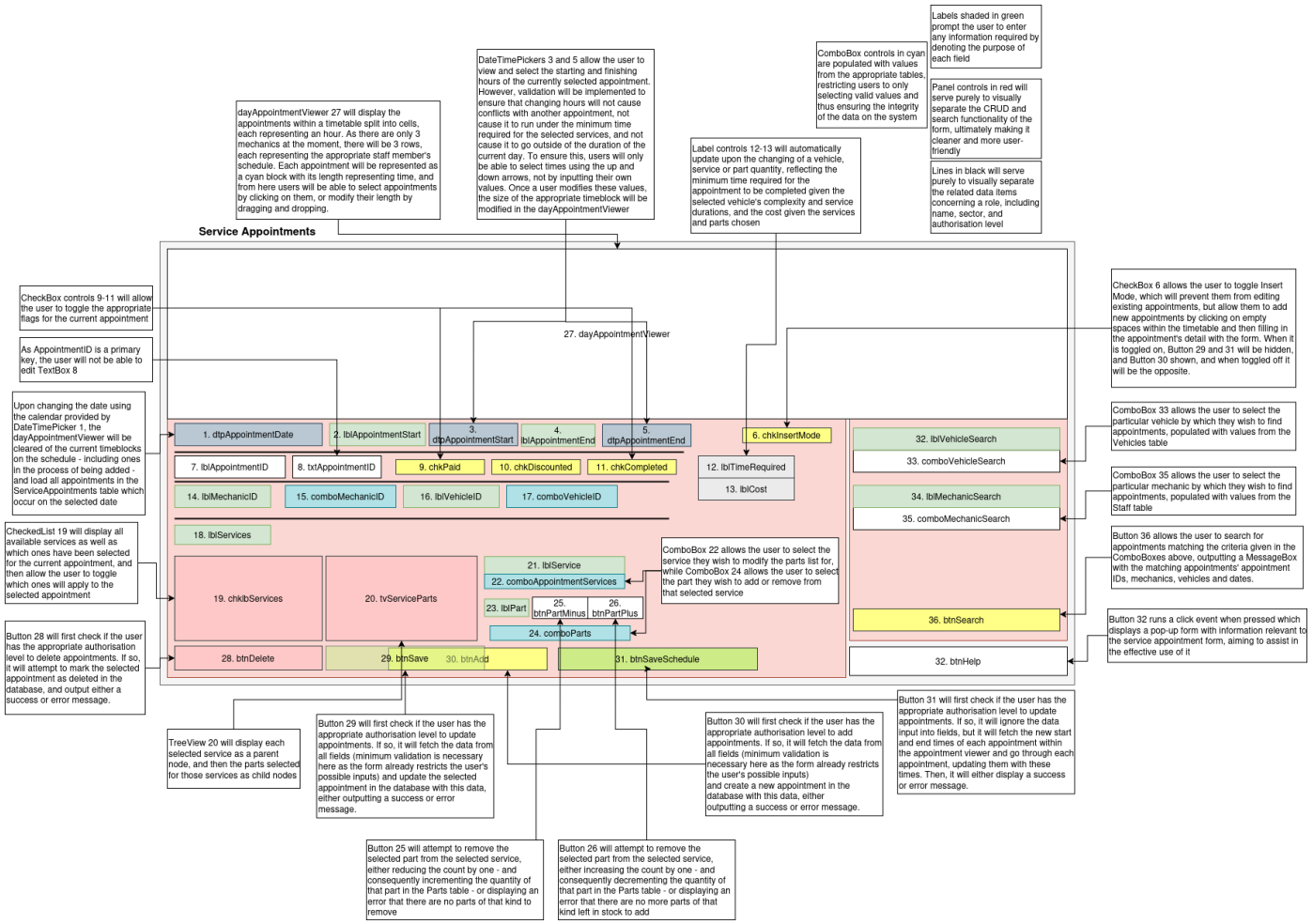
Customer ID: Shane O'Neill - 2 | Sell price: 2200

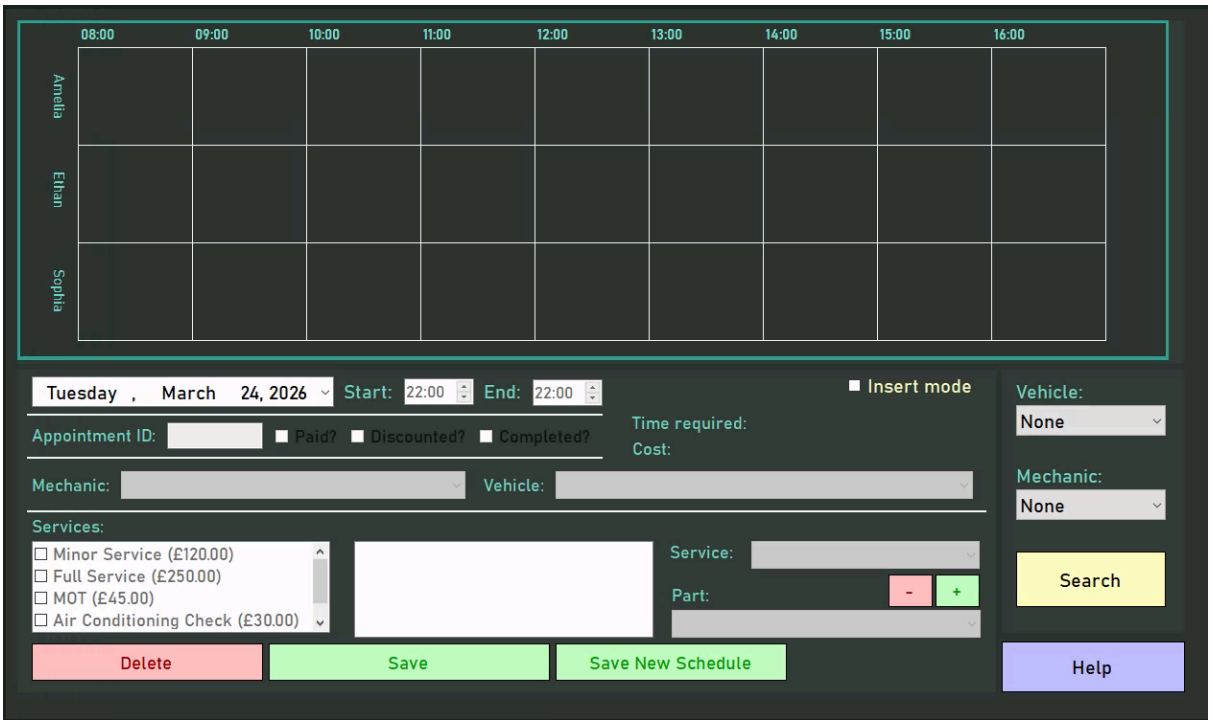
Storage Location: None | Available? | Inspected? | Reset | Help

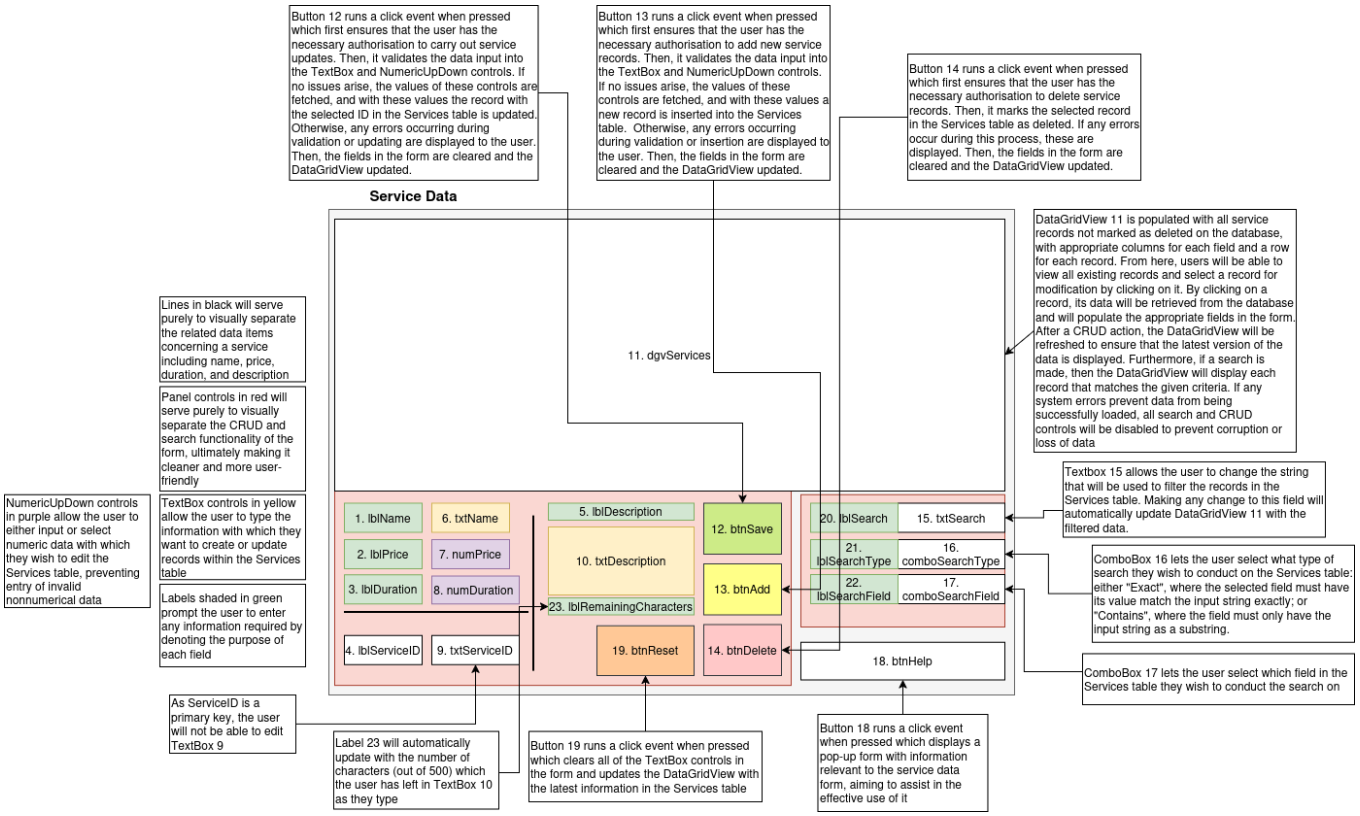


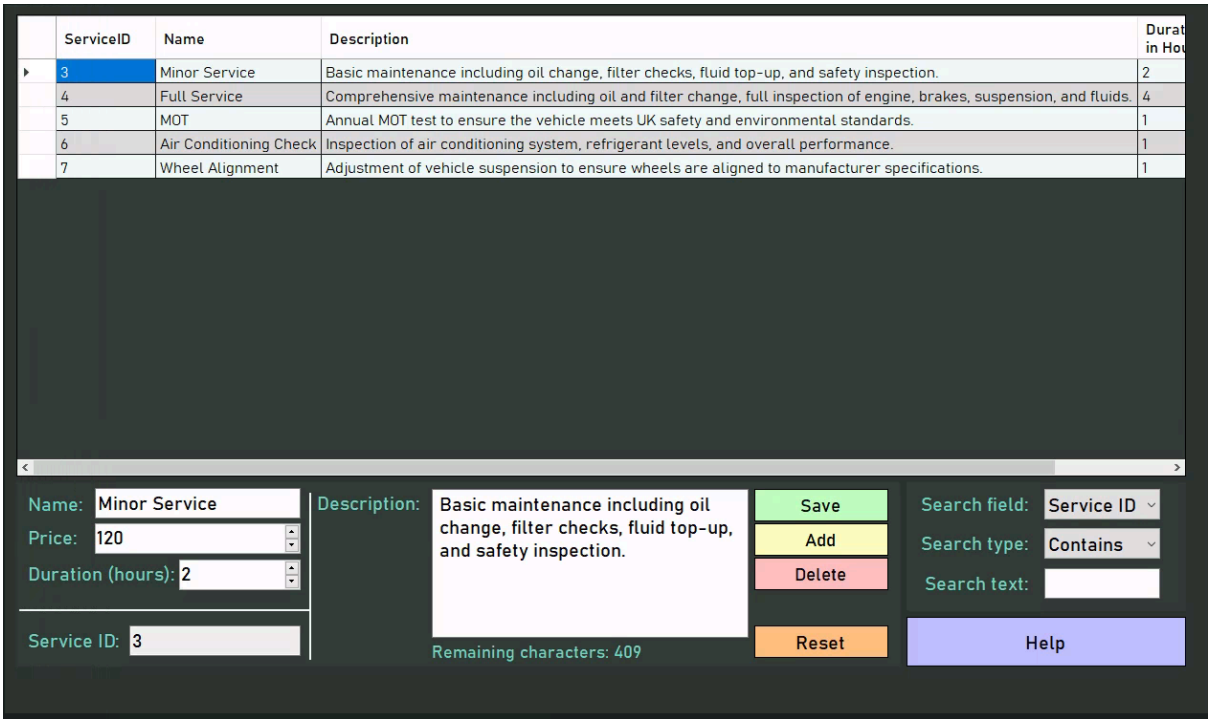
	Vehicle Type ID	Vehicle Type Name	Vehicle Complexity
▸	1	Sedan	1.0
	2	Hatchback	0.9
	3	Coupe	1.0
	4	SUV	1.2
	5	Pickup Truck	1.3
	6	Van	1.3
	7	Minivan	1.2
	8	Convertible	1.1
	9	Wagon	1.1
	10	Luxury Sedan	1.1
	11	Sports Car	1.2
	12	Motorbike	0.7
	13	Electric Vehicle	1.1
	14	Hybrid	1.1
	15	Truck	1.4
	16	Bus	1.5
	17	Moped	0.5

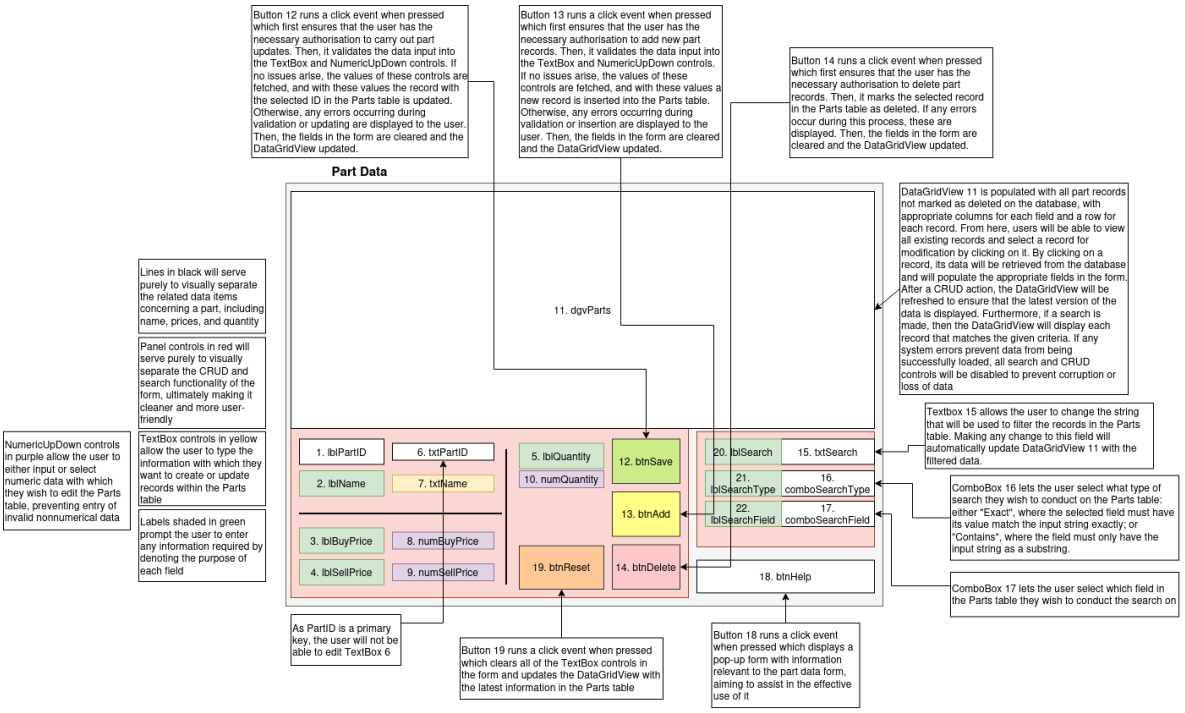
Vehicle Type ID: 1
Name: Sedan
Complexity: 1.0
Reset
Save
Add
Delete
Search text:
Search type: Contains
Search field: Vehicle Type ID
Help

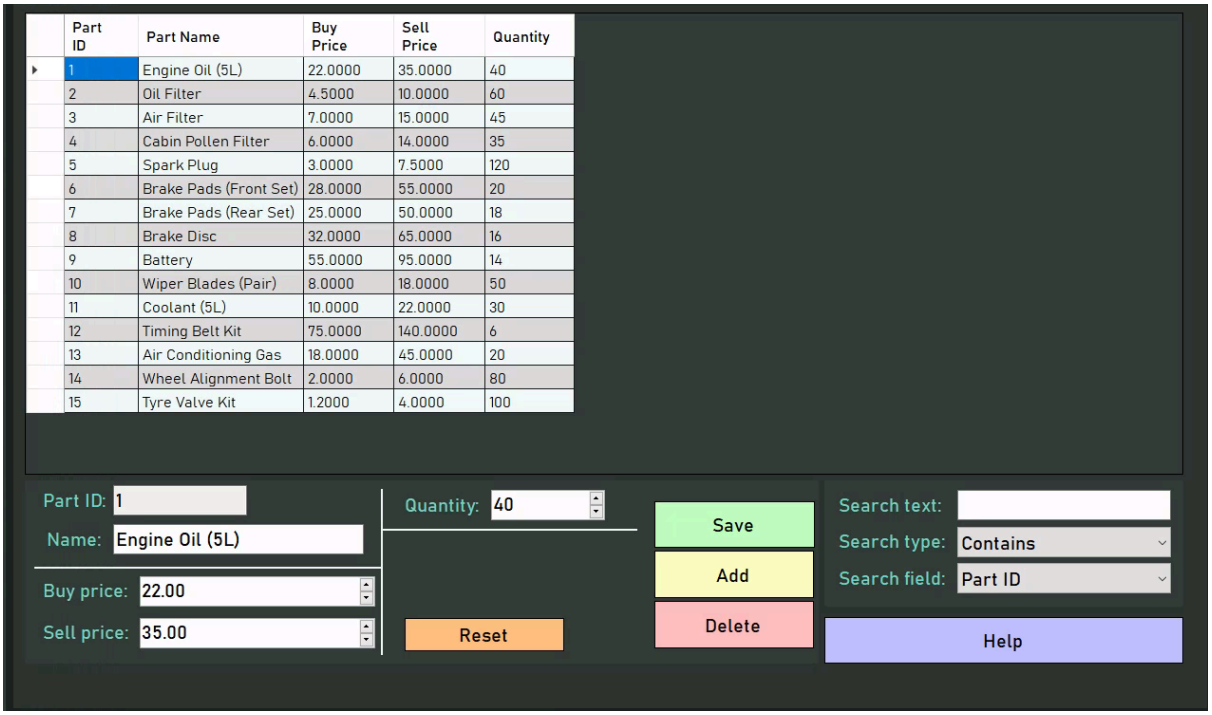

Service Appointments
dayAppointmentViewer 27 will display the appointments within a timetable split into cells, each representing an hour. As there are only 3 mechanics at the moment, there will be 3 rows, each representing the appropriate staff member's schedule. Each appointment will be represented as a cyan block with its length representing time, and from here users will be able to select appointments by clicking on them, or modify their length by dragging and dropping.
DateTimePickers 3 and 5 allow the user to view and select the starting and finishing hours of the currently selected appointment. However, validation will be implemented to ensure that changing hours will not cause conflicts with another appointment, not cause it to run under the minimum time required for the selected services, and not cause it to go outside of the duration of the current day. To ensure this, users will only be able to select times using the up and down arrows, not by inputting their own values. Once a user modifies these values, the size of the appropriate timeblock will be modified in the dayAppointmentViewer
ComboBox controls in cyan are populated with values from the appropriate tables, restricting users to only selecting valid values and thus ensuring the integrity of the data on the system
Labels shaded in green prompt the user to enter any information required by denoting the purpose of each field
Panel controls in red will serve purely to visually separate the CRUD and search functionality of the form, ultimately making it cleaner and more user-friendly
Lines in black will serve purely to visually separate the related data items concerning a role, including name, sector, and authorisation level
Label controls 12-13 will automatically update upon the changing of a vehicle, service or part quantity, reflecting the minimum time required for the appointment to be completed given the selected vehicle's complexity and service durations, and the cost given the services and parts chosen
CheckBox controls 9-11 will allow the user to toggle the appropriate flags for the current appointment
As AppointmentID is a primary key, the user will not be able to edit TextBox 8
Upon changing the date using the calendar provided by DateTimePicker 1, the dayAppointmentViewer will be cleared of the current timeblocks on the schedule - including ones in the process of being added - and load all appointments in the ServiceAppointments table which occur on the selected date
CheckedList 19 will display all available services as well as which ones have been selected for the current appointment, and then allow the user to toggle which ones will apply to the selected appointment
Button 28 will first check if the user has the appropriate authorisation level to delete appointments. If so, it will attempt to mark the selected appointment as deleted in the database, and output either a success or error message.
27. dayAppointmentViewer
1. dtpAppointmentDate
2. lblAppointmentStart
3. dtpAppointmentStart
4. lblAppointmentEnd
5. dtpAppointmentEnd
6. chkInsertMode
7. lblAppointmentID
8. txtAppointmentID
9. chkPaid
10. chkDiscounted
11. chkCompleted
12. lblTimeRequired
13. lblCost
14. lblMechanicID
15. comboMechanicID
16. lblVehicleID
17. comboVehicleID
18. lblServices
19. chklbServices
20. tvServiceParts
21. lblService
22. comboAppointmentServices
23. lblPart
24. comboParts
25. btnPartMinus
26. btnPartPlus
28. btnDelete
29. btnSave
30. btnAdd
31. btnSaveSchedule
32. lblVehicleSearch
33. comboVehicleSearch
34. lblMechanicSearch
35. comboMechanicSearch
36. btnSearch
32. btnHelp
CheckBox 6 allows the user to toggle Insert Mode, which will prevent them from editing existing appointments, but allow them to add new appointments by clicking on empty spaces within the timetable and then filling in the appointment's detail with the form. When it is toggled on, Button 29 and 31 will be hidden, and Button 30 shown, and when toggled off it will be the opposite.
ComboBox 33 allows the user to select the particular vehicle by which they wish to find appointments, populated with values from the Vehicles table
ComboBox 35 allows the user to select the particular mechanic by which they wish to find appointments, populated with values from the Staff table
Button 36 allows the user to search for appointments matching the criteria given in the ComboBoxes above, outputting a MessageBox with the matching appointments' appointment IDs, mechanics, vehicles and dates.
Button 32 runs a click event when pressed which displays a pop-up form with information relevant to the service appointment form, aiming to assist in the effective use of it
ComboBox 22 allows the user to select the service they wish to modify the parts list for, while ComboBox 24 allows the user to select the part they wish to add or remove from that selected service
TreeView 20 will display each selected service as a parent node, and then the parts selected for those services as child nodes
Button 29 will first check if the user has the appropriate authorisation level to update appointments. If so, it will fetch the data from all fields (minimum validation is necessary here as the form already restricts the user's possible inputs) and update the selected appointment in the database with this data, either outputting a success or error message.
Button 30 will first check if the user has the appropriate authorisation level to add appointments. If so, it will fetch the data from all fields (minimum validation is necessary here as the form already restricts the user's possible inputs) and create a new appointment in the database with this data, either outputting a success or error message.
Button 31 will first check if the user has the appropriate authorisation level to update appointments. If so, it will ignore the data input into fields, but it will fetch the new start and end times of each appointment within the appointment viewer and go through each appointment, updating them with these times. Then, it will either display a success or error message.
Button 25 will attempt to remove the selected part from the selected service, either reducing the count by one - and consequently incrementing the quantity of that part in the Parts table - or displaying an error that there are no parts of that kind to remove
Button 26 will attempt to remove the selected part from the selected service, either increasing the count by one - and consequently decrementing the quantity of that part in the Parts table - or displaying an error that there are no more parts of that kind left in stock to add

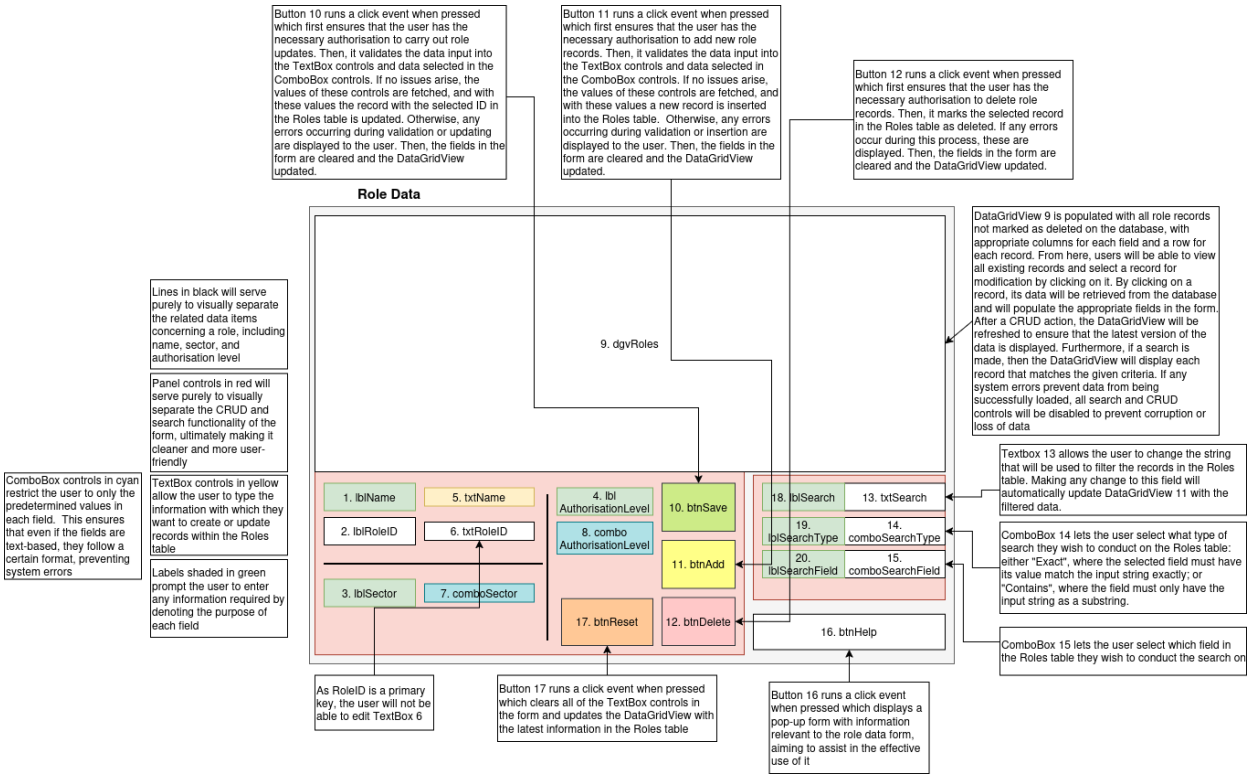

08:00
09:00
10:00
11:00
12:00
13:00
14:00
15:00
16:00
Amelia
Ethan
Sophia
Tuesday , March 24, 2026
Start: 22:00
End: 22:00
Insert mode
Appointment ID:
Paid?
Discounted?
Completed?
Time required:
Cost:
Mechanic:
Vehicle:
Services:
Minor Service (£120.00)
Full Service (£250.00)
MOT (£45.00)
Air Conditioning Check (£30.00)
Service:
Part:
-
+
Vehicle:
None
Mechanic:
None
Search
Delete
Save
Save New Schedule
Help

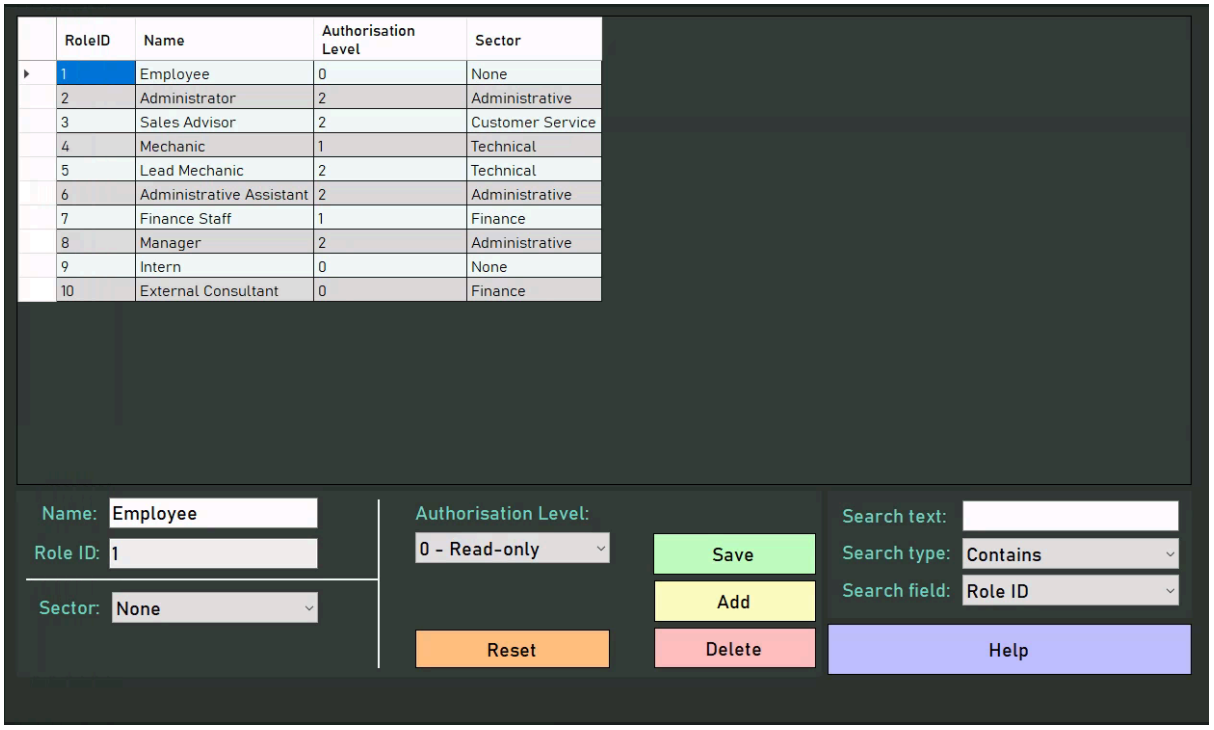

Button 12 runs a click event when pressed which first ensures that the user has the necessary authorisation to carry out service updates. Then, it validates the data input into the TextBox and NumericUpDown controls. If no issues arise, the values of these controls are fetched, and with these values the record with the selected ID in the Services table is updated. Otherwise, any errors occurring during validation or updating are displayed to the user. Then, the fields in the form are cleared and the DataGridView updated.
Button 13 runs a click event when pressed which first ensures that the user has the necessary authorisation to add new service records. Then, it validates the data input into the TextBox and NumericUpDown controls. If no issues arise, the values of these controls are fetched, and with these values a new record is inserted into the Services table. Otherwise, any errors occurring during validation or insertion are displayed to the user. Then, the fields in the form are cleared and the DataGridView updated.
Button 14 runs a click event when pressed which first ensures that the user has the necessary authorisation to delete service records. Then, it marks the selected record in the Services table as deleted. If any errors occur during this process, these are displayed. Then, the fields in the form are cleared and the DataGridView updated.
Service Data
DataGridView 11 is populated with all service records not marked as deleted on the database, with appropriate columns for each field and a row for each record. From here, users will be able to view all existing records and select a record for modification by clicking on it. By clicking on a record, its data will be retrieved from the database and will populate the appropriate fields in the form. After a CRUD action, the DataGridView will be refreshed to ensure that the latest version of the data is displayed. Furthermore, if a search is made, then the DataGridView will display each record that matches the given criteria. If any system errors prevent data from being successfully loaded, all search and CRUD controls will be disabled to prevent corruption or loss of data
Lines in black will serve purely to visually separate the related data items concerning a service including name, price, duration, and description
11. dgvServices
Panel controls in red will serve purely to visually separate the CRUD and search functionality of the form, ultimately making it cleaner and more user-friendly
Textbox 15 allows the user to change the string that will be used to filter the records in the Services table. Making any change to this field will automatically update DataGridView 11 with the filtered data.
NumericUpDown controls in purple allow the user to either input or select numeric data with which they wish to edit the Services table, preventing entry of invalid nonnumerical data
TextBox controls in yellow allow the user to type the information with which they want to create or update records within the Services table
1. lblName
6. txtName
5. lblDescription
12. btnSave
20. lblSearch
15. txtSearch
2. lblPrice
7. numPrice
10. txtDescription
21. lblSearchType
16. comboSearchType
ComboBox 16 lets the user select what type of search they wish to conduct on the Services table: either "Exact", where the selected field must have its value match the input string exactly; or "Contains", where the field must only have the input string as a substring.
3. lblDuration
8. numDuration
13. btnAdd
22. lblSearchField
17. comboSearchField
Labels shaded in green prompt the user to enter any information required by denoting the purpose of each field
23. lblRemainingCharacters
4. lblServiceID
9. txtServiceID
19. btnReset
14. btnDelete
18. btnHelp
ComboBox 17 lets the user select which field in the Services table they wish to conduct the search on
As ServiceID is a primary key, the user will not be able to edit TextBox 9
Label 23 will automatically update with the number of characters (out of 500) which the user has left in TextBox 10 as they type
Button 19 runs a click event when pressed which clears all of the TextBox controls in the form and updates the DataGridView with the latest information in the Services table
Button 18 runs a click event when pressed which displays a pop-up form with information relevant to the service data form, aiming to assist in the effective use of it

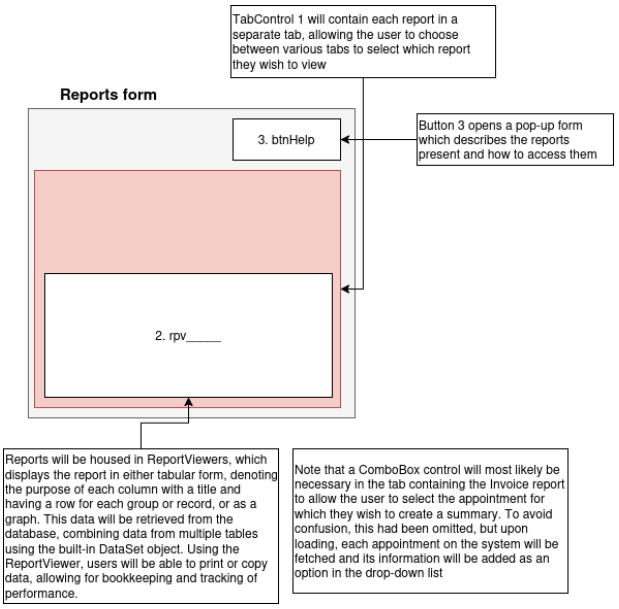

ServiceID
Name
Description
Durat in Ho
3
Minor Service
Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.
2
4
Full Service
Comprehensive maintenance including oil and filter change, full inspection of engine, brakes, suspension, and fluids.
4
5
MOT
Annual MOT test to ensure the vehicle meets UK safety and environmental standards.
1
6
Air Conditioning Check
Inspection of air conditioning system, refrigerant levels, and overall performance.
1
7
Wheel Alignment
Adjustment of vehicle suspension to ensure wheels are aligned to manufacturer specifications.
1
Name: Minor Service
Price: 120
Duration (hours): 2
Service ID: 3
Description:
Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.
Remaining characters: 409
Save
Add
Delete
Reset
Search field: Service ID
Search type: Contains
Search text:
Help

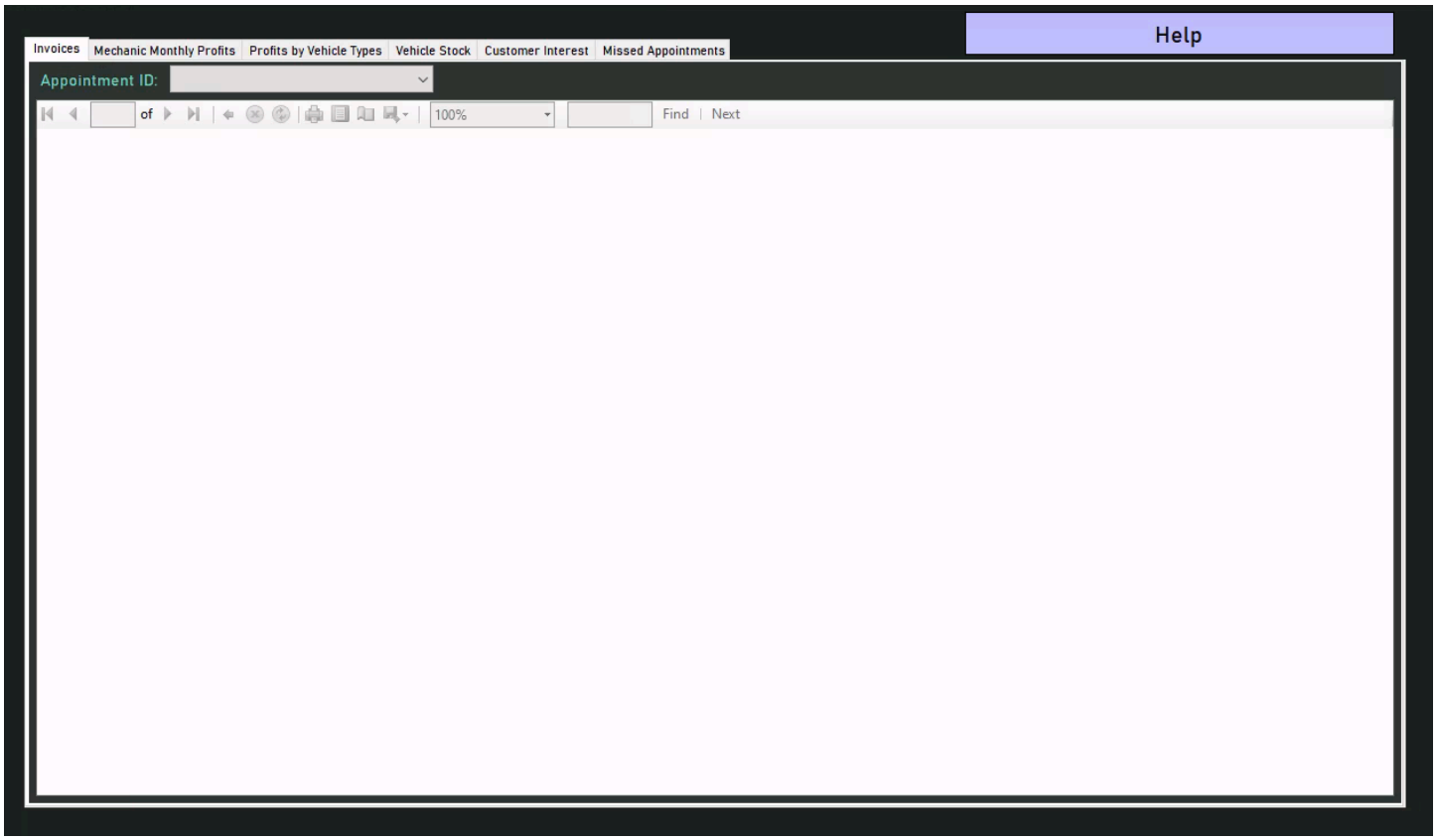

Button 12 runs a click event when pressed which first ensures that the user has the necessary authorisation to carry out part updates. Then, it validates the data input into the TextBox and NumericUpDown controls. If no issues arise, the values of these controls are fetched, and with these values the record with the selected ID in the Parts table is updated. Otherwise, any errors occurring during validation or updating are displayed to the user. Then, the fields in the form are cleared and the DataGridView updated.
Button 13 runs a click event when pressed which first ensures that the user has the necessary authorisation to add new part records. Then, it validates the data input into the TextBox and NumericUpDown controls. If no issues arise, the values of these controls are fetched, and with these values a new record is inserted into the Parts table. Otherwise, any errors occurring during validation or insertion are displayed to the user. Then, the fields in the form are cleared and the DataGridView updated.
Button 14 runs a click event when pressed which first ensures that the user has the necessary authorisation to delete part records. Then, it marks the selected record in the Parts table as deleted. If any errors occur during this process, these are displayed. Then, the fields in the form are cleared and the DataGridView updated.
Part Data
DataGridView 11 is populated with all part records not marked as deleted on the database, with appropriate columns for each field and a row for each record. From here, users will be able to view all existing records and select a record for modification by clicking on it. By clicking on a record, its data will be retrieved from the database and will populate the appropriate fields in the form. After a CRUD action, the DataGridView will be refreshed to ensure that the latest version of the data is displayed. Furthermore, if a search is made, then the DataGridView will display each record that matches the given criteria. If any system errors prevent data from being successfully loaded, all search and CRUD controls will be disabled to prevent corruption or loss of data
11. dgvParts
Lines in black will serve purely to visually separate the related data items concerning a part, including name, prices, and quantity
Panel controls in red will serve purely to visually separate the CRUD and search functionality of the form, ultimately making it cleaner and more user-friendly
NumericUpDown controls in purple allow the user to either input or select numeric data with which they wish to edit the Parts table, preventing entry of invalid nonnumerical data
TextBox controls in yellow allow the user to type the information with which they want to create or update records within the Parts table
Labels shaded in green prompt the user to enter any information required by denoting the purpose of each field
1. lblPartID
6. txtPartID
2. lblName
7. txtName
3. lblBuyPrice
8. numBuyPrice
4. lblSellPrice
9. numSellPrice
5. lblQuantity
10. numQuantity
12. btnSave
13. btnAdd
19. btnReset
14. btnDelete
20. lblSearch
15. txtSearch
21. lblSearchType
16. comboSearchType
22. lblSearchField
17. comboSearchField
18. btnHelp
Textbox 15 allows the user to change the string that will be used to filter the records in the Parts table. Making any change to this field will automatically update DataGridView 11 with the filtered data.
ComboBox 16 lets the user select what type of search they wish to conduct on the Parts table: either "Exact", where the selected field must have its value match the input string exactly; or "Contains", where the field must only have the input string as a substring.
ComboBox 17 lets the user select which field in the Parts table they wish to conduct the search on
As PartID is a primary key, the user will not be able to edit TextBox 6
Button 19 runs a click event when pressed which clears all of the TextBox controls in the form and updates the DataGridView with the latest information in the Parts table
Button 18 runs a click event when pressed which displays a pop-up form with information relevant to the part data form, aiming to assist in the effective use of it

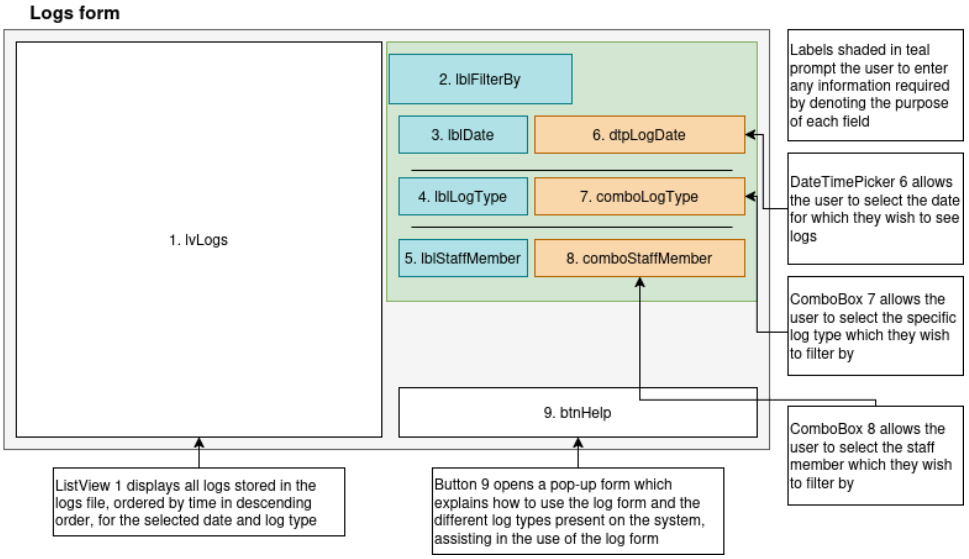

Part ID
Part Name
Buy Price
Sell Price
Quantity
1 Engine Oil (5L) 22.0000 35.0000 40
2 Oil Filter 4.5000 10.0000 60
3 Air Filter 7.0000 15.0000 45
4 Cabin Pollen Filter 6.0000 14.0000 35
5 Spark Plug 3.0000 7.5000 120
6 Brake Pads (Front Set) 28.0000 55.0000 20
7 Brake Pads (Rear Set) 25.0000 50.0000 18
8 Brake Disc 32.0000 65.0000 16
9 Battery 55.0000 95.0000 14
10 Wiper Blades (Pair) 8.0000 18.0000 50
11 Coolant (5L) 10.0000 22.0000 30
12 Timing Belt Kit 75.0000 140.0000 6
13 Air Conditioning Gas 18.0000 45.0000 20
14 Wheel Alignment Bolt 2.0000 6.0000 80
15 Tyre Valve Kit 1.2000 4.0000 100
Part ID: 1
Name: Engine Oil (5L)
Buy price: 22.00
Sell price: 35.00
Quantity: 40
Reset
Save
Add
Delete
Search text:
Search type: Contains
Search field: Part ID
Help

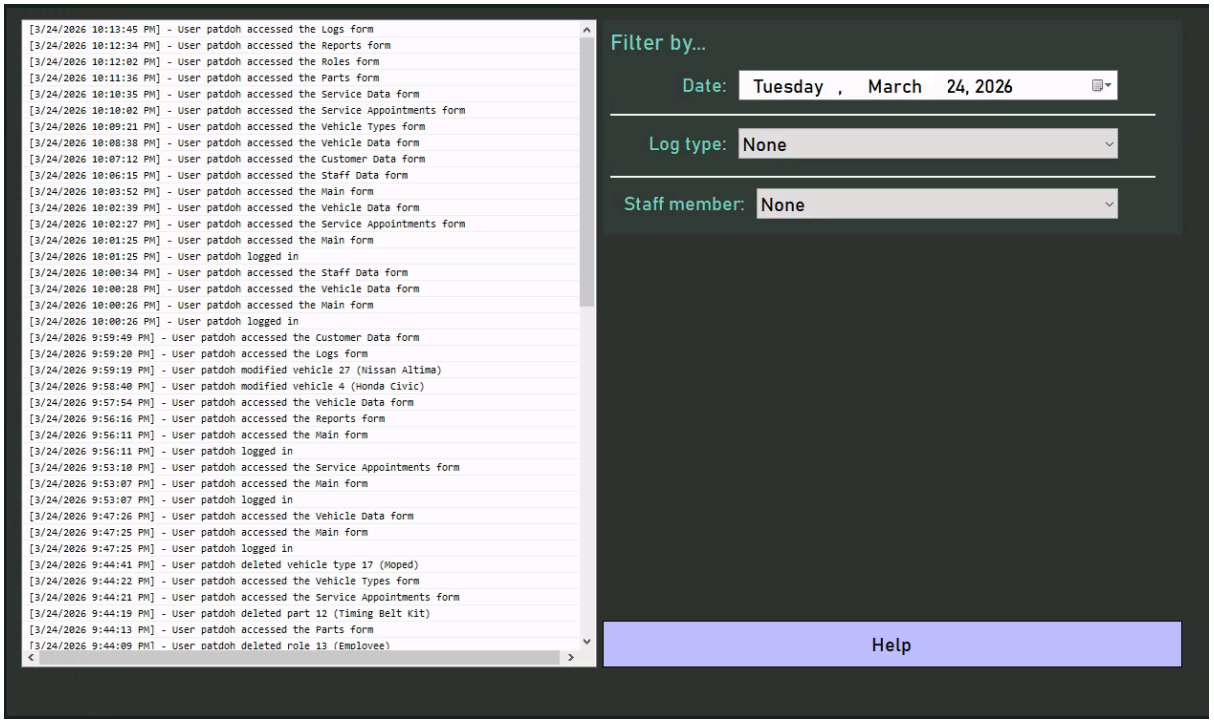

Button 10 runs a click event when pressed which first ensures that the user has the necessary authorisation to carry out role updates. Then, it validates the data input into the TextBox controls and data selected in the ComboBox controls. If no issues arise, the values of these controls are fetched, and with these values the record with the selected ID in the Roles table is updated. Otherwise, any errors occurring during validation or updating are displayed to the user. Then, the fields in the form are cleared and the DataGridView updated.
Button 11 runs a click event when pressed which first ensures that the user has the necessary authorisation to add new role records. Then, it validates the data input into the TextBox controls and data selected in the ComboBox controls. If no issues arise, the values of these controls are fetched, and with these values a new record is inserted into the Roles table. Otherwise, any errors occurring during validation or insertion are displayed to the user. Then, the fields in the form are cleared and the DataGridView updated.
Button 12 runs a click event when pressed which first ensures that the user has the necessary authorisation to delete role records. Then, it marks the selected record in the Roles table as deleted. If any errors occur during this process, these are displayed. Then, the fields in the form are cleared and the DataGridView updated.
Role Data
DataGridView 9 is populated with all role records not marked as deleted on the database, with appropriate columns for each field and a row for each record. From here, users will be able to view all existing records and select a record for modification by clicking on it. By clicking on a record, its data will be retrieved from the database and will populate the appropriate fields in the form. After a CRUD action, the DataGridView will be refreshed to ensure that the latest version of the data is displayed. Furthermore, if a search is made, then the DataGridView will display each record that matches the given criteria. If any system errors prevent data from being successfully loaded, all search and CRUD controls will be disabled to prevent corruption or loss of data
Lines in black will serve purely to visually separate the related data items concerning a role, including name, sector, and authorisation level
9. dgvRoles
Panel controls in red will serve purely to visually separate the CRUD and search functionality of the form, ultimately making it cleaner and more user-friendly
ComboBox controls in cyan restrict the user to only the predetermined values in each field. This ensures that even if the fields are text-based, they follow a certain format, preventing system errors
TextBox controls in yellow allow the user to type the information with which they want to create or update records within the Roles table
Labels shaded in green prompt the user to enter any information required by denoting the purpose of each field
1. lblName
5. txtName
4. lbl AuthorisationLevel
10. btnSave
2. lblRoleID
6. txtRoleID
8. combo AuthorisationLevel
11. btnAdd
3. lblSector
7. comboSector
17. btnReset
12. btnDelete
18. lblSearch
13. txtSearch
19. lblSearchType
14. comboSearchType
20. lblSearchField
15. comboSearchField
16. btnHelp
Textbox 13 allows the user to change the string that will be used to filter the records in the Roles table. Making any change to this field will automatically update DataGridView 11 with the filtered data.
ComboBox 14 lets the user select what type of search they wish to conduct on the Roles table: either "Exact", where the selected field must have its value match the input string exactly; or "Contains", where the field must only have the input string as a substring.
ComboBox 15 lets the user select which field in the Roles table they wish to conduct the search on
As RoleID is a primary key, the user will not be able to edit TextBox 6
Button 17 runs a click event when pressed which clears all of the TextBox controls in the form and updates the DataGridView with the latest information in the Roles table
Button 16 runs a click event when pressed which displays a pop-up form with information relevant to the role data form, aiming to assist in the effective use of it

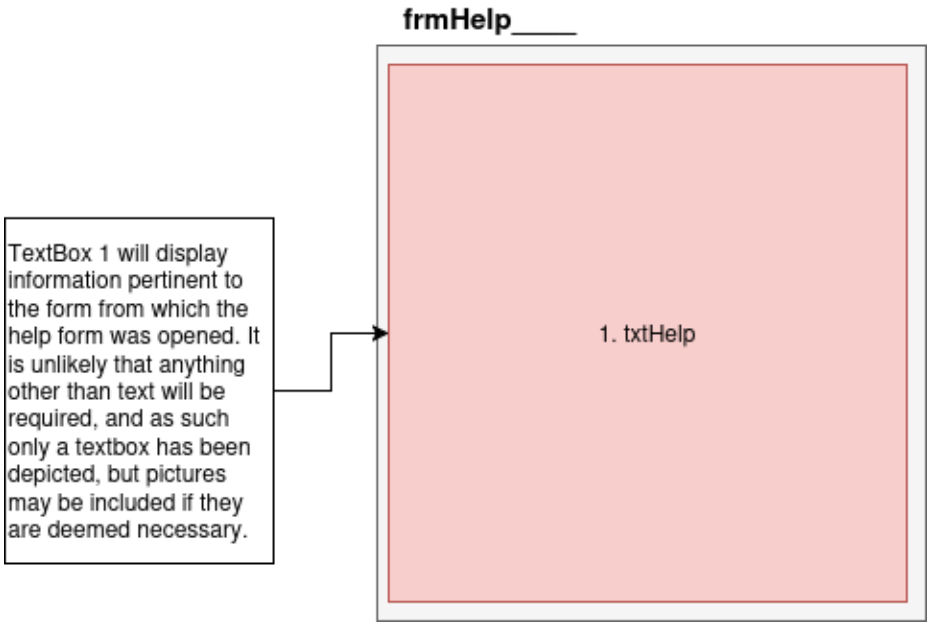

RoleID
Name
Authorisation Level
Sector
1 Employee 0 None
2 Administrator 2 Administrative
3 Sales Advisor 2 Customer Service
4 Mechanic 1 Technical
5 Lead Mechanic 2 Technical
6 Administrative Assistant 2 Administrative
7 Finance Staff 1 Finance
8 Manager 2 Administrative
9 Intern 0 None
10 External Consultant 0 Finance
Name: Employee
Role ID: 1
Sector: None
Authorisation Level:
0 - Read-only
Reset
Save
Add
Delete
Search text:
Search type: Contains
Search field: Role ID
Help

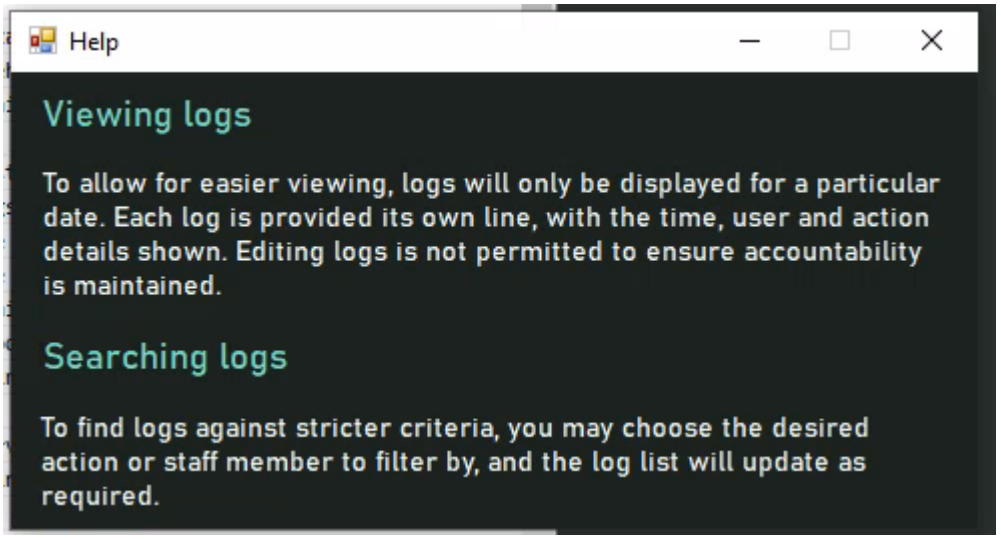

TabControl 1 will contain each report in a separate tab, allowing the user to choose between various tabs to select which report they wish to view
Reports form
3. btnHelp
Button 3 opens a pop-up form which describes the reports present and how to access them
2. rpv_____
Reports will be housed in ReportViewers, which displays the report in either tabular form, denoting the purpose of each column with a title and having a row for each group or record, or as a graph. This data will be retrieved from the database, combining data from multiple tables using the built-in DataSet object. Using the ReportViewer, users will be able to print or copy data, allowing for bookkeeping and tracking of performance.
Note that a ComboBox control will most likely be necessary in the tab containing the Invoice report to allow the user to select the appointment for which they wish to create a summary. To avoid confusion, this had been omitted, but upon loading, each appointment on the system will be fetched and its information will be added as an option in the drop-down list


Invoices
Mechanic Monthly Profits
Profits by Vehicle Types
Vehicle Stock
Customer Interest
Missed Appointments
Help
Appointment ID:
of
100%
Find
Next


Logs form
1. lvLogs
2. lblFilterBy
3. lblDate
6. dtpLogDate
4. lblLogType
7. comboLogType
5. lblStaffMember
8. comboStaffMember
9. btnHelp
Labels shaded in teal prompt the user to enter any information required by denoting the purpose of each field
DateTimePicker 6 allows the user to select the date for which they wish to see logs
ComboBox 7 allows the user to select the specific log type which they wish to filter by
ComboBox 8 allows the user to select the staff member which they wish to filter by
ListView 1 displays all logs stored in the logs file, ordered by time in descending order, for the selected date and log type
Button 9 opens a pop-up form which explains how to use the log form and the different log types present on the system, assisting in the use of the log form


[3/24/2026 10:13:45 PM] - User patdoh accessed the Logs form
[3/24/2026 10:12:34 PM] - User patdoh accessed the Reports form
[3/24/2026 10:12:02 PM] - User patdoh accessed the Roles form
[3/24/2026 10:11:36 PM] - User patdoh accessed the Parts form
[3/24/2026 10:10:35 PM] - User patdoh accessed the Service Data form
[3/24/2026 10:10:02 PM] - User patdoh accessed the Service Appointments form
[3/24/2026 10:09:21 PM] - User patdoh accessed the Vehicle Types form
[3/24/2026 10:08:38 PM] - User patdoh accessed the Vehicle Data form
[3/24/2026 10:07:12 PM] - User patdoh accessed the Customer Data form
[3/24/2026 10:06:15 PM] - User patdoh accessed the Staff Data form
[3/24/2026 10:03:52 PM] - User patdoh accessed the Main form
[3/24/2026 10:02:39 PM] - User patdoh accessed the Vehicle Data form
[3/24/2026 10:02:27 PM] - User patdoh accessed the Service Appointments form
[3/24/2026 10:01:25 PM] - User patdoh accessed the Main form
[3/24/2026 10:01:25 PM] - User patdoh logged in
[3/24/2026 10:00:34 PM] - User patdoh accessed the Staff Data form
[3/24/2026 10:00:28 PM] - User patdoh accessed the Vehicle Data form
[3/24/2026 10:00:26 PM] - User patdoh accessed the Main form
[3/24/2026 10:00:26 PM] - User patdoh logged in
[3/24/2026 9:59:49 PM] - User patdoh accessed the Customer Data form
[3/24/2026 9:59:20 PM] - User patdoh accessed the Logs form
[3/24/2026 9:59:19 PM] - User patdoh modified vehicle 27 (Nissan Altima)
[3/24/2026 9:58:40 PM] - User patdoh modified vehicle 4 (Honda Civic)
[3/24/2026 9:57:54 PM] - User patdoh accessed the Vehicle Data form
[3/24/2026 9:56:16 PM] - User patdoh accessed the Reports form
[3/24/2026 9:56:11 PM] - User patdoh accessed the Main form
[3/24/2026 9:56:11 PM] - User patdoh logged in
[3/24/2026 9:53:10 PM] - User patdoh accessed the Service Appointments form
[3/24/2026 9:53:07 PM] - User patdoh accessed the Main form
[3/24/2026 9:53:07 PM] - User patdoh logged in
[3/24/2026 9:47:26 PM] - User patdoh accessed the Vehicle Data form
[3/24/2026 9:47:25 PM] - User patdoh accessed the Main form
[3/24/2026 9:47:25 PM] - User patdoh logged in
[3/24/2026 9:44:41 PM] - User patdoh deleted vehicle type 17 (Moped)
[3/24/2026 9:44:22 PM] - User patdoh accessed the Vehicle Types form
[3/24/2026 9:44:21 PM] - User patdoh accessed the Service Appointments form
[3/24/2026 9:44:19 PM] - User patdoh deleted part 12 (Timing Belt Kit)
[3/24/2026 9:44:13 PM] - User patdoh accessed the Parts form
[3/24/2026 9:44:09 PM] - User patdoh deleted role 13 (Employee)
Filter by...
Date: Tuesday , March 24, 2026
Log type: None
Staff member: None
Help


frmHelp____
TextBox 1 will display information pertinent to the form from which the help form was opened. It is unlikely that anything other than text will be required, and as such only a textbox has been depicted, but pictures may be included if they are deemed necessary.
1. txtHelp


Help
Viewing logs
To allow for easier viewing, logs will only be displayed for a particular date. Each log is provided its own line, with the time, user and action details shown. Editing logs is not permitted to ensure accountability is maintained.
Searching logs
To find logs against stricter criteria, you may choose the desired action or staff member to filter by, and the log list will update as required.

Testing process

Overall, the testing process proved to be highly successful, uncovering various flaws within the system, particularly in the areas of logging in and maintaining the user's details in memory. The testing process began with a structured approach, starting with the identification of key test items for each form and the development of numerous test cases designed to evaluate all plausible execution paths. By establishing a comprehensive testing framework prior to execution, the process ended up being efficient and methodical, leading to the creation of a greater number of relevant test cases than would have been possible if testing and case development were conducted simultaneously. Additionally, documenting descriptions for each test item and comparing them against the expected outcomes allowed for the detection of a deviation from intended functionality.

However, white-box testing alone would not be able to ensure that the system was fit for purpose. As such, providing key end users with the opportunity to participate in a black-box usage session and fill out a questionnaire at the end provided crucial data as to the strengths and weaknesses of the system and its general usability, thereby ensuring its usability. Though the rather simplistic nature of the provided questionnaire did aid in response time, it made it more difficult to analyse the reasons for user dissatisfaction, which may have slightly hindered the effectiveness of the process. Regardless, the questionnaire has been deemed a success.

Ultimately, the testing process played a crucial role in ensuring the robustness of the system. Tests were designed to verify functionality at multiple levels - unit, subsystem, and full system integration through acceptance testing - ensuring consistent and reliable operation under normal conditions. The inclusion of diverse test data reinforced system stability by validating its ability to handle a wide range of inputs without failure. In cases where errors were identified, debugging tools such as breakpoints were employed to analyse the code, pinpoint faulty lines, and implement necessary modifications. These corrective actions were documented, illustrating the system's iterative evolution and continuous improvement.

Personal and technical skills

Though the success of the project does rely significantly on the approaches and processes involved throughout the development lifecycle, much of it also depends on my skills, both technical and interpersonal.

In terms of technical skills, I believe that I was fit for the project, with much prior development experience, both curricular and extra-curricular. As a result, the C# sections of the program were implemented rather smoothly, contributing to the final system's robustness and quality. Having much experience with LINQ methods with my background in competitive programming, I was able to effectively process and filter data from the database using code, helping translate an unfamiliar domain - that of database management - into one within which I had more experience, aiding in the development experience. One of the more complex sections of the program is the service appointment viewer, with many interdependent parts - including database, drawing, validation and event logic - all occurring simultaneously. Yet, my experience with the WinForms Graphics library proved to be sufficient for creating a working solution to this problem, highlighting my technical suitability for this project.

However, something my background did not prepare me for was the difference in thinking required when programming in SQL, with its focus on sets rather than procedures. Most notably, I was not conscious of the need for transactions when carrying out complex operations, which would slightly hinder the robustness of the final solution, but the C# logic surrounding this is sufficient to prevent the database entering an invalid state. Despite this, a considerable flaw in the project is the requirement for preexisting records in the service appointments form due to my lack of oversight when considering the way the system will operate: at the present, the mechanics for appointments are fetched using those that are presently stored in the table. As a result, if a new mechanic is added to the system without them having a service appointment already present in the table, they will not be shown. To mitigate this, a flag could have been attached to staff records indicating that they are to be displayed in service appointments, preventing the need for preexisting data. As such, for future projects, I will ensure that I have enough experience with a query language to comfortably understand the limitations and problems of certain approaches and thus ensure that the client is satisfied with the one chosen by me.

With regard to personal skills, I believe that the biggest issue was communication. Although the Waterfall model did not necessitate constant communication with the client for feedback, I admittedly did not provide them with sufficient updates throughout the development process of the project, potentially leaving them uncertain or concerned as to the stage the project is at, which may have reduced client satisfaction. In the future, I will aim to maintain more frequent communication with clients - whether through email or phone - to ensure that all stakeholders are up-to-date as to the status of the project, thus maintaining coherence.

Adherence to project plan

Task No.	Task	Planned Start Date	Actual Start Date	Duration (days)	Actual End Date	Planned End Date	Resources	Comments
Analysis								
1	Initial client meeting/problem discussion	1-Sep	1-Sep	1	1-Sep	1-Sep	Client	The project initiation and analysis phase was executed admittedly cleanly, but this was mostly due in part to the dealership already having outlined the issues with their current system before meeting with me. Had this not been the case, more time may have been required during the initial meeting and during the identification of needs, as often clients have difficulty expressing their initial requirements. The constant interaction with the client at the start was effective in determining the requirements for the rest of the system, as it provided a chance for me to correct any flaws in my interpretation of the business' issues, thus ensuring the efficacy of this stage. The timeline was easily adhered to due to the provision of the problem specification, thus providing additional time for the subsequent phases, and this ultimately improved the quality of the system by ensuring no tasks were rushed.
2	Identification of client needs & current system issues	2-Sep	2-Sep	2	3-Sep	4-Sep	Client	
3	Definition of functional requirements	5-Sep	4-Sep	3	6-Sep	7-Sep	—	
4	Definition of non-functional requirements	5-Sep	4-Sep	2	5-Sep	6-Sep	—	
5	Initial sketches based on requirements	8-Sep	7-Sep	4	10-Sep	11-Sep	—	

6	Client review & sign-off	12-Sep	11-Sep	1	11-Sep	12-Sep	Client	
Design								
7	Definition of system scope & methodology	13-Sep	12-Sep	3	14-Sep	15-Sep	—	First seen in this phase is the difficulty in estimating the duration of tasks. Though some slack was present from the initiation phase, it was quickly absorbed by the delay caused by drawing use cases and data flow diagrams. This could have been easily avoided by either using more efficient UML software such as PlantUML rather than the admittedly cumbersome draw.io to reduce the time taken to draw the same diagrams, or by drawing fewer diagrams altogether, as it was found that these diagrams did not prove critical in later phases of the project. Regardless, the other tasks were completed on schedule, and SQL Server proved to be an effective tool in creating the initial design of the database, helping with maintaining the schedule of the project and providing certainty as to the direction of development. As such, it will be a tool regularly used in future projects.
8	Creation of final sketches	16-Sep	15-Sep	7	21-Sep	22-Sep	—	
9	Use case and data flow diagrams	21-Sep	20-Sep	4	23-Sep	23-Sep	—	
10	Identification of data entities & relationships	24-Sep	24-Sep	3	26-Sep	26-Sep	—	
11	Database schema design	26-Sep	26-Sep	4	29-Sep	29-Sep	SQL Server	
Implementation								
12	Project setup in Visual Studio and database setup	30-Sep	30-Sep	1	30-Sep	30-Sep	Visual Studio, SQL Server	The implementation phase - as expected - was the most turbulent. With constant fluctuations in the task completion times - with several being completed

13	Implementation of tables & relationships	1-Oct	1-Oct	4	4-Oct	5-Oct	Visual Studio, SQL Server	early, several being delayed - there was an uncertainty as to how long the phase would actually take. Though ultimately it resulted in the project being delivered earlier than expected, in future projects, buffer periods would have helped ensure that any unexpected changes to the schedule would not impact the project's delivery time Many external resources also helped, as many unexpected errors or unfamiliar functionalities were quickly amended through the use of websites such as the official SQL and C# documentation when faced with unfamiliar features, or Stack Overflow when faced with troubling issues. Lastly, the ease of use of both Visual Studio and SQL Server proved to be crucial in the relatively smooth development of the project, as many of the features - such as automatic event generation and pop-up windows in Visual Studio and automatic query generation and simple data input in SQL Server - aided in tasks that would have otherwise been extremely arduous. However, although these applications are useful, a more modern development approach suitable for a short timescale - such as a Ruby On Rails web app - may have proven more effective for a small-scale operation like *Doherty Cars*, and consequently a different IDE such as RubyMine required.
14	Seeding of test data	6-Oct	5-Oct	2	6-Oct	8-Oct	SQL Server	
15	Implementation of frmLogin (validation, hashing)	9-Oct	7-Oct	13	19-Oct	20-Oct	Visual Studio, SQL Server	
16	Role-based access logic	9-Oct	7-Oct	10	16-Oct	16-Oct	Visual Studio	
17	Logging of user actions	21-Oct	20-Oct	4	23-Oct	24-Oct	Visual Studio, SQL Server	
18	Validation class implementation	21-Oct	20-Oct	3	22-Oct	23-Oct	Visual Studio	
19	Implementation of frmMainMenu	24-Oct	23-Oct	5	27-Oct	28-Oct	Visual Studio	
20	Implementation of frmStaffData	29-Oct	28-Oct	16	12-Nov	15-Nov	Visual Studio, SQL Server	
21	Implementation of frmRoleData	16-Nov	13-Nov	5	17-Nov	20-Nov	Visual Studio, SQL Server	

22	Implementation of frmCustomerData	16-Nov	13-Nov	11	23-Nov	27-Nov	Visual Studio, SQL Server	
23	Implementation of frmVehicleData	16-Nov	13-Nov	17	29-Nov	29-Nov	Visual Studio, SQL Server	
24	Implementation of frmVehicleTypeData	30-Nov	30-Nov	5	4-Dec	4-Dec	Visual Studio, SQL Server	
25	Implementation of frmServiceData	30-Nov	30-Nov	6	5-Dec	5-Dec	Visual Studio, SQL Server	
26	Implementation of frmPartData	30-Nov	30-Nov	6	5-Dec	6-Dec	Visual Studio, SQL Server	
27	ServiceAppointments (display logic)	7-Dec	6-Dec	24	29-Dec	27-Dec	Visual Studio, SQL Server	
28	ServiceAppointments (CRUD logic)	28-Dec	30-Dec	16	14-Jan	17-Jan	Visual Studio, SQL Server	
29	Implementation of frmReports	18-Jan	15-Jan	7	21-Jan	24-Jan	Visual Studio, SQL Server	

30	Implementation of frmLogs	25-Jan	22-Jan	4	25-Jan	28-Jan	Visual Studio, SQL Server	
31	Implementation of frmHelp(s)	25-Jan	22-Jan	4	25-Jan	27-Jan	Visual Studio	
32	Final clean-up	28-Jan	26-Jan	12	6-Feb	6-Feb	Visual Studio	
Testing								
33	Unit testing	31-Jan	29-Jan	12	9-Feb	9-Feb	Visual Studio, SQL Server	The testing phase was unremarkable, as each task was completed within the allocated time. Only tasks 34-35 had a slight but negligible fluctuation. This was mostly due in part to the effectiveness of the debugging software provided by Visual Studio, helping ensure that all the functionality could be easily tested, and the source of any issues promptly identified and amended. The relatively small test user base also helped ensure that the project was delivered within the allocated time, as little effort was required to locate and interact with them.
34	Integration testing	9-Feb	9-Feb	7	15-Feb	16-Feb	Visual Studio, SQL Server	
35	System testing	16-Feb	15-Feb	7	21-Feb	21-Feb	Visual Studio, SQL Server	
36	Acceptance testing	21-Feb	21-Feb	6	26-Feb	26-Feb	Client, Test Users	
Evaluation								
37	Distribution of questionnaires	23-Feb	23-Feb	4	26-Feb	28-Feb	Test Users	Due to the scale of the project, it was found that the evaluation phase took longer than anticipated, highlighting errors in my estimation and time management skills and forcing me to rush closure. Although this made little impact on the quality and
38	Creation of user guide	26-Feb	26-Feb	7	4-Mar	7-Mar	—	
39	Project evaluation & reflection	8-Mar	5-Mar	9	13-Mar	12-Mar	—	

40	Closure	13-Mar	14-Mar	2	15-Mar	15-Mar	—	delivery date of the final system, it could have been interpreted as unprofessional by the client. Thus, greater care needs to be taken when estimating times in the future, and preventative measures such as buffer periods could be used to mitigate such delays, as, had it not been for the questionnaires and user guide taking less time than expected, the critical path would have been delayed.

As evidenced by the table above, many tasks were started and completed earlier than anticipated. As a significant portion of these were on the critical path (i.e., had no slack, so that their completion time determines the completion time of the project), this ultimately resulted in there being slack (i.e., time by which the tasks could be delayed and still be delivered on time) which impacted the project by providing flexibility - compensating for the tasks that eventually ended up being completed later than anticipated - and improvements on the system's quality - providing some tasks such as testing additional time to ensure that they were carried out more thoroughly.

Regardless, these changes to the project plan made no major impact on the delivery of the project, as it was still closed on the time determined during project initiation with all tasks having been completed. These variations highlight the inherent difficulty in accurately estimating task durations, but ultimately demonstrate the project plan was sufficiently robust to accommodate change without negatively affecting the final outcome. and instead these variations were to be expected due to the difficulty in estimating the duration of tasks ahead of time.

User feedback

As a final step, an interview was conducted with Mr Doherty to assess his overall satisfaction with the system and, more importantly, to identify any areas where improvements could be made.

During the interview, Mr Doherty interacted with the system in real time, providing feedback as he explored its features and functionality. This hands-on evaluation allowed for a more detailed and practical assessment of the system's usability and effectiveness. The following is a summary of his points:

frmLogin

- Mr Doherty expressed satisfaction with the colour scheme, typography, and overall layout of the form, noting that these design choices effectively set the tone for the rest of the system, making it suitable for a business. Similar feedback was provided for other forms, though it has been omitted here for brevity.
- However, he did express annoyance at the form's validation messages, as they only displayed issues one at a time. He stated that he would've preferred errors to be listed as they do in CRUD forms, but due to the insignificance of the issue said this can be ignored.

frmMainMenu

- Mr Doherty was impressed with the expanding sidebar provided in the main menu, saying it gave the impression of sleekness and aided in the optimal navigation of the system.
- Mr Doherty was pleased with the sector access logic implemented, saying it would definitely help tackle the issue of staff members accessing data not applicable to them.
- He did comment that the starting main menu form was rather bland, containing only basic information about the user. To expand on this, he suggested that a clock or company news feed could be implemented to fill the whitespace. However, due to the aesthetic nature of the suggestion, he said this was not essential.

frm_____Data

As has previously been done before, for the sake of brevity all comments regarding data forms will be grouped under one category

- Mr Doherty was satisfied with the current implementation of CRUD processes. Due to the rather cookie-cutter nature of CRUD implementation as a whole, no comments unique to the system were made regarding this.
- He noted that the ubiquitous presence of validation, as well as the list of errors that appears when invalid data is submitted, would ensure the integrity of any information provided to the system, thus preventing the issues regarding consistent bookkeeping and valid data that plagued him initially.
- Specifically for the staff data form, he was confident in the security of the system as a result of the password hashing implemented, further protecting the business data from erroneous or malicious manipulation.
- Specifically for vehicles, he had the critique that the validation for the VIN did not follow the logic required for a VIN (i.e., 17 characters) but noted that the field still served its purpose as a unique physical identifier and thus did not matter too much for the purpose of the system.
- Specifically for vehicle types, he had the critique that the use of a scalar multiplier for determining the time for servicing a vehicle was rather reductionist, and would have preferred if mechanics were able to use their professional judgement. Regardless, the current solution to the problem has been deemed effective for now.
- Mr Doherty was particularly pleased with the presence of the “Reset” buttons, noting that they would save significant amounts of time when editing, deleting or saving multiple records, ensuring that previous form states do not interfere with future records.
- Furthermore, he was satisfied with the help forms provided, deeming them informative enough for use by staff.
- Mr Doherty - much like I did - pointed out the rather small selection of search options for the database. However, with the small scale of the business, he said that this was something that could be overlooked.

frmServiceAppointments

- From the outset, Mr Doherty was pleased with the appointment viewer control, saying that it would assist significantly in the creation and viewing of appointments on the system in a centralised manner, replacing the inconsistent and scattered paper calendars previously implemented.
- On interacting with the appointment viewer for a bit, Mr Doherty was very happy with the drag-and-drop nature of editing appointment times, saying that it even made the process almost “fun.” Furthermore, he greatly appreciated the validation implemented which prevented appointments from intersecting or going out of bounds, something that greatly irritated him with the way appointments were previously arranged.
- He was pleased with the subtle differences when creating appointments in “Insert Mode”, noting that it very explicitly separated the functionality of editing and adding appointments as to ensure that changes were not accidentally made to appointments when adding new ones.
- Similarly, he was very happy with the service and part adding systems, noting that it was easy to use and provided significant visual feedback as to the effect that adding a service and part would have.
- He thought that the schedule saving button - although not essential - was a good feature, ensuring that reordering a day did not necessitate multiple save operations.
- However, he did deem the searching system rather rudimentary, as he thought a message box would provide very transient information, and would have instead opted for a DataGridView as seen in other forms. However, with the small scale of operations so far, he deemed this okay.

frmLogs

- He was very satisfied with the implementation of this form, saying that it would greatly mitigate the issues in accountability caused by the current paper-based system. Notably, he was pleased with the inclusion of usernames attached to each action, ensuring that the person associated could be easily located.
- He was happy with the range of log types and the ability to search by staff or log type, but he did note that it would have been better if logs also stated the exact changes that were made to records, as at the present they are identified solely by record ID and old identifier.
- He did note that he would have preferred to be able to search logs by substring as in the frm____Data forms.

frmReports

- Overall, he was happy with the reports provided by the system. Though he did note that some were rather simple, he appreciated the insights they provided, ensuring that the management of the business would be able to make decisions based on the data provided.

Conclusion

Mr Doherty expressed overall satisfaction with the final system and did not request any modifications or reengineering for the current development cycle. His feedback indicates that the system meets the initial requirements and expectations, validating its completeness and functionality.

Extensive evaluation shows that the core issues identified at the project's outset - and the user requirements derived from them - have been successfully resolved. While minor issues remain, the system is overall effective and ready for deployment at Doherty Cars to support their continued growth.

Appendix

SQL to create tables

Customers

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Customers](
	[CustomerID] [int] IDENTITY(1,1) NOT NULL,
	[Firstname] [nvarchar](50) NOT NULL,
	[Surname] [nvarchar](50) NOT NULL,
	[Email] [nvarchar](50) NULL,
	[PhoneNumber] [varchar](15) NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[CustomerID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

Parts

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Parts](
	[PartID] [int] IDENTITY(1,1) NOT NULL,
	[Name] [nvarchar](50) NOT NULL,
	[BuyPrice] [money] NOT NULL,
	[SellPrice] [money] NOT NULL,
	[Quantity] [int] NOT NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[PartID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

Roles

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Roles](
	[RoleID] [int] IDENTITY(1,1) NOT NULL,
	[Name] [nvarchar](50) NOT NULL,
	[AuthorisationLevel] [int] NOT NULL,
	[Sector] [nvarchar](50) NOT NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[RoleID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

ServiceAppointments

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[ServiceAppointments](
	[AppointmentID] [int] IDENTITY(1,1) NOT NULL,
	[VehicleID] [int] NOT NULL,
	[StaffID] [int] NOT NULL,
	[AppointmentStart] [datetime] NOT NULL,
	[AppointmentEnd] [datetime] NOT NULL,
	[Paid] [bit] NOT NULL,
	[Completed] [bit] NOT NULL,
	[Discounted] [bit] NOT NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[AppointmentID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

ServiceAppointmentsServices

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[ServiceAppointmentsServices](
	[AppointmentID] [int] NOT NULL,
	[ServiceID] [int] NOT NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[AppointmentID] ASC,
	[ServiceID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

ServiceAppointmentsServicesParts

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[ServiceAppointmentsServicesParts](
	[AppointmentID] [int] NOT NULL,
	[ServiceID] [int] NOT NULL,
	[PartID] [int] NOT NULL,
	[Quantity] [int] NOT NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[AppointmentID] ASC,
	[ServiceID] ASC,
	[PartID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

Services

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Services](
	[ServiceID] [int] IDENTITY(1,1) NOT NULL,
	[Name] [nvarchar](50) NOT NULL,
	[Description] [nvarchar](500) NOT NULL,
	[DurationHours] [int] NOT NULL,
	[Price] [decimal](10, 2) NOT NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[ServiceID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

Staff

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Staff](
	[StaffID] [int] IDENTITY(1,1) NOT NULL,
	[Firstname] [nvarchar](50) NOT NULL,
	[Surname] [nvarchar](50) NOT NULL,
	[Username] [nvarchar](50) NOT NULL,
	[Password] [nvarchar](255) NOT NULL,
	[RoleID] [int] NOT NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[StaffID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

Vehicles

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[Vehicles](
	[VehicleID] [int] IDENTITY(1,1) NOT NULL,
	[VehicleTypeID] [int] NOT NULL,
	[CustomerID] [int] NULL,
	[Make] [nvarchar](50) NOT NULL,
	[Model] [nvarchar](50) NOT NULL,
	[Year] [int] NULL,
	[VIN] [nvarchar](50) NOT NULL,
	[LicensePlate] [nvarchar](20) NULL,
	[Condition] [nvarchar](50) NOT NULL,
	[Available] [bit] NOT NULL,
	[Inspected] [bit] NOT NULL,
	[BuyPrice] [decimal](10, 2) NULL,
	[SellPrice] [decimal](10, 2) NULL,
	[StorageLocation] [nvarchar](50) NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[VehicleID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

VehicleTypes

```
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [dbo].[VehicleTypes](
	[VehicleTypeID] [int] IDENTITY(1,1) NOT NULL,
	[Name] [nvarchar](50) NOT NULL,
	[Complexity] [decimal](2, 1) NOT NULL,
	[DateDeleted] [datetime] NULL,
PRIMARY KEY CLUSTERED
(
	[VehicleTypeID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

SQL to populate tables

Customers

```
SET IDENTITY_INSERT [dbo].[Customers] ON
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (1, N'John', N'Yossarian', N'catch22@pianosa.com', N'07221984421', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (2, N'Shane', N'O''Neill', N'igbro@greennet.com', N'07435633115', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (3, N'Lukas', N'Miller', N'lukas.m87@gmail.com', N'07701234567', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (4, N'Raoul', N'Duke', N'fear.n.loathing@vegas.com', N'07802345678', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (5, N'Dr.', N'Gonzo', N'dr.gonzo@samoa.net', N'07903456789', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (6, N'Winston', N'Smith', N'winston.smith1984@bigbrother.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (7, N'Noah', N'Johnson', N'noah.johnson01@mail.com', N'07404567890', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (8, N'Ava', N'Roberts', NULL, N'07305678901', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (9, N'Lucas', N'Evans', N'lucasevans77@outlook.com', N'07506789012', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (10, N'Mia', N'Davies', N'miah.davies@gmail.com', N'07707890123', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (11, N'Liam', N'White', NULL, NULL, NULL)
GO
```

```
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (12, N'Isabella', N'Harris',
N'bella.harris_3@yahoo.com', N'07808901234', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (13, N'James', N'Martin',
N'james.martin88@gmail.com', N'07909012345', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (14, N'Charlotte', N'Thompson',
N'charlotte.thompson@icloud.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (15, N'Benjamin', N'Garcia',
N'b.garcia23@hotmail.com', N'07410123456', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (16, N'Amelia', N'Martinez',
N'amelia.m_92@gmail.com', N'07311234567', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (17, N'Alexander', N'Clark', NULL,
N'07712345678', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (18, N'Ella', N'Rodriguez',
N'ella.r_03@yahoo.co.uk', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (19, N'William', N'Lewis',
N'will.lewis19@gmail.com', N'07813456789', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (20, N'Grace', N'Lee',
N'grace_lee@hotmail.com', N'07914567890', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (21, N'Henry', N'Walker',
N'h.walker7@gmail.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (22, N'Sofia', N'Hall', NULL, N'07415678901',
NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (23, N'Daniel', N'Allen',
N'd.allen84@outlook.com', N'07316789012', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (24, N'Chloe', N'Young',
N'chloe.young99@gmail.com', NULL, NULL)
```

```
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (25, N'Matthew', N'Hernandez',
N'matthewh_77@yahoo.com', N'07717890123', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (26, N'Lily', N'King', N'lily.king4@gmail.com',
N'07818901234', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (27, N'Joseph', N'Wright', NULL,
N'07919012345', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (28, N'Zoe', N'Lopez',
N'zoe_lopez21@hotmail.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (29, N'David', N'Hill',
N'david.hill_88@gmail.com', N'07420123456', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (30, N'Hannah', N'Scott',
N'hannahscott_01@outlook.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (31, N'Samuel', N'Green',
N'sam.green23@gmail.com', N'07321234567', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (32, N'Aria', N'Adams',
N'aria.adams5@yahoo.co.uk', N'07722345678', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (33, N'Jackson', N'Baker', NULL,
N'07823456789', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (34, N'Maya', N'Gonzalez',
N'maya.gonzalez_02@gmail.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (35, N'Logan', N'Nelson',
N'logan.nelson7@hotmail.com', N'07924567890', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (36, N'Scarlett', N'Carter',
N'scarlett.c_3@gmail.com', NULL, NULL)
GO
```

```
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (37, N'Owen', N'Mitchell', NULL, N'07425678901',
NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (38, N'Victoria', N'Perez',
N'victoria.perez12@outlook.com', N'07326789012', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (39, N'Leo', N'Robinson',
N'leo.robinson_9@gmail.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (40, N'Madison', N'Cook',
N'madison.cook77@yahoo.com', N'07727890123', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (41, N'Elijah', N'Morgan', NULL, N'07828901234',
NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (42, N'Layla', N'Bell',
N'layla.bell1996@gmail.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (43, N'Nathan', N'Murphy',
N'nathan.murphy_3@hotmail.com', N'07929012345', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (44, N'Holly', N'Bailey', NULL, N'07430123456',
NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (45, N'Ryan', N'Rivera',
N'ryan.rivera88@gmail.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (46, N'Lucy', N'Cooper',
N'lucy.cooper_2@outlook.com', N'07331234567', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (47, N'Aaron', N'Richardson', NULL,
N'07732345678', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (48, N'Zara', N'Cox',
N'zara.cox1997@gmail.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email],
[PhoneNumber], [DateDeleted]) VALUES (49, N'Isaac', N'Howard',
N'isaac.h_05@hotmail.com', N'07833456789', NULL)
```

```
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (50, N'Molly', N'Ward', NULL, N'07934567890', NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (51, N'Gabriel', N'Torres', N'gabriel.torres88@gmail.com', NULL, NULL)
GO
INSERT [dbo].[Customers] ([CustomerID], [Firstname], [Surname], [Email], [PhoneNumber], [DateDeleted]) VALUES (52, N'Emma', N'Peterson', N'emma.peterson_07@outlook.com', N'07435678901', NULL)
GO
SET IDENTITY_INSERT [dbo].[Customers] OFF
GO
```

Parts

```
SET IDENTITY_INSERT [dbo].[Parts] ON
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (1, N'Engine Oil (5L)', 22.0000, 35.0000, 40, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (2, N'Oil Filter', 4.5000, 10.0000, 60, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (3, N'Air Filter', 7.0000, 15.0000, 45, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (4, N'Cabin Pollen Filter', 6.0000, 14.0000, 35, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (5, N'Spark Plug', 3.0000, 7.5000, 120, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (6, N'Brake Pads (Front Set)', 28.0000, 55.0000, 20, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (7, N'Brake Pads (Rear Set)', 25.0000, 50.0000, 18, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (8, N'Brake Disc', 32.0000, 65.0000, 16, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (9, N'Battery', 55.0000, 95.0000, 14, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity], [DateDeleted]) VALUES (10, N'Wiper Blades (Pair)', 8.0000, 18.0000, 50, NULL)
GO
```

```
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity],
[DateDeleted]) VALUES (11, N'Coolant (5L)', 10.0000, 22.0000, 30, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity],
[DateDeleted]) VALUES (12, N'Timing Belt Kit', 75.0000, 140.0000, 6, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity],
[DateDeleted]) VALUES (13, N'Air Conditioning Gas', 18.0000, 45.0000, 20, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity],
[DateDeleted]) VALUES (14, N'Wheel Alignment Bolt', 2.0000, 6.0000, 80, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity],
[DateDeleted]) VALUES (15, N'Tyre Valve Kit', 1.2000, 4.0000, 100, NULL)
GO
INSERT [dbo].[Parts] ([PartID], [Name], [BuyPrice], [SellPrice], [Quantity],
[DateDeleted]) VALUES (16, N'Blinker Fluid', 0.5000, 9.9900, 3,
CAST(N'2026-01-27T22:45:33.510' AS DateTime))
GO
SET IDENTITY_INSERT [dbo].[Parts] OFF
GO
```

Roles

```
SET IDENTITY_INSERT [dbo].[Roles] ON
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (1, N'Employee', 0, N'None', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (2, N'Administrator', 2, N'Administrative', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (3, N'Sales Advisor', 2, N'Customer Service', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (4, N'Mechanic', 1, N'Technical', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (5, N'Lead Mechanic', 2, N'Technical', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (6, N'Administrative Assistant', 2, N'Administrative', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (7, N'Finance Staff', 1, N'Finance', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (8, N'Manager', 2, N'Administrative', NULL)
GO
```

```
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (9, N'Intern', 0, N'None', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (10, N'External Consultant', 0, N'Finance', NULL)
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (11, N'HR Worker', 0, N'HR', CAST(N'2025-11-12T15:42:00.000' AS DateTime))
GO
INSERT [dbo].[Roles] ([RoleID], [Name], [AuthorisationLevel], [Sector], [DateDeleted])
VALUES (12, N'HR Manager', 1, N'HR', CAST(N'2025-11-06T10:14:00.000' AS DateTime))
GO
SET IDENTITY_INSERT [dbo].[Roles] OFF
GO
```

ServiceAppointments

```
SET IDENTITY_INSERT [dbo].[ServiceAppointments] ON
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (17, 4, 2, CAST(N'2026-02-02T09:00:00.000' AS DateTime),
CAST(N'2026-02-02T11:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (18, 26, 3, CAST(N'2026-02-02T11:00:00.000' AS DateTime),
CAST(N'2026-02-02T16:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (19, 27, 6, CAST(N'2026-02-03T09:00:00.000' AS DateTime),
CAST(N'2026-02-03T12:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (20, 28, 2, CAST(N'2026-02-03T12:00:00.000' AS DateTime),
CAST(N'2026-02-03T17:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (21, 29, 3, CAST(N'2026-02-04T09:00:00.000' AS DateTime),
CAST(N'2026-02-04T11:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (22, 30, 6, CAST(N'2026-02-04T11:00:00.000' AS DateTime),
CAST(N'2026-02-04T17:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
```

[DateDeleted]) VALUES (23, 31, 2, CAST(N'2026-02-06T09:00:00.000' AS DateTime), CAST(N'2026-02-06T17:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (24, 32, 3, CAST(N'2026-02-06T09:00:00.000' AS DateTime), CAST(N'2026-02-06T11:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (25, 33, 6, CAST(N'2026-02-09T09:00:00.000' AS DateTime), CAST(N'2026-02-09T15:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (26, 36, 2, CAST(N'2026-02-09T15:00:00.000' AS DateTime), CAST(N'2026-02-09T17:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (27, 44, 3, CAST(N'2026-02-11T09:00:00.000' AS DateTime), CAST(N'2026-02-11T12:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (28, 49, 6, CAST(N'2026-02-11T12:00:00.000' AS DateTime), CAST(N'2026-02-11T17:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (30, 49, 6, CAST(N'2026-02-11T09:00:00.000' AS DateTime), CAST(N'2026-02-11T12:00:00.000' AS DateTime), 0, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (31, 4, 2, CAST(N'2025-09-03T09:00:00.000' AS DateTime), CAST(N'2025-09-03T12:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (32, 26, 3, CAST(N'2025-10-07T08:00:00.000' AS DateTime), CAST(N'2025-10-07T12:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (33, 27, 6, CAST(N'2025-09-15T10:00:00.000' AS DateTime), CAST(N'2025-09-15T12:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],

[DateDeleted]) VALUES (34, 28, 2, CAST(N'2025-11-02T08:30:00.000' AS DateTime),
CAST(N'2025-11-02T15:30:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (35, 29, 3, CAST(N'2025-09-25T13:00:00.000' AS DateTime),
CAST(N'2025-09-25T15:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (36, 30, 6, CAST(N'2025-12-05T09:00:00.000' AS DateTime),
CAST(N'2025-12-05T12:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (37, 31, 2, CAST(N'2026-01-10T08:00:00.000' AS DateTime),
CAST(N'2026-01-10T16:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (38, 32, 3, CAST(N'2025-10-20T11:00:00.000' AS DateTime),
CAST(N'2025-10-20T13:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (39, 33, 6, CAST(N'2025-11-18T09:00:00.000' AS DateTime),
CAST(N'2025-11-18T15:00:00.000' AS DateTime), 0, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (40, 36, 2, CAST(N'2025-09-09T14:00:00.000' AS DateTime),
CAST(N'2025-09-09T17:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (41, 44, 3, CAST(N'2025-12-12T08:00:00.000' AS DateTime),
CAST(N'2025-12-12T12:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (42, 47, 6, CAST(N'2025-11-25T10:00:00.000' AS DateTime),
CAST(N'2025-11-25T12:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],
[DateDeleted]) VALUES (43, 49, 2, CAST(N'2026-02-01T09:00:00.000' AS DateTime),
CAST(N'2026-02-01T12:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID],
[AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted],

[DateDeleted]) VALUES (44, 50, 3, CAST(N'2025-10-30T09:00:00.000' AS DateTime), CAST(N'2025-10-30T13:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (45, 28, 6, CAST(N'2026-01-20T08:00:00.000' AS DateTime), CAST(N'2026-01-20T15:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (46, 29, 2, CAST(N'2025-09-29T13:00:00.000' AS DateTime), CAST(N'2025-09-29T15:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (47, 30, 3, CAST(N'2025-12-03T09:00:00.000' AS DateTime), CAST(N'2025-12-03T13:00:00.000' AS DateTime), 0, 0, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (48, 31, 6, CAST(N'2025-11-06T10:00:00.000' AS DateTime), CAST(N'2025-11-06T16:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (49, 32, 2, CAST(N'2025-09-18T11:00:00.000' AS DateTime), CAST(N'2025-09-18T16:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
INSERT [dbo].[ServiceAppointments] ([AppointmentID], [VehicleID], [StaffID], [AppointmentStart], [AppointmentEnd], [Paid], [Completed], [Discounted], [DateDeleted]) VALUES (50, 33, 3, CAST(N'2026-02-10T09:00:00.000' AS DateTime), CAST(N'2026-02-10T12:00:00.000' AS DateTime), 1, 1, 0, NULL)
GO
SET IDENTITY_INSERT [dbo].[ServiceAppointments] OFF
GO

ServiceAppointmentsServices

```
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (17, 3, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (18, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (19, 5, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (19, 6, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (20, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (21, 5, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (22, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (23, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (23, 7, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (24, 3, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (25, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (26, 3, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (27, 5, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (27, 6, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (28, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID], [DateDeleted]) VALUES (31, 3, NULL)
GO
```

```
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (31, 5, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (32, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (33, 3, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (34, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (34, 7, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (35, 5, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (35, 6, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (36, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (37, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (37, 7, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (38, 3, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (39, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (40, 3, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (40, 5, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (41, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (42, 3, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (43, 4, NULL)
GO
```

```
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (44, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (45, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (45, 6, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (46, 3, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (47, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (48, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (49, 4, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (50, 5, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID],
[DateDeleted]) VALUES (50, 7, NULL)
GO
```

ServiceAppointmentsServicesParts

```
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (17, 3, 1, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (17, 3, 2, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (17, 3, 3, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (18, 4, 1, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (18, 4, 2, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (18, 4, 3, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (18, 4, 4, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (18, 4, 10, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (18, 4, 11, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (19, 5, 10, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (19, 6, 13, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (20, 4, 1, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (20, 4, 2, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (20, 4, 3, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (20, 4, 4, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (21, 5, 15, 4, NULL)
GO
```

```
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (22, 4, 1, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (22, 4, 2, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (22, 4, 5, 6, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (22, 4, 10, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (23, 4, 1, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (23, 4, 2, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (23, 4, 3, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (23, 7, 14, 2, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (24, 3, 1, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (24, 3, 2, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (25, 4, 1, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (25, 4, 2, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (25, 4, 6, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (25, 4, 7, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (26, 3, 1, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (27, 5, 10, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (27, 6, 13, 1, NULL)
GO
```

```
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (28, 4, 9, 1, NULL)
GO
INSERT [dbo].[ServiceAppointmentsServicesParts] ([AppointmentID], [ServiceID], [PartID], [Quantity], [DateDeleted]) VALUES (28, 4, 10, 1, NULL)
GO
```

Services

```
SET IDENTITY_INSERT [dbo].[Services] ON
GO
INSERT [dbo].[Services] ([ServiceID], [Name], [Description], [DurationHours], [Price], [DateDeleted]) VALUES (2, N'Blinker Fluid Change', N'Change the fluid in the blinkers of the car. May need spirit level bubbles.', 1, CAST(20.00 AS Decimal(10, 2)), CAST(N'2026-01-27T22:46:11.533' AS DateTime))
GO
INSERT [dbo].[Services] ([ServiceID], [Name], [Description], [DurationHours], [Price], [DateDeleted]) VALUES (3, N'Minor Service', N'Basic maintenance including oil change, filter checks, fluid top-up, and safety inspection.', 2, CAST(120.00 AS Decimal(10, 2)), NULL)
GO
INSERT [dbo].[Services] ([ServiceID], [Name], [Description], [DurationHours], [Price], [DateDeleted]) VALUES (4, N'Full Service', N'Comprehensive maintenance including oil and filter change, full inspection of engine, brakes, suspension, and fluids.', 4, CAST(250.00 AS Decimal(10, 2)), NULL)
GO
INSERT [dbo].[Services] ([ServiceID], [Name], [Description], [DurationHours], [Price], [DateDeleted]) VALUES (5, N'MOT', N'Annual MOT test to ensure the vehicle meets UK safety and environmental standards.', 1, CAST(45.00 AS Decimal(10, 2)), NULL)
GO
INSERT [dbo].[Services] ([ServiceID], [Name], [Description], [DurationHours], [Price], [DateDeleted]) VALUES (6, N'Air Conditioning Check', N'Inspection of air conditioning system, refrigerant levels, and overall performance.', 1, CAST(30.00 AS Decimal(10, 2)), NULL)
GO
INSERT [dbo].[Services] ([ServiceID], [Name], [Description], [DurationHours], [Price], [DateDeleted]) VALUES (7, N'Wheel Alignment', N'Adjustment of vehicle suspension to ensure wheels are aligned to manufacturer specifications.', 1, CAST(35.00 AS Decimal(10, 2)), NULL)
GO
SET IDENTITY_INSERT [dbo].[Services] OFF
GO
```

Staff

```
SET IDENTITY_INSERT [dbo].[Staff] ON
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (1, N'Patrick', N'Doherty', N'patdoh', N'wwmgrLMQlq3GDymT1xDuBjL1ExxO+we/FdbH1X1hMh33eLKs', 2, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (2, N'Amelia', N'Davies', N'amelia_d', N'dFR4JgPdTl/5u5+NpjrScqvHfJUbJsjxzMvsMnwG4xCfxdrO', 4, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (3, N'Ethan', N'Harrison', N'ethanh1', N'Cf6exrQtOCjvLbtx7w/0WewP5fMhRE01wtOQLjEcvrjsspgN', 4, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (6, N'Sophia', N'Diffin', N'sophiac', N'Q4DcPY5KGRfN4Ra6wqGhPbht/0pgP4MiGLdfmEtJipQu/bDv', 5, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (7, N'Harry', N'Turner', N'harryt', N'ZrD7a/kUNHysHTrRWlhyjPr4/EqmOy+4AoRXovMbtcJcA/tx', 3, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (8, N'Charlotte', N'Foster', N'charfor', N'yxfxvLan7fJxtC0ddamUDDrC7CR7I3c4GK7PPHwOnNuYTj9P', 3, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (13, N'Ella', N'Reid', N'ellar', N'OjBk+2Ex87lGejh0RqVjTR1HlxDiu/+6Jm943zkjGyyF0ynb', 7, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (14, N'James', N'Watson', N'jamesw', N'6T5K5Ax7/Q2OlGiuQYw6Vv4A3C/a5oR/EWKcYKGqQ4wPEPqT', 6, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (15, N'Noah', N'Wright', N'noahw', N'B8PMTtmHxm1JerQu+35FVh4NyJJ4o58GPAQY2nbucIx25YQ+', 8, NULL)
GO
INSERT [dbo].[Staff] ([StaffID], [Firstname], [Surname], [Username], [Password], [RoleID], [DateDeleted]) VALUES (16, N'Isla', N'Bennett', N'islab', N'w92fsSH6VdDz+3DNqCK1J7oB8YCyZhb4QWFjqfhQvsjuSDSb', 9, CAST(N'2025-12-11T09:54:00.000' AS DateTime))
GO
SET IDENTITY_INSERT [dbo].[Staff] OFF
GO
```

Vehicles

```
SET IDENTITY_INSERT [dbo].[Vehicles] ON
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (4, 1, 2, N'Honda', N'Civic', 2006, N'2C3CCAAG2EH118886', N'VU02GFI', N'Fair', 0, 1, CAST(1200.00 AS Decimal(10, 2)), CAST(2200.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (26, 3, NULL, N'Ford', N'Focus', 2021, N'WF0DXXGCDG7000052', NULL, N'Good', 1, 0, CAST(12000.00 AS Decimal(10, 2)), CAST(14500.00 AS Decimal(10, 2)), N'Internal Storage', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (27, 1, 6, N'Nissan', N'Altima', 2017, N'1N4AL3AP5HC000006', N'ALT17NIS', N'Fair', 0, 0, CAST(12000.00 AS Decimal(10, 2)), CAST(14500.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (28, 5, NULL, N'Toyota', N'RAV4', 2020, N'JTMBFREV7KJ000053', NULL, N'Excellent', 1, 1, CAST(22000.00 AS Decimal(10, 2)), CAST(26000.00 AS Decimal(10, 2)), N'Main Forecourt', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (29, 1, 1, N'Toyota', N'Camry', 2018, N'JT4RN12P7J5000001', N'YX18ABC', N'Good', 0, 1, CAST(15000.00 AS Decimal(10, 2)), CAST(18000.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (30, 12, 10, N'Porsche', N'911 Carrera', 2021, N'WP0AA2A94MS000010', N'MI21PCA', N'Excellent', 0, 1, CAST(75000.00 AS Decimal(10, 2)), CAST(85000.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (31, 16, 5, N'Ford', N'F-150', 2019, N'1FTFW1E58KFA00005', N'GONZO1', N'Good', 0, 1, CAST(28000.00 AS Decimal(10, 2)), CAST(34000.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (32, 1, NULL, N'Toyota', N'Camry', 2020, N'JT4RN12P7J5000069', NULL, N'Good', 1, 0, CAST(16000.00 AS Decimal(10, 2)), CAST(19000.00 AS Decimal(10, 2)), N'Main Forecourt', NULL)
```

```
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (33, 5, 4, N'Chevrolet', N'Suburban', 2020, N'3GNEC16T9XG100004', N'DUK23LV', N'Excellent', 0, 1, CAST(35000.00 AS Decimal(10, 2)), CAST(42000.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (34, 3, 2, N'Ford', N'Focus', 2016, N'WF0DXXGCDG7000002', N'AB16DEF', N'Fair', 0, 1, CAST(9000.00 AS Decimal(10, 2)), CAST(11000.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (35, 1, NULL, N'Toyota', N'Corolla', 2022, N'JTDBL40E399000051', NULL, N'New', 1, 1, CAST(15000.00 AS Decimal(10, 2)), CAST(18000.00 AS Decimal(10, 2)), N'Main Forecourt', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (36, 2, 12, N'Honda', N'PCX125', 2020, N'JH2KF3500LK000012', N'MOP20HON', N'Excellent', 0, 1, CAST(2500.00 AS Decimal(10, 2)), CAST(3200.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (37, 12, NULL, N'McLaren', N'720S', 2021, N'SBM14DCA0MW000056', NULL, N'Excellent', 1, 1, CAST(280000.00 AS Decimal(10, 2)), CAST(320000.00 AS Decimal(10, 2)), N'Service Garage', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (38, 1, NULL, N'Hyundai', N'Elantra', 2019, N'KMHD84LF5KU000057', NULL, N'Good', 1, 0, CAST(14000.00 AS Decimal(10, 2)), CAST(17000.00 AS Decimal(10, 2)), N'Main Forecourt', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (39, 5, NULL, N'Jeep', N'Grand Cherokee', 2020, N'1C4RJFBG9LC000061', NULL, N'Excellent', 1, 0, CAST(36000.00 AS Decimal(10, 2)), CAST(42000.00 AS Decimal(10, 2)), N'Main Forecourt', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (40, 1, 3, N'Honda', N'Accord', 2019, N'1HGCV1F14KA000062', N'AC19HON2', N'Good', 0, 1, CAST(16000.00 AS Decimal(10, 2)), CAST(19000.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice],
```

[SellPrice], [StorageLocation], [DateDeleted]) VALUES (41, 3, 7, N'Volkswagen', N'Golf', 2018, N'WVWZZZ1KZJW000063', N'NJ18JKL2', N'Good', 0, 1, CAST(13000.00 AS Decimal(10, 2)), CAST(15500.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (42, 6, NULL, N'Ford', N'Ranger', 2019, N'1FTEX1C86KFB000054', NULL, N'Good', 1, 0, CAST(28000.00 AS Decimal(10, 2)), CAST(33000.00 AS Decimal(10, 2)), N'Internal Storage', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (43, 7, NULL, N'Fiat', N'Ducato', 2018, N'ZFA25000000000055', NULL, N'Fair', 1, 1, CAST(20000.00 AS Decimal(10, 2)), CAST(24000.00 AS Decimal(10, 2)), N'Service Garage', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (44, 9, NULL, N'Mazda', N'MX-5', 2020, N'JMZND18ZL01000059', NULL, N'Excellent', 1, 1, CAST(20000.00 AS Decimal(10, 2)), CAST(25000.00 AS Decimal(10, 2)), N'Service Garage', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (45, 1, NULL, N'Audi', N'A4', 2017, N'WAUZZZ8K1HA000060', NULL, N'Good', 1, 1, CAST(18000.00 AS Decimal(10, 2)), CAST(21000.00 AS Decimal(10, 2)), N'Main Forecourt', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (46, 5, 9, N'Toyota', N'RAV4', 2019, N'JTMBFREV7KJ000064', N'LK19MNO2', N'Good', 0, 1, CAST(22000.00 AS Decimal(10, 2)), CAST(26000.00 AS Decimal(10, 2)), NULL, NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (47, 2, NULL, N'Honda', N'PCX125', 2021, N'JH2KF3500LK000066', NULL, N'Excellent', 1, 1, CAST(2500.00 AS Decimal(10, 2)), CAST(3200.00 AS Decimal(10, 2)), N'Internal Storage', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (48, 13, NULL, N'Yamaha', N'MT-07', 2019, N'JYARM07F0KA000067', NULL, N'Good', 1, 0, CAST(6000.00 AS Decimal(10, 2)), CAST(7500.00 AS Decimal(10, 2)), N'Internal Storage', NULL)
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (49, 14, NULL, N'Tesla', N'Model 3', 2021, N'5YJ3E1EA7MF000068', NULL, N'Excellent', 1, 1, CAST(35000.00 AS Decimal(10, 2)), CAST(42000.00 AS Decimal(10, 2)), N'Main Forecourt', NULL)

```
GO
INSERT [dbo].[Vehicles] ([VehicleID], [VehicleTypeID], [CustomerID], [Make], [Model], [Year], [VIN], [LicensePlate], [Condition], [Available], [Inspected], [BuyPrice], [SellPrice], [StorageLocation], [DateDeleted]) VALUES (50, 5, NULL, N'Nissan', N'Murano', 2020, N'JN8AZ2NE9K9000070', NULL, N'Excellent', 1, 1, CAST(27000.00 AS Decimal(10, 2)), CAST(32000.00 AS Decimal(10, 2)), N'Main Forecourt', NULL)
GO
SET IDENTITY_INSERT [dbo].[Vehicles] OFF
GO
```

VehicleTypes

```
SET IDENTITY_INSERT [dbo].[VehicleTypes] ON
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (1, N'Sedan', CAST(1.0 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (2, N'Hatchback', CAST(0.9 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (3, N'Coupe', CAST(1.0 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (4, N'SUV', CAST(1.2 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (5, N'Pickup Truck', CAST(1.3 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (6, N'Van', CAST(1.3 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (7, N'Minivan', CAST(1.2 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (8, N'Convertible', CAST(1.1 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (9, N'Wagon', CAST(1.1 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (10, N'Luxury Sedan', CAST(1.1 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (11, N'Sports Car', CAST(1.2 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted]) VALUES (12, N'Motorbike', CAST(0.7 AS Decimal(2, 1)), NULL)
GO
```

```
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted])
VALUES (13, N'Electric Vehicle', CAST(1.1 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted])
VALUES (14, N'Hybrid', CAST(1.1 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted])
VALUES (15, N'Truck', CAST(1.4 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted])
VALUES (16, N'Bus', CAST(1.5 AS Decimal(2, 1)), NULL)
GO
INSERT [dbo].[VehicleTypes] ([VehicleTypeID], [Name], [Complexity], [DateDeleted])
VALUES (17, N'Moped', CAST(0.5 AS Decimal(2, 1)), NULL)
GO
SET IDENTITY_INSERT [dbo].[VehicleTypes] OFF
GO
SET ANSI_PADDING ON
GO
```

SQL for additional constraints

Staff

```
ALTER TABLE [dbo].[Staff] ADD UNIQUE NONCLUSTERED
(
	[Username] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF, IGNORE_DUP_KEY = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
SET ANSI_PADDING ON
GO
```

Vehicles

```
ALTER TABLE [dbo].[Vehicles] ADD UNIQUE NONCLUSTERED
(
	[VIN] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF, IGNORE_DUP_KEY = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
```

Customers

```
ALTER TABLE [dbo].[Customers] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
```

Parts

```
ALTER TABLE [dbo].[Parts] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
```

Roles

```
ALTER TABLE [dbo].[Roles] ADD  DEFAULT ((0)) FOR [AuthorisationLevel]
GO
ALTER TABLE [dbo].[Roles] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
```

ServiceAppointments

```
ALTER TABLE [dbo].[ServiceAppointments] ADD  DEFAULT ((0)) FOR [Paid]
GO
ALTER TABLE [dbo].[ServiceAppointments] ADD  DEFAULT ((0)) FOR [Completed]
GO
ALTER TABLE [dbo].[ServiceAppointments] ADD  DEFAULT ((0)) FOR [Discounted]
GO
ALTER TABLE [dbo].[ServiceAppointments] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
ALTER TABLE [dbo].[ServiceAppointments]  WITH CHECK ADD  CONSTRAINT
[FK_ServiceAppointments_Staff] FOREIGN KEY([StaffID])
REFERENCES [dbo].[Staff] ([StaffID])
GO
ALTER TABLE [dbo].[ServiceAppointments] CHECK CONSTRAINT
[FK_ServiceAppointments_Staff]
GO
ALTER TABLE [dbo].[ServiceAppointments]  WITH CHECK ADD  CONSTRAINT
[FK_ServiceAppointments_Vehicle] FOREIGN KEY([VehicleID])
REFERENCES [dbo].[Vehicles] ([VehicleID])
GO
ALTER TABLE [dbo].[ServiceAppointments] CHECK CONSTRAINT
[FK_ServiceAppointments_Vehicle]
GO
```

ServiceAppointmentsServices

```
ALTER TABLE [dbo].[ServiceAppointmentsServices] ADD  DEFAULT (NULL) FOR
[DateDeleted]
GO
ALTER TABLE [dbo].[ServiceAppointmentsServices]  WITH CHECK ADD  CONSTRAINT
[FK_ServiceAppointmentsServices_AppointmentID] FOREIGN KEY([AppointmentID])
REFERENCES [dbo].[ServiceAppointments] ([AppointmentID])
GO
ALTER TABLE [dbo].[ServiceAppointmentsServices] CHECK CONSTRAINT
[FK_ServiceAppointmentsServices_AppointmentID]
GO
ALTER TABLE [dbo].[ServiceAppointmentsServices]  WITH CHECK ADD  CONSTRAINT
[FK_ServiceAppointmentsServices_ServiceID] FOREIGN KEY([ServiceID])
REFERENCES [dbo].[Services] ([ServiceID])
GO
ALTER TABLE [dbo].[ServiceAppointmentsServices] CHECK CONSTRAINT
[FK_ServiceAppointmentsServices_ServiceID]
GO
```

ServiceAppointmentsParts

```
ALTER TABLE [dbo].[ServiceAppointmentsServicesParts] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
ALTER TABLE [dbo].[ServiceAppointmentsServicesParts]  WITH CHECK ADD CONSTRAINT [FK_PartID] FOREIGN KEY([PartID])
REFERENCES [dbo].[Parts] ([PartID])
GO
ALTER TABLE [dbo].[ServiceAppointmentsServicesParts] CHECK CONSTRAINT [FK_PartID]
GO
ALTER TABLE [dbo].[ServiceAppointmentsServicesParts]  WITH CHECK ADD CONSTRAINT [FK_ServiceAppointmentsServices_AppointmentID_ServiceID] FOREIGN KEY([AppointmentID], [ServiceID])
REFERENCES [dbo].[ServiceAppointmentsServices] ([AppointmentID], [ServiceID])
GO
ALTER TABLE [dbo].[ServiceAppointmentsServicesParts] CHECK CONSTRAINT [FK_ServiceAppointmentsServices_AppointmentID_ServiceID]
GO
```

Services

```
ALTER TABLE [dbo].[Services] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
```

Staff

```
ALTER TABLE [dbo].[Staff] ADD  DEFAULT ((1)) FOR [RoleID]
GO
ALTER TABLE [dbo].[Staff] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
ALTER TABLE [dbo].[Staff]  WITH CHECK ADD  CONSTRAINT [FK_Staff_RoleID] FOREIGN KEY([RoleID])
REFERENCES [dbo].[Roles] ([RoleID])
GO
ALTER TABLE [dbo].[Staff] CHECK CONSTRAINT [FK_Staff_RoleID]
GO
```

Vehicles

```
ALTER TABLE [dbo].[Vehicles] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
ALTER TABLE [dbo].[Vehicles]  WITH CHECK ADD  CONSTRAINT
[FK_Vehicles_Customer] FOREIGN KEY([CustomerID])
REFERENCES [dbo].[Customers] ([CustomerID])
GO
ALTER TABLE [dbo].[Vehicles] CHECK CONSTRAINT [FK_Vehicles_Customer]
GO
ALTER TABLE [dbo].[Vehicles]  WITH CHECK ADD  CONSTRAINT
[FK_Vehicles_VehicleType] FOREIGN KEY([VehicleTypeID])
REFERENCES [dbo].[VehicleTypes] ([VehicleTypeID])
GO
ALTER TABLE [dbo].[Vehicles] CHECK CONSTRAINT [FK_Vehicles_VehicleType]
GO
```

VehicleTypes

```
ALTER TABLE [dbo].[VehicleTypes] ADD  DEFAULT ((1)) FOR [Complexity]
GO
ALTER TABLE [dbo].[VehicleTypes] ADD  DEFAULT (NULL) FOR [DateDeleted]
GO
```